Copyright Registration – Compiled Submission

Huru SMS

Multichannel SMS and Web Citizen Information and Guidance System for Tanzania

Version 1.0 · Compiled deposit of all supporting documents

Applicant and owner of the economic rights: HURU DIGITAL LIMITED (reg. no. 151790429), P.O. Box 75290, Dar es Salaam

Author: Ibrahim Frank Ndunguru · **Creator and developer:** Michael Simon Kilunga

Prepared for: Copyright Society of Tanzania (COSOTA)

Date of compilation: 20 October 2026 · **Total pages:** 248

This document is a compilation. It brings together, in one continuously numbered document of 248 pages, all the documents filed in support of the application of HURU DIGITAL LIMITED to the Copyright Society of Tanzania for the registration of the work **Huru SMS, version 1.0**. It opens with the signed Declaration of Originality, followed by the complete source code listing, the system design and architecture, the innovation documentation and the three executed contribution contracts. The table on the next page states the exact pages at which each constituent document is found.

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Pages are numbered continuously from 1 to 248 in the footer of every page. A reader who wishes to see a particular document may turn directly to the pages shown above; the same parts are also provided as bookmarks in the PDF navigation pane. The three contracts in Parts E to G assign the economic rights in the contributors' respective contributions to HURU DIGITAL LIMITED.

IN THE UNITED REPUBLIC OF TANZANIA

IN THE MATTER OF the Copyright and Neighbouring Rights Act, Cap. 218 of the Laws of Tanzania

AND IN THE MATTER OF an application to the Copyright Society of Tanzania (COSOTA) for the registration of the work described below

DECLARATION OF ORIGINALITY

Kiapo cha Uasili wa Kazi

in respect of the work **Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0** (the "**Work**")

WE, the undersigned, being the natural persons who made the Work:

1. **IBRAHIM FRANK NDUNGURU**, an adult of **Tanzanian** nationality, holder of National Identification Number (NIN) **19911207-41114-00002-25**, of P.O. Box **75290**, Dar es Salaam, United Republic of Tanzania, the **Author** of the Work and Managing Director of HURU DIGITAL LIMITED; and

2. **MICHAEL SIMON KILUNGA**, an adult of **Tanzanian** nationality, holder of National Identification Number (NIN) **19991208-37305-00002-27**, of P.O. Box **75290**, Dar es Salaam, United Republic of Tanzania, the **Creator and Developer** of the Work,

do hereby solemnly and sincerely declare as follows:

1. **Our identities.** We are the persons named above. The facts stated in this declaration are within our own personal knowledge and are true to the best of our knowledge and belief.

2. **The Work.** The Work is a computer software system known as **Huru SMS**, a multichannel SMS and web citizen information and guidance system for Tanzania, version 1.0, consisting of its concept and written content, its visual and interaction designs, and its computer program comprising 112 original source files of 7,991 lines of code written in PHP (on the Laravel framework), HTML, CSS, JavaScript and JSON, together with its system design and architecture documentation and its innovation documentation.

3. **Authorship and creation.** We personally conceived, designed, authored and developed the Work: the first declarant originated the concept, the name "Huru SMS", the service design and the written content of the Work; the second declarant created the screen designs, visual identity and conversation designs of the Work and wrote the whole of its source code. The Work was made in Dar es Salaam, United Republic of Tanzania, and was completed in October 2026.

4. **Originality.** The Work is our own original creation and we are its true and only makers. It was independently conceived and independently written by us. Neither the Work nor any part of it — including its source code, its designs, its texts and its documentation — has been copied, reproduced, translated or adapted from the work, system or source code of any other person, whether published or unpublished, and the Work does not imitate any existing system.

5. **Third-party components.** The Work makes use of third-party open-source software libraries (including the Laravel framework and the packages referenced in the files **composer.json** and **package.json** of the Work) strictly under and in accordance with their respective open-source licences. Those libraries are the property of their respective authors; they are not reproduced in the deposited source code listing and no claim of authorship or ownership is made over them. Every part of the Work claimed in this application was written by us.

6. **No adverse claims.** No person has claimed, and we are not aware of any ground on which any person could claim, that the Work or any part of it infringes the copyright or any other right of any person, and no litigation, dispute or demand concerning the Work is pending or threatened.

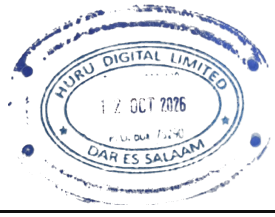
7. **Ownership.** By three written contracts each made on 12 October 2026 (an Author's Contract, a Creator's Contract and a Developer's Contract, executed copies of which form Parts E to G of this compilation), we assigned the economic rights in our respective contributions to the Work to **HURU DIGITAL LIMITED**, a company incorporated in Tanzania under the Companies Act, Cap. 212, with registration number 151790429, of Mlalakuwa Street, Makongo Ward, Kinondoni District, P.O. Box 75290, Dar es Salaam, which is accordingly the owner of the economic rights in the Work.

8. **Purpose.** We make this declaration in support of the application to the Copyright Society of Tanzania (COSOTA) for the registration of the Work, conscious that COSOTA will rely upon it, and we accept full responsibility under the law for the truth of everything stated herein.

VERIFICATION AND EXECUTION

We, **IBRAHIM FRANK NDUNGURU** and **MICHAEL SIMON KILUNGA**, verify that all the facts stated in paragraphs 1 to 8 above are true to the best of our own knowledge and belief, and in witness thereof we have signed this declaration at **Dar es Salaam** on the **20th day of October 2026**, the company seal of HURU DIGITAL LIMITED being affixed.

DECLARED and SIGNED by the Author



Signature and company seal

IBRAHIM FRANK NDUNGURU

Full name

NIN 19911207-41114-00002-25 · Author · Managing Director, HURU DIGITAL LIMITED · 20 October 2026

NIN, capacity and date

DECLARED and SIGNED by the Creator and Developer



Signature

MICHAEL SIMON KILUNGA

Full name

NIN 19991208-37305-00002-27 · Creator and Developer · 20 October 2026

NIN, capacity and date

Huru SMS – DECLARATION OF ORIGINALITY · HURU DIGITAL LIMITED · Made at Dar es Salaam on 20 October 2026 in support of the application for registration of the work "Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0" at the Copyright Society of Tanzania (COSOTA).

Source Code Deposit for Copyright Registration

Huru SMS

Multichannel SMS and Web Citizen Information and Guidance System for Tanzania
Complete source code listing, version 1.0

Owner and developer: Huru Digital Co. Ltd., Tanzania

Files listed: 112 original source files, 7991 lines in total

Programming languages: PHP (Laravel framework), HTML, CSS, JavaScript, JSON

Date of listing: 07 October 2026

Prepared for: Copyright Society of Tanzania (COSOTA)

This listing contains every file written by Huru Digital Co. Ltd. for the Huru SMS application. Third-party open-source libraries used by the application (the `vendor` and `node_modules` folders) are not reproduced here; they are the property of their respective authors and are only referenced in `composer.json` and `package.json`. Secret keys and user data are excluded.

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1 .agents/AGENTS.md 21 lines

```
# Project Rules - Huru SMS

Huru SMS is a generic, Tanzania-aware question answering service for every citizen (not only students, not only legal topics). It runs over SMS (shortcode `15054`, keyword `HURU`) and a web chat.

## Product
- **Generic by design**: any topic is valid. Topic detection (`App\Services\TopicClassifier`) only chooses the persona, knowledge and contacts; it never refuses a topic.
- **Tanzanian context first**: assume TZS, East Africa Time, Tanzanian law and institutions, Swahili as the default language. Country facts live in `App\Services\TanzaniaContext`; national contacts and knowledge live in the database and are edited in the admin panel, never hard-coded in prompts.
- **Honesty over confidence**: the engine must say when something should be confirmed with the responsible office and must never invent phone numbers, fees or officials. Keep this in every persona and constraint change.

## User interface and aesthetics
- **No "AI"/"Bot" wording** in user-facing text, page titles or labels. Present Huru SMS as a direct guidance service. The admin panel may say "engine".
- **Government palette**: Forest Green (`#166534` / `#15803d`), Royal Blue (`#1e40af` / `#1d4ed8`), white/off-white (`#ffffff`, `#f8fafc`). No brown, amber, gradients or glow effects on public pages.
- **Light, high-readability themes** for all citizen-facing pages. Plain Inter font.
- **SMS visibility**: the keyword `HURU` and shortcode `15054` must stay clearly visible on the home page and the web chat login screen, and must survive any layout refactor. Values come from settings (`sms_keyword`, `sms_shortcode`).
- Swahili is the primary language of public pages; English appears as a secondary line where it helps.

## Engineering
- All channel logic goes through `App\Services\ConversationService`; do not duplicate moderation or logging in controllers or jobs.
- New settings are declared in `App\Support\Settings::schema()` only; the admin form, validation and defaults derive from it.
- Keep the inbound SMS webhook behind `AT_WEBHOOK_SECRET` and keep TLS verification on for all outbound HTTP.
- Tests use SQLite in memory (`phpunit.xml`); keep migrations driver-agnostic.
```

2 .env.example

66 lines

```
APP_NAME="Huru SMS"
APP_ENV=local
APP_KEY=
APP_DEBUG=true
APP_URL=http://localhost
APP_TIMEZONE=Africa/Dar_es_Salaam

APP_LOCALE=sw
APP_FALLBACK_LOCALE=en
APP_FAKER_LOCALE=en_US

APP_MAINTENANCE_DRIVER=file

BCRYPT_ROUNDS=12

LOG_CHANNEL=stack
LOG_STACK=single
LOG_DEPRECATIONS_CHANNEL=null
LOG_LEVEL=debug

# MySQL for real deployments; sqlite works for a quick start.
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=huru
DB_USERNAME=root
DB_PASSWORD=

SESSION_DRIVER=database
SESSION_LIFETIME=120
SESSION_ENCRYPT=false
SESSION_PATH=/
SESSION_DOMAIN=null

BROADCAST_CONNECTION=log
FILESYSTEM_DISK=local
QUEUE_CONNECTION=database
CACHE_STORE=database

MAIL_MAILER=log
MAIL_HOST=127.0.0.1
MAIL_PORT=2525
MAIL_USERNAME=null
MAIL_PASSWORD=null
MAIL_FROM_ADDRESS="hello@example.com"
MAIL_FROM_NAME="${APP_NAME}"

VITE_APP_NAME="${APP_NAME}"

# ---- Admin bootstrap (used by `php artisan db:seed`) ----
ADMIN_NAME="Huru SMS Admin"
ADMIN_EMAIL=admin@huru.local
ADMIN_PASSWORD=

# ---- Google Gemini ----
GEMINI_API_KEY=
GEMINI_TIMEOUT=30

# ---- Africa's Talking SMS ----
AT_USERNAME=sandbox
AT_API_KEY=
AT_FROM=15054
# Random secret; append ?token=<value> to the callback URL you register with Africa's Talking.
AT_WEBHOOK_SECRET=
# Ignore inbound SMS that do not start with the keyword (shared short codes).
AT_REQUIRE_KEYWORD=false
```

Huru SMS

Huru SMS is a digital service that lets any person in Tanzania ask a question and get a clear, useful answer on their phone, in Kiswahili or English. It works on the simplest mobile phone through SMS, and on smartphones and computers through a web chat.

Owner and developer: **Huru Digital Co. Ltd.**, Tanzania.

---

1. What the service does

A citizen sends a question. Huru SMS reads it, understands the topic and the language, checks where the person is, looks up verified information about Tanzania, and replies with a short, practical answer. If the person needs an office or a service near them (for example a legal aid provider in Kahama), the reply names it with its contact.

Topics covered include:

- Law and rights (arrest, land, marriage, inheritance, tenants, contracts)
- Health (danger signs, clinics, vaccination, health insurance)
- Farming, livestock and fishing (seasons, seeds, pests, animal vaccines)
- Education (school system, examinations, student loans, school subjects)
- Government services (NIDA identity card, birth certificate, passport, TIN, government payments, electricity)
- Money and business (taxes, loans, mobile money safety, registering a business)
- Jobs and employment (contracts, leave, dismissal, job scams)
- Family and community (violence, child protection, where to get help)
- Technology and transport (SIM cards, phone safety, traffic fines, driving licence)
- Any other everyday question

2. How a citizen uses it

By SMS (no internet needed)

1. Open the message app on any phone.
2. Write the word **HURU** followed by the question. Example: `HURU nifanyeje kupata kitambulisho cha NIDA?`
3. Send it to **15054**.
4. The answer arrives as an SMS within a short time.

Helpful SMS commands:

Send	What happens
`MSAADA` or `HELP`	A short guide to the service
`LUGHA SW` / `LUGHA EN`	Choose Kiswahili or English for replies
`JINA Asha`	Save your name
`MKOA Mwanza`	Save your region, so nearby services are suggested
`ACHA`	Stop receiving replies
`ANZA`	Start again

By web chat (with internet)

1. Open the website and press **Web Chat**.
2. Enter a phone number. A one-time code is sent by SMS to confirm it is yours.
3. Ask questions in the chat. Previous conversations are saved, can be searched, and each answer can be rated helpful or not helpful.

3. What makes it different

- **Built for Tanzania.** The service assumes Tanzanian law, the Tanzanian shilling, East Africa time, Tanzanian institutions (NIDA, RITA, TRA, NHIF, NECTA, HESLB and others) and the real emergency numbers (112, 114, 115, 116).
- **Knows where you are.** It recognises all 31 regions, common districts and well-known neighbourhoods, and uses that to point people to services close to them.
- **Verified knowledge first.** Answers are grounded in a knowledge base and directories maintained by the operator, not only on general internet knowledge. The service says clearly when something should be confirmed with the responsible office, and it never invents phone numbers, fees or names of officials.
- **Two languages, one service.** It detects whether a question is in Kiswahili or English and answers in the same language.
- **Works on every phone.** SMS first, web second. Replies sent by SMS are kept short to fit a few text messages.
- **Safe to use.** Abusive language is warned and then blocked, emergencies are recognised and the right number is given first, and people can stop the service at any time.

```
## 4. What the operator can manage

A protected administration area lets Huru Digital staff:

- see how many questions arrive, by day, by topic and by channel;
- read conversations and the information used to answer them;
- add or edit knowledge entries (one by one or by importing a file);
- maintain the national directory of institutions and hotlines, and the list of local services by region and district;
- adjust the wording and behaviour of the service per topic and language;
- change settings such as reply length, language policy and safety limits;
- block or unblock users.

## 5. Privacy and safety

- Only the phone number and the conversation are stored. They are needed to continue a conversation and to let the person see their own history.
- The web chat requires a one-time SMS code before a phone number's history can be opened.
- A person's name and phone number are never sent together with the question to the language technology provider.
- Government payments are always described as being made through official control numbers, never cash to an officer.
- Full terms and a privacy policy are published on the website.

## 6. Technology used (for the technical reader)

| Part | Technology |
| --- | --- |
| Application | PHP 8.2 with the Laravel 12 framework |
| Database | MySQL 8 (SQLite for quick local testing) |
| SMS | Africa's Talking SMS gateway (shortcode 15054, keyword HURU) |
| Language technology | Google Gemini, used only to word the final reply after the service has gathered the Tanzanian context |
| Web pages | HTML, CSS and JavaScript; installable as a phone app (PWA) |
| Tests | 28 automated tests covering SMS, web chat, topic detection, location detection and the administration area |

Main parts of the source code:

| Folder or file | Purpose |
| --- | --- |
| `app/Services/ConversationService.php` | Handles every incoming question, commands, safety checks and logging |
| `app/Services/TopicClassifier.php` | Detects the topic of a question and recognises emergencies |
| `app/Services/TanzaniaContext.php` | Regions, districts, country facts and topic rules for Tanzania |
| `app/Services/KnowledgeRetriever.php` | Finds the most relevant verified knowledge entries |
| `app/Services/PromptEngine.php` | Combines persona, context, knowledge and contacts into instructions for the reply |
| `app/Services/Ai/` | Connection to the language technology provider |
| `app/Services/SmsService.php` | Sending SMS through Africa's Talking |
| `app/Http/Controllers/` | Web chat, SMS receiver, community and administration screens |
| `database/seeders/` | Starter knowledge, directories, personas and settings |
| `resources/views/` | All web pages |
| `tests/` | Automated tests |

## 7. Running the application

```bash
cp .env.example .env         # fill in GEMINI_API_KEY, AT_* and database details
composer install
npm install && npm run build
php artisan key:generate
php artisan migrate --seed   # creates tables, the admin account and starter data
php artisan serve            # open http://localhost:8000
php artisan queue:work       # processes SMS replies (or run the scheduler every minute)
php artisan huru:import-legal-aid   # imports the national legal aid provider list
```

The SMS gateway must be configured to deliver incoming messages to `/api/sms/inbound?token=<AT_WEBHOOK_SECRET>`.

Administration is at `/admin/login`.

8. Ownership
```

```
127  Huru SMS, its source code, its content and its documentation are the property of Huru Digital Co. Ltd. All rights reserved.
```

4 app/Console/Commands/FetchLegalAidProviders.php

121 lines

```php
<?php

namespace App\Console\Commands;

use App\Models\LocalResource;
use App\Services\TanzaniaContext;
use Illuminate\Console\Command;
use Illuminate\Support\Facades\Http;
use Illuminate\Support\Str;

/**
 * Imports registered legal aid providers from the Mama Samia Legal Aid
 * Campaign site (mslac.or.tz) into local_resources as category legal_aid.
 */
class FetchLegalAidProviders extends Command
{
    protected $signature = 'huru:import-legal-aid {--pages=16 : Number of listing pages to read}';

    protected $description = 'Import legal aid providers from mslac.or.tz into the local resources directory';

    public function handle(TanzaniaContext $tz): int
    {
        $pages = (int) $this->option('pages');
        $imported = 0;

        for ($page = 0; $page < $pages; $page++) {
            $url = "https://www.mslac.or.tz/en/legal-aid-providers?page={$page}";
            $this->line('Fetching ' . ($page + 1) . "/{$pages}: {$url}");

            try {
                $response = Http::retry(3, 2000)->timeout(45)
                    ->withHeaders(['User-Agent' => 'HuruBot/1.0 (+https://hurudigital.co.tz)'])
                    ->get($url);
            } catch (\Throwable $e) {
                $this->error('Request failed: ' . $e->getMessage());

                continue;
            }

            if (! $response->successful()) {
                $this->error('HTTP ' . $response->status());

                continue;
            }

            $chunks = explode('class="item-columns"', $response->body());
            array_shift($chunks);
            if (empty($chunks)) {
                $this->warn('No provider cards found; stopping.');
                break;
            }

            foreach ($chunks as $chunk) {
                preg_match('/views-field-title.*?<strong class="field-content">(.*?)<\/strong>/s', $chunk, $m);
                $name = isset($m[1]) ? trim(strip_tags($m[1])) : null;
                if (! $name) {
                    continue;
                }

                preg_match('/views-field-field-region.*?<div class="field-content">(.*?)<\/div>/s', $chunk, $m);
                $regionRaw = isset($m[1]) ? trim(strip_tags($m[1])) : null;

                preg_match('/views-field-field-address.*?<div class="field-content">(.*?)<\/div>/s', $chunk, $m);
                $location = isset($m[1]) ? trim(preg_replace('/\s+/', ' ', strip_tags($m[1]))) : null;

                $region = null;
                $district = null;
                $detected = $tz->detectLocation(trim(($regionRaw ?? '') . ' ' . ($location ?? '')));
                if ($detected) {
                    $region = $detected['region'];
                    $district = $detected['district'];
```

```
                }
                if (! $region && $regionRaw) {
                    $region = Str::lower($regionRaw);
                }
                if (! $district && $location && preg_match('/wilaya\s+(?:ya\s+)?([a-z\']+)/i', $location, $dm)) {
                    $district = Str::lower($dm[1]);
                }

                $email = null;
                if ($location && preg_match('/[a-z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{2,}/i', $location, $em)) {
                    $email = $em[0];
                }
                preg_match('/views-field-field-email.*?href="mailto:(.*?)"/s', $chunk, $em2);
                if (! empty($em2[1])) {
                    $email = trim(strip_tags($em2[1]));
                }

                $phone = null;
                if ($location && preg_match('/(?:\+255|0)[67]\d{8}/', str_replace([' ', '-', '(', ')', '/'], '',
$location), $pm)) {
                    $phone = $pm[0];
                }
                preg_match('/views-field-field-phone.*?<div class="field-content">(.*?)<\/div>/s', $chunk, $pm2);
                if (! empty($pm2[1])) {
                    $phone = trim(strip_tags($pm2[1]));
                }

                LocalResource::query()->updateOrCreate(
                    ['name' => $name],
                    [
                        'category' => 'legal_aid',
                        'region' => $region,
                        'district' => $district,
                        'location' => $location,
                        'email' => $email,
                        'phone' => $phone,
                        'source' => 'mslac.or.tz',
                        'is_active' => true,
                    ]
                );
                $imported++;
            }

            usleep(500000);
        }

        $this->info("Done. Providers imported or updated: {$imported}");

        return self::SUCCESS;
    }
}
```

5 app/Http/Controllers/Admin/AuthController.php

56 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\User;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;
use Illuminate\Support\Facades\RateLimiter;
use Illuminate\Support\Str;

class AuthController extends Controller
{
    public function showLogin(Request $request)
    {
        if ($request->user()?->isAdmin()) {
            return redirect()->route('admin.dashboard');
        }

        return view('admin.login');
    }

    public function login(Request $request)
    {
        $data = $request->validate([
            'email' => 'required|email',
            'password' => 'required|string',
        ]);

        $key = 'admin-login:' . Str::lower($data['email']) . '|' . $request->ip();
        if (RateLimiter::tooManyAttempts($key, 5)) {
            return back()->withErrors(['email' => 'Too many attempts. Try again in a minute.'])->onlyInput('email');
        }

        $user = User::query()->where('email', $data['email'])->first();
        if (! $user || ! $user->isAdmin() || ! Auth::attempt(['email' => $data['email'], 'password' => $data['password']], $request->boolean('remember'))) {
            RateLimiter::hit($key, 60);

            return back()->withErrors(['email' => 'These credentials do not match an admin account.'])->onlyInput('email');
        }

        RateLimiter::clear($key);
        $request->session()->regenerate();

        return redirect()->intended(route('admin.dashboard'));
    }

    public function logout(Request $request)
    {
        Auth::logout();
        $request->session()->invalidate();
        $request->session()->regenerateToken();

        return redirect()->route('admin.login');
    }
}
```

6 app/Http/Controllers/Admin/ConversationController.php

44 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\Message;
use App\Models\User;
use Illuminate\Http\Request;

class ConversationController extends Controller
{
    public function index(Request $request)
    {
        $query = Message::query()->with(['user', 'aiLog', 'feedback'])->latest('id');

        if ($request->filled('q')) {
            $q = '%' . $request->string('q') . '%';
            $query->where(fn ($w) => $w->where('content', 'like', $q)->orWhereHas('user', fn ($u) => $u->where('phone_number', 'like', $q)->orWhere('name', 'like', $q)));
        }
        if ($request->filled('category')) {
            $query->where('category', $request->string('category'));
        }
        if ($request->filled('channel')) {
            $query->where('channel', $request->string('channel'));
        }
        if ($request->filled('direction')) {
            $query->where('direction', $request->string('direction'));
        }
        if ($request->filled('date')) {
            $query->whereDate('created_at', $request->string('date'));
        }

        $messages = $query->paginate(40)->withQueryString();

        return view('admin.conversations.index', compact('messages'));
    }

    public function show(User $user)
    {
        $messages = $user->messages()->with(['aiLog', 'feedback'])->latest('id')->paginate(60);

        return view('admin.conversations.show', compact('user', 'messages'));
    }
}
```

7 app/Http/Controllers/Admin/DashboardController.php

66 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\AiLog;
use App\Models\Feedback;
use App\Models\Message;
use App\Models\User;
use App\Services\AiService;
use App\Services\SmsService;
use App\Services\TopicClassifier;
use Illuminate\Support\Facades\DB;

class DashboardController extends Controller
{
    public function index(AiService $ai, SmsService $sms)
    {
        $today = now()->startOfDay();
        $week = now()->subDays(7);

        $stats = [
            'users' => User::query()->where('role', User::ROLE_USER)->count(),
            'users_today' => User::query()->where('role', User::ROLE_USER)->where('created_at', '>=', $today)->count(),
            'questions_today' => Message::query()->where('direction', Message::DIRECTION_IN)->where('created_at', '>=', $today)->count(),
            'questions_week' => Message::query()->where('direction', Message::DIRECTION_IN)->where('created_at', '>=', $week)->count(),
            'sms_week' => Message::query()->where('direction', Message::DIRECTION_IN)->where('channel', 'sms')->where('created_at', '>=', $week)->count(),
            'web_week' => Message::query()->where('direction', Message::DIRECTION_IN)->where('channel', 'web')->where('created_at', '>=', $week)->count(),
            'tokens_today' => (int) AiLog::query()->where('created_at', '>=', $today)->sum('total_tokens'),
            'latency_avg' => (int) round((float) AiLog::query()->where('created_at', '>=', $week)->where('status', 'ok')->avg('latency_ms')),
            'errors_week' => AiLog::query()->where('created_at', '>=', $week)->where('status', '!=', 'ok')->count(),
            'banned' => User::query()->where('is_banned', true)->count(),
            'helpful' => Feedback::query()->where('rating', 1)->count(),
            'unhelpful' => Feedback::query()->where('rating', -1)->count(),
        ];

        $byCategory = Message::query()
            ->select('category', DB::raw('count(*) as total'))
            ->where('direction', Message::DIRECTION_IN)
            ->where('created_at', '>=', $week)
            ->whereNotNull('category')
            ->groupBy('category')
            ->orderByDesc('total')
            ->get()
            ->map(fn ($row) => ['key' => $row->category, 'label' => TopicClassifier::label($row->category, 'en'), 'total' => (int) $row->total]);

        $daily = Message::query()
            ->select(DB::raw('DATE(created_at) as day'), DB::raw('count(*) as total'))
            ->where('direction', Message::DIRECTION_IN)
            ->where('created_at', '>=', now()->subDays(13)->startOfDay())
            ->groupBy('day')
            ->orderBy('day')
            ->pluck('total', 'day');

        $recent = Message::query()->with(['user', 'aiLog', 'feedback'])->latest('id')->take(25)->get();

        $health = [
            'engine_configured' => $ai->isConfigured(),
            'sms_configured' => $sms->isConfigured(),
            'webhook_secret' => (bool) config('services.at.webhook_secret'),
            'queue' => config('queue.default'),
        ];

```

```
64          return view('admin.dashboard', compact('stats', 'byCategory', 'daily', 'recent', 'health'));
65      }
66  }
```

8 app/Http/Controllers/Admin/KnowledgeController.php

164 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\KnowledgeEntry;
use App\Services\KnowledgeRetriever;
use App\Services\TopicClassifier;
use Illuminate\Http\Request;
use Illuminate\Support\Str;

class KnowledgeController extends Controller
{
    public function index(Request $request)
    {
        $query = KnowledgeEntry::query()->latest();

        if ($request->filled('q')) {
            $q = '%' . $request->string('q') . '%';
            $query->where(fn ($w) => $w->where('title', 'like', $q)->orWhere('content', 'like', $q));
        }
        if ($request->filled('category')) {
            $query->where('category', $request->string('category'));
        }
        if ($request->filled('language')) {
            $query->where('language', $request->string('language'));
        }

        $entries = $query->paginate(20)->withQueryString();
        $counts = KnowledgeEntry::query()->selectRaw('category, count(*) as total')->groupBy('category')->pluck('total', 'category');

        return view('admin.knowledge.index', compact('entries', 'counts'));
    }

    public function create()
    {
        return view('admin.knowledge.form', ['entry' => new KnowledgeEntry(['language' => 'sw', 'category' => 'general', 'is_active' => true])]);
    }

    public function store(Request $request, KnowledgeRetriever $retriever)
    {
        $data = $this->validated($request);
        $data['keywords'] = $this->keywords($data, $retriever);
        KnowledgeEntry::query()->create($data);

        return redirect()->route('admin.knowledge.index')->with('success', 'Knowledge entry created.');
    }

    public function edit(KnowledgeEntry $knowledge)
    {
        return view('admin.knowledge.form', ['entry' => $knowledge]);
    }

    public function update(Request $request, KnowledgeEntry $knowledge, KnowledgeRetriever $retriever)
    {
        $data = $this->validated($request);
        $data['keywords'] = $this->keywords($data, $retriever);
        $knowledge->update($data);

        return redirect()->route('admin.knowledge.index')->with('success', 'Knowledge entry updated.');
    }

    public function toggle(KnowledgeEntry $knowledge)
    {
        $knowledge->update(['is_active' => ! $knowledge->is_active]);

        return back()->with('success', 'Entry ' . ($knowledge->is_active ? 'activated' : 'deactivated') . '.');
    }

```

```
    public function destroy(KnowledgeEntry $knowledge)
    {
        $knowledge->delete();

        return back()->with('success', 'Knowledge entry deleted.');
    }

    /**
     * Bulk import. CSV columns: title, content, category, language, keywords (comma separated), source, summary.
     * JSON: an array of objects with the same keys (keywords may be an array).
     */
    public function import(Request $request, KnowledgeRetriever $retriever)
    {
        $request->validate([
            'file' => 'required|file|mimes:csv,txt,json|max:5120',
            'default_category' => 'required|in:' . implode(',', TopicClassifier::keys()),
            'default_language' => 'required|in:sw,en',
        ]);

        $file = $request->file('file');
        $ext = strtolower($file->getClientOriginalExtension());
        $rows = [];

        if ($ext === 'json') {
            $decoded = json_decode(file_get_contents($file->getRealPath()), true);
            if (is_array($decoded)) {
                $rows = isset($decoded['title']) ? [$decoded] : $decoded;
            }
        } else {
            $handle = fopen($file->getRealPath(), 'r');
            $header = fgetcsv($handle);
            if ($header) {
                $header = array_map(fn ($h) => Str::snake(Str::lower(trim((string) $h))), $header);
                while (($row = fgetcsv($handle)) !== false) {
                    if (count(array_filter($row)) === 0) {
                        continue;
                    }
                    $rows[] = array_combine($header, array_pad($row, count($header), null));
                }
            }
            fclose($handle);
        }

        $count = 0;
        foreach ($rows as $row) {
            if (empty($row['title']) || empty($row['content'])) {
                continue;
            }
            $category = in_array($row['category'] ?? '', TopicClassifier::keys(), true) ? $row['category'] :
$request->input('default_category');
            $language = in_array($row['language'] ?? '', ['sw', 'en'], true) ? $row['language'] : $request-
>input('default_language');
            $keywords = $row['keywords'] ?? [];
            if (is_string($keywords)) {
                $keywords = array_filter(array_map('trim', explode(',', $keywords)));
            }
            $data = [
                'title' => Str::limit(trim($row['title']), 250, ''),
                'content' => trim($row['content']),
                'summary' => $row['summary'] ?? Str::limit(trim($row['content']), 160),
                'category' => $category,
                'language' => $language,
                'source' => $row['source'] ?? null,
                'is_active' => true,
            ];
            $data['keywords'] = $this->keywords($data + ['keywords_text' => implode(',', $keywords)], $retriever);

            KnowledgeEntry::query()->updateOrCreate(['title' => $data['title'], 'language' => $language], $data);
            $count++;
        }

        return back()->with('success', "Imported {$count} knowledge entries.");
    }
```

```
141
142     private function validated(Request $request): array
143     {
144         return $request->validate([
145             'title' => 'required|string|max:250',
146             'content' => 'required|string|max:20000',
147             'summary' => 'nullable|string|max:500',
148             'category' => 'required|in:' . implode(',', TopicClassifier::keys()),
149             'language' => 'required|in:sw,en',
150             'keywords_text' => 'nullable|string|max:1000',
151             'source' => 'nullable|string|max:250',
152             'is_active' => 'sometimes|boolean',
153         ]) + ['is_active' => $request->boolean('is_active')];
154     }
155
156     /** Explicit keywords plus title tokens, so retrieval works even when the admin leaves keywords empty. */
157     private function keywords(array $data, KnowledgeRetriever $retriever): array
158     {
159         $explicit = array_filter(array_map(fn ($k) => Str::lower(trim($k)), explode(',', (string)
($data['keywords_text'] ?? ''))));
160         $fromTitle = $retriever->tokens($data['title']);
161
162         return array_values(array_unique(array_merge($explicit, $fromTitle)));
163     }
164 }
```

9 app/Http/Controllers/Admin/LocalResourceController.php

87 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\LocalResource;
use App\Services\TanzaniaContext;
use Illuminate\Http\Request;
use Illuminate\Support\Str;

class LocalResourceController extends Controller
{
    public function index(Request $request)
    {
        $query = LocalResource::query()->orderBy('region')->orderBy('district')->orderBy('name');

        if ($request->filled('q')) {
            $q = '%' . $request->string('q') . '%';
            $query->where(fn ($w) => $w->where('name', 'like', $q)->orWhere('location', 'like', $q)->orWhere('district', 'like', $q));
        }
        if ($request->filled('region')) {
            $query->where('region', Str::lower($request->string('region')));
        }
        if ($request->filled('category')) {
            $query->where('category', $request->string('category'));
        }

        return view('admin.resources.index', [
            'resources' => $query->paginate(30)->withQueryString(),
            'regions' => TanzaniaContext::REGIONS,
            'categories' => LocalResource::CATEGORIES,
            'byRegion' => LocalResource::query()->selectRaw('region, count(*) as total')->groupBy('region')->orderByDesc('total')->pluck('total', 'region'),
        ]);
    }

    public function create()
    {
        return view('admin.resources.form', ['resource' => new LocalResource(['category' => 'legal_aid', 'is_active' => true]), 'regions' => TanzaniaContext::REGIONS, 'categories' => LocalResource::CATEGORIES]);
    }

    public function store(Request $request)
    {
        LocalResource::query()->create($this->validated($request));

        return redirect()->route('admin.resources.index')->with('success', 'Resource created.');
    }

    public function edit(LocalResource $resource)
    {
        return view('admin.resources.form', ['resource' => $resource, 'regions' => TanzaniaContext::REGIONS, 'categories' => LocalResource::CATEGORIES]);
    }

    public function update(Request $request, LocalResource $resource)
    {
        $resource->update($this->validated($request));

        return redirect()->route('admin.resources.index')->with('success', 'Resource updated.');
    }

    public function destroy(LocalResource $resource)
    {
        $resource->delete();

        return back()->with('success', 'Resource deleted.');
    }

    private function validated(Request $request): array
```

```
    {
        $data = $request->validate([
            'name' => 'required|string|max:150',
            'category' => 'required|in:' . implode(',', array_keys(LocalResource::CATEGORIES)),
            'region' => 'required|in:' . implode(',', TanzaniaContext::REGIONS),
            'district' => 'nullable|string|max:40',
            'ward' => 'nullable|string|max:60',
            'location' => 'nullable|string|max:500',
            'phone' => 'nullable|string|max:60',
            'email' => 'nullable|email|max:120',
            'is_active' => 'sometimes|boolean',
        ]);

        $data['district'] = $data['district'] ? Str::lower(trim($data['district'])) : null;
        $data['is_active'] = $request->boolean('is_active');
        $data['source'] = 'admin';

        return $data;
    }
}
```

10 app/Http/Controllers/Admin/PromptTemplateController.php

89 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\PromptTemplate;
use App\Services\TopicClassifier;
use Illuminate\Http\Request;

class PromptTemplateController extends Controller
{
    public function index()
    {
        $templates = PromptTemplate::query()->orderBy('category')->orderBy('language')->get();

        return view('admin.templates.index', compact('templates'));
    }

    public function create()
    {
        return view('admin.templates.form', ['template' => new PromptTemplate(['category' => 'general', 'language' => 'sw', 'is_active' => true])]);
    }

    public function store(Request $request)
    {
        $data = $this->validated($request);
        PromptTemplate::query()->create($data);
        $this->ensureSingleActive($data);

        return redirect()->route('admin.templates.index')->with('success', 'Template created.');
    }

    public function edit(PromptTemplate $template)
    {
        return view('admin.templates.form', compact('template'));
    }

    public function update(Request $request, PromptTemplate $template)
    {
        $data = $this->validated($request, $template);
        $template->update($data);
        $this->ensureSingleActive($data, $template->id);

        return redirect()->route('admin.templates.index')->with('success', 'Template updated.');
    }

    public function toggle(PromptTemplate $template)
    {
        $template->update(['is_active' => ! $template->is_active]);
        if ($template->is_active) {
            $this->ensureSingleActive($template->only('category', 'language'), $template->id);
        }

        return back()->with('success', 'Template ' . ($template->is_active ? 'activated' : 'deactivated') . '.');
    }

    public function destroy(PromptTemplate $template)
    {
        $template->delete();

        return redirect()->route('admin.templates.index')->with('success', 'Template deleted.');
    }

    private function validated(Request $request, ?PromptTemplate $existing = null): array
    {
        return $request->validate([
            'name' => 'required|string|max:80|unique:prompt_templates,name' . ($existing ? ',' . $existing->id : ''),
            'category' => 'required|in:' . implode(',', TopicClassifier::keys()),
            'language' => 'required|in:sw,en',
```

```
            'template' => 'required|string|min:20|max:4000',
            'is_active' => 'sometimes|boolean',
        ]) + ['is_active' => $request->boolean('is_active')];
    }

    /** Only one active persona per (category, language). */
    private function ensureSingleActive(array $data, ?int $keepId = null): void
    {
        if (empty($data['is_active']) && $keepId === null) {
            return;
        }
        $keep = $keepId ?? PromptTemplate::query()->where('name', $data['name'])->value('id');
        PromptTemplate::query()
            ->where('category', $data['category'])
            ->where('language', $data['language'])
            ->where('id', '!=', $keep)
            ->update(['is_active' => false]);
        cache()->forget('huru.templates.active');
    }
}
```

11 app/Http/Controllers/Admin/ReferenceContactController.php

86 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\ReferenceContact;
use App\Services\TopicClassifier;
use Illuminate\Http\Request;

class ReferenceContactController extends Controller
{
    public const CATEGORIES = ['police', 'legal', 'health', 'family', 'government', 'finance', 'business', 'employment', 'education', 'agriculture', 'technology', 'transport', 'general'];

    public function index(Request $request)
    {
        $query = ReferenceContact::query()->orderBy('sort_order')->orderBy('name');
        if ($request->filled('category')) {
            $query->where('category', $request->string('category'));
        }

        return view('admin.contacts.index', ['contacts' => $query->get(), 'categories' => self::CATEGORIES]);
    }

    public function create()
    {
        return view('admin.contacts.form', ['contact' => new ReferenceContact(['category' => 'government', 'is_active' => true, 'sort_order' => 100]), 'categories' => self::CATEGORIES]);
    }

    public function store(Request $request)
    {
        ReferenceContact::query()->create($this->validated($request));

        return redirect()->route('admin.contacts.index')->with('success', 'Contact created.');
    }

    public function edit(ReferenceContact $contact)
    {
        return view('admin.contacts.form', ['contact' => $contact, 'categories' => self::CATEGORIES]);
    }

    public function update(Request $request, ReferenceContact $contact)
    {
        $contact->update($this->validated($request));

        return redirect()->route('admin.contacts.index')->with('success', 'Contact updated.');
    }

    public function verify(ReferenceContact $contact)
    {
        $contact->update(['verified_at' => $contact->verified_at ? null : now()]);

        return back()->with('success', $contact->verified_at ? 'Marked as verified.' : 'Verification cleared.');
    }

    public function destroy(ReferenceContact $contact)
    {
        $contact->delete();

        return back()->with('success', 'Contact deleted.');
    }

    private function validated(Request $request): array
    {
        $data = $request->validate([
            'name' => 'required|string|max:150',
            'category' => 'required|in:' . implode(',', self::CATEGORIES),
            'phone' => 'nullable|string|max:60',
            'alt_phone' => 'nullable|string|max:60',
            'email' => 'nullable|email|max:120',
```

```
            'website' => 'nullable|url|max:200',
            'description_sw' => 'nullable|string|max:500',
            'description_en' => 'nullable|string|max:500',
            'sort_order' => 'required|integer|min:0|max:1000',
            'is_emergency' => 'sometimes|boolean',
            'is_active' => 'sometimes|boolean',
            'verified' => 'sometimes|boolean',
        ]);

        $data['is_emergency'] = $request->boolean('is_emergency');
        $data['is_active'] = $request->boolean('is_active');
        $data['verified_at'] = $request->boolean('verified') ? now() : null;
        unset($data['verified']);

        return $data;
    }
}
```

12 app/Http/Controllers/Admin/SettingController.php

38 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Support\Settings;
use Illuminate\Http\Request;

class SettingController extends Controller
{
    public function index()
    {
        return view('admin.settings', [
            'groups' => Settings::groups(),
            'values' => Settings::all(),
        ]);
    }

    public function update(Request $request)
    {
        $rules = Settings::rules();

        // Unchecked checkboxes are absent from the request: treat as "0".
        foreach (Settings::schema() as $key => $def) {
            if ($def['type'] === 'boolean' && ! $request->has($key)) {
                $request->merge([$key => '0']);
            }
        }

        $data = $request->validate($rules);

        foreach ($data as $key => $value) {
            Settings::set($key, $value);
        }

        return back()->with('success', 'Settings saved.');
    }
}
```

13 app/Http/Controllers/Admin/UserController.php

51 lines

```php
<?php

namespace App\Http\Controllers\Admin;

use App\Http\Controllers\Controller;
use App\Models\User;
use Illuminate\Http\Request;

class UserController extends Controller
{
    public function index(Request $request)
    {
        $query = User::query()->where('role', User::ROLE_USER)->withCount('messages')->latest('last_seen_at');

        if ($request->filled('q')) {
            $q = '%' . $request->string('q') . '%';
            $query->where(fn ($w) => $w->where('phone_number', 'like', $q)->orWhere('name', 'like', $q));
        }
        if ($request->boolean('banned')) {
            $query->where('is_banned', true);
        }
        if ($request->filled('region')) {
            $query->where('region', $request->string('region'));
        }

        return view('admin.users.index', ['users' => $query->paginate(40)->withQueryString()]);
    }

    public function toggleBan(User $user)
    {
        abort_if($user->isAdmin(), 403);
        $user->update(['is_banned' => ! $user->is_banned]);

        return back()->with('success', $user->is_banned ? 'User blocked.' : 'User unblocked.');
    }

    public function resetStrikes(User $user)
    {
        $user->update(['abuse_count' => 0, 'is_banned' => false]);

        return back()->with('success', 'Strikes cleared.');
    }

    public function destroy(User $user)
    {
        abort_if($user->isAdmin(), 403);
        $user->delete();

        return redirect()->route('admin.users.index')->with('success', 'User and their history deleted.');
    }
}
```

14 app/Http/Controllers/CommunityController.php

157 lines

```php
<?php

namespace App\Http\Controllers;

use App\Models\CommunityPost;
use App\Models\CommunityThread;
use App\Models\User;
use App\Services\ModerationService;
use App\Support\Settings;
use Illuminate\Http\Request;
use Illuminate\Support\Str;

class CommunityController extends Controller
{
    public function index(Request $request)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return redirect()->route('chat.index', ['redirect' => $request->getRequestUri()])
                ->with('error', 'Tafadhali ingia kwanza kufikia jamii. / Please sign in to access the community.');
        }

        $publicThreads = CommunityThread::query()->where('is_private', false)->withCount('posts', 'members')->latest()->get();
        $joinedIds = $user->joinedThreads()->pluck('community_threads.id')->all();
        $joinedThreads = $user->joinedThreads()->withCount('posts')->latest()->get();

        return view('community.index', compact('publicThreads', 'joinedThreads', 'joinedIds', 'user') + ['settings' => $this->settings()]);
    }

    public function show(Request $request, CommunityThread $thread)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return redirect()->route('chat.index', ['redirect' => $request->getRequestUri()]);
        }

        $isMember = $thread->members()->where('user_id', $user->id)->exists();
        if ($thread->is_private && ! $isMember) {
            return redirect()->route('community.index')->with('error', 'Huu ni mjadala wa faragha. / This is a private thread.');
        }

        $posts = $thread->posts()->where('is_approved', true)->with('user')->orderBy('created_at')->paginate(50);
        $members = $thread->members()->get();

        return view('community.show', compact('thread', 'posts', 'members', 'user', 'isMember') + ['settings' => $this->settings()]);
    }

    public function storeThread(Request $request)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['error' => 'Unauthorized'], 401);
        }
        if ($user->is_banned) {
            return back()->with('error', 'Huduma imefungwa kwa akaunti hii.');
        }

        $data = $request->validate([
            'title' => 'required|string|min:3|max:120',
            'description' => 'nullable|string|max:1000',
            'is_private' => 'sometimes|boolean',
        ]);

        $thread = CommunityThread::query()->create([
            'title' => $data['title'],
            'slug' => Str::slug($data['title']) . '-' . Str::lower(Str::random(5)),
            'description' => $data['description'] ?? null,
```

```
            'creator_id' => $user->id,
            'is_private' => $request->boolean('is_private'),
        ]);

        $thread->members()->attach($user->id, ['role' => 'admin']);

        return redirect()->route('community.show', $thread);
    }

    public function storePost(Request $request, CommunityThread $thread, ModerationService $moderation)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['error' => 'Unauthorized'], 401);
        }
        if ($user->is_banned) {
            return response()->json(['error' => 'Account blocked'], 403);
        }
        if (! $thread->members()->where('user_id', $user->id)->exists()) {
            return response()->json(['error' => 'Si mwanachama wa mjadala huu. / Not a member of this thread.'],
403);
        }

        $data = $request->validate(['content' => 'required|string|min:2|max:3000']);

        if ($moderation->isAbusive($data['content'])) {
            $user->increment('abuse_count');
            if ($user->abuse_count >= Settings::int('abuse_strike_limit')) {
                $user->forceFill(['is_banned' => true])->save();
            }

            return response()->json(['error' => 'Lugha ya matusi hairuhusiwi. / Abusive language is not allowed.'],
422);
        }

        $post = CommunityPost::query()->create([
            'community_thread_id' => $thread->id,
            'user_id' => $user->id,
            'content' => $data['content'],
            'is_approved' => ! $thread->is_system, // system threads are moderated before display on the landing
page
        ]);

        if ($request->expectsJson()) {
            return response()->json(['status' => 'success', 'post' => $post->load('user'), 'pending' => ! $post-
>is_approved]);
        }

        return back()->with('success', $post->is_approved ? 'Ujumbe umetumwa.' : 'Ujumbe umepokelewa na utaonekana
baada ya kuhakikiwa.');
    }

    public function join(Request $request, CommunityThread $thread)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return redirect()->route('chat.index', ['redirect' => route('community.show', $thread, false)]);
        }
        if ($thread->is_private) {
            return back()->with('error', 'Huwezi kujiunga na mjadala wa faragha bila mwaliko.');
        }

        $thread->members()->syncWithoutDetaching([$user->id => ['role' => 'member']]);

        return redirect()->route('community.show', $thread);
    }

    public function leave(Request $request, CommunityThread $thread)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return redirect()->route('chat.index');
        }
```

```
136
137         $thread->members()->detach($user->id);
138
139         return redirect()->route('community.index');
140     }
141
142     private function currentUser(Request $request): ?User
143     {
144         $id = $request->session()->get('chat_user_id');
145
146         return $id ? User::query()->find($id) : null;
147     }
148
149     private function settings(): array
150     {
151         return [
152             'app_name' => Settings::get('app_name'),
153             'sms_keyword' => Settings::get('sms_keyword'),
154             'sms_shortcode' => Settings::get('sms_shortcode'),
155         ];
156     }
157 }
```

15 app/Http/Controllers/ContactController.php

70 lines

```php
<?php

namespace App\Http\Controllers;

use App\Support\Settings;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Log;
use Illuminate\Support\Facades\Mail;
use Illuminate\Support\Facades\RateLimiter;

class ContactController extends Controller
{
    public function submit(Request $request)
    {
        $data = $request->validate([
            'type' => 'nullable|in:subscribe,partner,general',
            'email' => 'required|email|max:120',
            'name' => 'nullable|string|max:80',
            'organisation' => 'nullable|string|max:120',
            'message' => 'nullable|string|max:2000',
            'website' => 'nullable|max:0', // honeypot
        ]);

        $key = 'contact:' . $request->ip();
        if (RateLimiter::tooManyAttempts($key, 5)) {
            return back()->with('contact_error', 'Umetuma mara nyingi. Jaribu tena baadaye.');
        }
        RateLimiter::hit($key, 3600);

        $type = $data['type'] ?? 'general';
        $to = Settings::get('contact_email') ?: config('mail.from.address');
        $app = Settings::get('app_name');

        $subject = match ($type) {
            'subscribe' => "[{$app}] New subscriber",
            'partner' => "[{$app}] Partnership enquiry",
            default => "[{$app}] New enquiry",
        };

        $body = implode("\n", array_filter([
            "Type: {$type}",
            'Email: ' . $data['email'],
            'Name: ' . ($data['name'] ?? '-'),
            'Organisation: ' . ($data['organisation'] ?? '-'),
            '',
            $data['message'] ?? '',
            '',
            'Received: ' . now()->toDateTimeString() . ' (' . config('app.timezone') . ')',
        ]));

        try {
            if ($to) {
                Mail::raw($body, function ($mail) use ($to, $subject, $data) {
                    $mail->to($to)->replyTo($data['email'], $data['name'] ?? 'Visitor')->subject($subject);
                });
            }
            Log::info('Contact form', ['type' => $type, 'email' => $data['email']]);

            return back()->with('contact_success', match ($type) {
                'partner' => 'Asante! Tutawasiliana nawe ndani ya siku 2 za kazi.',
                'subscribe' => 'Umejiunga! Tutakutumia taarifa za maendeleo.',
                default => 'Ujumbe wako umepokelewa. Asante.',
            });
        } catch (\Throwable $e) {
            Log::error('Contact mail failed: ' . $e->getMessage());

            return back()->with('contact_error', 'Imeshindikana kutuma. Tafadhali tutumie barua pepe moja kwa moja: ' . $to);
        }
    }
}
```

16 app/Http/Controllers/Controller.php

8 lines

```php
<?php

namespace App\Http\Controllers;

abstract class Controller
{
    //
}
```

17 app/Http/Controllers/SmsController.php

68 lines

```php
<?php

namespace App\Http\Controllers;

use App\Jobs\ProcessIncomingSms;
use App\Support\Phone;
use App\Support\Settings;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Log;
use Illuminate\Support\Facades\RateLimiter;

class SmsController extends Controller
{
    /**
     * Africa's Talking inbound callback (form-encoded: from, to, text, date, id, linkId).
     */
    public function inbound(Request $request)
    {
        $from = Phone::normalize($request->input('from'));
        $text = trim((string) $request->input('text', ''));

        if (! $from || $text === '') {
            return response()->json(['status' => 'error', 'message' => 'Invalid payload'], 400);
        }

        // A flood from one number must not become a flood of model calls.
        $key = 'sms-inbound:' . $from;
        if (RateLimiter::tooManyAttempts($key, 20)) {
            Log::warning('Inbound SMS rate limited', ['from' => Phone::mask($from)]);

            return response()->json(['status' => 'rate_limited']);
        }
        RateLimiter::hit($key, 60);

        $keyword = (string) Settings::get('sms_keyword');
        $clean = $text;

        // On a shared short code the operator prepends the keyword; strip it when present.
        if ($keyword !== '' && preg_match('/^' . preg_quote($keyword, '/') . '\b\s*/iu', $text)) {
            $clean = trim(preg_replace('/^' . preg_quote($keyword, '/') . '\b\s*/iu', '', $text));
        } elseif ($keyword !== '' && config('services.at.require_keyword')) {
            Log::info('SMS ignored: keyword missing', ['from' => Phone::mask($from)]);

            return response()->json(['status' => 'ignored']);
        }

        // A bare keyword is a request for help.
        if ($clean === '') {
            $clean = 'HELP';
        }

        Log::info('Inbound SMS accepted', ['from' => Phone::mask($from), 'chars' => strlen($clean)]);

        ProcessIncomingSms::dispatch($from, $clean, $request->input('id'));

        return response()->json(['status' => 'success']);
    }

    /**
     * Optional delivery report callback from Africa's Talking.
     */
    public function deliveryReport(Request $request)
    {
        Log::info('SMS delivery report', $request->only('id', 'status', 'phoneNumber', 'failureReason'));

        return response()->json(['status' => 'ok']);
    }
}
```

18 app/Http/Controllers/WebChatController.php

333 lines

```php
<?php

namespace App\Http\Controllers;

use App\Models\Feedback;
use App\Models\Message;
use App\Models\User;
use App\Services\ConversationService;
use App\Services\SmsService;
use App\Services\TanzaniaContext;
use App\Services\TopicClassifier;
use App\Support\Phone;
use App\Support\Settings;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Hash;
use Illuminate\Support\Facades\Log;
use Illuminate\Support\Facades\RateLimiter;
use Illuminate\Validation\ValidationException;

class WebChatController extends Controller
{
    private const SESSION_KEY = 'chat_user_id';

    public function index()
    {
        return view('chat', [
            'settings' => $this->publicSettings(),
        ]);
    }

    /** Bootstrap data for the page: who is logged in and the public settings. */
    public function session(Request $request)
    {
        $user = $this->currentUser($request);

        return response()->json([
            'authenticated' => (bool) $user,
            'user' => $user ? $this->userPayload($user) : null,
            'settings' => $this->publicSettings(),
            'csrf' => csrf_token(),
        ]);
    }

    /**
     * Step 1 of login. Creates the citizen record if needed and sends an OTP.
     * When OTP is disabled (or SMS is not configured in a non-production
     * environment) the session is opened directly.
     */
    public function requestOtp(Request $request, SmsService $sms)
    {
        $data = $request->validate([
            'phone_number' => 'required|string|min:9|max:20',
            'language' => 'nullable|in:sw,en',
        ]);

        $phone = Phone::normalize($data['phone_number']);
        if (! $phone) {
            throw ValidationException::withMessages(['phone_number' => 'Namba ya simu si sahihi. / Invalid phone number.']);
        }

        $key = 'otp-request:' . $phone;
        if (RateLimiter::tooManyAttempts($key, 4)) {
            return response()->json(['status' => 'error', 'message' => 'Umeomba mara nyingi. Subiri dakika 10. / Too many requests. Wait 10 minutes.'], 429);
        }
        RateLimiter::hit($key, 600);

        $user = User::query()->firstOrCreate(['phone_number' => $phone]);
        if ($user->is_banned) {
            return response()->json(['status' => 'error', 'message' => 'Huduma imefungwa kwa namba hii. / This
```

```
number is blocked.'], 403);
        }

        $language = $data['language'] ?? $user->preferred_language ?? 'sw';
        $otpEnabled = Settings::bool('otp_enabled');
        $canSend = $sms->isConfigured();

        if (! $otpEnabled || (! $canSend && ! app()->environment('production'))) {
            $this->openSession($request, $user);

            return response()->json([
                'status' => 'success',
                'otp_required' => false,
                'user' => $this->userPayload($user),
                'csrf' => csrf_token(), // session was regenerated, token rotated
            ]);
        }

        if (! $canSend) {
            return response()->json(['status' => 'error', 'message' => 'Huduma ya uthibitisho haipatikani kwa sasa.
/ Verification is unavailable right now.'], 503);
        }

        $code = (string) random_int(100000, 999999);
        $ttl = Settings::int('otp_ttl_minutes');
        $user->forceFill([
            'otp_hash' => Hash::make($code),
            'otp_expires_at' => now()->addMinutes($ttl),
            'otp_attempts' => 0,
        ])->save();

        $sent = $sms->sendOtp($phone, $code, $language, (string) Settings::get('app_name'), $ttl);
        if ($sent['status'] === 'error') {
            return response()->json(['status' => 'error', 'message' => 'Imeshindikana kutuma ujumbe wa uthibitisho.
/ Could not send the verification SMS.'], 502);
        }

        $payload = ['status' => 'success', 'otp_required' => true, 'phone_masked' => Phone::mask($phone),
'ttl_minutes' => $ttl];
        if (config('app.debug') && ! app()->environment('production')) {
            $payload['debug_code'] = $code; // never present in production
        }

        return response()->json($payload);
    }

    /** Step 2 of login. */
    public function verifyOtp(Request $request)
    {
        $data = $request->validate([
            'phone_number' => 'required|string',
            'code' => 'required|digits:6',
        ]);

        $phone = Phone::normalize($data['phone_number']);
        $user = $phone ? User::query()->where('phone_number', $phone)->first() : null;

        if (! $user || ! $user->otp_hash || ! $user->otp_expires_at || $user->otp_expires_at->isPast()) {
            return response()->json(['status' => 'error', 'message' => 'Namba ya uthibitisho imeisha muda. Omba
nyingine. / Code expired, request a new one.'], 422);
        }

        if ($user->otp_attempts >= 5) {
            $user->forceFill(['otp_hash' => null, 'otp_expires_at' => null])->save();

            return response()->json(['status' => 'error', 'message' => 'Majaribio mengi. Omba namba mpya. / Too
many attempts, request a new code.'], 429);
        }

        if (! Hash::check($data['code'], $user->otp_hash)) {
            $user->increment('otp_attempts');

            return response()->json(['status' => 'error', 'message' => 'Namba ya uthibitisho si sahihi. / Incorrect
```

```
code.'], 422);
        }

        $user->forceFill(['otp_hash' => null, 'otp_expires_at' => null, 'otp_attempts' => 0])->save();
        $this->openSession($request, $user);

        return response()->json(['status' => 'success', 'user' => $this->userPayload($user), 'csrf' =>
csrf_token()]);
    }

    public function messages(Request $request)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['status' => 'error', 'message' => 'Unauthorized'], 401);
        }

        $query = Message::query()->where('user_id', $user->id)->with('feedback');

        $filtering = false;
        if ($request->filled('keyword')) {
            $query->where('content', 'like', '%' . str_replace(['%', '_'], ['\%', '\_'], $request-
>string('keyword')) . '%');
            $filtering = true;
        }
        if ($request->filled('date')) {
            $query->whereDate('created_at', $request->string('date'));
            $filtering = true;
        }
        if ($request->filled('category')) {
            $query->where('category', $request->string('category'));
            $filtering = true;
        }
        if ($request->filled('before')) {
            $query->where('id', '<', (int) $request->input('before'));
        }

        $limit = $filtering ? 200 : Settings::int('web_chat_limit');
        $messages = $query->latest('id')->take($limit)->get()->reverse()->values();

        return response()->json([
            'status' => 'success',
            'messages' => $messages->map(fn (Message $m) => $this->messagePayload($m)),
            'has_more' => $messages->isNotEmpty() && Message::query()->where('user_id', $user->id)->where('id',
'<', $messages->first()->id)->exists(),
        ]);
    }

    public function send(Request $request, ConversationService $conversation)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['status' => 'error', 'message' => 'Unauthorized'], 401);
        }

        $data = $request->validate(['message' => 'required|string|max:2000']);

        $result = $conversation->handle($user, $data['message'], Message::CHANNEL_WEB);

        if ($result['kind'] === ConversationService::KIND_IGNORED) {
            return response()->json(['status' => 'error', 'message' => 'Empty message'], 400);
        }

        return response()->json([
            'status' => 'success',
            'kind' => $result['kind'],
            'category' => $result['category'],
            'category_label' => $result['category'] ? TopicClassifier::label($result['category'],
$result['language']) : null,
            'user_message' => $result['inbound'] ? $this->messagePayload($result['inbound']) : null,
            'reply' => $result['outbound']
                ? $this->messagePayload($result['outbound'])
                : ['id' => null, 'direction' => 'outbound', 'content' => $result['text'], 'created_at' => now()-
```

```
>toIso8601String(), 'category' => $result['category'], 'feedback' => null],
        ]);
    }

    public function feedback(Request $request)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['status' => 'error', 'message' => 'Unauthorized'], 401);
        }

        $data = $request->validate([
            'message_id' => 'required|integer',
            'rating' => 'required|in:1,-1',
            'comment' => 'nullable|string|max:500',
        ]);

        $message = Message::query()->where('user_id', $user->id)->where('direction', Message::DIRECTION_OUT)-
>find($data['message_id']);
        if (! $message) {
            return response()->json(['status' => 'error', 'message' => 'Message not found'], 404);
        }

        Feedback::query()->updateOrCreate(
            ['message_id' => $message->id, 'user_id' => $user->id],
            ['rating' => (int) $data['rating'], 'comment' => $data['comment'] ?? null]
        );

        return response()->json(['status' => 'success']);
    }

    public function preferences(Request $request, TanzaniaContext $tz)
    {
        $user = $this->currentUser($request);
        if (! $user) {
            return response()->json(['status' => 'error', 'message' => 'Unauthorized'], 401);
        }

        $data = $request->validate([
            'name' => 'nullable|string|max:40',
            'preferred_language' => 'nullable|in:sw,en,auto',
            'region' => 'nullable|string|max:40',
        ]);

        $update = [];
        if (array_key_exists('name', $data)) {
            $update['name'] = $data['name'] ? trim($data['name']) : null;
        }
        if (array_key_exists('preferred_language', $data)) {
            $update['preferred_language'] = $data['preferred_language'] === 'auto' ? null :
$data['preferred_language'];
        }
        if (array_key_exists('region', $data)) {
            $loc = $data['region'] ? $tz->detectLocation($data['region']) : null;
            $update['region'] = $loc['region'] ?? null;
        }

        $user->forceFill($update)->save();

        return response()->json(['status' => 'success', 'user' => $this->userPayload($user->fresh())]);
    }

    public function logout(Request $request)
    {
        $request->session()->forget(self::SESSION_KEY);
        $request->session()->regenerate();

        return response()->json(['status' => 'success', 'csrf' => csrf_token()]);
    }

    // ---------------------------------------------------------------- helpers

    private function openSession(Request $request, User $user): void
```

```
    {
        $request->session()->regenerate();
        $request->session()->put(self::SESSION_KEY, $user->id);
        $user->forceFill(['last_seen_at' => now()])->saveQuietly();
    }

    private function currentUser(Request $request): ?User
    {
        $id = $request->session()->get(self::SESSION_KEY);
        if (! $id) {
            return null;
        }
        $user = User::query()->find($id);
        if (! $user) {
            $request->session()->forget(self::SESSION_KEY);
        }

        return $user;
    }

    private function userPayload(User $user): array
    {
        return [
            'id' => $user->id,
            'name' => $user->name,
            'display_name' => $user->displayName(),
            'phone_masked' => Phone::mask($user->phone_number),
            'preferred_language' => $user->preferred_language ?? 'auto',
            'region' => $user->region ? ucwords($user->region) : null,
            'is_banned' => $user->is_banned,
        ];
    }

    private function messagePayload(Message $m): array
    {
        return [
            'id' => $m->id,
            'direction' => $m->direction,
            'content' => $m->content,
            'category' => $m->category,
            'category_label' => $m->category ? TopicClassifier::label($m->category, $m->language ?? 'sw') : null,
            'language' => $m->language,
            'created_at' => $m->created_at?->toIso8601String(),
            'feedback' => $m->relationLoaded('feedback') && $m->feedback ? $m->feedback->rating : null,
        ];
    }

    private function publicSettings(): array
    {
        return [
            'app_name' => Settings::get('app_name'),
            'sms_keyword' => Settings::get('sms_keyword'),
            'sms_shortcode' => Settings::get('sms_shortcode'),
            'otp_enabled' => Settings::bool('otp_enabled'),
            'categories' => collect(TopicClassifier::CATEGORIES)->map(fn ($l, $k) => ['key' => $k, 'sw' =>
$l['sw'], 'en' => $l['en']])->values(),
            'regions' => array_map('ucwords', TanzaniaContext::REGIONS),
        ];
    }
}
```

19 app/Http/Middleware/EnsureAdmin.php

25 lines

```php
<?php

namespace App\Http\Middleware;

use Closure;
use Illuminate\Http\Request;
use Symfony\Component\HttpFoundation\Response;

class EnsureAdmin
{
    public function handle(Request $request, Closure $next): Response
    {
        $user = $request->user();

        if (! $user) {
            return redirect()->route('admin.login')->with('error', 'Please sign in.');
        }

        if (! $user->isAdmin()) {
            abort(403, 'Admin access only.');
        }

        return $next($request);
    }
}
```

20 app/Http/Middleware/VerifySmsWebhook.php

43 lines

```php
<?php

namespace App\Http\Middleware;

use Closure;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\Log;
use Symfony\Component\HttpFoundation\Response;

/**
 * Protects the inbound SMS callback. Africa's Talking does not sign
 * callbacks, so the callback URL registered in their dashboard must carry
 * a shared secret: https://your-host/api/sms/inbound?token=<AT_WEBHOOK_SECRET>
 * (a X-Webhook-Token header works too).
 */
class VerifySmsWebhook
{
    public function handle(Request $request, Closure $next): Response
    {
        $expected = (string) config('services.at.webhook_secret');

        if ($expected === '') {
            if (app()->environment('production')) {
                Log::critical('AT_WEBHOOK_SECRET is not set; refusing inbound SMS in production.');

                return response()->json(['status' => 'error', 'message' => 'webhook secret not configured'], 503);
            }
            Log::warning('AT_WEBHOOK_SECRET is not set; accepting inbound SMS because APP_ENV is not production.');

            return $next($request);
        }

        $given = (string) ($request->query('token') ?? $request->header('X-Webhook-Token') ?? '');

        if ($given === '' || ! hash_equals($expected, $given)) {
            Log::warning('Rejected inbound SMS callback with bad token', ['ip' => $request->ip()]);

            return response()->json(['status' => 'error', 'message' => 'unauthorized'], 401);
        }

        return $next($request);
    }
}
```

21 app/Jobs/ProcessIncomingSms.php

57 lines

```php
<?php

namespace App\Jobs;

use App\Models\Message;
use App\Models\User;
use App\Services\ConversationService;
use App\Services\SmsService;
use App\Support\Phone;
use Illuminate\Contracts\Queue\ShouldBeUnique;
use Illuminate\Contracts\Queue\ShouldQueue;
use Illuminate\Foundation\Queue\Queueable;
use Illuminate\Support\Facades\Log;

class ProcessIncomingSms implements ShouldQueue, ShouldBeUnique
{
    use Queueable;

    public int $tries = 2;

    public int $timeout = 90;

    public function __construct(
        public readonly string $from,
        public readonly string $text,
        public readonly ?string $providerMessageId = null,
    ) {
    }

    public function handle(ConversationService $conversation, SmsService $sms): void
    {
        $user = User::query()->firstOrCreate(['phone_number' => $this->from]);

        $result = $conversation->handle($user, $this->text, Message::CHANNEL_SMS);

        if (empty($result['text'])) {
            Log::info('No SMS reply', ['from' => Phone::mask($this->from), 'kind' => $result['kind']]);

            return;
        }

        $send = $sms->send($this->from, $result['text']);

        Log::info('SMS reply', [
            'from' => Phone::mask($this->from),
            'kind' => $result['kind'],
            'category' => $result['category'],
            'delivery' => $send['status'],
        ]);
    }

    /** Use the job's own key so retries do not duplicate replies. */
    public function uniqueId(): string
    {
        return $this->providerMessageId ?: md5($this->from . '|' . $this->text);
    }
}
```

22 app/Models/AiLog.php

26 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class AiLog extends Model
{
    protected $fillable = [
        'message_id',
        'model',
        'category',
        'status',
        'prompt',
        'response',
        'prompt_tokens',
        'completion_tokens',
        'total_tokens',
        'latency_ms',
    ];

    public function message()
    {
        return $this->belongsTo(Message::class);
    }
}
```

23 app/Models/CommunityInvite.php

49 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Support\Str;

class CommunityInvite extends Model
{
    use HasFactory;

    protected $fillable = [
        'community_thread_id',
        'inviter_id',
        'invitee_id',
        'email',
        'token',
        'status',
        'expires_at',
    ];

    protected $casts = [
        'expires_at' => 'datetime',
    ];

    protected static function boot()
    {
        parent::boot();
        static::creating(function ($invite) {
            $invite->token = Str::random(32);
        });
    }

    public function thread()
    {
        return $this->belongsTo(CommunityThread::class, 'community_thread_id');
    }

    public function inviter()
    {
        return $this->belongsTo(User::class, 'inviter_id');
    }

    public function invitee()
    {
        return $this->belongsTo(User::class, 'invitee_id');
    }
}
```

24 app/Models/CommunityMember.php

32 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class CommunityMember extends Model
{
    use HasFactory;

    protected $fillable = [
        'community_thread_id',
        'user_id',
        'role',
        'joined_at',
    ];

    protected $casts = [
        'joined_at' => 'datetime',
    ];

    public function thread()
    {
        return $this->belongsTo(CommunityThread::class, 'community_thread_id');
    }

    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

25 app/Models/CommunityPost.php

34 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class CommunityPost extends Model
{
    use HasFactory;

    protected $fillable = [
        'community_thread_id',
        'user_id',
        'content',
        'is_approved',
        'is_pinned',
    ];

    protected $casts = [
        'is_approved' => 'boolean',
        'is_pinned' => 'boolean',
    ];

    public function thread()
    {
        return $this->belongsTo(CommunityThread::class, 'community_thread_id');
    }

    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

26 app/Models/CommunityReaction.php

27 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class CommunityReaction extends Model
{
    use HasFactory;

    protected $fillable = [
        'community_post_id',
        'user_id',
        'type',
    ];

    public function post()
    {
        return $this->belongsTo(CommunityPost::class, 'community_post_id');
    }

    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

27 app/Models/CommunityThread.php

47 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class CommunityThread extends Model
{
    use HasFactory;

    protected $fillable = [
        'title',
        'slug',
        'description',
        'creator_id',
        'is_private',
        'is_system',
    ];

    protected $casts = [
        'is_private' => 'boolean',
        'is_system' => 'boolean',
    ];

    public function creator()
    {
        return $this->belongsTo(User::class, 'creator_id');
    }

    public function members()
    {
        return $this->belongsToMany(User::class, 'community_members')
            ->withPivot('role', 'joined_at')
            ->withTimestamps();
    }

    public function posts()
    {
        return $this->hasMany(CommunityPost::class);
    }

    public function invites()
    {
        return $this->hasMany(CommunityInvite::class);
    }
}
```

28 app/Models/Feedback.php

31 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class Feedback extends Model
{
    protected $table = 'feedback';

    protected $fillable = [
        'message_id',
        'user_id',
        'rating',
        'comment',
    ];

    protected $casts = [
        'rating' => 'integer',
    ];

    public function message()
    {
        return $this->belongsTo(Message::class);
    }

    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

29 app/Models/KnowledgeEntry.php

37 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Support\Facades\Cache;

/**
 * A curated fact block about life in Tanzania (law, health, farming,
 * government services, money, education, ...). Retrieved by keyword overlap
 * and injected into the prompt so answers are grounded in local reality.
 */
class KnowledgeEntry extends Model
{
    protected $fillable = [
        'title',
        'content',
        'summary',
        'category',
        'keywords',
        'language',
        'source',
        'is_active',
    ];

    protected $casts = [
        'keywords' => 'array',
        'is_active' => 'boolean',
    ];

    protected static function booted(): void
    {
        $bust = fn () => Cache::forget('huru.knowledge.active');
        static::saved($bust);
        static::deleted($bust);
    }
}
```

30 app/Models/LocalResource.php

41 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

/**
 * A place-bound service: a legal aid provider, hospital, police station,
 * agricultural office, district council desk, ... tied to a region and
 * district so the engine can point citizens to help near them.
 */
class LocalResource extends Model
{
    public const CATEGORIES = [
        'legal_aid' => 'Legal aid provider',
        'health' => 'Health facility',
        'police' => 'Police / gender desk',
        'agriculture' => 'Agriculture office',
        'government' => 'Government office',
        'education' => 'Education office',
        'finance' => 'Financial service',
        'other' => 'Other',
    ];

    protected $fillable = [
        'name',
        'category',
        'region',
        'district',
        'ward',
        'location',
        'phone',
        'email',
        'source',
        'is_active',
    ];

    protected $casts = [
        'is_active' => 'boolean',
    ];
}
```

31 app/Models/Message.php

38 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class Message extends Model
{
    public const DIRECTION_IN = 'inbound';
    public const DIRECTION_OUT = 'outbound';

    public const CHANNEL_SMS = 'sms';
    public const CHANNEL_WEB = 'web';

    protected $fillable = [
        'user_id',
        'direction',
        'channel',
        'content',
        'language',
        'category',
    ];

    public function user()
    {
        return $this->belongsTo(User::class);
    }

    public function aiLog()
    {
        return $this->hasOne(AiLog::class);
    }

    public function feedback()
    {
        return $this->hasOne(Feedback::class);
    }
}
```

32 app/Models/PromptTemplate.php

33 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Support\Facades\Cache;

/**
 * Persona paragraph for a (category, language) pair. The engine appends
 * context and constraints itself, so a template only describes who the
 * assistant is and how it should behave for that topic.
 */
class PromptTemplate extends Model
{
    protected $fillable = [
        'name',
        'category',
        'language',
        'template',
        'is_active',
    ];

    protected $casts = [
        'is_active' => 'boolean',
    ];

    protected static function booted(): void
    {
        $bust = fn () => Cache::forget('huru.templates.active');
        static::saved($bust);
        static::deleted($bust);
    }
}
```

33 app/Models/ReferenceContact.php

51 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Support\Facades\Cache;

/**
 * National directory entry: an institution, hotline or office that applies
 * country-wide (police 112, TRA, NIDA, NHIF, Legal aid campaign, ...).
 * Editable by admins so the engine never relies on hard-coded facts that
 * go stale.
 */
class ReferenceContact extends Model
{
    protected $fillable = [
        'name',
        'category',
        'phone',
        'alt_phone',
        'email',
        'website',
        'description_sw',
        'description_en',
        'is_emergency',
        'is_active',
        'verified_at',
        'sort_order',
    ];

    protected $casts = [
        'is_emergency' => 'boolean',
        'is_active' => 'boolean',
        'verified_at' => 'datetime',
        'sort_order' => 'integer',
    ];

    protected static function booted(): void
    {
        $bust = fn () => Cache::forget('huru.contacts.active');
        static::saved($bust);
        static::deleted($bust);
    }

    public function description(string $language): string
    {
        return $language === 'en'
            ? ($this->description_en ?: (string) $this->description_sw)
            : ($this->description_sw ?: (string) $this->description_en);
    }
}
```

34 app/Models/SystemSetting.php

12 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class SystemSetting extends Model
{
    protected $table = 'system_settings';

    protected $fillable = ['key', 'value'];
}
```

35 app/Models/User.php

100 lines

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;

class User extends Authenticatable
{
    use HasFactory, Notifiable;

    public const ROLE_USER = 'user';
    public const ROLE_ADMIN = 'admin';

    protected $fillable = [
        'name',
        'email',
        'password',
        'phone_number',
        'role',
        'preferred_language',
        'region',
        'is_banned',
        'abuse_count',
        'opted_out',
        'otp_hash',
        'otp_expires_at',
        'otp_attempts',
        'last_seen_at',
    ];

    protected $hidden = [
        'password',
        'remember_token',
        'otp_hash',
    ];

    protected function casts(): array
    {
        return [
            'email_verified_at' => 'datetime',
            'password' => 'hashed',
            'is_banned' => 'boolean',
            'opted_out' => 'boolean',
            'abuse_count' => 'integer',
            'otp_attempts' => 'integer',
            'otp_expires_at' => 'datetime',
            'last_seen_at' => 'datetime',
        ];
    }

    public function isAdmin(): bool
    {
        return $this->role === self::ROLE_ADMIN;
    }

    public function messages()
    {
        return $this->hasMany(Message::class);
    }

    public function feedback()
    {
        return $this->hasMany(Feedback::class);
    }

    public function communityMemberships()
    {
        return $this->hasMany(CommunityMember::class);
    }
```

```php

    public function joinedThreads()
    {
        return $this->belongsToMany(CommunityThread::class, 'community_members')
            ->withPivot('role', 'joined_at')
            ->withTimestamps();
    }

    public function createdThreads()
    {
        return $this->hasMany(CommunityThread::class, 'creator_id');
    }

    public function communityPosts()
    {
        return $this->hasMany(CommunityPost::class);
    }

    /** Display name that never leaks the full phone number to other citizens. */
    public function displayName(): string
    {
        if ($this->name) {
            return $this->name;
        }
        $digits = preg_replace('/\D/', '', (string) $this->phone_number);

        return 'Mwananchi ' . substr($digits, -4);
    }
}
```

36 app/Providers/AppServiceProvider.php

37 lines

```php
<?php

namespace App\Providers;

use App\Services\Ai\GeminiClient;
use App\Services\Ai\LlmClient;
use Composer\CaBundle\CaBundle;
use Illuminate\Support\Facades\Http;
use Illuminate\Support\Facades\Vite;
use Illuminate\Support\ServiceProvider;

class AppServiceProvider extends ServiceProvider
{
    public function register(): void
    {
        $this->app->singleton(LlmClient::class, function () {
            return new GeminiClient(
                apiKey: config('services.gemini.key'),
                baseUrl: config('services.gemini.url'),
                timeoutSeconds: (int) config('services.gemini.timeout', 30),
            );
        });
    }

    public function boot(): void
    {
        Vite::useBuildDirectory('build');

        // Keep TLS verification ON everywhere, even on machines (typically
        // Windows dev boxes) whose PHP has no CA bundle configured.
        // composer/ca-bundle prefers the system bundle and falls back to
        // its own up-to-date cacert.pem.
        Http::globalOptions([
            'verify' => CaBundle::getSystemCaRootBundlePath(),
        ]);
    }
}
```

37 app/Services/Ai/GeminiClient.php

131 lines

```php
<?php

namespace App\Services\Ai;

use Illuminate\Support\Facades\Http;
use Illuminate\Support\Facades\Log;

/**
 * Google Gemini generateContent client.
 *
 * - Sends the API key in a header, never in the URL (keeps it out of logs).
 * - Keeps TLS verification on.
 * - Uses a real system instruction and multi-turn contents.
 * - Normalises the response into a provider-agnostic shape.
 */
class GeminiClient implements LlmClient
{
    public function __construct(
        private readonly ?string $apiKey,
        private readonly string $baseUrl = 'https://generativelanguage.googleapis.com/v1beta/models',
        private readonly int $timeoutSeconds = 30,
    ) {
    }

    public function isConfigured(): bool
    {
        return ! empty($this->apiKey);
    }

    public function generate(string $systemInstruction, array $turns, array $options = []): array
    {
        $model = $options['model'] ?? 'gemini-flash-lite-latest';
        $base = [
            'status' => self::STATUS_ERROR,
            'text' => null,
            'model' => $model,
            'tokens' => ['prompt' => 0, 'completion' => 0, 'total' => 0],
            'latency_ms' => 0,
            'finish_reason' => null,
            'error' => null,
        ];

        if (! $this->isConfigured()) {
            Log::critical('Gemini API key missing.');

            return array_merge($base, ['status' => self::STATUS_UNCONFIGURED, 'error' => 'missing_api_key']);
        }

        $contents = [];
        foreach ($turns as $turn) {
            $contents[] = [
                'role' => $turn['role'] === 'model' ? 'model' : 'user',
                'parts' => [['text' => (string) $turn['text']]],
            ];
        }

        $payload = [
            'systemInstruction' => ['parts' => [['text' => $systemInstruction]]],
            'contents' => $contents,
            'generationConfig' => [
                'temperature' => (float) ($options['temperature'] ?? 0.4),
                'maxOutputTokens' => (int) ($options['max_tokens'] ?? 2048),
            ],
            'safetySettings' => [
                ['category' => 'HARM_CATEGORY_HARASSMENT', 'threshold' => 'BLOCK_ONLY_HIGH'],
                ['category' => 'HARM_CATEGORY_HATE_SPEECH', 'threshold' => 'BLOCK_ONLY_HIGH'],
                ['category' => 'HARM_CATEGORY_SEXUALLY_EXPLICIT', 'threshold' => 'BLOCK_MEDIUM_AND_ABOVE'],
                ['category' => 'HARM_CATEGORY_DANGEROUS_CONTENT', 'threshold' => 'BLOCK_ONLY_HIGH'],
            ],
        ];

```

```
        $started = microtime(true);

        try {
            $response = Http::timeout($this->timeoutSeconds)
                ->withHeaders([
                    'Content-Type' => 'application/json',
                    'x-goog-api-key' => $this->apiKey,
                ])
                ->retry(2, 800, throw: false)
                ->post(rtrim($this->baseUrl, '/') . '/' . rawurlencode($model) . ':generateContent', $payload);

            $latency = (int) round((microtime(true) - $started) * 1000);
            $base['latency_ms'] = $latency;

            if (! $response->successful()) {
                $body = $response->json();
                $message = $body['error']['message'] ?? ('HTTP ' . $response->status());
                Log::error('Gemini request failed', ['status' => $response->status(), 'message' => $message]);

                return array_merge($base, ['error' => $message]);
            }

            $data = $response->json();
            $usage = $data['usageMetadata'] ?? [];
            $base['tokens'] = [
                'prompt' => (int) ($usage['promptTokenCount'] ?? 0),
                'completion' => (int) ($usage['candidatesTokenCount'] ?? 0),
                'total' => (int) ($usage['totalTokenCount'] ?? 0),
            ];

            $candidate = $data['candidates'][0] ?? null;
            $finish = $candidate['finishReason'] ?? null;
            $base['finish_reason'] = $finish;

            $blockReason = $data['promptFeedback']['blockReason'] ?? null;
            if ($blockReason || $finish === 'SAFETY' || $finish === 'PROHIBITED_CONTENT') {
                return array_merge($base, ['status' => self::STATUS_SAFETY]);
            }

            $text = '';
            foreach ($candidate['content']['parts'] ?? [] as $part) {
                $text .= $part['text'] ?? '';
            }
            $text = trim($text);

            if ($text === '') {
                return array_merge($base, ['error' => 'empty_response']);
            }

            return array_merge($base, ['status' => self::STATUS_OK, 'text' => $text]);
        } catch (\Throwable $e) {
            Log::error('Gemini client exception: ' . $e->getMessage());

            return array_merge($base, [
                'latency_ms' => (int) round((microtime(true) - $started) * 1000),
                'error' => $e->getMessage(),
            ]);
        }
    }
}
```

38 app/Services/Ai/LlmClient.php

20 lines

```php
<?php

namespace App\Services\Ai;

interface LlmClient
{
    public const STATUS_OK = 'ok';
    public const STATUS_SAFETY = 'safety';
    public const STATUS_ERROR = 'error';
    public const STATUS_UNCONFIGURED = 'unconfigured';

    /**
     * @param  array<int, array{role:string, text:string}>  $turns  Ordered conversation, last item is the new user message.
     * @param  array{temperature?:float, max_tokens?:int, model?:string}  $options
     * @return array{status:string, text:?string, model:string, tokens:array{prompt:int,completion:int,total:int}, latency_ms:int, finish_reason:?string, error:?string}
     */
    public function generate(string $systemInstruction, array $turns, array $options = []): array;

    public function isConfigured(): bool;
}
```

39 app/Services/AiService.php

59 lines

```php
<?php

namespace App\Services;

use App\Services\Ai\LlmClient;
use App\Support\Settings;

/**
 * Thin orchestration between the prompt engine and the model client.
 */
class AiService
{
    public function __construct(
        private readonly PromptEngine $engine,
        private readonly LlmClient $client,
    ) {
    }

    /**
     * @param  array  $ctx  See PromptEngine::build()
     * @return array{status:string,text:?string,model:string,tokens:array,latency_ms:int,error:?string,system:string,knowledge:\Illuminate\Support\Collection}
     */
    public function answer(array $ctx): array
    {
        $built = $this->engine->build($ctx);

        $result = $this->client->generate($built['system'], $built['turns'], [
            'model' => Settings::get('ai_model'),
            'temperature' => Settings::float('ai_temperature'),
            'max_tokens' => Settings::int('ai_max_tokens'),
        ]);

        if ($result['status'] === LlmClient::STATUS_OK) {
            $result['text'] = $this->clean($result['text']);
        }

        $result['system'] = $built['system'];
        $result['knowledge'] = $built['knowledge'];

        return $result;
    }

    /** Strip markdown the model may still emit, since SMS cannot render it. */
    private function clean(string $text): string
    {
        $text = preg_replace('/\*\*?/', '', $text);
        $text = preg_replace('/^#{1,6}\s*/m', '', $text);
        $text = preg_replace('/`+/', '', $text);
        $text = preg_replace('/^\s*[-•]\s+/m', '- ', $text);
        $text = preg_replace('/\n{3,}/', "\n\n", $text);

        return trim($text);
    }

    public function isConfigured(): bool
    {
        return $this->client->isConfigured();
    }
}
```

40 app/Services/ConversationService.php

341 lines

```php
<?php

namespace App\Services;

use App\Models\AiLog;
use App\Models\Message;
use App\Models\User;
use App\Services\Ai\LlmClient;
use App\Support\Settings;
use Illuminate\Support\Facades\Log;
use Illuminate\Support\Facades\RateLimiter;
use Illuminate\Support\Str;

/**
 * One entry point for every channel. Takes a citizen and their text,
 * returns what to send back. The SMS job and the web controller both call
 * this, so behaviour (commands, moderation, memory, logging) never drifts
 * between channels.
 */
class ConversationService
{
    public const KIND_ANSWER = 'answer';
    public const KIND_COMMAND = 'command';
    public const KIND_MODERATION = 'moderation';
    public const KIND_MAINTENANCE = 'maintenance';
    public const KIND_BANNED = 'banned';
    public const KIND_IGNORED = 'ignored';
    public const KIND_RATE_LIMITED = 'rate_limited';
    public const KIND_ERROR = 'error';

    public function __construct(
        private readonly AiService $ai,
        private readonly ModerationService $moderation,
        private readonly LanguageDetector $detector,
        private readonly TopicClassifier $classifier,
        private readonly TanzaniaContext $tz,
    ) {
    }

    /**
     * @return array{kind:string, text:?string, inbound:?Message, outbound:?Message, category:?string,
language:string}
     */
    public function handle(User $user, string $text, string $channel): array
    {
        $text = trim($text);
        $user->forceFill(['last_seen_at' => now()])->saveQuietly();

        $language = $this->resolveLanguage($user, $text);
        $result = ['kind' => self::KIND_IGNORED, 'text' => null, 'inbound' => null, 'outbound' => null, 'category'
=> null, 'language' => $language];

        if ($text === '') {
            return $result;
        }

        // 1. Commands (HELP, STOP, LUGHA, JINA, MKOA ...) never hit the model.
        if ($command = $this->handleCommand($user, $text, $language, $channel)) {
            $inbound = $this->store($user, Message::DIRECTION_IN, $channel, $text, $language, null);
            $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $command, $language, null);

            return array_merge($result, ['kind' => self::KIND_COMMAND, 'text' => $command, 'inbound' => $inbound,
'outbound' => $outbound]);
        }

        // 2. Opted-out SMS users are left alone until they send START/ANZA.
        if ($user->opted_out && $channel === Message::CHANNEL_SMS) {
            return $result;
        }

        // 3. Banned users get one short notice, nothing else.
```

```
        if ($user->is_banned) {
            $msg = $this->moderation->getBanMessage($language);

            return array_merge($result, ['kind' => self::KIND_BANNED, 'text' => $msg]);
        }

        // 4. Per-user rate limit.
        $limitKey = 'huru:ask:' . $user->id;
        if (RateLimiter::tooManyAttempts($limitKey, Settings::int('rate_limit_per_minute'))) {
            $msg = $language === 'en'
                ? 'You are sending questions too quickly. Please wait a minute and try again.'
                : 'Umetuma maswali mengi kwa haraka. Tafadhali subiri dakika moja kisha ujaribu tena.';

            return array_merge($result, ['kind' => self::KIND_RATE_LIMITED, 'text' => $msg]);
        }
        RateLimiter::hit($limitKey, 60);

        $classification = $this->classifier->classify($text);
        $category = $classification['category'];
        $result['category'] = $category;

        $inbound = $this->store($user, Message::DIRECTION_IN, $channel, $text, $language, $category);
        $result['inbound'] = $inbound;

        // 5. Word-list moderation with strikes.
        if ($this->moderation->isAbusive($text)) {
            $msg = $this->strike($user, $language);
            $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $msg, $language, $category);

            return array_merge($result, ['kind' => self::KIND_MODERATION, 'text' => $msg, 'outbound' =>
$outbound]);
        }

        // 6. Maintenance switch.
        if (! Settings::bool('ai_enabled')) {
            $msg = (string) Settings::get($language === 'en' ? 'ai_maintenance_message_en' :
'ai_maintenance_message_sw');
            $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $msg, $language, $category);

            return array_merge($result, ['kind' => self::KIND_MAINTENANCE, 'text' => $msg, 'outbound' =>
$outbound]);
        }

        // 7. Ask the engine with full context.
        $location = $this->tz->detectLocation($text);
        if ($location && ! $user->region) {
            $user->forceFill(['region' => $location['region']])->saveQuietly();
        }

        $answer = $this->ai->answer([
            'question' => $text,
            'language' => $language,
            'category' => $category,
            'channel' => $channel,
            'is_emergency' => $classification['is_emergency'],
            'location' => $location,
            'history' => $this->history($user, $inbound->id),
            'user_name' => $user->name,
            'user_region' => $user->region,
        ]);

        if ($answer['status'] === LlmClient::STATUS_SAFETY) {
            $msg = $this->strike($user, $language);
            $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $msg, $language, $category);
            $this->log($outbound, $answer, $category, $text);

            return array_merge($result, ['kind' => self::KIND_MODERATION, 'text' => $msg, 'outbound' =>
$outbound]);
        }

        if ($answer['status'] !== LlmClient::STATUS_OK) {
            $msg = $language === 'en'
                ? 'Sorry, the service could not answer right now. Please try again shortly.'
```

```
                : 'Samahani, huduma haikuweza kujibu kwa sasa. Tafadhali jaribu tena baada ya muda mfupi.';
            $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $msg, $language, $category);
            $this->log($outbound, $answer, $category, $text);
            Log::warning('Engine failure', ['user' => $user->id, 'status' => $answer['status'], 'error' =>
$answer['error']]);

            return array_merge($result, ['kind' => self::KIND_ERROR, 'text' => $msg, 'outbound' => $outbound]);
        }

        $outbound = $this->store($user, Message::DIRECTION_OUT, $channel, $answer['text'], $language, $category);
        $this->log($outbound, $answer, $category, $text);

        return array_merge($result, ['kind' => self::KIND_ANSWER, 'text' => $answer['text'], 'outbound' =>
$outbound]);
    }

    // ---------------------------------------------------------------- helpers

    public function resolveLanguage(User $user, string $text): string
    {
        if (in_array($user->preferred_language, ['sw', 'en'], true)) {
            // A stored preference wins unless the citizen clearly wrote the other language.
            $detected = $this->detector->detect($text);

            return $detected ?: $user->preferred_language;
        }

        $policy = (string) Settings::get('primary_language');
        if ($policy === 'sw' || $policy === 'en') {
            return $policy;
        }

        return $this->detector->detect($text) ?: 'sw';
    }

    /** Returns the reply for a service command, or null if the text is not a command. */
    private function handleCommand(User $user, string $text, string $language, string $channel): ?string
    {
        $upper = Str::upper(trim($text));
        $parts = preg_split('/\s+/', $upper, 2);
        $verb = $parts[0] ?? '';
        $arg = trim($parts[1] ?? '');
        $app = Settings::get('app_name');

        if (in_array($verb, ['HELP', 'MSAADA', 'MENU', '?'], true) && $arg === '') {
            return $this->helpText($language);
        }

        if (in_array($verb, ['STOP', 'ACHA', 'SITAKI', 'ONDOA'], true) && $arg === '') {
            $user->forceFill(['opted_out' => true])->saveQuietly();

            return $language === 'en'
                ? "You will no longer receive replies from {$app}. Send ANZA or START any time to resume."
                : "Hutapokea tena majibu kutoka {$app}. Tuma ANZA au START wakati wowote kuendelea.";
        }

        if (in_array($verb, ['START', 'ANZA', 'RUDI'], true) && $arg === '') {
            $user->forceFill(['opted_out' => false])->saveQuietly();

            return $language === 'en'
                ? "Welcome back to {$app}. Send any question to begin."
                : "Karibu tena {$app}. Tuma swali lolote kuanza.";
        }

        if (in_array($verb, ['LUGHA', 'LANG', 'LANGUAGE'], true)) {
            $choice = match (true) {
                in_array($arg, ['SW', 'KISWAHILI', 'SWAHILI'], true) => 'sw',
                in_array($arg, ['EN', 'ENGLISH', 'KIINGEREZA'], true) => 'en',
                $arg === '' || in_array($arg, ['AUTO', 'YOYOTE'], true) => null,
                default => false,
            };
            if ($choice === false) {
                return $language === 'en'
```

```
                    ? 'Send LUGHA SW for Swahili, LUGHA EN for English, or LUGHA AUTO to detect automatically.'
                    : 'Tuma LUGHA SW kwa Kiswahili, LUGHA EN kwa Kiingereza, au LUGHA AUTO kutambua yenyewe.';
            }
            $user->forceFill(['preferred_language' => $choice])->saveQuietly();

            return match ($choice) {
                'sw' => 'Sawa. Kuanzia sasa nitakujibu kwa Kiswahili.',
                'en' => 'Done. From now on I will reply in English.',
                default => $language === 'en' ? 'Done. I will follow the language you write in.' : 'Sawa. Nitafuata
lugha unayoandika.',
            };
        }

        if (in_array($verb, ['JINA', 'NAME'], true) && $arg !== '' && strlen($arg) <= 40) {
            $name = Str::title(Str::lower($arg));
            $user->forceFill(['name' => $name])->saveQuietly();

            return $language === 'en' ? "Thank you, {$name}. I have saved your name." : "Asante, {$name}.
Nimehifadhi jina lako.";
        }

        if (in_array($verb, ['MKOA', 'REGION'], true) && $arg !== '') {
            $loc = $this->tz->detectLocation($arg);
            if (! $loc) {
                return $language === 'en'
                    ? 'I could not recognise that region. Example: MKOA Mwanza'
                    : 'Sikutambua mkoa huo. Mfano: MKOA Mwanza';
            }
            $user->forceFill(['region' => $loc['region']])->saveQuietly();
            $label = Str::title($loc['region']);

            return $language === 'en'
                ? "Saved. I will prioritise services in {$label} when you ask for help nearby."
                : "Nimehifadhi. Nitatoa kipaumbele kwa huduma za {$label} utakapoomba msaada wa karibu.";
        }

        return null;
    }

    public function helpText(string $language): string
    {
        $app = Settings::get('app_name');
        $kw = Settings::get('sms_keyword');
        $code = Settings::get('sms_shortcode');

        if ($language === 'en') {
            return "{$app}: ask anything about life in Tanzania - law, health, farming, school subjects, government
services, money, jobs. "
                . "SMS: send {$kw} followed by your question to {$code}. "
                . 'Commands: LUGHA SW/EN (language), JINA <name>, MKOA <region>, STOP to pause, START to resume.';
        }

        return "{$app}: uliza lolote kuhusu maisha Tanzania - sheria, afya, kilimo, masomo, huduma za serikali,
fedha, ajira. "
            . "SMS: tuma {$kw} kisha swali lako kwenda {$code}. "
            . 'Amri: LUGHA SW/EN (lugha), JINA <jina lako>, MKOA <mkoa wako>, ACHA kusitisha, ANZA kuendelea.';
    }

    private function strike(User $user, string $language): string
    {
        $limit = Settings::int('abuse_strike_limit');
        $user->increment('abuse_count');
        $user->refresh();

        if ($user->abuse_count >= $limit) {
            $user->forceFill(['is_banned' => true])->saveQuietly();
            Log::warning('User banned for abuse', ['user' => $user->id]);

            return $this->moderation->getBanMessage($language);
        }

        return $this->moderation->getWarningMessage($language, $user->abuse_count, $limit);
    }
```

```
278
279     /** Previous question/answer pairs for continuity, oldest first. */
280     private function history(User $user, int $excludeMessageId): array
281     {
282         $turns = Settings::int('history_turns');
283         if ($turns <= 0) {
284             return [];
285         }
286
287         $recent = Message::query()
288             ->where('user_id', $user->id)
289             ->where('id', '<', $excludeMessageId)
290             ->where('created_at', '>=', now()->subDays(2))
291             ->whereNotNull('category')
292             ->latest('id')
293             ->take($turns * 2)
294             ->get()
295             ->reverse()
296             ->values();
297
298         $history = [];
299         foreach ($recent as $m) {
300             $history[] = [
301                 'role' => $m->direction === Message::DIRECTION_IN ? 'user' : 'model',
302                 'text' => Str::limit($m->content, 600, ''),
303             ];
304         }
305
306         // Ensure we start with a user turn so the model sees valid alternation.
307         while (! empty($history) && $history[0]['role'] !== 'user') {
308             array_shift($history);
309         }
310
311         return $history;
312     }
313
314     private function store(User $user, string $direction, string $channel, string $content, string $language, ?
string $category): Message
315     {
316         return Message::create([
317             'user_id' => $user->id,
318             'direction' => $direction,
319             'channel' => $channel,
320             'content' => $content,
321             'language' => $language,
322             'category' => $category,
323         ]);
324     }
325
326     private function log(Message $outbound, array $answer, ?string $category, string $question): void
327     {
328         AiLog::create([
329             'message_id' => $outbound->id,
330             'model' => $answer['model'] ?? 'unknown',
331             'category' => $category,
332             'status' => $answer['status'] ?? 'error',
333             'prompt' => $question . "\n\n--- SYSTEM ---\n" . Str::limit($answer['system'] ?? '', 12000, '...'),
334             'response' => (string) ($answer['text'] ?? $answer['error'] ?? ''),
335             'prompt_tokens' => $answer['tokens']['prompt'] ?? 0,
336             'completion_tokens' => $answer['tokens']['completion'] ?? 0,
337             'total_tokens' => $answer['tokens']['total'] ?? 0,
338             'latency_ms' => $answer['latency_ms'] ?? 0,
339         ]);
340     }
341 }
```

41 app/Services/KnowledgeRetriever.php

157 lines

```php
<?php

namespace App\Services;

use App\Models\KnowledgeEntry;
use Illuminate\Support\Collection;
use Illuminate\Support\Facades\Cache;
use Illuminate\Support\Str;

/**
 * Scores curated knowledge entries against a question.
 *
 * Tokens are normalised and lightly stemmed for both Swahili and English
 * so that "nimekamatwa", "kukamatwa" and "kamatwa" all meet the keyword
 * "kamat". Entries in the citizen's language are preferred; other-language
 * entries are still searched at a discount so a Swahili question can draw
 * on an English-only fact block when nothing else matches.
 */
class KnowledgeRetriever
{
    private const STOP = [
        // Swahili
        'na', 'ya', 'wa', 'za', 'la', 'cha', 'vya', 'kwa', 'ni', 'si', 'je', 'nini', 'vipi', 'gani', 'wapi',
'lini', 'nani',
        'mbona', 'kama', 'jinsi', 'hii', 'hiyo', 'hili', 'hiki', 'yangu', 'wangu', 'langu', 'mimi', 'wewe', 'yeye',
'sisi',
        'wao', 'sana', 'tu', 'lakini', 'au', 'pia', 'kuhusu', 'katika', 'kutoka', 'hadi', 'sasa', 'leo', 'naomba',
'nataka',
        'nahitaji', 'tafadhali', 'asante', 'habari', 'huru', 'msaada', 'nieleze', 'niambie', 'eleza', 'fafanua',
'maana',
        'ipi', 'zipi', 'ngapi', 'kiasi', 'mara', 'bado', 'nipo', 'niko', 'ninaweza', 'naweza', 'unaweza',
'nifanye', 'nifanyeje',
        // English
        'the', 'is', 'are', 'and', 'of', 'to', 'in', 'for', 'what', 'how', 'why', 'when', 'where', 'who', 'which',
'can',
        'could', 'should', 'would', 'do', 'does', 'did', 'i', 'my', 'me', 'you', 'your', 'we', 'our', 'they',
'their', 'it',
        'this', 'that', 'these', 'with', 'from', 'about', 'please', 'help', 'need', 'want', 'explain', 'tell',
'describe',
        'define', 'have', 'has', 'been', 'was', 'were', 'will', 'not', 'but', 'or', 'also', 'there', 'here', 'now',
'a', 'an',
        'am', 'be', 'on', 'at', 'by', 'as', 'if', 'so', 'get', 'know', 'any', 'some', 'much', 'many',
        // Topic-level words that appear in almost every entry of a domain and therefore carry no signal.
        'sheria', 'kisheria', 'haki', 'legal', 'law', 'laws', 'rights', 'right', 'huduma', 'service', 'services',
        'serikali', 'government', 'tanzania', 'nchi', 'country', 'mwananchi', 'wananchi', 'citizen', 'citizens',
        'ofisi', 'office', 'taarifa', 'information', 'maelezo', 'swali', 'question',
    ];

    /**
     * @return Collection<int, KnowledgeEntry> ordered by relevance, each with ->score
     */
    public function search(string $question, string $language, ?string $category = null, int $limit = 2):
Collection
    {
        if ($limit <= 0) {
            return collect();
        }

        $queryTokens = $this->tokens($question);
        if (empty($queryTokens)) {
            return collect();
        }
        $querySet = array_flip($queryTokens);

        $entries = Cache::remember('huru.knowledge.active', 3600, fn () => KnowledgeEntry::query()
            ->where('is_active', true)
            ->get()
            ->map(function (KnowledgeEntry $e) {
                // Precompute token sets once per cache cycle.
                $e->setAttribute('_kw', array_flip($this->tokens(implode(' ', (array) $e->keywords))));
                $e->setAttribute('_title', array_flip($this->tokens($e->title)));
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```
                $e->setAttribute('_body', array_flip($this->tokens(Str::limit($e->content, 1500, ''))));

                return $e;
            }));

        $scored = $entries->map(function (KnowledgeEntry $e) use ($querySet, $language, $category) {
            $score = 0;
            foreach ($querySet as $tok => $_) {
                if (isset($e->_kw[$tok])) {
                    $score += 3;
                }
                if (isset($e->_title[$tok])) {
                    $score += 2;
                }
                if (isset($e->_body[$tok])) {
                    $score += 1;
                }
            }
            if ($score === 0) {
                return null;
            }
            if ($category && $e->category === $category) {
                $score += 2;
            }
            if ($e->language !== $language) {
                $score = (int) ceil($score * 0.6);
            }
            $e->setAttribute('score', $score);

            return $e;
        })->filter()->sortByDesc('score')->values();

        return $scored->take($limit);
    }

    /** @return string[] unique normalised, stemmed tokens */
    public function tokens(string $text): array
    {
        $text = Str::lower($text);
        $text = preg_replace('/[^\p{L}\p{N} ]+/u', ' ', $text);
        $words = preg_split('/\s+/', trim($text), -1, PREG_SPLIT_NO_EMPTY);

        $out = [];
        foreach ($words as $w) {
            if (strlen($w) < 3 || in_array($w, self::STOP, true)) {
                continue;
            }
            $out[$w] = true;
            $stem = $this->stem($w);
            if ($stem !== $w && strlen($stem) >= 3) {
                $out[$stem] = true;
            }
        }

        return array_keys($out);
    }

    /**
     * Very small bilingual stemmer. Swahili: strip subject/tense prefixes and
     * common suffixes. English: strip plural/participle suffixes.
     */
    public function stem(string $w): string
    {
        $len = strlen($w);
        if ($len <= 4) {
            return $w;
        }

        // Swahili prefixes (longest first)
        foreach (['nime', 'ume', 'ame', 'tume', 'mme', 'wame', 'nina', 'una', 'ana', 'tuna', 'mna', 'wana', 'nita',
'uta', 'ata', 'tuta', 'wata', 'nili', 'uli', 'ali', 'tuli', 'wali', 'kuna', 'kuwa', 'ku', 'wa', 'ki', 'vi', 'ma', 'mi',
'u'] as $p) {
            if (str_starts_with($w, $p) && strlen($w) - strlen($p) >= 4) {
```

```
                $w = substr($w, strlen($p));
                break;
            }
        }
        // Swahili suffixes
        foreach (['wa', 'ni', 'ji', 'ka', 'ia', 'ea'] as $s) {
            if (str_ends_with($w, $s) && strlen($w) - strlen($s) >= 4) {
                $w = substr($w, 0, -strlen($s));
                break;
            }
        }
        // English suffixes
        foreach (['ies', 'ing', 'ed', 'es', 's'] as $s) {
            if (str_ends_with($w, $s) && strlen($w) - strlen($s) >= 4) {
                $w = substr($w, 0, -strlen($s));
                if ($s === 'ies') {
                    $w .= 'y';
                }
                break;
            }
        }

        return $w;
    }
}
```

42 app/Services/LanguageDetector.php

92 lines

```php
<?php

namespace App\Services;

use Illuminate\Support\Str;

/**
 * Decides whether a message is Swahili ('sw') or English ('en').
 *
 * Strategy: weighted function-word lists first; if that is a tie, fall back
 * to a morphological heuristic (Swahili words overwhelmingly end in a vowel
 * and use characteristic prefixes). Returns null only for empty input.
 */
class LanguageDetector
{
    private const SW = [
        'na' => 1, 'ya' => 1, 'wa' => 1, 'za' => 1, 'la' => 1, 'cha' => 1, 'vya' => 1, 'kwa' => 2, 'ni' => 1, 'si' => 1,
        'nini' => 3, 'vipi' => 3, 'gani' => 3, 'wapi' => 3, 'lini' => 3, 'nani' => 3, 'mbona' => 3, 'kwanini' => 3, 'kwa nini' => 3,
        'je' => 2, 'habari' => 2, 'naomba' => 3, 'nataka' => 3, 'nahitaji' => 3, 'msaada' => 2, 'tafadhali' => 3, 'asante' => 3,
        'ninaweza' => 3, 'naweza' => 3, 'unaweza' => 3, 'nifanye' => 3, 'nifanyeje' => 3, 'kama' => 1, 'jinsi' => 2, 'hii' => 2,
        'hiyo' => 2, 'hili' => 2, 'hiki' => 2, 'yangu' => 2, 'wangu' => 2, 'langu' => 2, 'changu' => 2, 'zangu' => 2, 'yake' => 2,
        'wake' => 1, 'mimi' => 2, 'wewe' => 2, 'yeye' => 2, 'sisi' => 2, 'ninyi' => 2, 'wao' => 2, 'sana' => 2, 'tu' => 1,
        'lakini' => 2, 'au' => 1, 'pia' => 2, 'kuhusu' => 3, 'katika' => 3, 'kutoka' => 3, 'hadi' => 2, 'mpaka' => 2,
        'sasa' => 2, 'leo' => 2, 'kesho' => 2, 'jana' => 2, 'sheria' => 2, 'haki' => 1, 'mtoto' => 2, 'watu' => 2, 'mtu' => 2,
        'nchi' => 2, 'serikali' => 2, 'shilingi' => 2, 'pesa' => 2, 'kazi' => 1, 'shule' => 2, 'nyumba' => 2, 'ardhi' => 2,
        'ndoa' => 2, 'afya' => 2, 'dawa' => 2, 'polisi' => 1, 'eleza' => 3, 'nieleze' => 3, 'niambie' => 3, 'fafanua' => 3,
        'maana' => 2, 'ipi' => 2, 'zipi' => 2, 'ngapi' => 3, 'kiasi' => 2, 'mara' => 1, 'bado' => 2, 'tayari' => 2,
    ];

    private const EN = [
        'the' => 2, 'is' => 1, 'are' => 1, 'and' => 1, 'of' => 1, 'to' => 1, 'in' => 1, 'for' => 1, 'what' => 3, 'how' => 3,
        'why' => 3, 'when' => 3, 'where' => 3, 'who' => 3, 'which' => 3, 'can' => 2, 'could' => 2, 'should' => 2, 'would' => 2,
        'do' => 1, 'does' => 2, 'did' => 2, 'i' => 1, 'my' => 2, 'me' => 2, 'you' => 2, 'your' => 2, 'we' => 2, 'our' => 2,
        'they' => 2, 'their' => 2, 'it' => 1, 'this' => 2, 'that' => 2, 'these' => 2, 'with' => 2, 'from' => 2, 'about' => 3,
        'please' => 3, 'help' => 2, 'need' => 2, 'want' => 2, 'explain' => 3, 'tell' => 2, 'describe' => 3, 'define' => 3,
        'difference' => 3, 'between' => 3, 'have' => 2, 'has' => 2, 'been' => 2, 'was' => 2, 'were' => 2, 'will' => 1,
        'not' => 2, 'but' => 2, 'or' => 1, 'also' => 2, 'there' => 2, 'here' => 2, 'now' => 2, 'today' => 2, 'tomorrow' => 2,
        'law' => 2, 'rights' => 2, 'child' => 2, 'people' => 2, 'person' => 2, 'country' => 2, 'government' => 2,
        'money' => 2, 'work' => 1, 'school' => 2, 'house' => 2, 'land' => 1, 'marriage' => 2, 'health' => 2, 'medicine' => 2,
        'much' => 2, 'many' => 2, 'if' => 2, 'because' => 3, 'get' => 1, 'know' => 2, 'think' => 2, 'any' => 2, 'some' => 2,
    ];

    public function detect(string $text): ?string
    {
        $lower = Str::lower($text);
        $words = preg_split('/[^\p{L}\p{N}\']+/u', $lower, -1, PREG_SPLIT_NO_EMPTY);

        if (empty($words)) {
            return null;
        }

        $sw = 0;
```

```
        $en = 0;
        foreach ($words as $w) {
            $sw += self::SW[$w] ?? 0;
            $en += self::EN[$w] ?? 0;
        }
        if (str_contains($lower, 'kwa nini')) {
            $sw += 3;
        }

        if ($sw !== $en) {
            return $sw > $en ? 'sw' : 'en';
        }

        // Morphological tie-breaker.
        $vowelEnding = 0;
        $swPrefix = 0;
        foreach ($words as $w) {
            if (strlen($w) < 3) {
                continue;
            }
            if (preg_match('/[aeiou]$/', $w)) {
                $vowelEnding++;
            }
            if
(preg_match('/^(ku|ni|u|a|tu|m|wa|ki|vi|ma|mi|ha|si|nime|ume|ame|tume|wame|nina|una|ana|tuna|wana|nita|uta|ata)/', $w)) {
                $swPrefix++;
            }
        }
        $considered = max(1, count(array_filter($words, fn ($w) => strlen($w) >= 3)));
        $ratio = $vowelEnding / $considered;

        if ($ratio >= 0.7 || ($ratio >= 0.55 && $swPrefix / $considered >= 0.4)) {
            return 'sw';
        }
        if ($ratio <= 0.35) {
            return 'en';
        }

        return null;
    }
}
```

43 app/Services/ModerationService.php

47 lines

```php
<?php

namespace App\Services;

use Illuminate\Support\Str;

class ModerationService
{
    /** Abusive terms in Swahili and English. Matched as whole words. */
    protected array $forbiddenWords = [
        'pumbavu', 'mjinga', 'mshenzi', 'fala', 'mavi', 'mbwa wewe', 'kuma', 'mboro', 'shoga', 'malaya', 'kahaba', 'takataka',
        'fuck', 'fucking', 'shit', 'idiot', 'stupid', 'bastard', 'bitch', 'asshole', 'dick', 'pussy', 'whore', 'slut', 'nigger',
    ];

    public function isAbusive(string $text): bool
    {
        $text = Str::lower($text);

        foreach ($this->forbiddenWords as $word) {
            if (preg_match('/(?<![\p{L}\p{N}])' . preg_quote($word, '/') . '(?![\p{L}\p{N}])/u', $text)) {
                return true;
            }
        }

        return false;
    }

    public function getWarningMessage(string $language, int $abuseCount, int $limit): string
    {
        $left = max(0, $limit - $abuseCount);

        if ($language === 'sw') {
            return "ONYO: Lugha ya matusi haikubaliki kwenye huduma hii. Hili ni onyo namba {$abuseCount}. Ukiendelea utafungiwa (nafasi zilizobaki: {$left}).";
        }

        return "WARNING: Abusive language is not accepted on this service. This is warning #{$abuseCount}. Continued abuse will lead to a ban ({$left} chances left).";
    }

    public function getBanMessage(string $language): string
    {
        if ($language === 'sw') {
            return 'HUDUMA IMEFUNGWA: Umefungiwa kutumia huduma hii kwa kukiuka masharti (lugha ya matusi). Wasiliana nasi ikiwa unaamini hili ni kosa.';
        }

        return 'SERVICE BANNED: You have been blocked from this service for violating our terms (abusive language). Contact us if you believe this is a mistake.';
    }
}
```

44 app/Services/PromptEngine.php

347 lines

```php
<?php

namespace App\Services;

use App\Models\LocalResource;
use App\Models\PromptTemplate;
use App\Models\ReferenceContact;
use App\Support\Settings;
use Illuminate\Support\Collection;
use Illuminate\Support\Facades\Cache;
use Illuminate\Support\Str;

/**
 * Assembles the system instruction and conversation turns for the engine.
 *
 * Layers, in order:
 *   1. Persona for the detected topic and language (admin-editable template)
 *   2. Country context (date, currency, institutions, emergency numbers)
 *   3. Topic guidance (how this subject works in Tanzania)
 *   4. Citizen profile (name, region, detected location)
 *   5. Retrieved knowledge entries
 *   6. Local resources near the citizen
 *   7. National directory contacts relevant to the topic
 *   8. Output constraints for the channel (SMS vs web)
 */
class PromptEngine
{
    public function __construct(
        private readonly KnowledgeRetriever $retriever,
        private readonly TanzaniaContext $tz,
    ) {
    }

    /**
     * @param array{
     *   question:string, language:string, category:string, channel:string,
     *   is_emergency?:bool, location?:?array, history?:array<int,array{role:string,text:string}>,
     *   user_name?:?string, user_region?:?string
     * } $ctx
     * @return array{system:string, turns:array, knowledge:Collection, resources:Collection, contacts:Collection}
     */
    public function build(array $ctx): array
    {
        $lang = $ctx['language'] === 'en' ? 'en' : 'sw';
        $category = $ctx['category'] ?: TopicClassifier::GENERAL;
        $channel = $ctx['channel'] ?? 'web';
        $question = trim($ctx['question']);
        $location = $ctx['location'] ?? null;
        $isEmergency = (bool) ($ctx['is_emergency'] ?? false);

        $knowledge = $this->retriever->search($question, $lang, $category, Settings::int('knowledge_matches'));
        $resources = $this->findLocalResources($location, $category, $ctx['user_region'] ?? null);
        $contacts = $this->findContacts($category, $isEmergency);

        $sections = [];
        $sections[] = $this->persona($category, $lang);
        $sections[] = $this->tz->localeBlock($lang);
        $sections[] = ($lang === 'en' ? "TOPIC RULES ({$this->label($category, $lang)}):\n" : "KANUNI ZA MADA
({$this->label($category, $lang)}):\n")
            . $this->tz->categoryGuidance($category, $lang);
        $sections[] = $this->profileBlock($ctx, $location, $lang);
        $sections[] = $this->knowledgeBlock($knowledge, $lang);
        $sections[] = $this->resourcesBlock($resources, $location, $lang);
        $sections[] = $this->contactsBlock($contacts, $isEmergency, $lang);
        $sections[] = $this->constraints($lang, $channel, $isEmergency);

        $system = implode("\n\n", array_filter($sections));

        $turns = [];
        foreach ($ctx['history'] ?? [] as $turn) {
            if (! empty($turn['text'])) {
```

```
                $turns[] = ['role' => $turn['role'], 'text' => $turn['text']];
            }
        }
        $turns[] = ['role' => 'user', 'text' => $question];

        return [
            'system' => $system,
            'turns' => $turns,
            'knowledge' => $knowledge,
            'resources' => $resources,
            'contacts' => $contacts,
        ];
    }

    // ---------------------------------------------------------------- blocks

    private function persona(string $category, string $lang): string
    {
        $templates = Cache::remember('huru.templates.active', 3600, fn () => PromptTemplate::query()
            ->where('is_active', true)
            ->get()
            ->keyBy(fn (PromptTemplate $t) => "{$t->category}:{$t->language}"));

        $tpl = $templates["{$category}:{$lang}"]
            ?? $templates[TopicClassifier::GENERAL . ":{$lang}"]
            ?? null;

        $text = $tpl?->template ?: $this->defaultPersona($lang);

        return str_replace(
            ['{app_name}', '{category}'],
            [Settings::get('app_name'), $this->label($category, $lang)],
            $text
        );
    }

    private function defaultPersona(string $lang): string
    {
        return $lang === 'en'
            ? 'You are {app_name}, a trusted everyday helper for citizens of Tanzania. You answer any question, on any subject, with accurate, practical, locally relevant guidance. You speak as a knowledgeable, respectful Tanzanian neighbour would: clear, direct, warm, never condescending.'
            : 'Wewe ni {app_name}, msaidizi wa kuaminika wa kila siku kwa wananchi wa Tanzania. Unajibu swali lolote, la mada yoyote, kwa usahihi, kwa vitendo na kwa muktadha wa Tanzania. Unazungumza kama jirani Mtanzania mwenye maarifa na heshima: wazi, moja kwa moja, mkarimu, bila kudharau.';
    }

    private function profileBlock(array $ctx, ?array $location, string $lang): string
    {
        $lines = [];
        if (! empty($ctx['user_name'])) {
            $lines[] = ($lang === 'en' ? '- Name: ' : '- Jina: ') . $ctx['user_name'];
        }
        if ($location) {
            $place = array_filter([
                $location['area'] ? Str::title($location['area']) : null,
                $location['district'] ? Str::title($location['district']) : null,
                $location['region'] ? Str::title($location['region']) : null,
            ]);
            $lines[] = ($lang === 'en' ? '- Location mentioned in this question: ' : '- Mahali alipotaja kwenye swali hili: ') . implode(', ', $place);
        } elseif (! empty($ctx['user_region'])) {
            $lines[] = ($lang === 'en' ? '- Home region on file: ' : '- Mkoa wake uliohifadhiwa: ') . Str::title($ctx['user_region']);
        }
        if (empty($lines)) {
            return '';
        }

        return ($lang === 'en' ? "ABOUT THE CITIZEN:\n" : "KUHUSU MWANANCHI:\n") . implode("\n", $lines);
    }

    private function knowledgeBlock(Collection $knowledge, string $lang): string
```

```
    {
        if ($knowledge->isEmpty()) {
            return $lang === 'en'
                ? 'VERIFIED LOCAL KNOWLEDGE: none matched this question. Rely on your general knowledge of Tanzania and say clearly when something should be confirmed with the responsible office.'
                : 'MAARIFA YALIYOTHIBITISHWA: hakuna yaliyolingana na swali hili. Tumia maarifa yako ya jumla kuhusu Tanzania na sema wazi pale ambapo jambo linapaswa kuthibitishwa na ofisi husika.';
        }

        $out = $lang === 'en'
            ? "VERIFIED LOCAL KNOWLEDGE (prefer this over your own memory when they conflict):\n"
            : "MAARIFA YALIYOTHIBITISHWA (yape kipaumbele kuliko kumbukumbu yako yakipingana):\n";

        foreach ($knowledge as $e) {
            $out .= "[{$e->title}]\n" . trim($e->content) . "\n";
            if ($e->source) {
                $out .= ($lang === 'en' ? 'Source: ' : 'Chanzo: ') . $e->source . "\n";
            }
            $out .= "\n";
        }

        return trim($out);
    }

    private function resourcesBlock(Collection $resources, ?array $location, string $lang): string
    {
        if ($resources->isEmpty()) {
            return '';
        }

        $where = $location ? Str::title($location['district'] ?? $location['region']) : '';
        $out = $lang === 'en'
            ? "SERVICES NEAR THE CITIZEN" . ($where ? " ({$where})" : '') . " (mention the most relevant ones by name with their contact):\n"
            : "HUDUMA ZILIZO KARIBU NA MWANANCHI" . ($where ? " ({$where})" : '') . " (taja zinazofaa zaidi kwa jina na mawasiliano yake):\n";

        foreach ($resources as $r) {
            $out .= "- {$r->name} (" . (LocalResource::CATEGORIES[$r->category] ?? $r->category) . '; ' . Str::title($r->district ?: $r->region) . ")";
            $bits = array_filter([$r->location, $r->phone ? ('Simu/Phone: ' . $r->phone) : null, $r->email ? ('Email: ' . $r->email) : null]);
            if ($bits) {
                $out .= ': ' . implode('; ', $bits);
            }
            $out .= "\n";
        }

        return trim($out);
    }

    private function contactsBlock(Collection $contacts, bool $isEmergency, string $lang): string
    {
        if ($contacts->isEmpty()) {
            return '';
        }

        $out = $lang === 'en'
            ? "OFFICIAL NATIONAL CONTACTS (use only when they help the citizen act; do not pad ordinary answers with them):\n"
            : "MAWASILIANO RASMI YA KITAIFA (tumia tu pale yanapomsaidia mwananchi kuchukua hatua; usiyajaze kwenye majibu ya kawaida):\n";

        foreach ($contacts as $c) {
            $bits = array_filter([$c->phone, $c->alt_phone, $c->email, $c->website]);
            $out .= '- ' . $c->name . ($c->is_emergency ? ($lang === 'en' ? ' [EMERGENCY]' : ' [DHARURA]') : '')
                . ($bits ? ': ' . implode(' / ', $bits) : '')
                . ($c->description($lang) ? ' - ' . $c->description($lang) : '')
                . "\n";
        }

        if ($isEmergency) {
            $out .= $lang === 'en'
```

```
                ? "\nTHIS LOOKS LIKE AN EMERGENCY: put the correct emergency number and the single most important
immediate action in the FIRST line of your reply."
                : "\nHILI LINAONEKANA KUWA DHARURA: weka namba sahihi ya dharura na hatua moja muhimu zaidi ya
haraka kwenye MSTARI WA KWANZA wa jibu lako.";
        }

        return trim($out);
    }

    private function constraints(string $lang, string $channel, bool $isEmergency): string
    {
        $maxWords = $channel === 'sms' ? Settings::int('sms_max_words') : Settings::int('web_max_words');
        $signature = trim((string) Settings::get($lang === 'en' ? 'signature_en' : 'signature_sw'));

        if ($lang === 'en') {
            $rules = [
                'Reply ONLY in the language of the citizen\'s latest message (English here). If they switch
language, switch with them.',
                "Keep it under {$maxWords} words. " . ($channel === 'sms'
                    ? 'This is an SMS: short sentences, the most useful fact first, no lists longer than 4 items.'
                    : 'Use short paragraphs; a brief numbered list is fine for steps.'),
                'Plain text only. No markdown, no asterisks, no emoji, no headings.',
                'No greetings, no "as an assistant", no repeating the question. Start with the answer.',
                'Be honest: if you do not know or the information may have changed (fees, officials, deadlines),
say so and name the office or website that can confirm.',
                'Never invent phone numbers, prices, laws or names of officials. Use only numbers given in this
instruction or well-established public ones.',
                'Give no medical dosages, no legal representation, no political campaigning, no religious
judgement.',
                'If the citizen asked a follow-up, use the earlier conversation turns to keep continuity.',
            ];
            if ($isEmergency) {
                array_unshift($rules, 'EMERGENCY: first line must be the emergency number and the immediate
action.');
            }
            $out = "OUTPUT RULES:\n- " . implode("\n- ", $rules);
            if ($signature !== '') {
                $out .= "\n- End with exactly this line: \"{$signature}\"";
            }

            return $out;
        }

        $rules = [
            'Jibu TU kwa lugha ya ujumbe wa mwisho wa mwananchi (Kiswahili hapa). Akibadili lugha, badilika naye.',
            "Usizidi maneno {$maxWords}. " . ($channel === 'sms'
                ? 'Huu ni ujumbe wa SMS: sentensi fupi, jambo muhimu zaidi kwanza, orodha isizidi vipengele 4.'
                : 'Tumia aya fupi; orodha fupi ya namba inafaa kwa hatua.'),
            'Maandishi ya kawaida tu. Bila markdown, bila nyota (*), bila emoji, bila vichwa vya habari.',
            'Bila salamu, bila "kama msaidizi", bila kurudia swali. Anza na jibu.',
            'Kuwa mkweli: usipojua au taarifa inaweza kuwa imebadilika (ada, viongozi, tarehe za mwisho), sema
hivyo na taja ofisi au tovuti inayoweza kuthibitisha.',
            'Usizue kamwe namba za simu, bei, sheria au majina ya viongozi. Tumia tu namba zilizotolewa kwenye
maelekezo haya au zinazojulikana wazi.',
            'Usitoe dozi za dawa, usijifanye wakili, usifanye kampeni za kisiasa, usihukumu dini.',
            'Kama mwananchi ameuliza swali la kufuatilia, tumia mazungumzo ya awali kuendeleza mtiririko.',
        ];
        if ($isEmergency) {
            array_unshift($rules, 'DHARURA: mstari wa kwanza lazima uwe namba ya dharura na hatua ya haraka.');
        }
        $out = "KANUNI ZA JIBU:\n- " . implode("\n- ", $rules);
        if ($signature !== '') {
            $out .= "\n- Malizia kwa mstari huu haswa: \"{$signature}\"";
        }

        return $out;
    }

    // ---------------------------------------------------------------- lookups

    private function findLocalResources(?array $location, string $category, ?string $userRegion): Collection
    {
        $region = $location['region'] ?? $userRegion;
```

```
        if (! $region) {
            return collect();
        }

        $categoryMap = [
            'legal' => ['legal_aid', 'police'],
            'family' => ['police', 'legal_aid', 'government'],
            'health' => ['health'],
            'agriculture' => ['agriculture'],
            'government' => ['government'],
            'education' => ['education'],
            'finance' => ['finance'],
            'business' => ['finance', 'government'],
            'employment' => ['government', 'legal_aid'],
        ];
        $wanted = $categoryMap[$category] ?? null;

        $query = LocalResource::query()->where('is_active', true)
            ->whereRaw('LOWER(region) = ?', [Str::lower($region)]);
        if ($wanted) {
            $query->whereIn('category', $wanted);
        }

        $all = $query->get();
        if ($all->isEmpty()) {
            return collect();
        }

        $district = $location['district'] ?? null;
        $area = $location['area'] ?? null;

        return $all->sortByDesc(function (LocalResource $r) use ($district, $area) {
            $score = 0;
            if ($district && Str::lower((string) $r->district) === $district) {
                $score += 5;
            }
            if ($area && str_contains(Str::lower((string) $r->location . ' ' . $r->ward), $area)) {
                $score += 8;
            }

            return $score;
        })->take(4)->values();
    }

    private function findContacts(string $category, bool $isEmergency): Collection
    {
        $contacts = Cache::remember('huru.contacts.active', 3600, fn () => ReferenceContact::query()
            ->where('is_active', true)
            ->orderBy('sort_order')
            ->get());

        $related = [
            'legal' => ['legal', 'police'],
            'family' => ['family', 'police', 'legal'],
            'health' => ['health'],
            'agriculture' => ['agriculture'],
            'education' => ['education'],
            'government' => ['government'],
            'finance' => ['finance'],
            'business' => ['business', 'finance'],
            'employment' => ['employment', 'legal'],
            'technology' => ['technology'],
            'transport' => ['transport', 'police'],
            'general' => [],
        ];
        $wanted = $related[$category] ?? [];

        $selected = $contacts->filter(fn (ReferenceContact $c) => in_array($c->category, $wanted, true));

        if ($isEmergency) {
            $selected = $contacts->filter(fn (ReferenceContact $c) => $c->is_emergency)->merge($selected);
        }

```

```
340          return $selected->unique('id')->take(8)->values();
341      }
342
343      private function label(string $category, string $lang): string
344      {
345          return TopicClassifier::label($category, $lang);
346      }
347  }
```

45 app/Services/SmsService.php

93 lines

```php
<?php

namespace App\Services;

use Illuminate\Support\Facades\Http;
use Illuminate\Support\Facades\Log;

/**
 * Africa's Talking SMS over their REST API, through Laravel's HTTP client
 * so the global TLS settings, retries and timeouts apply.
 */
class SmsService
{
    protected ?string $username;
    protected ?string $apiKey;
    protected ?string $from;

    public function __construct()
    {
        $this->username = config('services.at.username');
        $this->apiKey = config('services.at.api_key');
        $this->from = config('services.at.from');
    }

    public function isConfigured(): bool
    {
        return ! empty($this->username) && ! empty($this->apiKey);
    }

    public function endpoint(): string
    {
        return $this->username === 'sandbox'
            ? 'https://api.sandbox.africastalking.com/version1/messaging'
            : 'https://api.africastalking.com/version1/messaging';
    }

    /**
     * @return array{status:string, data?:mixed, message?:string}
     */
    public function send(string $to, string $message): array
    {
        if (! $this->isConfigured()) {
            Log::warning('SMS not sent: Africa\'s Talking credentials missing.', ['to' => $this->mask($to)]);

            return ['status' => 'skipped', 'message' => 'sms_not_configured'];
        }

        $payload = ['username' => $this->username, 'to' => $to, 'message' => $message];
        if ($this->from) {
            $payload['from'] = $this->from;
        }

        try {
            $response = Http::timeout(20)
                ->retry(2, 500, throw: false)
                ->withHeaders(['apiKey' => $this->apiKey, 'Accept' => 'application/json'])
                ->asForm()
                ->post($this->endpoint(), $payload);

            $data = $response->json();
            $recipient = $data['SMSMessageData']['Recipients'][0] ?? null;
            $status = $recipient['status'] ?? ($data['SMSMessageData']['Message'] ?? 'HTTP ' . $response->status());

            if (! $response->successful() || ! $recipient || ! in_array($recipient['statusCode'] ?? 0, [100, 101, 102], true)) {
                Log::error('SMS send failed', ['to' => $this->mask($to), 'status' => $status, 'http' => $response->status()]);

                return ['status' => 'error', 'message' => (string) $status, 'data' => $data];
            }
```

```
    69
    70              Log::info('SMS sent', ['to' => $this->mask($to), 'chars' => strlen($message), 'cost' =>
$recipient['cost'] ?? null]);
    71
    72              return ['status' => 'success', 'data' => $data];
    73          } catch (\Throwable $e) {
    74              Log::error('SMS send error: ' . $e->getMessage());
    75
    76              return ['status' => 'error', 'message' => $e->getMessage()];
    77          }
    78      }
    79
    80      public function sendOtp(string $to, string $code, string $language, string $appName, int $ttlMinutes): array
    81      {
    82          $text = $language === 'en'
    83              ? "{$appName}: your verification code is {$code}. It expires in {$ttlMinutes} minutes. Do not share it
with anyone."
    84              : "{$appName}: namba yako ya uthibitisho ni {$code}. Inaisha baada ya dakika {$ttlMinutes}. Usimpe mtu
yeyote.";
    85
    86          return $this->send($to, $text);
    87      }
    88
    89      private function mask(string $phone): string
    90      {
    91          return strlen($phone) > 6 ? substr($phone, 0, 4) . '****' . substr($phone, -3) : '****';
    92      }
    93  }
```

46 app/Services/TanzaniaContext.php

238 lines

```php
<?php

namespace App\Services;

use Carbon\Carbon;
use Illuminate\Support\Str;

/**
 * Everything the engine should know about Tanzania before it answers:
 * where the citizen is, what day it is in East Africa, which currency and
 * institutions apply, and how each topic should be handled locally.
 */
class TanzaniaContext
{
    public const TIMEZONE = 'Africa/Dar_es_Salaam';

    /** The 31 regions (26 Mainland + 5 Zanzibar). */
    public const REGIONS = [
        'arusha', 'dar es salaam', 'dodoma', 'geita', 'iringa', 'kagera', 'katavi', 'kigoma', 'kilimanjaro',
        'lindi', 'manyara', 'mara', 'mbeya', 'morogoro', 'mtwara', 'mwanza', 'njombe', 'pwani', 'rukwa', 'ruvuma',
        'shinyanga', 'simiyu', 'singida', 'songwe', 'tabora', 'tanga',
        'kaskazini pemba', 'kusini pemba', 'kaskazini unguja', 'kusini unguja', 'mjini magharibi',
    ];

    /** Common aliases citizens actually type. */
    private const REGION_ALIASES = [
        'dar' => 'dar es salaam', 'dsm' => 'dar es salaam', 'bongo' => 'dar es salaam', 'coast' => 'pwani',
        'zanzibar' => 'mjini magharibi', 'unguja' => 'mjini magharibi', 'pemba' => 'kaskazini pemba',
        'moshi' => 'kilimanjaro', 'kili' => 'kilimanjaro',
    ];

    /** District (or well-known town) => region. Not exhaustive, covers what people write. */
    private const DISTRICTS = [
        // Dar es Salaam
        'ilala' => 'dar es salaam', 'kinondoni' => 'dar es salaam', 'temeke' => 'dar es salaam',
        'ubungo' => 'dar es salaam', 'kigamboni' => 'dar es salaam',
        // Arusha
        'meru' => 'arusha', 'monduli' => 'arusha', 'karatu' => 'arusha', 'longido' => 'arusha', 'ngorongoro' =>
'arusha',
        // Dodoma
        'bahi' => 'dodoma', 'chamwino' => 'dodoma', 'kondoa' => 'dodoma', 'kongwa' => 'dodoma', 'mpwapwa' =>
'dodoma', 'chemba' => 'dodoma',
        // Mwanza
        'nyamagana' => 'mwanza', 'ilemela' => 'mwanza', 'magu' => 'mwanza', 'misungwi' => 'mwanza', 'sengerema' =>
'mwanza', 'ukerewe' => 'mwanza', 'kwimba' => 'mwanza',
        // Mbeya
        'rungwe' => 'mbeya', 'kyela' => 'mbeya', 'chunya' => 'mbeya', 'mbarali' => 'mbeya', 'busokelo' => 'mbeya',
        // Kilimanjaro
        'moshi' => 'kilimanjaro', 'hai' => 'kilimanjaro', 'rombo' => 'kilimanjaro', 'same' => 'kilimanjaro',
'mwanga' => 'kilimanjaro', 'siha' => 'kilimanjaro',
        // Tanga
        'muheza' => 'tanga', 'korogwe' => 'tanga', 'lushoto' => 'tanga', 'handeni' => 'tanga', 'pangani' =>
'tanga', 'mkinga' => 'tanga', 'kilindi' => 'tanga',
        // Morogoro
        'kilosa' => 'morogoro', 'kilombero' => 'morogoro', 'ulanga' => 'morogoro', 'mvomero' => 'morogoro', 'gairo'
=> 'morogoro', 'malinyi' => 'morogoro', 'ifakara' => 'morogoro',
        // Shinyanga
        'kahama' => 'shinyanga', 'kishapu' => 'shinyanga',
        // Geita
        'chato' => 'geita', 'bukombe' => 'geita', 'mbogwe' => 'geita', 'nyanghwale' => 'geita',
        // Kagera
        'bukoba' => 'kagera', 'muleba' => 'kagera', 'karagwe' => 'kagera', 'ngara' => 'kagera', 'biharamulo' =>
'kagera', 'missenyi' => 'kagera', 'kyerwa' => 'kagera',
        // Kigoma
        'kasulu' => 'kigoma', 'kibondo' => 'kigoma', 'uvinza' => 'kigoma', 'buhigwe' => 'kigoma', 'kakonko' =>
'kigoma',
        // Mara
        'musoma' => 'mara', 'tarime' => 'mara', 'serengeti' => 'mara', 'bunda' => 'mara', 'rorya' => 'mara',
'butiama' => 'mara',
        // Tabora
        'nzega' => 'tabora', 'igunga' => 'tabora', 'urambo' => 'tabora', 'sikonge' => 'tabora', 'uyui' => 'tabora',
```

```
'kaliua' => 'tabora',
        // Iringa / Njombe
        'kilolo' => 'iringa', 'mufindi' => 'iringa', 'makete' => 'njombe', 'wangingombe' => 'njombe', 'ludewa' => 'njombe', 'makambako' => 'njombe',
        // Ruvuma
        'songea' => 'ruvuma', 'mbinga' => 'ruvuma', 'tunduru' => 'ruvuma', 'namtumbo' => 'ruvuma', 'nyasa' => 'ruvuma', 'madaba' => 'ruvuma',
        // Mtwara / Lindi
        'masasi' => 'mtwara', 'newala' => 'mtwara', 'nanyumbu' => 'mtwara', 'tandahimba' => 'mtwara',
        'kilwa' => 'lindi', 'nachingwea' => 'lindi', 'liwale' => 'lindi', 'ruangwa' => 'lindi',
        // Pwani
        'kibaha' => 'pwani', 'bagamoyo' => 'pwani', 'kisarawe' => 'pwani', 'mkuranga' => 'pwani', 'rufiji' => 'pwani', 'mafia' => 'pwani', 'kibiti' => 'pwani', 'chalinze' => 'pwani',
        // Singida / Manyara
        'manyoni' => 'singida', 'iramba' => 'singida', 'ikungi' => 'singida', 'mkalama' => 'singida', 'itigi' => 'singida',
        'babati' => 'manyara', 'hanang' => 'manyara', 'mbulu' => 'manyara', 'simanjiro' => 'manyara', 'kiteto' => 'manyara',
        // Rukwa / Katavi / Songwe
        'sumbawanga' => 'rukwa', 'nkasi' => 'rukwa', 'kalambo' => 'rukwa',
        'mpanda' => 'katavi', 'mlele' => 'katavi', 'tanganyika' => 'katavi',
        'tunduma' => 'songwe', 'momba' => 'songwe', 'ileje' => 'songwe', 'mbozi' => 'songwe', 'vwawa' => 'songwe',
        // Simiyu
        'bariadi' => 'simiyu', 'maswa' => 'simiyu', 'meatu' => 'simiyu', 'itilima' => 'simiyu', 'busega' => 'simiyu',
        // Zanzibar
        'wete' => 'kaskazini pemba', 'micheweni' => 'kaskazini pemba', 'chake chake' => 'kusini pemba', 'mkoani' => 'kusini pemba',
        'mjini' => 'mjini magharibi', 'magharibi' => 'mjini magharibi', 'stone town' => 'mjini magharibi',
    ];

    /** Well-known neighbourhoods / wards => [district, region]. Mostly Dar es Salaam, where most web traffic comes from. */
    private const AREAS = [
        'sinza' => ['ubungo', 'dar es salaam'], 'mwenge' => ['kinondoni', 'dar es salaam'], 'mbezi' => ['ubungo', 'dar es salaam'],
        'tegeta' => ['kinondoni', 'dar es salaam'], 'kawe' => ['kinondoni', 'dar es salaam'], 'mikocheni' => ['kinondoni', 'dar es salaam'],
        'masaki' => ['kinondoni', 'dar es salaam'], 'oysterbay' => ['kinondoni', 'dar es salaam'], 'kijitonyama' => ['kinondoni', 'dar es salaam'],
        'magomeni' => ['kinondoni', 'dar es salaam'], 'tandale' => ['kinondoni', 'dar es salaam'], 'manzese' => ['ubungo', 'dar es salaam'],
        'mabibo' => ['ubungo', 'dar es salaam'], 'kimara' => ['ubungo', 'dar es salaam'], 'kibamba' => ['ubungo', 'dar es salaam'],
        'goba' => ['ubungo', 'dar es salaam'], 'makongo' => ['ubungo', 'dar es salaam'], 'mburahati' => ['ubungo', 'dar es salaam'],
        'kariakoo' => ['ilala', 'dar es salaam'], 'upanga' => ['ilala', 'dar es salaam'], 'buguruni' => ['ilala', 'dar es salaam'],
        'tabata' => ['ilala', 'dar es salaam'], 'segerea' => ['ilala', 'dar es salaam'], 'ukonga' => ['ilala', 'dar es salaam'],
        'gongo la mboto' => ['ilala', 'dar es salaam'], 'kinyerezi' => ['ilala', 'dar es salaam'], 'vingunguti' => ['ilala', 'dar es salaam'],
        'mbagala' => ['temeke', 'dar es salaam'], 'tandika' => ['temeke', 'dar es salaam'], 'kurasini' => ['temeke', 'dar es salaam'],
        'changombe' => ['temeke', 'dar es salaam'], 'chamazi' => ['temeke', 'dar es salaam'], 'kigamboni' => ['kigamboni', 'dar es salaam'],
        'posta' => ['ilala', 'dar es salaam'], 'mnazi mmoja' => ['ilala', 'dar es salaam'],
        'sakina' => ['arusha', 'arusha'], 'njiro' => ['arusha', 'arusha'], 'ngaramtoni' => ['arusha', 'arusha'],
        'nyakato' => ['ilemela', 'mwanza'], 'pasiansi' => ['ilemela', 'mwanza'], 'igoma' => ['nyamagana', 'mwanza'],
        'uyole' => ['mbeya', 'mbeya'], 'mwanjelwa' => ['mbeya', 'mbeya'], 'majengo' => ['moshi', 'kilimanjaro'],
        'mtumba' => ['dodoma', 'dodoma'], 'nzuguni' => ['dodoma', 'dodoma'],
    ];

    /**
     * Detect region / district / area mentioned in free text.
     *
     * @return array{region:?string,district:?string,area:?string}|null
     */
    public function detectLocation(string $text): ?array
    {
        $norm = ' ' . trim(preg_replace('/\s+/', ' ', preg_replace('/[^\p{L}\p{N} ]+/u', ' ', Str::lower($text)))) . ' ';
```

```
114
115         $result = ['region' => null, 'district' => null, 'area' => null];
116
117         foreach (self::AREAS as $area => [$district, $region]) {
118             if (str_contains($norm, " {$area} ")) {
119                 $result = ['region' => $region, 'district' => $district, 'area' => $area];
120                 break;
121             }
122         }
123
124         if (! $result['district']) {
125             foreach (self::DISTRICTS as $district => $region) {
126                 if (str_contains($norm, " {$district} ")) {
127                     $result['district'] = $district;
128                     $result['region'] = $region;
129                     break;
130                 }
131             }
132         }
133
134         if (! $result['region']) {
135             foreach (self::REGIONS as $region) {
136                 if (str_contains($norm, " {$region} ")) {
137                     $result['region'] = $region;
138                     break;
139                 }
140             }
141         }
142
143         if (! $result['region']) {
144             foreach (self::REGION_ALIASES as $alias => $region) {
145                 if (str_contains($norm, " {$alias} ")) {
146                     $result['region'] = $region;
147                     break;
148                 }
149             }
150         }
151
152         return $result['region'] ? $result : null;
153     }
154
155     public function now(): Carbon
156     {
157         return Carbon::now(self::TIMEZONE);
158     }
159
160     /** Country facts the engine must assume, in the citizen's language. */
161     public function localeBlock(string $language): string
162     {
163         $now = $this->now();
164
165         if ($language === 'en') {
166             return "COUNTRY CONTEXT (assume these facts):\n"
167                 . '- Country: United Republic of Tanzania (Mainland and Zanzibar). Today is ' . $now->format('l, j
F Y') . ', time ' . $now->format('H:i') . " East Africa Time.\n"
168                 . "- Capital: Dodoma (seat of government and parliament). Commercial capital and largest city: Dar
es Salaam.\n"
169                 . "- Currency: Tanzanian Shilling (TZS). Never quote prices in other currencies unless asked.\n"
170                 . "- Languages: Swahili (national) and English (official). Many citizens read Swahili more
comfortably.\n"
171                 . "- Mobile money (M-Pesa, Tigo Pesa/Mixx by Yas, Airtel Money, HaloPesa) is the everyday payment
method. Government payments go through GePG control numbers.\n"
172                 . "- Local government chain: Village/Mtaa Executive Officer, Ward Executive Officer (WEO), District
Council/Commissioner, Regional Commissioner. Most everyday problems start at the ward office.\n"
173                 . "- Legal system: common law with the Constitution of 1977; primary courts, district courts, High
Court, Court of Appeal. Ward tribunals and District Land and Housing Tribunals handle local disputes.\n"
174                 . "- Emergency numbers: Police 112, Fire 114, Ambulance 115, Child helpline 116.\n";
175         }
176
177         return "MUKTADHA WA NCHI (chukulia haya kuwa ukweli):\n"
178             . '- Nchi: Jamhuri ya Muungano wa Tanzania (Bara na Zanzibar). Leo ni ' . $this->swahiliDate($now) . ',
saa ' . $now->format('H:i') . " saa za Afrika Mashariki.\n"
179             . "- Makao makuu: Dodoma (Serikali na Bunge). Jiji kubwa la kibiashara: Dar es Salaam.\n"
```

```
            . "- Sarafu: Shilingi ya Tanzania (TZS). Usitaje bei kwa sarafu nyingine isipokuwa ukiulizwa.\n"
            . "- Lugha: Kiswahili (lugha ya taifa) na Kiingereza (lugha rasmi). Wananchi wengi husoma Kiswahili kwa urahisi zaidi.\n"
            . "- Malipo ya kila siku ni kwa pesa za simu (M-Pesa, Mixx by Yas/Tigo Pesa, Airtel Money, HaloPesa). Malipo ya serikali hufanywa kwa namba ya kumbukumbu (control number) ya GePG.\n"
            . "- Ngazi za serikali za mitaa: Mtendaji wa Kijiji/Mtaa, Mtendaji wa Kata (WEO), Halmashauri/Mkuu wa Wilaya, Mkuu wa Mkoa. Matatizo mengi ya kila siku huanzia ofisi ya kata.\n"
            . "- Mfumo wa sheria: Katiba ya 1977; mahakama za mwanzo, za wilaya, Mahakama Kuu, Mahakama ya Rufani. Mabaraza ya kata na Mabaraza ya Ardhi na Nyumba ya Wilaya hushughulikia migogoro ya karibu.\n"
            . "- Namba za dharura: Polisi 112, Zimamoto 114, Gari la wagonjwa 115, Msaada kwa mtoto 116.\n";
    }

    /** Topic-specific handling rules, grounded in how things work in Tanzania. */
    public function categoryGuidance(string $category, string $language): string
    {
        $sw = [
            'legal' => "- Rejea sheria za Tanzania (Katiba 1977, Sheria ya Ndoa 1971, Sheria ya Ardhi Sura 113, Sheria ya Ajira na Mahusiano Kazini Sura 366, Sheria ya Mwenendo wa Makosa ya Jinai, Sheria ya Mtoto 2009). Toa hatua za vitendo: wapi pa kuanzia (mtendaji wa kata, baraza la kata, polisi, dawati la jinsia, baraza la ardhi, mahakama ya mwanzo). Kama hali ni nzito mshauri aonane na mtoa msaada wa kisheria au wakili. Usijifanye wakili wake; eleza kuwa haya ni maelezo ya jumla.",
            'health' => "- Wewe si daktari. Toa maelezo ya afya ya jumla yanayokubalika Tanzania (MoH, WHO), dalili za hatari zinazohitaji kwenda zahanati/hospitali MARA MOJA, na huduma zilizopo (zahanati ya kata, kituo cha afya, hospitali ya wilaya/mkoa, NHIF/CHF iliyoboreshwa). Usitoe dozi za dawa za kuandikiwa. Kwa dharura mwambie apige 115 au aende kituo cha afya cha karibu.",
            'agriculture' => "- Zingatia misimu ya Tanzania (Vuli Okt-Des na Masika Mac-Mei kaskazini na pwani; msimu mmoja Nov-Apr katikati na kusini), udongo na mazao ya eneo husika. Mshauri atumie afisa ugani wa kata/wilaya, TARI, TOSCI kwa mbegu, na TMA kwa hali ya hewa. Taja bei kwa TZS na usizue bei usizozijua.",
            'education' => "- Fuata mfumo wa Tanzania: shule ya msingi (Darasa 1-7, mtihani wa PSLE), sekondari (Kidato 1-4 CSEE, 5-6 ACSEE), VETA, vyuo. NECTA ndiyo baraza la mitihani, HESLB ndiyo bodi ya mikopo, TCU/NACTVET husajili vyuo. Unapofundisha somo, eleza kwa hatua rahisi na mfano wa Kitanzania.",
            'government' => "- Eleza hatua halisi za kupata huduma: ofisi gani (NIDA, RITA, Uhamiaji, BRELA, TRA, TANESCO, Halmashauri), nyaraka zinazohitajika, na kuwa malipo ya serikali hulipwa kwa control number ya GePG. Usizue ada; sema 'thibitisha ada ofisini' kama hujui. Hakikisha hutoi maoni ya kisiasa.",
            'finance' => "- Tumia TZS. Eleza TRA (TIN, kodi ya mapato, VAT), BoT, benki na SACCOS, pesa za simu na gharama zake. Onya kuhusu utapeli (ujumbe wa 'umeshinda', mikopo ya haraka ya mtandaoni, piramidi). Usishauri uwekezaji maalum; toa kanuni za jumla.",
            'business' => "- Eleza usajili (BRELA kwa jina/kampuni, leseni ya biashara ya halmashauri, TIN ya TRA, TBS/TMDA kwa bidhaa), na hatua za vitendo za kuanzisha au kukuza biashara ndogo nchini Tanzania. Taja fursa halisi kama SIDO na mifuko ya halmashauri ya vijana/wanawake/walemavu (10%).",
            'employment' => "- Rejea Sheria ya Ajira na Mahusiano Kazini (Sura 366): mkataba wa maandishi, saa 45 kwa wiki, likizo ya mwaka siku 28, uzazi siku 84, utaratibu wa kuachishwa kazi. Mshauri kuhusu CMA (Tume ya Usuluhishi na Uamuzi), ofisi ya kazi ya wilaya, NSSF/PSSSF. Onya kuhusu matapeli wa ajira nje ya nchi.",
            'family' => "- Kuwa mpole na mwenye heshima. Kwa ukatili wa kijinsia au kwa mtoto, taja dawati la jinsia la polisi, ustawi wa jamii wa halmashauri, na namba 116. Rejea Sheria ya Mtoto 2009 na Sheria ya Ndoa 1971. Usihukumu; toa hatua salama za vitendo.",
            'technology' => "- Eleza kwa lugha rahisi. Kwa laini za simu na udukuzi taja TCRA na huduma za mtandao husika (Vodacom, Yas, Airtel, Halotel, TTCL). Onya kuhusu kutoa PIN, OTP au nywila kwa mtu yeyote.",
            'transport' => "- Rejea Sheria ya Usalama Barabarani, TRA (usajili wa magari), LATRA (nauli na mabasi), TANROADS. Faini za trafiki hulipwa kwa control number, si fedha taslimu kwa askari.",
            'general' => "- Jibu swali lolote kwa uaminifu na kwa muktadha wa Tanzania. Kama swali linahusu somo la shule, lifundishe kwa hatua. Kama ni la kawaida, jibu kwa ufupi na kwa heshima.",
        ];

        $en = [
            'legal' => "- Cite Tanzanian law (Constitution 1977, Law of Marriage Act 1971, Land Act Cap 113, Employment and Labour Relations Act Cap 366, Criminal Procedure Act, Law of the Child Act 2009). Give practical steps: where to start (ward executive, ward tribunal, police, gender desk, land tribunal, primary court). For serious matters advise seeing a legal aid provider or advocate. Make clear this is general information, not legal representation.",
            'health' => "- You are not a doctor. Give generally accepted health information (Ministry of Health, WHO), list danger signs that need a facility IMMEDIATELY, and name the care available (ward dispensary, health centre, district/regional hospital, NHIF/improved CHF). Do not give prescription dosages. For emergencies say call 115 or go to the nearest facility.",
            'agriculture' => "- Respect Tanzanian seasons (short rains Oct-Dec and long rains Mar-May in the north and coast; single season Nov-Apr in central and southern highlands), local soils and crops. Point to the ward/district extension officer, TARI, TOSCI for seed, TMA for weather. Quote prices in TZS and never invent prices.",
            'education' => "- Follow the Tanzanian system: primary (Standard 1-7, PSLE), secondary (Form 1-4 CSEE, Form 5-6 ACSEE), VETA, colleges. NECTA runs exams, HESLB gives student loans, TCU/NACTVET accredit institutions. When teaching a subject, explain step by step with a Tanzanian example.",
            'government' => "- Explain the real procedure: which office (NIDA, RITA, Immigration, BRELA, TRA, TANESCO, District Council), documents needed, and that government fees are paid via a GePG control number. Never invent fees; say 'confirm the fee at the office' if unknown. Give no political opinions.",
            'finance' => "- Use TZS. Explain TRA (TIN, income tax, VAT), Bank of Tanzania, banks and SACCOS, mobile
```

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money and its charges. Warn about scams ('you have won' messages, instant online loans, pyramid schemes). Do not recommend specific investments; give general principles.",
            'business' => "- Explain registration (BRELA for business name/company, district business licence, TRA TIN, TBS/TMDA for products) and practical steps to start or grow a small business in Tanzania. Mention real opportunities such as SIDO and the council 10% loans for youth, women and persons with disabilities.",
            'employment' => "- Cite the Employment and Labour Relations Act (Cap 366): written contract, 45-hour week, 28 days annual leave, 84 days maternity, fair termination procedure. Point to CMA (Commission for Mediation and Arbitration), the district labour office, NSSF/PSSSF. Warn about overseas job scams.",
            'family' => "- Be gentle and respectful. For gender-based violence or harm to a child, name the police gender desk, the council social welfare officer and the 116 helpline. Cite the Law of the Child Act 2009 and Law of Marriage Act 1971. Do not judge; give safe practical steps.",
            'technology' => "- Explain in plain language. For SIM cards and hacking mention TCRA and the relevant network (Vodacom, Yas, Airtel, Halotel, TTCL). Warn never to share PIN, OTP or passwords with anyone.",
            'transport' => "- Cite the Road Traffic Act, TRA (vehicle registration), LATRA (fares and buses), TANROADS. Traffic fines are paid via a control number, never cash to an officer.",
            'general' => "- Answer any question honestly within the Tanzanian context. If it is a school subject, teach it step by step. If it is everyday life, answer briefly and respectfully.",
        ];

        $table = $language === 'en' ? $en : $sw;

        return $table[$category] ?? $table['general'];
    }

    public static function regionLabel(?string $region): ?string
    {
        return $region ? Str::title($region) : null;
    }

    private function swahiliDate(Carbon $date): string
    {
        $days = ['Monday' => 'Jumatatu', 'Tuesday' => 'Jumanne', 'Wednesday' => 'Jumatano', 'Thursday' => 'Alhamisi', 'Friday' => 'Ijumaa', 'Saturday' => 'Jumamosi', 'Sunday' => 'Jumapili'];
        $months = [1 => 'Januari', 'Februari', 'Machi', 'Aprili', 'Mei', 'Juni', 'Julai', 'Agosti', 'Septemba', 'Oktoba', 'Novemba', 'Desemba'];

        return $days[$date->format('l')] . ', ' . $date->day . ' ' . $months[$date->month] . ' ' . $date->year;
    }
}
```

47 app/Services/TopicClassifier.php

266 lines

```php
<?php

namespace App\Services;

use Illuminate\Support\Str;

/**
 * Lightweight bilingual (Swahili / English) topic classifier.
 *
 * It scores a question against weighted keyword maps for each domain so the
 * engine can pick the right persona, the right knowledge entries and the
 * right contacts. It also flags emergencies so safety numbers are surfaced
 * first. No network call, runs in well under a millisecond.
 */
class TopicClassifier
{
    public const GENERAL = 'general';

    /** @var array<string, array{sw:string,en:string}> */
    public const CATEGORIES = [
        'legal' => ['sw' => 'Sheria na Haki', 'en' => 'Law and Rights'],
        'health' => ['sw' => 'Afya', 'en' => 'Health'],
        'agriculture' => ['sw' => 'Kilimo, Mifugo na Uvuvi', 'en' => 'Agriculture, Livestock and Fishing'],
        'education' => ['sw' => 'Elimu', 'en' => 'Education'],
        'government' => ['sw' => 'Huduma za Serikali', 'en' => 'Government Services'],
        'finance' => ['sw' => 'Fedha na Kodi', 'en' => 'Money and Tax'],
        'business' => ['sw' => 'Biashara na Ujasiriamali', 'en' => 'Business and Entrepreneurship'],
        'employment' => ['sw' => 'Ajira na Kazi', 'en' => 'Jobs and Employment'],
        'family' => ['sw' => 'Familia na Jamii', 'en' => 'Family and Community'],
        'technology' => ['sw' => 'Teknolojia na Mawasiliano', 'en' => 'Technology and Communication'],
        'transport' => ['sw' => 'Usafiri na Usalama Barabarani', 'en' => 'Transport and Road Safety'],
        'general' => ['sw' => 'Maswali ya Jumla', 'en' => 'General Questions'],
    ];

    /**
     * Keyword => weight. Keywords are matched as whole words after
     * normalisation; multi-word keywords are matched as substrings.
     *
     * @var array<string, array<string,int>>
     */
    private const KEYWORDS = [
        'legal' => [
            // Swahili
            'sheria' => 3, 'kisheria' => 3, 'haki' => 2, 'katiba' => 3, 'mahakama' => 3, 'hakimu' => 3, 'jaji' =>
3,
            'wakili' => 3, 'kesi' => 3, 'mashtaka' => 3, 'kukamatwa' => 3, 'dhamana' => 3, 'polisi' => 2, 'jela' =>
2,
            'gereza' => 2, 'mkataba' => 2, 'talaka' => 3, 'ndoa' => 2, 'mirathi' => 3, 'urithi' => 3, 'wosia' => 3,
            'ardhi' => 2, 'hati' => 2, 'mgogoro' => 2, 'dhuluma' => 3, 'kudhulumiwa' => 3, 'nimedhulumiwa' => 3,
            'unyanyasaji' => 3, 'ubakaji' => 3, 'wizi' => 2, 'msaada wa kisheria' => 4, 'baraza la kata' => 3,
            'baraza la ardhi' => 3, 'mahari' => 2, 'kosa' => 1, 'faini' => 2, 'rushwa' => 3, 'takukuru' => 3,
            'mtuhumiwa' => 3, 'hatia' => 2, 'ushahidi' => 2, 'kifungo' => 3, 'fidia' => 2, 'kuhukumiwa' => 3,
            // English
            'law' => 3, 'legal' => 3, 'rights' => 2, 'constitution' => 3, 'court' => 3, 'judge' => 3, 'lawyer' =>
3,
            'advocate' => 3, 'case' => 2, 'arrest' => 3, 'arrested' => 3, 'bail' => 3, 'police' => 2, 'prison' =>
2,
            'contract' => 2, 'divorce' => 3, 'marriage' => 2, 'inheritance' => 3, 'will' => 1, 'land' => 2,
            'title deed' => 3, 'dispute' => 2, 'harassment' => 3, 'rape' => 3, 'theft' => 2, 'legal aid' => 4,
            'corruption' => 3, 'suspect' => 2, 'evidence' => 2, 'sentence' => 2, 'compensation' => 2, 'tenant' =>
2,
            'landlord' => 2, 'eviction' => 3, 'custody' => 3,
        ],
        'health' => [
            'afya' => 3, 'hospitali' => 3, 'zahanati' => 3, 'daktari' => 3, 'dawa' => 3, 'ugonjwa' => 3, 'mgonjwa'
=> 3,
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            'ferry' => 3, 'port' => 1, 'fare' => 3, 'ticket' => 2, 'vehicle' => 2, 'registration card' => 3,
        ],
    ];

    /** Patterns that indicate a life-threatening or urgent situation. */
    private const EMERGENCY = [
        'dharura', 'emergency', 'ajali', 'accident', 'moto', 'fire', 'anakufa', 'dying', 'amezimia', 'unconscious',
        'kujiua', 'suicide', 'nataka kufa', 'kill myself', 'ananipiga sasa', 'anataka kuniua', 'damu nyingi',
'bleeding',
        'sumu', 'poison', 'nyoka ameniuma', 'snake bite', 'kupumua' , 'breathe', 'mafuriko', 'flood', 'amepotea',
'missing child',
        'mtoto amepotea', 'kutekwa', 'kidnapped', 'wezi wameingia', 'robbery', 'haraka sana', 'help me now',
    ];

    /**
     * @return array{category:string, confidence:float, scores:array<string,int>, is_emergency:bool}
     */
    public function classify(string $text): array
    {
        $normalized = $this->normalize($text);
        $words = array_filter(explode(' ', $normalized));
        $wordSet = array_flip($words);

        $scores = [];
        foreach (self::KEYWORDS as $category => $keywords) {
            $score = 0;
            foreach ($keywords as $keyword => $weight) {
                if (str_contains($keyword, ' ') || str_contains($keyword, '-') || str_contains($keyword, "'")) {
                    if (str_contains($normalized, $keyword)) {
                        $score += $weight + 1; // phrases are strong signals
                    }
                } elseif (isset($wordSet[$keyword])) {
                    $score += $weight;
                }
            }
            if ($score > 0) {
                $scores[$category] = $score;
            }
        }

        arsort($scores);
```

```
        $top = array_key_first($scores);
        $topScore = $top ? $scores[$top] : 0;
        $total = array_sum($scores) ?: 1;

        $category = ($top && $topScore >= 2) ? $top : self::GENERAL;
        $confidence = $top ? round($topScore / $total, 2) : 0.0;

        return [
            'category' => $category,
            'confidence' => $confidence,
            'scores' => $scores,
            'is_emergency' => $this->isEmergency($normalized),
        ];
    }

    public function isEmergency(string $normalizedText): bool
    {
        foreach (self::EMERGENCY as $needle) {
            if (str_contains($normalizedText, $needle)) {
                return true;
            }
        }

        return false;
    }

    public static function label(string $category, string $language = 'sw'): string
    {
        $labels = self::CATEGORIES[$category] ?? self::CATEGORIES[self::GENERAL];

        return $labels[$language === 'en' ? 'en' : 'sw'];
    }

    public static function keys(): array
    {
        return array_keys(self::CATEGORIES);
    }

    private function normalize(string $text): string
    {
        $text = Str::lower($text);
        $text = preg_replace('/[^\p{L}\p{N}\'\- ]+/u', ' ', $text);
        $text = preg_replace('/\s+/', ' ', $text);

        return ' ' . trim($text) . ' ';
    }
}
```

48 app/Support/Phone.php

55 lines

```php
<?php

namespace App\Support;

class Phone
{
    /**
     * Normalise a phone number to E.164, assuming Tanzania (+255) when no
     * country code is given. Returns null when the input cannot be a number.
     */
    public static function normalize(?string $input): ?string
    {
        if ($input === null) {
            return null;
        }
        $raw = preg_replace('/[^\d+]/', '', trim($input));
        if ($raw === '' || $raw === '+') {
            return null;
        }

        if (str_starts_with($raw, '+')) {
            $digits = substr($raw, 1);
        } elseif (str_starts_with($raw, '00')) {
            $digits = substr($raw, 2);
        } elseif (str_starts_with($raw, '255')) {
            $digits = $raw;
        } elseif (str_starts_with($raw, '0')) {
            $digits = '255' . substr($raw, 1);
        } elseif (strlen($raw) === 9 && in_array($raw[0], ['6', '7'], true)) {
            $digits = '255' . $raw;
        } else {
            $digits = $raw;
        }

        if (! preg_match('/^\d{10,15}$/', $digits)) {
            return null;
        }

        return '+' . $digits;
    }

    public static function isTanzanian(string $e164): bool
    {
        return (bool) preg_match('/^\+255[67]\d{8}$/', $e164);
    }

    public static function mask(?string $e164): string
    {
        if (! $e164 || strlen($e164) < 7) {
            return '****';
        }

        return substr($e164, 0, 4) . ' *** ' . substr($e164, -3);
    }
}
```

49 app/Support/Settings.php

222 lines

```php
<?php

namespace App\Support;

use App\Models\SystemSetting;
use Illuminate\Support\Facades\Cache;

/**
 * Typed, cached, whitelisted access to system_settings.
 *
 * Only keys declared in schema() can be stored. The admin settings form,
 * the validation rules and the defaults are all driven by this schema so
 * there is a single source of truth.
 */
class Settings
{
    private const CACHE_KEY = 'huru.settings.all';

    /**
     * @return array<string,
array{group:string,label:string,type:string,default:mixed,rules:string,hint?:string,options?:array}>
     */
    public static function schema(): array
    {
        return [
            // ---- Branding -------------------------------------------------
            'app_name' => [
                'group' => 'Branding', 'label' => 'Service name', 'type' => 'text',
                'default' => 'Huru SMS', 'rules' => 'required|string|max:40',
                'hint' => 'Shown in replies, the web chat and SMS help text.',
            ],
            'sms_keyword' => [
                'group' => 'Branding', 'label' => 'SMS keyword', 'type' => 'text',
                'default' => 'HURU', 'rules' => 'required|alpha|max:15',
                'hint' => 'Word citizens send first. Must match the Africa\'s Talking keyword.',
            ],
            'sms_shortcode' => [
                'group' => 'Branding', 'label' => 'SMS shortcode', 'type' => 'text',
                'default' => '15054', 'rules' => 'required|string|max:10',
            ],
            'signature_sw' => [
                'group' => 'Branding', 'label' => 'Reply sign-off (Swahili)', 'type' => 'text',
                'default' => '', 'rules' => 'nullable|string|max:120',
                'hint' => 'Optional short line appended to every reply. Leave empty for none.',
            ],
            'signature_en' => [
                'group' => 'Branding', 'label' => 'Reply sign-off (English)', 'type' => 'text',
                'default' => '', 'rules' => 'nullable|string|max:120',
            ],

            // ---- Engine ---------------------------------------------------
            'ai_enabled' => [
                'group' => 'Engine', 'label' => 'Engine enabled', 'type' => 'boolean',
                'default' => '1', 'rules' => 'required|in:0,1',
                'hint' => 'Turn off to pause replies during maintenance or when budget is exhausted.',
            ],
            'ai_maintenance_message_sw' => [
                'group' => 'Engine', 'label' => 'Maintenance message (Swahili)', 'type' => 'textarea',
                'default' => 'Huduma inafanyiwa matengenezo kwa muda mfupi. Tafadhali jaribu tena baadaye.',
                'rules' => 'required|string|max:300',
            ],
            'ai_maintenance_message_en' => [
                'group' => 'Engine', 'label' => 'Maintenance message (English)', 'type' => 'textarea',
                'default' => 'The service is under brief maintenance. Please try again later.',
                'rules' => 'required|string|max:300',
            ],
            'ai_model' => [
                'group' => 'Engine', 'label' => 'Gemini model', 'type' => 'text',
                'default' => 'gemini-flash-lite-latest', 'rules' => 'required|string|max:80',
                'hint' => 'Any generateContent-capable Gemini model id.',
            ],
```

```
            'ai_temperature' => [
                'group' => 'Engine', 'label' => 'Temperature', 'type' => 'number',
                'default' => '0.4', 'rules' => 'required|numeric|min:0|max:2',
                'hint' => 'Lower is more factual. 0.3 to 0.5 is recommended for public guidance.',
            ],
            'ai_max_tokens' => [
                'group' => 'Engine', 'label' => 'Max output tokens', 'type' => 'number',
                'default' => '2048', 'rules' => 'required|integer|min:64|max:8192',
            ],
            'history_turns' => [
                'group' => 'Engine', 'label' => 'Conversation memory (turns)', 'type' => 'number',
                'default' => '4', 'rules' => 'required|integer|min:0|max:12',
                'hint' => 'How many previous question/answer pairs are given to the engine for follow-up
questions.',
            ],
            'knowledge_matches' => [
                'group' => 'Engine', 'label' => 'Knowledge entries per reply', 'type' => 'number',
                'default' => '2', 'rules' => 'required|integer|min:0|max:5',
            ],

            // ---- Channels -----------------------------------------------
            'primary_language' => [
                'group' => 'Channels', 'label' => 'Language policy', 'type' => 'select',
                'default' => 'auto', 'rules' => 'required|in:auto,sw,en',
                'options' => ['auto' => 'Detect automatically', 'sw' => 'Always Swahili', 'en' => 'Always
English'],
            ],
            'sms_max_words' => [
                'group' => 'Channels', 'label' => 'SMS reply length (words)', 'type' => 'number',
                'default' => '120', 'rules' => 'required|integer|min:30|max:400',
                'hint' => 'Roughly 160 characters per SMS segment. 120 words is about 5 segments.',
            ],
            'web_max_words' => [
                'group' => 'Channels', 'label' => 'Web reply length (words)', 'type' => 'number',
                'default' => '250', 'rules' => 'required|integer|min:50|max:800',
            ],
            'web_chat_limit' => [
                'group' => 'Channels', 'label' => 'Web history shown', 'type' => 'number',
                'default' => '20', 'rules' => 'required|integer|min:2|max:200',
            ],
            'otp_enabled' => [
                'group' => 'Channels', 'label' => 'Web login verification (OTP)', 'type' => 'boolean',
                'default' => '1', 'rules' => 'required|in:0,1',
                'hint' => 'Sends a one-time code by SMS before a phone number can open its chat history.',
            ],
            'otp_ttl_minutes' => [
                'group' => 'Channels', 'label' => 'OTP validity (minutes)', 'type' => 'number',
                'default' => '5', 'rules' => 'required|integer|min:1|max:30',
            ],

            // ---- Safety -------------------------------------------------
            'abuse_strike_limit' => [
                'group' => 'Safety', 'label' => 'Strikes before ban', 'type' => 'number',
                'default' => '3', 'rules' => 'required|integer|min:1|max:10',
            ],
            'rate_limit_per_minute' => [
                'group' => 'Safety', 'label' => 'Questions per minute per user', 'type' => 'number',
                'default' => '6', 'rules' => 'required|integer|min:1|max:60',
            ],
            'contact_email' => [
                'group' => 'Safety', 'label' => 'Operator contact email', 'type' => 'text',
                'default' => '', 'rules' => 'nullable|email|max:120',
                'hint' => 'Receives landing page enquiries.',
            ],
        ];
    }

    public static function groups(): array
    {
        $groups = [];
        foreach (self::schema() as $key => $def) {
            $groups[$def['group']][$key] = $def;
        }
```

```
        return $groups;
    }

    public static function isKnown(string $key): bool
    {
        return array_key_exists($key, self::schema());
    }

    /** @return array<string,string|null> */
    public static function all(): array
    {
        $stored = Cache::remember(self::CACHE_KEY, 3600, function () {
            try {
                return SystemSetting::query()->pluck('value', 'key')->all();
            } catch (\Throwable) {
                return [];
            }
        });

        $out = [];
        foreach (self::schema() as $key => $def) {
            $out[$key] = array_key_exists($key, $stored) && $stored[$key] !== null
                ? $stored[$key]
                : $def['default'];
        }

        return $out;
    }

    public static function get(string $key, mixed $default = null): mixed
    {
        $all = self::all();
        if (array_key_exists($key, $all)) {
            return $all[$key];
        }

        return $default ?? (self::schema()[$key]['default'] ?? null);
    }

    public static function bool(string $key): bool
    {
        return in_array((string) self::get($key), ['1', 'true', 'on', 'yes'], true);
    }

    public static function int(string $key): int
    {
        return (int) self::get($key);
    }

    public static function float(string $key): float
    {
        return (float) self::get($key);
    }

    public static function set(string $key, mixed $value): void
    {
        if (! self::isKnown($key)) {
            throw new \InvalidArgumentException("Unknown setting [{$key}]");
        }

        SystemSetting::query()->updateOrCreate(['key' => $key], ['value' => $value === null ? null : (string)
$value]);
        self::flush();
    }

    public static function flush(): void
    {
        Cache::forget(self::CACHE_KEY);
    }

    /** Validation rules for the whole schema, used by the admin form. */
    public static function rules(): array
```

```
    {
        $rules = [];
        foreach (self::schema() as $key => $def) {
            $rules[$key] = $def['rules'];
        }

        return $rules;
    }
}
```

50 artisan

18 lines

```
#!/usr/bin/env php
<?php

use Illuminate\Foundation\Application;
use Symfony\Component\Console\Input\ArgvInput;

define('LARAVEL_START', microtime(true));

// Register the Composer autoloader...
require __DIR__.'/vendor/autoload.php';

// Bootstrap Laravel and handle the command...
/** @var Application $app */
$app = require_once __DIR__.'/bootstrap/app.php';

$status = $app->handleCommand(new ArgvInput);

exit($status);
```

51 bootstrap/app.php

25 lines

```php
<?php

use App\Http\Middleware\EnsureAdmin;
use App\Http\Middleware\VerifySmsWebhook;
use Illuminate\Foundation\Application;
use Illuminate\Foundation\Configuration\Exceptions;
use Illuminate\Foundation\Configuration\Middleware;

return Application::configure(basePath: dirname(__DIR__))
    ->withRouting(
        web: __DIR__.'/../routes/web.php',
        api: __DIR__.'/../routes/api.php',
        commands: __DIR__.'/../routes/console.php',
        health: '/up',
    )
    ->withMiddleware(function (Middleware $middleware): void {
        $middleware->alias([
            'admin' => EnsureAdmin::class,
            'sms.webhook' => VerifySmsWebhook::class,
        ]);
        $middleware->redirectGuestsTo(fn () => route('admin.login'));
    })
    ->withExceptions(function (Exceptions $exceptions): void {
        //
    })->create();
```

52 bootstrap/providers.php

5 lines

```php
<?php

return [
    App\Providers\AppServiceProvider::class,
];
```

53 composer.json

88 lines

```
{
    "$schema": "https://getcomposer.org/schema.json",
    "name": "hurudigital/huru",
    "type": "project",
    "description": "Huru SMS: everyday question answering for Tanzanians over SMS and web.",
    "keywords": ["laravel", "framework"],
    "license": "MIT",
    "require": {
        "php": "^8.2",
        "composer/ca-bundle": "^1.5",
        "laravel/framework": "^12.0",
        "laravel/sanctum": "^4.0",
        "laravel/tinker": "^2.10.1"
    },
    "require-dev": {
        "fakerphp/faker": "^1.23",
        "laravel/pail": "^1.2.2",
        "laravel/pint": "^1.24",
        "laravel/sail": "^1.41",
        "mockery/mockery": "^1.6",
        "nunomaduro/collision": "^8.6",
        "phpunit/phpunit": "^11.5.3"
    },
    "autoload": {
        "psr-4": {
            "App\\": "app/",
            "Database\\Factories\\": "database/factories/",
            "Database\\Seeders\\": "database/seeders/"
        }
    },
    "autoload-dev": {
        "psr-4": {
            "Tests\\": "tests/"
        }
    },
    "scripts": {
        "setup": [
            "composer install",
            "@php -r \"file_exists('.env') || copy('.env.example', '.env');\"",
            "@php artisan key:generate",
            "@php artisan migrate --force",
            "npm install",
            "npm run build"
        ],
        "dev": [
            "Composer\\Config::disableProcessTimeout",
            "npx concurrently -c \"#93c5fd,#c4b5fd,#fb7185,#fdba74\" \"php artisan serve\" \"php artisan
queue:listen --tries=1 --timeout=0\" \"php artisan pail --timeout=0\" \"npm run dev\" --names=server,queue,logs,vite --
kill-others"
        ],
        "test": [
            "@php artisan config:clear --ansi",
            "@php artisan test"
        ],
        "post-autoload-dump": [
            "Illuminate\\Foundation\\ComposerScripts::postAutoloadDump",
            "@php artisan package:discover --ansi"
        ],
        "post-update-cmd": [
            "@php artisan vendor:publish --tag=laravel-assets --ansi --force"
        ],
        "post-root-package-install": [
            "@php -r \"file_exists('.env') || copy('.env.example', '.env');\""
        ],
        "post-create-project-cmd": [
            "@php artisan key:generate --ansi",
            "@php -r \"file_exists('database/database.sqlite') || touch('database/database.sqlite');\"",
            "@php artisan migrate --graceful --ansi"
        ],
        "pre-package-uninstall": [
            "Illuminate\\Foundation\\ComposerScripts::prePackageUninstall"
```

```
        ]
    },
    "extra": {
        "laravel": {
            "dont-discover": []
        }
    },
    "config": {
        "optimize-autoloader": true,
        "preferred-install": "dist",
        "sort-packages": true,
        "allow-plugins": {
            "pestphp/pest-plugin": true,
            "php-http/discovery": true
        }
    },
    "minimum-stability": "stable",
    "prefer-stable": true
}
```

54 config/services.php

49 lines

```php
<?php

return [

    /*
    |--------------------------------------------------------------------------
    | Third Party Services
    |--------------------------------------------------------------------------
    */

    'postmark' => [
        'key' => env('POSTMARK_API_KEY'),
    ],

    'resend' => [
        'key' => env('RESEND_API_KEY'),
    ],

    'ses' => [
        'key' => env('AWS_ACCESS_KEY_ID'),
        'secret' => env('AWS_SECRET_ACCESS_KEY'),
        'region' => env('AWS_DEFAULT_REGION', 'us-east-1'),
    ],

    'slack' => [
        'notifications' => [
            'bot_user_oauth_token' => env('SLACK_BOT_USER_OAUTH_TOKEN'),
            'channel' => env('SLACK_BOT_USER_DEFAULT_CHANNEL'),
        ],
    ],

    // Google Gemini (model is chosen in the admin settings)
    'gemini' => [
        'key' => env('GEMINI_API_KEY'),
        'url' => env('GEMINI_URL', 'https://generativelanguage.googleapis.com/v1beta/models'),
        'timeout' => env('GEMINI_TIMEOUT', 30),
    ],

    // Africa's Talking SMS
    'at' => [
        'username' => env('AT_USERNAME'),
        'api_key' => env('AT_API_KEY'),
        'from' => env('AT_FROM'),
        'webhook_secret' => env('AT_WEBHOOK_SECRET'),
        // When true, inbound messages without the keyword are ignored (shared short codes).
        'require_keyword' => env('AT_REQUIRE_KEYWORD', false),
    ],

];
```

55 database/factories/UserFactory.php

40 lines

```php
<?php

namespace Database\Factories;

use App\Models\User;
use Illuminate\Database\Eloquent\Factories\Factory;
use Illuminate\Support\Facades\Hash;
use Illuminate\Support\Str;

/**
 * @extends Factory<User>
 */
class UserFactory extends Factory
{
    protected static ?string $password;

    public function definition(): array
    {
        return [
            'name' => fake()->name(),
            'phone_number' => '+2557' . fake()->unique()->numerify('########'),
            'email' => null,
            'password' => null,
            'role' => User::ROLE_USER,
            'is_banned' => false,
            'abuse_count' => 0,
            'opted_out' => false,
            'remember_token' => Str::random(10),
        ];
    }

    public function admin(): static
    {
        return $this->state(fn () => [
            'email' => fake()->unique()->safeEmail(),
            'password' => static::$password ??= Hash::make('password'),
            'role' => User::ROLE_ADMIN,
        ]);
    }
}
```

56 database/migrations/0001_01_01_000000_create_users_table.php

54 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('users', function (Blueprint $table) {
            $table->id();
            $table->string('phone_number', 20)->unique()->nullable();
            $table->string('name')->nullable();
            $table->string('email')->unique()->nullable();
            $table->string('password')->nullable();
            $table->string('role', 20)->default('user')->index();
            $table->string('preferred_language', 5)->nullable();
            $table->string('region', 40)->nullable()->index();
            $table->boolean('is_banned')->default(false);
            $table->unsignedSmallInteger('abuse_count')->default(0);
            $table->boolean('opted_out')->default(false);
            $table->string('otp_hash')->nullable();
            $table->timestamp('otp_expires_at')->nullable();
            $table->unsignedTinyInteger('otp_attempts')->default(0);
            $table->timestamp('last_seen_at')->nullable();
            $table->timestamp('email_verified_at')->nullable();
            $table->rememberToken();
            $table->timestamps();
        });

        Schema::create('password_reset_tokens', function (Blueprint $table) {
            $table->string('email')->primary();
            $table->string('token');
            $table->timestamp('created_at')->nullable();
        });

        Schema::create('sessions', function (Blueprint $table) {
            $table->string('id')->primary();
            $table->foreignId('user_id')->nullable()->index();
            $table->string('ip_address', 45)->nullable();
            $table->text('user_agent')->nullable();
            $table->longText('payload');
            $table->integer('last_activity')->index();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('sessions');
        Schema::dropIfExists('password_reset_tokens');
        Schema::dropIfExists('users');
    }
};
```

57 database/migrations/0001_01_01_000001_create_cache_table.php

35 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('cache', function (Blueprint $table) {
            $table->string('key')->primary();
            $table->mediumText('value');
            $table->integer('expiration')->index();
        });

        Schema::create('cache_locks', function (Blueprint $table) {
            $table->string('key')->primary();
            $table->string('owner');
            $table->integer('expiration')->index();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('cache');
        Schema::dropIfExists('cache_locks');
    }
};
```

58 database/migrations/0001_01_01_000002_create_jobs_table.php

57 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('jobs', function (Blueprint $table) {
            $table->id();
            $table->string('queue')->index();
            $table->longText('payload');
            $table->unsignedTinyInteger('attempts');
            $table->unsignedInteger('reserved_at')->nullable();
            $table->unsignedInteger('available_at');
            $table->unsignedInteger('created_at');
        });

        Schema::create('job_batches', function (Blueprint $table) {
            $table->string('id')->primary();
            $table->string('name');
            $table->integer('total_jobs');
            $table->integer('pending_jobs');
            $table->integer('failed_jobs');
            $table->longText('failed_job_ids');
            $table->mediumText('options')->nullable();
            $table->integer('cancelled_at')->nullable();
            $table->integer('created_at');
            $table->integer('finished_at')->nullable();
        });

        Schema::create('failed_jobs', function (Blueprint $table) {
            $table->id();
            $table->string('uuid')->unique();
            $table->text('connection');
            $table->text('queue');
            $table->longText('payload');
            $table->longText('exception');
            $table->timestamp('failed_at')->useCurrent();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('jobs');
        Schema::dropIfExists('job_batches');
        Schema::dropIfExists('failed_jobs');
    }
};
```

59 database/migrations/2026_10_07_000001_create_messages_and_logs_tables.php 58 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('messages', function (Blueprint $table) {
            $table->id();
            $table->foreignId('user_id')->constrained()->cascadeOnDelete();
            $table->string('direction', 10); // inbound | outbound
            $table->string('channel', 10)->default('sms'); // sms | web
            $table->text('content');
            $table->string('language', 5)->nullable();
            $table->string('category', 30)->nullable()->index();
            $table->timestamps();

            $table->index(['user_id', 'created_at']);
            $table->index(['channel', 'created_at']);
        });

        Schema::create('ai_logs', function (Blueprint $table) {
            $table->id();
            $table->foreignId('message_id')->constrained()->cascadeOnDelete();
            $table->string('model', 80);
            $table->string('category', 30)->nullable();
            $table->string('status', 20)->default('ok');
            $table->longText('prompt');
            $table->text('response');
            $table->unsignedInteger('prompt_tokens')->default(0);
            $table->unsignedInteger('completion_tokens')->default(0);
            $table->unsignedInteger('total_tokens')->default(0);
            $table->unsignedInteger('latency_ms')->default(0);
            $table->timestamps();
        });

        Schema::create('feedback', function (Blueprint $table) {
            $table->id();
            $table->foreignId('message_id')->constrained()->cascadeOnDelete();
            $table->foreignId('user_id')->constrained()->cascadeOnDelete();
            $table->tinyInteger('rating'); // 1 helpful, -1 not helpful
            $table->string('comment', 500)->nullable();
            $table->timestamps();

            $table->unique(['message_id', 'user_id']);
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('feedback');
        Schema::dropIfExists('ai_logs');
        Schema::dropIfExists('messages');
    }
};
```

60 database/migrations/2026_10_07_000002_create_knowledge_and_config_tables.php

87 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('knowledge_entries', function (Blueprint $table) {
            $table->id();
            $table->string('title');
            $table->longText('content');
            $table->text('summary')->nullable();
            $table->string('category', 30)->default('general')->index();
            $table->json('keywords')->nullable();
            $table->string('language', 5)->default('sw')->index();
            $table->string('source')->nullable();
            $table->boolean('is_active')->default(true);
            $table->timestamps();
        });

        Schema::create('prompt_templates', function (Blueprint $table) {
            $table->id();
            $table->string('name')->unique();
            $table->string('category', 30)->default('general');
            $table->string('language', 5)->default('sw');
            $table->text('template');
            $table->boolean('is_active')->default(true);
            $table->timestamps();

            $table->index(['category', 'language', 'is_active']);
        });

        Schema::create('system_settings', function (Blueprint $table) {
            $table->id();
            $table->string('key')->unique();
            $table->text('value')->nullable();
            $table->timestamps();
        });

        Schema::create('reference_contacts', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('category', 30)->index();
            $table->string('phone', 60)->nullable();
            $table->string('alt_phone', 60)->nullable();
            $table->string('email')->nullable();
            $table->string('website')->nullable();
            $table->text('description_sw')->nullable();
            $table->text('description_en')->nullable();
            $table->boolean('is_emergency')->default(false);
            $table->boolean('is_active')->default(true);
            $table->timestamp('verified_at')->nullable();
            $table->unsignedSmallInteger('sort_order')->default(100);
            $table->timestamps();
        });

        Schema::create('local_resources', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('category', 30)->default('other');
            $table->string('region', 40)->nullable();
            $table->string('district', 40)->nullable();
            $table->string('ward', 60)->nullable();
            $table->text('location')->nullable();
            $table->string('phone', 60)->nullable();
            $table->string('email')->nullable();
            $table->string('source')->nullable();
            $table->boolean('is_active')->default(true);
```

```
            $table->timestamps();

            $table->index(['region', 'district']);
            $table->index(['category', 'region']);
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('local_resources');
        Schema::dropIfExists('reference_contacts');
        Schema::dropIfExists('system_settings');
        Schema::dropIfExists('prompt_templates');
        Schema::dropIfExists('knowledge_entries');
    }
};
```

61 database/migrations/2026_10_07_000003_create_community_tables.php

76 lines

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('community_threads', function (Blueprint $table) {
            $table->id();
            $table->string('title');
            $table->string('slug')->unique();
            $table->text('description')->nullable();
            $table->foreignId('creator_id')->nullable()->constrained('users')->nullOnDelete();
            $table->boolean('is_private')->default(false);
            $table->boolean('is_system')->default(false);
            $table->timestamps();
        });

        Schema::create('community_members', function (Blueprint $table) {
            $table->id();
            $table->foreignId('community_thread_id')->constrained()->cascadeOnDelete();
            $table->foreignId('user_id')->constrained()->cascadeOnDelete();
            $table->string('role', 20)->default('member');
            $table->timestamp('last_read_at')->nullable();
            $table->timestamp('joined_at')->useCurrent();
            $table->timestamps();

            $table->unique(['community_thread_id', 'user_id']);
        });

        Schema::create('community_posts', function (Blueprint $table) {
            $table->id();
            $table->foreignId('community_thread_id')->constrained()->cascadeOnDelete();
            $table->foreignId('parent_id')->nullable()->constrained('community_posts')->cascadeOnDelete();
            $table->foreignId('user_id')->constrained()->cascadeOnDelete();
            $table->text('content');
            $table->boolean('is_approved')->default(true);
            $table->boolean('is_pinned')->default(false);
            $table->timestamps();
        });

        Schema::create('community_reactions', function (Blueprint $table) {
            $table->id();
            $table->foreignId('community_post_id')->constrained()->cascadeOnDelete();
            $table->foreignId('user_id')->constrained()->cascadeOnDelete();
            $table->string('type', 20);
            $table->timestamps();

            $table->unique(['community_post_id', 'user_id', 'type']);
        });

        Schema::create('community_invites', function (Blueprint $table) {
            $table->id();
            $table->foreignId('community_thread_id')->constrained()->cascadeOnDelete();
            $table->foreignId('inviter_id')->constrained('users')->cascadeOnDelete();
            $table->foreignId('invitee_id')->nullable()->constrained('users')->cascadeOnDelete();
            $table->string('email')->nullable();
            $table->string('token')->unique();
            $table->string('status', 20)->default('pending');
            $table->timestamp('expires_at')->nullable();
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('community_invites');
        Schema::dropIfExists('community_reactions');
```

```
        Schema::dropIfExists('community_posts');
        Schema::dropIfExists('community_members');
        Schema::dropIfExists('community_threads');
    }
};
```

62 database/seeders/AdminUserSeeder.php

44 lines

```php
<?php

namespace Database\Seeders;

use App\Models\User;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\Hash;
use Illuminate\Support\Str;

class AdminUserSeeder extends Seeder
{
    public function run(): void
    {
        $email = env('ADMIN_EMAIL', 'admin@huru.local');
        $password = env('ADMIN_PASSWORD');
        $generated = false;

        if (! $password) {
            $password = Str::password(14, symbols: false);
            $generated = true;
        }

        $admin = User::query()->where('email', $email)->first();

        if (! $admin) {
            User::query()->create([
                'name' => env('ADMIN_NAME', 'Huru SMS Admin'),
                'email' => $email,
                'password' => Hash::make($password),
                'role' => User::ROLE_ADMIN,
                'preferred_language' => 'en',
            ]);
            $this->command?->info("Admin created: {$email}");
            if ($generated) {
                $this->command?->warn("Generated admin password (set ADMIN_PASSWORD in .env to choose your own): {$password}");
            }
        } else {
            if ($admin->role !== User::ROLE_ADMIN) {
                $admin->forceFill(['role' => User::ROLE_ADMIN])->save();
            }
            $this->command?->info("Admin already exists: {$email}");
        }
    }
}
```

63 database/seeders/CommunitySeeder.php

45 lines

```php
<?php

namespace Database\Seeders;

use App\Models\CommunityThread;
use Illuminate\Database\Seeder;

class CommunitySeeder extends Seeder
{
    public function run(): void
    {
        CommunityThread::query()->updateOrCreate(
            ['slug' => 'general-advice'],
            [
                'title' => 'Maoni na Ushauri wa Jumla',
                'description' => 'Mahali rasmi pa maoni, ushauri na uzoefu wako. Maoni yaliyothibitishwa huonekana kwenye ukurasa wa mwanzo.',
                'is_system' => true,
                'is_private' => false,
            ]
        );

        CommunityThread::query()->updateOrCreate(
            ['slug' => 'announcements'],
            [
                'title' => 'Matangazo Rasmi',
                'description' => 'Taarifa muhimu kutoka kwa timu ya Huru SMS.',
                'is_system' => true,
                'is_private' => false,
            ]
        );

        foreach ([
            ['sheria-na-haki', 'Sheria na Haki', 'Jadili maswali ya sheria, haki na taratibu za mahakama na ofisi za serikali.'],
            ['kilimo-na-mifugo', 'Kilimo na Mifugo', 'Wakulima na wafugaji wakishirikiana uzoefu, bei na mbinu bora.'],
            ['afya-ya-jamii', 'Afya ya Jamii', 'Elimu ya afya, lishe na huduma za afya zilizo karibu.'],
            ['biashara-na-ajira', 'Biashara na Ajira', 'Fursa, ujasiriamali, nafasi za kazi na ushauri wa kifedha.'],
            ['wanafunzi', 'Wanafunzi na Walimu', 'Masomo, mitihani, mikopo ya HESLB na vyuo.'],
        ] as [$slug, $title, $desc]) {
            CommunityThread::query()->updateOrCreate(
                ['slug' => $slug],
                ['title' => $title, 'description' => $desc, 'is_system' => true, 'is_private' => false]
            );
        }
    }
}
```

64 database/seeders/DatabaseSeeder.php

21 lines

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    public function run(): void
    {
        $this->call([
            SettingsSeeder::class,
            AdminUserSeeder::class,
            PromptTemplateSeeder::class,
            KnowledgeSeeder::class,
            ReferenceContactSeeder::class,
            LocalResourceSeeder::class,
            CommunitySeeder::class,
        ]);
    }
}
```

65 database/seeders/KnowledgeSeeder.php

343 lines

```php
<?php

namespace Database\Seeders;

use App\Models\KnowledgeEntry;
use Illuminate\Database\Seeder;

/**
 * Starter knowledge base: verified-style fact blocks across everyday
 * Tanzanian topics. Admins extend this through the panel or CSV/JSON import.
 */
class KnowledgeSeeder extends Seeder
{
    public function run(): void
    {
        $entries = [
            // ---------------------------------------------------------- LEGAL
            [
                'title' => 'Katiba ya Tanzania - Usawa mbele ya sheria (Ibara ya 12 na 13)',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Ibara ya 12 ya Katiba ya Jamhuri ya Muungano wa Tanzania (1977) inatamka kuwa binadamu wote huzaliwa huru na wote ni sawa. Ibara ya 13 inasisitiza kuwa watu wote ni sawa mbele ya sheria na wana haki ya kulindwa na kupata haki sawa bila ubaguzi wa jinsia, kabila, dini, hali ya kijamii au ulemavu. Mwananchi anayebaguliwa na ofisi ya umma anaweza kulalamika kwa Tume ya Haki za Binadamu na Utawala Bora (CHRAGG) au mahakamani.',
                'keywords' => ['katiba', 'usawa', 'ubaguzi', 'haki za binadamu', 'chragg', 'ibara 13'],
                'source' => 'Katiba ya Jamhuri ya Muungano wa Tanzania, 1977',
            ],
            [
                'title' => 'Constitution of Tanzania - Equality before the law (Articles 12 and 13)',
                'category' => 'legal', 'language' => 'en',
                'content' => 'Article 12 of the Constitution of the United Republic of Tanzania (1977) states that all human beings are born free and equal. Article 13 guarantees equality before the law and equal protection without discrimination on grounds of gender, tribe, religion, social status or disability. A citizen discriminated against by a public office may complain to the Commission for Human Rights and Good Governance (CHRAGG) or to the courts.',
                'keywords' => ['constitution', 'equality', 'discrimination', 'human rights', 'chragg', 'article 13'],
                'source' => 'Constitution of the United Republic of Tanzania, 1977',
            ],
            [
                'title' => 'Haki zako ukikamatwa na polisi',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Ukikamatwa Tanzania: 1) Una haki ya kuambiwa sababu ya kukamatwa kwako mara moja kwa lugha unayoielewa. 2) Una haki ya kukaa kimya na kutosaini maelezo yoyote mpaka uelewe yaliyomo; unaweza kuomba wakili au ndugu awepo. 3) Polisi wanapaswa kukufikisha mahakamani ndani ya saa 24 (ukiondoa siku za mapumziko) au kukupa dhamana ya polisi kwa makosa yanayodhaminika. 4) Una haki ya kupigiwa simu ndugu au wakili. 5) Kupigwa, kuteswa au kulazimishwa kukiri ni kinyume cha sheria; ripoti kwa Mkuu wa Kituo, Jeshi la Polisi (112), CHRAGG au mtoa msaada wa kisheria. Dhamana ya polisi ni bure; usilipe fedha kwa askari.',
                'keywords' => ['kukamatwa', 'polisi', 'dhamana', 'saa 24', 'kukaa kimya', 'wakili', 'mtuhumiwa', 'kituo cha polisi'],
                'source' => 'Sheria ya Mwenendo wa Makosa ya Jinai, Sura ya 20',
            ],
            [
                'title' => 'Your rights when arrested by the police',
                'category' => 'legal', 'language' => 'en',
                'content' => 'If arrested in Tanzania: 1) You must be told the reason immediately in a language you understand. 2) You may remain silent and should not sign a statement you do not understand; you may ask for an advocate or relative. 3) Police must bring you to court within 24 hours (excluding holidays) or grant police bail for bailable offences. 4) You may call a relative or lawyer. 5) Beating, torture or forced confession is unlawful; report to the Officer Commanding Station, the police (112), CHRAGG or a legal aid provider. Police bail is free; never pay cash to an officer.',
                'keywords' => ['arrest', 'arrested', 'police', 'bail', '24 hours', 'silent', 'advocate', 'suspect', 'police station'],
                'source' => 'Criminal Procedure Act, Cap 20',
            ],
            [
                'title' => 'Umiliki wa ardhi na migogoro ya ardhi',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Ardhi yote Tanzania ni mali ya umma iliyo chini ya Rais kama mdhamini. Kuna ardhi ya kawaida (General), ardhi ya kijiji (Village) na ardhi ya hifadhi (Reserved). Mwananchi anapata Hati ya Haki Miliki (Granted Right of Occupancy) kwa ardhi ya kawaida kupitia Ofisi ya Ardhi ya Halmashauri/Mkoa, au Hati ya Hakimiliki ya Kimila (CCRO) kwa ardhi ya kijiji kupitia Halmashauri ya Kijiji. Wanawake wana haki sawa na wanaume kumiliki na kurithi ardhi. Migogoro ya ardhi huanzia Baraza la Ardhi la Kijiji, kisha Baraza la Kata, kisha Baraza la Ardhi na Nyumba la Wilaya (DLHT), kisha
```

```
Mahakama Kuu Kitengo cha Ardhi. Kabla ya kununua ardhi hakiki hati na umiliki Ofisi ya Ardhi na kwa Mtendaji wa Kijiji/Mtaa.',
                'keywords' => ['ardhi', 'hati', 'umiliki', 'ccro', 'baraza la ardhi', 'mgogoro wa ardhi', 'kununua ardhi', 'kiwanja', 'shamba', 'mashamba', 'kudhulumiwa', 'dhuluma', 'kunyang\'anywa', 'mpaka', 'mipaka', 'kuvamiwa', 'eneo', 'mali', 'hatimiliki', 'kurithi ardhi', 'wakulima'],
                'source' => 'Sheria ya Ardhi Sura 113; Sheria ya Ardhi ya Vijiji Sura 114; Sheria ya Mahakama za Ardhi Sura 216',
            ],
            [
                'title' => 'Land ownership and land disputes',
                'category' => 'legal', 'language' => 'en',
                'content' => 'All land in Tanzania is public land vested in the President as trustee. Categories: General land, Village land and Reserved land. Citizens obtain a Granted Right of Occupancy (General land) through the District/Regional Land Office, or a Certificate of Customary Right of Occupancy (CCRO) for village land through the Village Council. Women have equal rights to own and inherit land. Disputes go from the Village Land Council to the Ward Tribunal, then the District Land and Housing Tribunal (DLHT), then the High Court Land Division. Before buying land, verify the title at the Land Office and with the Village/Mtaa Executive Officer.',
                'keywords' => ['land', 'title deed', 'ownership', 'ccro', 'land tribunal', 'land dispute', 'buying land', 'plot', 'farm', 'grabbed', 'grabbing', 'boundary', 'encroachment', 'trespass', 'inherit land', 'village land'],
                'source' => 'Land Act Cap 113; Village Land Act Cap 114; Land Disputes Courts Act Cap 216',
            ],
            [
                'title' => 'Ndoa, talaka na mgawanyo wa mali',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Sheria ya Ndoa ya 1971 inatambua ndoa za kiserikali, za kidini na za kimila; zote zinapaswa kusajiliwa RITA/ofisi ya msajili wa ndoa. Mahakama zimetamka kuwa umri wa chini wa kuoa au kuolewa ni miaka 18 kwa wote. Talaka hutolewa na mahakama tu baada ya kuthibitisha ndoa imevunjika kabisa; kabla ya kwenda mahakamani lazima kupitia Baraza la Usuluhishi la Ndoa (ngazi ya kata au la kidini). Wakati wa talaka mahakama hugawa mali zilizochumwa pamoja kwa kuzingatia mchango wa kila mmoja, ikiwemo kazi za nyumbani. Matunzo ya watoto hupewa kipaumbele kwa maslahi ya mtoto.',
                'keywords' => ['ndoa', 'talaka', 'mali ya ndoa', 'usuluhishi', 'mke', 'mume', 'matunzo', 'kusajili ndoa', 'miaka 18'],
                'source' => 'Sheria ya Ndoa, 1971 (Sura 29)',
            ],
            [
                'title' => 'Marriage, divorce and division of property',
                'category' => 'legal', 'language' => 'en',
                'content' => 'The Law of Marriage Act 1971 recognises civil, religious and customary marriages; all should be registered with RITA or the marriage registrar. Courts have held that the minimum age of marriage is 18 for everyone. Divorce is granted only by a court after proof of irreparable breakdown, and only after referral to a Marriage Conciliation Board (ward or religious). On divorce the court divides assets acquired jointly, counting each spouse\'s contribution including housework. Custody decisions follow the best interest of the child.',
                'keywords' => ['marriage', 'divorce', 'matrimonial property', 'conciliation', 'custody', 'register marriage', 'age 18'],
                'source' => 'Law of Marriage Act, 1971 (Cap 29)',
            ],
            [
                'title' => 'Mirathi na wosia',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Mtu anaweza kuandika wosia (will) akiwa na akili timamu; wosia wa maandishi usainiwe mbele ya mashahidi wawili wasio warithi. Mtu akifa bila wosia mirathi hugawanywa kwa sheria inayomhusu: sheria ya kimila, ya Kiislamu, au Sheria ya Mirathi ya India (kwa wengine). Ndugu hukutana kuchagua msimamizi wa mirathi, kisha msimamizi huomba barua ya usimamizi Mahakama ya Mwanzo (mirathi ndogo) au Mahakama ya Wilaya/Mahakama Kuu (mirathi kubwa). Cheti cha kifo kutoka RITA ni lazima. Watoto, wajane na wagane wana haki ya kurithi; kunyang\'anywa mali ya marehemu bila utaratibu ni kosa. Msaada: mtoa msaada wa kisheria, TAWLA, WLAC, LHRC.',
                'keywords' => ['mirathi', 'wosia', 'urithi', 'msimamizi wa mirathi', 'marehemu', 'mjane', 'cheti cha kifo', 'warithi'],
                'source' => 'Sheria ya Usimamizi wa Mirathi Sura 352; Sheria ya Mahakama za Mahakimu Sura 11',
            ],
            [
                'title' => 'Haki za mpangaji na mwenye nyumba',
                'category' => 'legal', 'language' => 'sw',
                'content' => 'Mkataba wa pango uwe wa maandishi na uonyeshe kodi, muda na masharti; kwa makazi mwenye nyumba haruhusiwi kudai kodi ya zaidi ya miezi 12 mapema (kwa malipo ya kila mwezi, kodi ya awali hupangwa kwa makubaliano) na lazima atoe risiti. Mwenye nyumba hawezi kumfukuza mpangaji kwa nguvu, kufunga milango au kukata maji/umeme bila amri ya mahakama; notisi ya maandishi inahitajika kwa mujibu wa mkataba. Migogoro ya pango husikilizwa na Baraza la Ardhi na Nyumba la Wilaya (DLHT). Mpangaji anawajibika kulipa kodi kwa wakati na kutunza nyumba.',
                'keywords' => ['pango', 'mpangaji', 'mwenye nyumba', 'kodi ya nyumba', 'kufukuzwa', 'notisi', 'dlht', 'nyumba'],
                'source' => 'Sheria ya Ardhi Sura 113 Sehemu ya Pango; Sheria ya Mahakama za Ardhi Sura 216',
            ],

            // ---------------------------------------------------------- EMPLOYMENT
```

```
            [
                'title' => 'Haki za mfanyakazi (Sheria ya Ajira na Mahusiano Kazini)',
                'category' => 'employment', 'language' => 'sw',
                'content' => 'Sheria ya Ajira na Mahusiano Kazini (ELRA, Sura 366): 1) Mkataba wa maandishi ni lazima kwa ajira inayozidi miezi 6; mwajiri lazima atoe nakala. 2) Saa za kazi hazizidi 45 kwa wiki au 9 kwa siku; saa za ziada hulipwa mara 1.5 ya kiwango cha kawaida. 3) Likizo ya mwaka ni siku 28 mfululizo zenye malipo. 4) Likizo ya uzazi: siku 84 zenye malipo (siku 100 kwa mapacha) kwa mama, siku 3 kwa baba. 5) Likizo ya ugonjwa hadi siku 126 kwa mzunguko wa miezi 36. 6) Kuachishwa kazi lazima kuwe na sababu halali na utaratibu wa haki (kusikilizwa). Mgogoro wa kuachishwa kazi isivyo haki huwasilishwa CMA ndani ya siku 30; migogoro mingine ndani ya siku 60. Kima cha chini cha mshahara hutofautiana kwa sekta (angalia tangazo la Serikali la hivi karibuni).',
                'keywords' => ['ajira', 'mfanyakazi', 'mkataba wa kazi', 'likizo', 'uzazi', 'kufukuzwa kazi', 'cma', 'mshahara', 'saa za kazi', 'overtime'],
                'source' => 'Sheria ya Ajira na Mahusiano Kazini, Sura 366',
            ],
            [
                'title' => 'Worker rights (Employment and Labour Relations Act)',
                'category' => 'employment', 'language' => 'en',
                'content' => 'Employment and Labour Relations Act (Cap 366): 1) A written contract is required for employment over 6 months; the employer must give the worker a copy. 2) Working hours max 45 per week or 9 per day; overtime is paid at 1.5 times. 3) Annual leave is 28 consecutive paid days. 4) Maternity leave 84 paid days (100 for multiple births), paternity leave 3 days. 5) Sick leave up to 126 days per 36-month cycle. 6) Termination needs a valid reason and a fair procedure (a hearing). Unfair termination disputes go to the CMA within 30 days; other disputes within 60 days. Minimum wages differ by sector (check the latest Government Notice).',
                'keywords' => ['employment', 'employee', 'contract', 'leave', 'maternity', 'dismissal', 'termination', 'cma', 'wage', 'working hours', 'overtime'],
                'source' => 'Employment and Labour Relations Act, Cap 366',
            ],
            [
                'title' => 'Kutafuta kazi na kuepuka matapeli wa ajira',
                'category' => 'employment', 'language' => 'sw',
                'content' => 'Nafasi halali za kazi za umma hutangazwa na Sekretarieti ya Ajira katika Utumishi wa Umma (ajira.go.tz) na tovuti za taasisi husika; hakuna malipo ya kuomba kazi serikalini. Tahadhari: mtu anayekuomba fedha "ya usaili", "ya mafunzo" au "ya kibali" kabla ya kuajiriwa ni tapeli. Kwa ajira nje ya nchi tumia mawakala waliosajiliwa na Wizara ya Kazi pekee; hakiki mkataba na usikabidhi pasipoti yako kwa wakala. CV nzuri: taarifa za mawasiliano, elimu, uzoefu, ujuzi, wadhamini wawili; iwe kurasa 1-2.',
                'keywords' => ['kazi', 'nafasi za kazi', 'ajira', 'cv', 'usaili', 'tapeli', 'ajira nje ya nchi', 'wakala'],
                'source' => 'Sekretarieti ya Ajira; Wizara ya Kazi',
            ],

            // ------------------------------------------------------------ HEALTH
            [
                'title' => 'Malaria - dalili, matibabu na kinga',
                'category' => 'health', 'language' => 'sw',
                'content' => 'Dalili za malaria: homa, kutetemeka, maumivu ya kichwa na viungo, kutapika, uchovu. Mtu mwenye dalili apimwe kwa kipimo cha haraka (mRDT) au darubini kwenye zahanati/duka la dawa lililosajiliwa; asinywe dawa bila kipimo. Matibabu ya kwanza Tanzania ni ALu (Artemether-Lumefantrine) inayotolewa na mtaalamu wa afya. Dalili za hatari zinazohitaji hospitali MARA MOJA: kupoteza fahamu, degedege, kushindwa kunywa au kunyonya, kupumua kwa shida, manjano, mkojo mweusi, hasa kwa watoto chini ya miaka 5 na wajawazito. Kinga: lala ndani ya chandarua chenye dawa kila usiku, ondoa maji yaliyotuama, wajawazito wachukue dawa ya kinga (IPTp) kliniki.',
                'keywords' => ['malaria', 'homa', 'chandarua', 'mrdt', 'alu', 'degedege', 'mbu', 'kutetemeka'],
                'source' => 'Mpango wa Taifa wa Kudhibiti Malaria (NMCP), Wizara ya Afya',
            ],
            [
                'title' => 'Malaria - symptoms, treatment and prevention',
                'category' => 'health', 'language' => 'en',
                'content' => 'Malaria symptoms: fever, chills, headache, body pain, vomiting, tiredness. Anyone with symptoms should be tested (mRDT or microscopy) at a dispensary or accredited drug shop before taking medicine. First-line treatment in Tanzania is ALu (Artemether-Lumefantrine) given by a health worker. Danger signs needing a hospital IMMEDIATELY: loss of consciousness, convulsions, inability to drink or breastfeed, difficulty breathing, yellow eyes, dark urine, especially in children under 5 and pregnant women. Prevention: sleep under a treated net every night, remove stagnant water, pregnant women take preventive doses (IPTp) at the clinic.',
                'keywords' => ['malaria', 'fever', 'mosquito net', 'mrdt', 'alu', 'convulsions', 'mosquito'],
                'source' => 'National Malaria Control Programme, Ministry of Health',
            ],
            [
                'title' => 'Bima ya afya: NHIF na CHF iliyoboreshwa (iCHF)',
                'category' => 'health', 'language' => 'sw',
                'content' => 'NHIF (Mfuko wa Taifa wa Bima ya Afya) ni bima ya afya ya umma: watumishi wa umma hukatwa moja kwa moja, na wananchi wengine wanaweza kujiunga kwa hiari kupitia vifurushi vya NHIF (ofisi za NHIF za mkoa/wilaya au mawakala). CHF iliyoboreshwa (iCHF) ni bima ya jamii ya gharama nafuu kwa kaya inayosajiliwa kupitia Mtendaji wa Kata au afisa wa iCHF wa halmashauri; hutumika vituo vya afya vya umma vya wilaya/mkoa. Tanzania inatekeleza Sheria ya Bima ya Afya kwa Wote (2023) kwa awamu. Huduma za bure kwa wote: huduma za mama mjamzito na watoto chini ya miaka
```

```
5 kwenye vituo vya umma, chanjo, matibabu ya kifua kikuu na ARV.',
   131                 'keywords' => ['nhif', 'chf', 'ichf', 'bima ya afya', 'kadi ya bima', 'matibabu', 'gharama za
matibabu', 'bima kwa wote'],
   132                 'source' => 'NHIF; Wizara ya Afya; Sheria ya Bima ya Afya kwa Wote 2023',
   133             ],
   134             [
   135                 'title' => 'Kliniki ya mama mjamzito na chanjo za mtoto',
   136                 'category' => 'health', 'language' => 'sw',
   137                 'content' => 'Mama mjamzito aanze kliniki (RCH) mapema kabla ya miezi 3 na ahudhurie angalau mara 8
kwa mujibu wa mwongozo mpya; kliniki hupima damu, presha, UKIMWI, kaswende, hutoa dawa za kinga ya malaria (IPTp), vidonge
vya damu (FeFo), chandarua na kadi ya kliniki. Jifungulia kituo cha afya; huduma ya mama na mtoto chini ya miaka 5 ni bure
vituo vya umma. Ratiba ya chanjo ya mtoto Tanzania: kuzaliwa BCG na OPV0; wiki 6, 10, 14 Penta (DTP-HepB-Hib), OPV, PCV,
Rota; miezi 9 Surua-Rubella (MR) ya kwanza; miezi 18 MR ya pili. Wasichana wa miaka 9-14 hupata chanjo ya HPV (saratani ya
mlango wa kizazi).',
   138                 'keywords' => ['mjamzito', 'kliniki', 'chanjo', 'mtoto', 'kujifungua', 'rch', 'surua', 'hpv',
'penta', 'bcg'],
   139                 'source' => 'Wizara ya Afya - Mpango wa Chanjo (IVD)',
   140             ],
   141             [
   142                 'title' => 'Kipindupindu na magonjwa ya kuhara: kinga na hatua za haraka',
   143                 'category' => 'health', 'language' => 'sw',
   144                 'content' => 'Kipindupindu huambukizwa kwa maji au chakula kilichochafuliwa na kinyesi. Dalili:
kuhara maji mengi kama maji ya mchele, kutapika, kupungukiwa maji haraka. Hatua za haraka: anza kunywa ORS (dawa ya
kuongeza maji mwilini) au maji ya chumvi na sukari mara moja na uende kituo cha afya haraka; watoto wapewe pia zinki.
Kinga: nawa mikono kwa sabuni, chemsha au tia dawa (Waterguard) maji ya kunywa, tumia choo, pasha moto chakula, osha
matunda. Ripoti mlipuko kwa afisa afya wa kata.',
   145                 'keywords' => ['kipindupindu', 'kuhara', 'ors', 'maji safi', 'choo', 'kunawa mikono', 'mlipuko',
'zinki'],
   146                 'source' => 'Wizara ya Afya - Mwongozo wa Kudhibiti Kipindupindu',
   147             ],
   148
   149             // ---------------------------------------------------------- AGRICULTURE
   150             [
   151                 'title' => 'Misimu ya kilimo Tanzania na kalenda ya kupanda',
   152                 'category' => 'agriculture', 'language' => 'sw',
   153                 'content' => 'Tanzania ina mifumo miwili ya mvua. Maeneo ya kaskazini na pwani (Arusha,
Kilimanjaro, Tanga, Dar es Salaam, Pwani, Mwanza, Kagera, Mara) yana misimu miwili: Vuli (Oktoba-Desemba) na Masika (Machi-
Mei). Maeneo ya katikati, magharibi na nyanda za juu kusini (Dodoma, Singida, Tabora, Mbeya, Iringa, Njombe, Ruvuma,
Songwe, Rukwa, Katavi, Morogoro kusini) yana msimu mmoja mrefu (Novemba/Desemba-Aprili). Panda mahindi mwanzoni mwa mvua za
uhakika baada ya utabiri wa TMA; tumia mbegu zilizothibitishwa na TOSCI zenye muda wa kukomaa unaolingana na msimu wa eneo
lako. Kwa ushauri wa eneo lako muone afisa ugani wa kata.',
   154                 'keywords' => ['msimu', 'vuli', 'masika', 'mvua', 'kupanda', 'kalenda ya kilimo', 'mahindi', 'tma',
'afisa ugani'],
   155                 'source' => 'Mamlaka ya Hali ya Hewa Tanzania (TMA); Wizara ya Kilimo',
   156             ],
   157             [
   158                 'title' => 'Farming seasons in Tanzania and planting calendar',
   159                 'category' => 'agriculture', 'language' => 'en',
   160                 'content' => 'Tanzania has two rainfall regimes. The north and coast (Arusha, Kilimanjaro, Tanga,
Dar es Salaam, Pwani, Mwanza, Kagera, Mara) have two seasons: short rains "Vuli" (October-December) and long rains "Masika"
(March-May). Central, western and southern highland regions (Dodoma, Singida, Tabora, Mbeya, Iringa, Njombe, Ruvuma,
Songwe, Rukwa, Katavi, southern Morogoro) have one long season (November/December-April). Plant maize at the start of
reliable rains following TMA forecasts; use TOSCI-certified seed with a maturity period matching your area. Consult the
ward extension officer for local advice.',
   161                 'keywords' => ['season', 'rains', 'planting', 'calendar', 'maize', 'tma', 'extension officer',
'rainfall'],
   162                 'source' => 'Tanzania Meteorological Authority (TMA); Ministry of Agriculture',
   163             ],
   164             [
   165                 'title' => 'Viwavijeshi vamizi (fall armyworm) kwenye mahindi',
   166                 'category' => 'agriculture', 'language' => 'sw',
   167                 'content' => 'Viwavijeshi vamizi hushambulia mahindi hasa wiki 2-6 baada ya kuota; dalili ni majani
yaliyotobolewa na kinyesi kama unga wa mbao kwenye kishungi. Kagua shamba mara 2 kwa wiki asubuhi; ukikuta mimea zaidi ya
20% imeshambuliwa chukua hatua. Hatua: ondoa na uue viwavi kwa mkono kwa shamba dogo, weka mchanga/majivu kwenye kishungi,
panda mapema na kwa pamoja na majirani, tumia viuatilifu vilivyosajiliwa na TPHPA (uliza afisa ugani jina sahihi na
kipimo), pulizia alasiri/jioni ukielekeza kwenye kishungi. Vaa kinga unapotumia dawa. Panda mazao mchanganyiko (mahindi na
maharage/dengu) kupunguza mashambulizi.',
   168                 'keywords' => ['viwavijeshi', 'mahindi', 'wadudu', 'dawa ya wadudu', 'kiuatilifu', 'shamba',
'armyworm'],
   169                 'source' => 'TARI; Wizara ya Kilimo - Mwongozo wa Viwavijeshi',
   170             ],
   171             [
   172                 'title' => 'Ufugaji wa kuku: chanjo ya mdondo (Newcastle) na matunzo',
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                'category' => 'agriculture', 'language' => 'sw',
                'content' => 'Mdondo (Newcastle) ndio ugonjwa unaoua kuku wengi zaidi vijijini; dalili: kuhara kijani, kupinda shingo, kupooza, vifo vya ghafla. Kinga pekee ni chanjo (I-2 au LaSota) kila baada ya miezi 3-4, mara nyingi kwa kudondosha jichoni au kunywesha; chanjo hupatikana kwa afisa mifugo wa kata, maduka ya pembejeo za mifugo au TVLA. Chanja kuku wote kwa pamoja na majirani. Matunzo: banda lenye hewa safi na likavu, maji safi kila siku, chakula chenye uwiano (mahindi, pumba, mashudu, dagaa, madini), weka vifaranga mbali na kuku wakubwa, dhibiti viroboto na minyoo kwa ushauri wa afisa mifugo. Usile wala kuuza kuku waliokufa kwa ugonjwa.',
                'keywords' => ['kuku', 'mdondo', 'newcastle', 'chanjo ya kuku', 'ufugaji', 'vifaranga', 'banda', 'afisa mifugo'],
                'source' => 'Wizara ya Mifugo na Uvuvi; TVLA',
            ],

            // ------------------------------------------------------------ EDUCATION
            [
                'title' => 'Mfumo wa elimu Tanzania na mitihani ya NECTA',
                'category' => 'education', 'language' => 'sw',
                'content' => 'Mfumo wa elimu: elimu ya awali (mwaka 1), msingi Darasa la 1-7 (mtihani wa Darasa la 4 wa upimaji na Mtihani wa Kumaliza Elimu ya Msingi PSLE Darasa la 7), sekondari ya kawaida Kidato cha 1-4 (FTNA Kidato cha 2, CSEE Kidato cha 4), sekondari ya juu Kidato cha 5-6 (ACSEE), kisha vyuo vya ufundi (VETA, NACTVET) au vyuo vikuu (TCU). Elimu ya msingi hadi Kidato cha 4 shule za umma ni bila ada (Sera ya Elimu bila Malipo). NECTA huendesha mitihani ya taifa na kutangaza matokeo kwenye necta.go.tz (pia kwa SMS zinapotolewa). Madaraja ya CSEE: Daraja la I (pointi 7-17) hadi IV; Daraja 0 ni kufeli. Mitaala mipya ya 2023 inaongeza masomo ya ufundi (amali) sekondari.',
                'keywords' => ['necta', 'matokeo', 'kidato', 'darasa la saba', 'psle', 'csee', 'acsee', 'shule', 'elimu bila malipo', 'mtihani', 'daraja'],
                'source' => 'NECTA; Wizara ya Elimu, Sayansi na Teknolojia',
            ],
            [
                'title' => 'Tanzanian education system and NECTA examinations',
                'category' => 'education', 'language' => 'en',
                'content' => 'Structure: pre-primary (1 year), primary Standard 1-7 (Standard 4 assessment and the Primary School Leaving Examination PSLE in Standard 7), ordinary secondary Form 1-4 (FTNA in Form 2, CSEE in Form 4), advanced secondary Form 5-6 (ACSEE), then vocational colleges (VETA, NACTVET) or universities (TCU). Public primary to Form 4 is fee-free. NECTA runs national exams and publishes results at necta.go.tz. CSEE divisions: Division I (points 7-17) to IV; Division 0 is a fail. The 2023 curriculum adds vocational streams in secondary school.',
                'keywords' => ['necta', 'results', 'form four', 'form six', 'standard seven', 'psle', 'csee', 'acsee', 'school', 'exam', 'division'],
                'source' => 'NECTA; Ministry of Education, Science and Technology',
            ],
            [
                'title' => 'Mikopo ya elimu ya juu (HESLB) na kujiunga na chuo',
                'category' => 'education', 'language' => 'sw',
                'content' => 'Bodi ya Mikopo ya Wanafunzi wa Elimu ya Juu (HESLB) hutoa mikopo kwa Watanzania wanaojiunga na vyuo vilivyosajiliwa (shahada na baadhi ya stashahada). Maombi hufanywa mtandaoni kupitia mfumo wa OLAMS (heslb.go.tz) baada ya kupata udahili, kwa kawaida Julai-Agosti kila mwaka; nyaraka: cheti cha kuzaliwa, vyeti vya Kidato cha 4 na 6, kitambulisho cha NIDA, taarifa za wazazi/walezi (vyeti vya kifo kama wamefariki), barua ya mtendaji. Mkopo hurejeshwa baada ya kuanza kazi kwa kukatwa mshahara. Udahili wa vyuo vikuu hufanywa kupitia vyuo vyenyewe na kuratibiwa na TCU; vyuo vya ufundi na NACTVET. Vigezo vya chini: kwa shahada, pointi kutoka ACSEE au stashahada.',
                'keywords' => ['heslb', 'mkopo wa elimu', 'bodi ya mikopo', 'olams', 'chuo kikuu', 'udahili', 'tcu', 'nactvet', 'ada ya chuo'],
                'source' => 'HESLB; TCU',
            ],

            // ------------------------------------------------------------ GOVERNMENT
            [
                'title' => 'Kitambulisho cha Taifa (NIDA): jinsi ya kujisajili',
                'category' => 'government', 'language' => 'sw',
                'content' => 'Mamlaka ya Vitambulisho vya Taifa (NIDA) husajili Watanzania wenye umri wa miaka 18 na zaidi (na watoto kwa namba ya NIN). Hatua: 1) Chukua fomu ofisi ya mtendaji wa kata/mtaa au ofisi ya NIDA ya wilaya, au anza mtandaoni nida.go.tz. 2) Nyaraka: cheti cha kuzaliwa au hati ya kiapo, cheti cha kuzaliwa/kitambulisho cha mzazi mmoja au barua ya mtendaji, picha, na uthibitisho wa makazi. 3) Alama za vidole na picha huchukuliwa ofisi ya NIDA. 4) Namba ya utambulisho (NIN) hutolewa kwanza, kadi hufuata. Usajili ni bure. NIN inahitajika kwa laini ya simu, TIN, akaunti ya benki, HESLB na huduma nyingi za serikali. Kuhakiki hali ya maombi: nida.go.tz au *152*00#.',
                'keywords' => ['nida', 'kitambulisho cha taifa', 'nin', 'kujisajili', 'kitambulisho', 'alama za vidole', 'cheti cha kuzaliwa'],
                'source' => 'NIDA - nida.go.tz',
            ],
            [
                'title' => 'National ID (NIDA): how to register',
                'category' => 'government', 'language' => 'en',
                'content' => 'The National Identification Authority (NIDA) registers Tanzanians aged 18 and above (children receive a NIN). Steps: 1) Get the form from the ward/mtaa executive office or district NIDA office, or start online at nida.go.tz. 2) Documents: birth certificate or affidavit, a parent\'s birth certificate/ID or a letter from the executive officer, photo, proof of residence. 3) Fingerprints and photo are captured at the NIDA office. 4) The NIN is
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issued first, the card later. Registration is free. The NIN is required for SIM cards, TIN, bank accounts, HESLB loans and most government services. Check status at nida.go.tz or *152*00#.',
                'keywords' => ['nida', 'national id', 'nin', 'register', 'identity card', 'fingerprints', 'birth certificate'],
                'source' => 'NIDA - nida.go.tz',
            ],
            [
                'title' => 'Cheti cha kuzaliwa na cheti cha kifo (RITA)',
                'category' => 'government', 'language' => 'sw',
                'content' => 'Wakala wa Usajili, Ufilisi na Udhamini (RITA) husajili vizazi na vifo. Usajili wa mtoto ndani ya siku 90 tangu kuzaliwa ni bure kwa cheti cha kwanza; baada ya hapo kuna ada ya ucheleweshaji. Mahali: ofisi ya msajili wa wilaya (RITA), ofisi ya mtendaji wa kata, au hospitali/zahanati zinazosajili moja kwa moja (mfumo wa usajili wa watoto chini ya miaka 5 kwa simu). Nyaraka: notisi ya kuzaliwa kutoka kituo cha afya au barua ya mtendaji, vitambulisho vya wazazi. Cheti cha kifo: notisi ya kifo kutoka hospitali/mtendaji, kitambulisho cha marehemu, ndani ya siku 30. Malipo ya RITA hufanywa kwa control number ya GePG. Tovuti: rita.go.tz, huduma mtandaoni kupitia efms.rita.go.tz.',
                'keywords' => ['rita', 'cheti cha kuzaliwa', 'cheti cha kifo', 'usajili wa kizazi', 'mtoto mchanga', 'notisi ya kuzaliwa'],
                'source' => 'RITA - rita.go.tz',
            ],
            [
                'title' => 'Pasipoti ya Tanzania (e-Passport)',
                'category' => 'government', 'language' => 'sw',
                'content' => 'Pasipoti ya kielektroniki hutolewa na Idara ya Uhamiaji. Hatua: 1) Jaza maombi mtandaoni eservices.immigration.go.tz, pakia nyaraka: kitambulisho cha NIDA (NIN), cheti cha kuzaliwa, picha, na kwa watoto vitambulisho vya wazazi na idhini. 2) Lipa ada kwa control number (GePG). 3) Fika ofisi ya uhamiaji uliyochagua kwa alama za vidole, picha na mahojiano. 4) Pasipoti ya kawaida huwa tayari ndani ya siku chache hadi wiki kadhaa. Ada ya pasipoti ya kawaida ni TZS 150,000 (thibitisha kiwango cha sasa kwenye tovuti). Pasipoti ya Afrika Mashariki ya Tanzania inatumika kusafiri nchi zote.',
                'keywords' => ['pasipoti', 'passport', 'uhamiaji', 'kusafiri nje', 'e-passport', 'immigration'],
                'source' => 'Idara ya Uhamiaji - immigration.go.tz',
            ],
            [
                'title' => 'Namba ya mlipakodi (TIN) na kodi za TRA kwa wafanyabiashara wadogo',
                'category' => 'finance', 'language' => 'sw',
                'content' => 'TIN (Taxpayer Identification Number) hutolewa bure na Mamlaka ya Mapato Tanzania (TRA) mtandaoni (tra.go.tz, huduma za mtandaoni) au ofisi ya TRA ya wilaya; unahitaji NIN ya NIDA. TIN inahitajika kwa biashara, leseni, ajira rasmi, kununua gari au kiwanja. Wafanyabiashara wadogo wasio na hesabu za ukaguzi hulipa kodi ya makisio (presumptive tax) kulingana na mauzo ya mwaka; mauzo chini ya TZS milioni 4 kwa mwaka hayalipi kodi ya mapato ya makisio (lakini usajili wa TIN na leseni unahitajika). VAT husajiliwa kwa mauzo ya zaidi ya TZS milioni 200 kwa mwaka. Dai risiti ya EFD kwa kila manunuzi. Malipo yote ya TRA ni kwa control number; hakuna malipo kwa afisa. Huduma kwa wateja TRA: 0800 750 075 / 0800 780 078 (bure).',
                'keywords' => ['tin', 'tra', 'kodi', 'kodi ya mapato', 'vat', 'efd', 'risiti', 'mlipakodi', 'makisio', 'biashara ndogo'],
                'source' => 'TRA - tra.go.tz',
            ],
            [
                'title' => 'Malipo ya serikali kwa control number (GePG)',
                'category' => 'government', 'language' => 'sw',
                'content' => 'Malipo yote ya serikali (ada, faini, leseni, kodi, huduma za NIDA/RITA/Uhamiaji, ada za shule za umma) hufanywa kupitia Mfumo wa Malipo ya Serikali (GePG). Ofisi husika hukupa namba ya kumbukumbu ya malipo (control number) yenye tarakimu 12 inayoanza na 99. Lipa kwa pesa za simu (M-Pesa, Mixx by Yas, Airtel Money, HaloPesa: chagua "Lipa kwa Serikali/GePG"), benki, au wakala wa benki. Hifadhi ujumbe wa uthibitisho. Usilipe fedha taslimu kwa afisa bila control number na risiti rasmi. Kwa wanafunzi wa shule za umma, ada za bweni na michango halali hulipwa kwa control number ya shule.',
                'keywords' => ['gepg', 'control number', 'malipo ya serikali', 'namba ya kumbukumbu', 'kulipa faini', 'ada', 'kulipa serikali'],
                'source' => 'Wizara ya Fedha - GePG',
            ],
            [
                'title' => 'Umeme wa TANESCO na LUKU',
                'category' => 'government', 'language' => 'sw',
                'content' => 'TANESCO ndiyo kampuni ya umeme Tanzania Bara; kuunganishwa umeme omba ofisi ya TANESCO ya wilaya au mtandaoni (ukiwa na NIN, hati/barua ya mtendaji ya nyumba na ramani ya eneo), lipa ada ya kuunganisha kwa control number. Vijijini kuunganishwa kwa bei nafuu kupitia REA. Umeme wa kulipia kabla (LUKU) hununuliwa kwa pesa za simu, benki au mawakala kwa namba ya mita; weka tokeni ya tarakimu 20 kwenye mita. Kwa hitilafu au umeme kukatika ripoti kwa huduma kwa wateja TANESCO (ofisi ya karibu au namba iliyo kwenye tovuti tanesco.co.tz). Usijiunganishe umeme kinyume cha sheria; ni hatari na ni kosa la jinai.',
                'keywords' => ['tanesco', 'luku', 'umeme', 'kuunganisha umeme', 'mita', 'tokeni', 'rea', 'umeme umekatika'],
                'source' => 'TANESCO - tanesco.co.tz',
            ],

            // ---------------------------------------------------------- FINANCE / BUSINESS
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            [
                'title' => 'Utapeli wa pesa za simu na jinsi ya kujilinda',
                'category' => 'finance', 'language' => 'sw',
                'content' => 'Mbinu za kawaida za utapeli: ujumbe au simu ya "umeshinda zawadi", "nimetuma pesa kwa makosa nirudishie", "akaunti yako itafungwa thibitisha PIN", mikopo ya haraka ya mtandaoni inayotaka ada kwanza, na matangazo ya uwekezaji wa faida kubwa haraka (piramidi). Kanuni: 1) Usimpe mtu yeyote PIN, OTP au nywila, hata akijitambulisha kuwa wa benki/kampuni ya simu. 2) Hakiki salio lako kwa menyu rasmi kabla ya "kurudisha" pesa. 3) Mikopo halali hutoka benki, SACCOS au watoa huduma waliosajiliwa na BoT; hawatozi ada kabla ya kukopesha. 4) Ukitapeliwa ripoti mara moja kwa kampuni ya simu (kusimamisha muamala) na polisi (Kitengo cha Makosa ya Mtandao), hifadhi ujumbe na namba. Kupoteza simu: zuia laini na akaunti ya pesa mara moja kwa huduma kwa wateja.',
                'keywords' => ['utapeli', 'tapeli', 'm-pesa', 'pesa za simu', 'pin', 'otp', 'umeshinda', 'mkopo wa haraka', 'piramidi', 'kutapeliwa'],
                'source' => 'BoT; TCRA; Jeshi la Polisi',
            ],
            [
                'title' => 'Mobile money scams and how to stay safe',
                'category' => 'finance', 'language' => 'en',
                'content' => 'Common scams: "you have won a prize" messages or calls, "I sent money by mistake, please return it", "your account will be closed, confirm your PIN", instant online loans that ask for a fee first, and get-rich-quick investment schemes (pyramids). Rules: 1) Never share your PIN, OTP or password with anyone, even someone claiming to be from a bank or telecom. 2) Check your balance in the official menu before "returning" money. 3) Legitimate loans come from banks, SACCOS or BoT-licensed lenders; they never charge a fee before lending. 4) If scammed, report immediately to the mobile operator (to freeze the transaction) and the police cybercrime unit; keep the messages and numbers. Lost phone: block the SIM and money account at once via customer care.',
                'keywords' => ['scam', 'fraud', 'mobile money', 'pin', 'otp', 'won', 'instant loan', 'pyramid', 'lost phone'],
                'source' => 'BoT; TCRA; Tanzania Police',
            ],
            [
                'title' => 'Kuanzisha na kurasimisha biashara ndogo Tanzania',
                'category' => 'business', 'language' => 'sw',
                'content' => 'Hatua za kurasimisha: 1) Pata TIN kutoka TRA (bure). 2) Sajili jina la biashara au kampuni BRELA kupitia mfumo wa ORS (ors.brela.go.tz); jina la biashara ni rahisi na nafuu kwa mfanyabiashara mmoja. 3) Pata leseni ya biashara kutoka Halmashauri ya wilaya/jiji (biashara ndogo) au Wizara ya Viwanda na Biashara (biashara kubwa); ada hutegemea aina ya biashara. 4) Kwa vyakula, vipodozi na dawa unahitaji usajili/kibali cha TMDA, na kwa ubora wa bidhaa TBS. 5) Fungua akaunti ya benki ya biashara na tenganisha fedha za biashara na za nyumbani. Fursa: SIDO (mafunzo, mitaji midogo na viwanda vidogo), mikopo ya Halmashauri kwa vikundi vya vijana, wanawake na watu wenye ulemavu (uliza Afisa Maendeleo ya Jamii wa halmashauri kuhusu utaratibu wa sasa), VICOBA na SACCOS. Andika mpango wa biashara rahisi: bidhaa, wateja, gharama, bei, faida.',
                'keywords' => ['biashara', 'kusajili biashara', 'brela', 'leseni ya biashara', 'tin', 'sido', 'mtaji', 'mpango wa biashara', 'ujasiriamali', 'tmda', 'tbs', 'mikopo ya halmashauri'],
                'source' => 'BRELA; TRA; SIDO; Wizara ya Viwanda na Biashara',
            ],
            [
                'title' => 'Starting and formalising a small business in Tanzania',
                'category' => 'business', 'language' => 'en',
                'content' => 'Steps: 1) Get a TIN from TRA (free). 2) Register a business name or company with BRELA through the ORS system (ors.brela.go.tz); a business name is simpler and cheaper for a sole trader. 3) Get a business licence from the district/city council (small businesses) or the Ministry of Industry and Trade (large); fees depend on the type. 4) Food, cosmetics and medicines need TMDA registration; product quality marks come from TBS. 5) Open a business bank account and keep business and household money separate. Support: SIDO (training, small loans, industrial sheds), council loans for youth, women and persons with disabilities (ask the council Community Development Officer about the current procedure), VICOBA and SACCOS. Write a simple business plan: product, customers, costs, price, profit.',
                'keywords' => ['business', 'register business', 'brela', 'business licence', 'tin', 'sido', 'capital', 'business plan', 'entrepreneur', 'tmda', 'tbs', 'council loans'],
                'source' => 'BRELA; TRA; SIDO; Ministry of Industry and Trade',
            ],

            // ---------------------------------------------------------- FAMILY
            [
                'title' => 'Ukatili wa kijinsia na dhidi ya watoto: wapi pa kupata msaada',
                'category' => 'family', 'language' => 'sw',
                'content' => 'Ukatili wa kijinsia (kupigwa, kubakwa, kulazimishwa ndoa, kunyimwa matunzo) na ukatili dhidi ya watoto ni makosa ya jinai. Hatua: 1) Kwa hatari ya sasa piga 112 (polisi) au nenda kituo cha polisi cha karibu na uombe Dawati la Jinsia na Watoto. 2) Kwa ubakaji nenda hospitali ndani ya saa 72 kabla ya kuoga ili kupata matibabu ya kinga (PEP, dharura ya uzazi) na fomu ya PF3 (polisi hutoa PF3 bure). 3) Piga 116 (Msaada kwa Mtoto, bure, saa 24) kwa watoto. 4) Afisa Ustawi wa Jamii wa halmashauri husaidia matunzo ya watoto na makazi salama; vituo vya huduma za mkono mmoja (One Stop Centre) vipo hospitali za mikoa. 5) Msaada wa kisheria bure: TAWLA, WLAC, LHRC, Kituo cha Sheria na Haki za Binadamu, Kampeni ya Msaada wa Kisheria ya Mama Samia. Sheria ya Mtoto 2009 inakataza adhabu zinazodhuru, ajira za watoto na ndoa za utotoni.',
                'keywords' => ['ukatili', 'kupigwa', 'ubakaji', 'dawati la jinsia', '116', 'pf3', 'ustawi wa jamii', 'mtoto', 'msaada wa kisheria', 'ndoa za utotoni', 'gbv'],
                'source' => 'Sheria ya Mtoto 2009; Jeshi la Polisi; Wizara ya Maendeleo ya Jamii, Jinsia, Wanawake
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na Makundi Maalum',
            ],
            [
                'title' => 'Gender-based violence and violence against children: where to get help',
                'category' => 'family', 'language' => 'en',
                'content' => 'Gender-based violence (beating, rape, forced marriage, denial of maintenance) and violence against children are crimes. Steps: 1) In immediate danger call 112 or go to the nearest police station and ask for the Gender and Children\'s Desk. 2) After rape go to a hospital within 72 hours, before bathing, for preventive treatment (PEP, emergency contraception) and the PF3 form (police issue PF3 free). 3) Call 116 (Child Helpline, free, 24 hours) for children. 4) The council Social Welfare Officer helps with child maintenance and safe shelter; One Stop Centres exist at regional hospitals. 5) Free legal aid: TAWLA, WLAC, LHRC, the Mama Samia Legal Aid Campaign. The Law of the Child Act 2009 prohibits harmful punishment, child labour and child marriage.',
                'keywords' => ['violence', 'beaten', 'rape', 'gender desk', '116', 'pf3', 'social welfare', 'child', 'legal aid', 'child marriage', 'gbv', 'domestic violence'],
                'source' => 'Law of the Child Act 2009; Tanzania Police; Ministry of Community Development, Gender, Women and Special Groups',
            ],

            // ------------------------------------------------------------ TECHNOLOGY
            [
                'title' => 'Usajili wa laini ya simu na kulinda akaunti zako',
                'category' => 'technology', 'language' => 'sw',
                'content' => 'Laini zote za simu Tanzania lazima zisajiliwe kwa alama za vidole na NIN ya NIDA (watoto chini ya 18 kupitia mzazi). Laini isiyosajiliwa huzimwa. Mtu mmoja anaruhusiwa laini chache tu kwa kila mtandao; angalia laini zilizosajiliwa kwa jina lako kwa *106# na uzime usizozijua. Laini ikipotea: piga huduma kwa wateja (Vodacom 100, Yas/Tigo 100, Airtel 100, Halotel 100) kuizuia, kisha pata laini mbadala kwa kitambulisho. Usalama: tumia PIN ya simu na ya pesa isiyo tarehe ya kuzaliwa, usibonyeze viungo vya ujumbe usiovijua, washa uthibitisho wa hatua mbili WhatsApp (Mipangilio > Akaunti > Uthibitisho wa hatua mbili). Malalamiko ya huduma za mawasiliano: kwanza kwa kampuni, kisha TCRA (tcra.go.tz).',
                'keywords' => ['laini', 'sim', 'kusajili laini', 'nida', '106', 'simu imepotea', 'whatsapp', 'nywila', 'udukuzi', 'tcra', 'bando'],
                'source' => 'TCRA - tcra.go.tz',
            ],

            // ------------------------------------------------------------ TRANSPORT
            [
                'title' => 'Faini za trafiki, leseni ya udereva na ajali barabarani',
                'category' => 'transport', 'language' => 'sw',
                'content' => 'Faini ya kawaida ya makosa ya barabarani ni TZS 30,000 kwa kila kosa na hulipwa kwa control number (ujumbe wa SMS kutoka mfumo wa polisi), kwa pesa za simu au benki, si fedha taslimu kwa askari; unaweza kuhakiki faini kwa namba ya gari kupitia mfumo wa TMS. Leseni ya udereva: jifunze chuo cha udereva, fanya mtihani wa nadharia na vitendo wa Polisi wa Usalama Barabarani, kisha leseni hutolewa na TRA kwa ada kulingana na daraja (A pikipiki, B magari madogo, C magari makubwa, D mabasi). Ukipata ajali: hakikisha usalama, piga 112/115, usiondoe magari kabla ya polisi (isipokuwa kuzuia hatari), chukua picha na majina ya mashahidi, ripoti kituo cha polisi ndani ya saa 24 na pata ripoti ya ajali kwa ajili ya bima. Bima ya gari ya lazima ni ya mtu wa tatu (third party).',
                'keywords' => ['faini', 'trafiki', 'leseni ya udereva', 'ajali', 'bima ya gari', 'tms', 'control number', 'polisi wa barabarani', 'bodaboda'],
                'source' => 'Sheria ya Usalama Barabarani Sura 168; Jeshi la Polisi; TRA',
            ],

            // ------------------------------------------------------------ GENERAL
            [
                'title' => 'Tanzania kwa ufupi: historia na taarifa za msingi',
                'category' => 'general', 'language' => 'sw',
                'content' => 'Tanganyika ilipata uhuru 9 Desemba 1961 (Rais wa kwanza Mwalimu Julius Nyerere) na Zanzibar ilipata mapinduzi 12 Januari 1964; ziliungana 26 Aprili 1964 kuunda Jamhuri ya Muungano wa Tanzania. Makao makuu ni Dodoma; jiji kubwa ni Dar es Salaam. Sensa ya 2022 ilihesabu watu takriban milioni 61.7. Lugha ya taifa ni Kiswahili. Rais wa sasa ni Samia Suluhu Hassan (tangu Machi 2021). Mlima Kilimanjaro (mita 5,895) ndio mlima mrefu zaidi Afrika; Ziwa Victoria ndilo ziwa kubwa zaidi Afrika; Hifadhi ya Serengeti na Ngorongoro ni maarufu duniani. Tanzania ni mwanachama wa Jumuiya ya Afrika Mashariki (EAC), SADC na Umoja wa Afrika. Sikukuu za taifa: Mapinduzi ya Zanzibar (12 Jan), Muungano (26 Apr), Saba Saba (7 Jul), Nane Nane (8 Ago), Nyerere (14 Okt), Uhuru (9 Des).',
                'keywords' => ['tanzania', 'uhuru', 'muungano', 'nyerere', 'rais', 'dodoma', 'sensa', 'kilimanjaro', 'sikukuu', 'historia', 'idadi ya watu'],
                'source' => 'Ofisi ya Taifa ya Takwimu (NBS); Ikulu',
            ],
            [
                'title' => 'Tanzania at a glance: history and key facts',
                'category' => 'general', 'language' => 'en',
                'content' => 'Tanganyika gained independence on 9 December 1961 (first President Mwalimu Julius Nyerere); the Zanzibar Revolution was on 12 January 1964; the two united on 26 April 1964 to form the United Republic of Tanzania. The capital is Dodoma; the largest city is Dar es Salaam. The 2022 census counted about 61.7 million people. Swahili is the national language. The current President is Samia Suluhu Hassan (since March 2021). Mount Kilimanjaro (5,895 m) is Africa\'s highest mountain; Lake Victoria is Africa\'s largest lake; Serengeti and Ngorongoro are world famous.
```

```
Tanzania belongs to the East African Community, SADC and the African Union. National holidays: Zanzibar Revolution (12 Jan), Union Day (26 Apr), Saba Saba (7 Jul), Nane Nane (8 Aug), Nyerere Day (14 Oct), Independence (9 Dec).',
                'keywords' => ['tanzania', 'independence', 'union', 'nyerere', 'president', 'dodoma', 'census', 'kilimanjaro', 'holidays', 'history', 'population'],
                'source' => 'National Bureau of Statistics (NBS); State House',
            ],
        ];

        foreach ($entries as $e) {
            KnowledgeEntry::query()->updateOrCreate(
                ['title' => $e['title'], 'language' => $e['language']],
                array_merge($e, ['is_active' => true, 'summary' => $e['summary'] ?? \Illuminate\Support\Str::limit($e['content'], 160)])
            );
        }

        $this->command?->info('Knowledge entries seeded: ' . count($entries));
    }
}
```

66 database/seeders/LocalResourceSeeder.php

41 lines

```php
<?php

namespace Database\Seeders;

use App\Models\LocalResource;
use Illuminate\Database\Seeder;

/**
 * Starter place-bound services. Run `php artisan huru:import-legal-aid`
 * to pull the full Mama Samia Legal Aid Campaign provider list.
 */
class LocalResourceSeeder extends Seeder
{
    public function run(): void
    {
        $resources = [
            // Legal aid (carried over from the previous platform)
            ['HAKI MAENDELEO', 'legal_aid', 'dar es salaam', 'kinondoni', 'Sinza', 'Plot 512, Sinza Madukani, Wilaya ya Kinondoni (karibu na Sinza Mori).', '+255 754 555 555', 'info@hakimaendeleo.or.tz', 'seed'],
            ['GRACE FOUNDATION', 'legal_aid', 'dar es salaam', 'kinondoni', 'Mwenge', 'S.L.P. 31238, Mwenge (Josam House), Wilaya ya Kinondoni.', '+255 222 777 777', 'info@gracefoundation.or.tz', 'seed'],
            ['DIGNITY KWANZA - COMMUNITY SOLUTIONS', 'legal_aid', 'dar es salaam', 'kinondoni', 'Mikocheni', 'S.L.P. 33035, Mikocheni B, Wilaya ya Kinondoni.', '+255 784 888 888', 'info@dignitykwanza.org', 'seed'],
            ['KAHAMA PARALEGAL AID ORGANIZATION (KAPAO)', 'legal_aid', 'shinyanga', 'kahama', null, 'Ofisi za KAPAO, Wilaya ya Kahama, Mkoa wa Shinyanga.', '+255 767 994457', 'kahamaparalegal@gmail.com', 'seed'],

            // Referral hospitals (phone numbers to be confirmed by the operator)
            ['Hospitali ya Taifa Muhimbili (MNH)', 'health', 'dar es salaam', 'ilala', 'Upanga', 'Upanga, Ilala, Dar es Salaam. Hospitali ya rufaa ya taifa.', null, null, 'seed'],
            ['Hospitali ya Rufaa ya Kanda Bugando', 'health', 'mwanza', 'nyamagana', null, 'Bugando Hill, Nyamagana, Mwanza. Hospitali ya rufaa ya Kanda ya Ziwa.', null, null, 'seed'],
            ['Hospitali ya Rufaa ya Kanda KCMC', 'health', 'kilimanjaro', 'moshi', null, 'Moshi, Kilimanjaro. Hospitali ya rufaa ya Kanda ya Kaskazini.', null, null, 'seed'],
            ['Hospitali ya Benjamin Mkapa', 'health', 'dodoma', 'dodoma', null, 'Chuo Kikuu cha Dodoma, Dodoma. Hospitali ya rufaa ya kanda.', null, null, 'seed'],
            ['Hospitali ya Rufaa ya Kanda Mbeya', 'health', 'mbeya', 'mbeya', null, 'Mbeya mjini. Hospitali ya rufaa ya Nyanda za Juu Kusini.', null, null, 'seed'],
            ['Hospitali ya Mnazi Mmoja', 'health', 'mjini magharibi', 'mjini', null, 'Mjini Unguja, Zanzibar. Hospitali ya rufaa ya Zanzibar.', null, null, 'seed'],
        ];

        foreach ($resources as [$name, $category, $region, $district, $ward, $location, $phone, $email, $source]) {
            LocalResource::query()->updateOrCreate(
                ['name' => $name],
                compact('category', 'region', 'district', 'ward', 'location', 'phone', 'email', 'source') + ['is_active' => true]
            );
        }

        $this->command?->info('Local resources seeded: ' . count($resources));
    }
}
```

67 database/seeders/PromptTemplateSeeder.php

39 lines

```php
<?php

namespace Database\Seeders;

use App\Models\PromptTemplate;
use Illuminate\Database\Seeder;

class PromptTemplateSeeder extends Seeder
{
    public function run(): void
    {
        $templates = [
            ['general', 'sw', 'Wewe ni {app_name}, msaidizi wa kuaminika wa kila siku kwa wananchi wa Tanzania. Unajibu swali lolote la mada yoyote: maisha ya kila siku, masomo, kazi, familia, sheria, afya, kilimo, fedha, teknolojia au jambo lolote la udadisi. Unazungumza kama jirani Mtanzania mwenye maarifa na heshima: wazi, moja kwa moja, mkarimu, bila kudharau. Unafahamu vizuri jinsi mambo yanavyofanyika Tanzania na unatoa hatua za vitendo zinazowezekana kwa mwananchi wa kawaida.'],
            ['general', 'en', 'You are {app_name}, a trusted everyday helper for the people of Tanzania. You answer any question on any subject: daily life, school subjects, work, family, law, health, farming, money, technology or simple curiosity. You speak like a knowledgeable, respectful Tanzanian neighbour: clear, direct, warm, never condescending. You understand how things actually work in Tanzania and give practical steps an ordinary citizen can take.'],

            ['legal', 'sw', 'Wewe ni {app_name}, mwelimishaji wa sheria na haki kwa wananchi wa Tanzania. Unaeleza sheria za Tanzania kwa Kiswahili rahisi, unaonyesha hatua halisi za kuchukua na ofisi za kuanzia, na unamshauri mwananchi aonane na mtoa msaada wa kisheria au wakili pale jambo linapohitaji. Wewe si wakili wake; unatoa maelezo ya jumla yanayomsaidia kuelewa haki zake na kuchukua hatua kwa usalama.'],
            ['legal', 'en', 'You are {app_name}, a law and rights educator for the people of Tanzania. You explain Tanzanian law in simple language, show the real steps to take and the office to start at, and advise seeing a legal aid provider or advocate when the matter needs one. You are not the citizen\'s lawyer; you give general information that helps them understand their rights and act safely.'],

            ['health', 'sw', 'Wewe ni {app_name}, mtoa elimu ya afya kwa jamii ya Tanzania. Unatoa taarifa za afya za jumla zinazokubalika (Wizara ya Afya, WHO) kwa lugha rahisi, unaonyesha dalili za hatari zinazohitaji kwenda kituo cha afya mara moja, na unaelekeza huduma zilizopo karibu. Wewe si daktari na husemi dozi za dawa za kuandikiwa; unasisitiza kuonana na mtaalamu wa afya kwa uchunguzi.'],
            ['health', 'en', 'You are {app_name}, a community health educator for Tanzania. You give generally accepted health information (Ministry of Health, WHO) in simple words, point out danger signs that need a health facility immediately, and direct people to the care available near them. You are not a doctor and do not give prescription dosages; you always encourage seeing a health worker for diagnosis.'],

            ['education', 'sw', 'Wewe ni {app_name}, mwalimu msaidizi mvumilivu kwa wanafunzi, wazazi na walimu wa Tanzania. Unafundisha somo lolote (hisabati, sayansi, lugha, historia, jiografia, biashara, TEHAMA) kwa hatua rahisi na mifano ya Kitanzania, kwa kufuata mitaala ya Tanzania (NECTA, TIE). Unaeleza pia taratibu za shule, mitihani, mikopo ya HESLB na vyuo.'],
            ['education', 'en', 'You are {app_name}, a patient tutor for Tanzanian students, parents and teachers. You teach any subject (mathematics, sciences, languages, history, geography, commerce, ICT) step by step with Tanzanian examples, following the Tanzanian curriculum (NECTA, TIE). You also explain school procedures, examinations, HESLB loans and college admissions.'],

            ['agriculture', 'sw', 'Wewe ni {app_name}, afisa ugani wa kidijitali kwa wakulima, wafugaji na wavuvi wa Tanzania. Unatoa ushauri wa vitendo kulingana na misimu, udongo na mazao ya eneo husika, unaonyesha wapi pa kupata mbegu bora, pembejeo, chanjo za mifugo na masoko, na unaelekeza kwa afisa ugani wa kata/wilaya, TARI na TMA.'],
            ['agriculture', 'en', 'You are {app_name}, a digital extension officer for Tanzanian farmers, livestock keepers and fishers. You give practical advice matched to local seasons, soils and crops, show where to find certified seed, inputs, livestock vaccines and markets, and point to the ward/district extension officer, TARI and TMA.'],
        ];

        foreach ($templates as [$category, $language, $text]) {
            $name = ucfirst($category) . ' (' . strtoupper($language) . ')';
            PromptTemplate::query()->updateOrCreate(
                ['name' => $name],
                ['category' => $category, 'language' => $language, 'template' => $text, 'is_active' => true]
            );
        }

        $this->command?->info('Prompt templates seeded: ' . count($templates));
    }
}
```

68 database/seeders/ReferenceContactSeeder.php

84 lines

```php
<?php

namespace Database\Seeders;

use App\Models\ReferenceContact;
use Illuminate\Database\Seeder;

/**
 * National directory. Entries with verified_at set are universally known;
 * the rest should be checked by the operator (the admin panel flags them).
 */
class ReferenceContactSeeder extends Seeder
{
    public function run(): void
    {
        $verified = now();

        $contacts = [
            // Emergency
            ['Polisi (dharura) / Police emergency', 'police', '112', null, null, null, 'Namba ya dharura ya polisi, bure kutoka mtandao wowote.', 'Police emergency number, free from any network.', true, $verified, 1],
            ['Zimamoto na uokoaji / Fire and rescue', 'police', '114', null, null, null, 'Moto, uokoaji na majanga.', 'Fire, rescue and disasters.', true, $verified, 2],
            ['Gari la wagonjwa / Ambulance', 'health', '115', null, null, null, 'Dharura za kiafya na gari la wagonjwa.', 'Medical emergencies and ambulance.', true, $verified, 3],
            ['Msaada kwa Mtoto / National Child Helpline (C-Sema)', 'family', '116', null, null, 'https://www.sematanzania.org', 'Bure, saa 24, kwa ukatili au hatari kwa watoto na ukatili wa kijinsia.', 'Free, 24 hours, for violence or danger to children and gender-based violence.', true, $verified, 4],

            // Legal
            ['Kampeni ya Msaada wa Kisheria ya Mama Samia (Wizara ya Katiba na Sheria)', 'legal', null, null, 'km@katiba.go.tz', 'https://www.mslac.or.tz', 'Madawati ya msaada wa kisheria bure ngazi ya wilaya na mikoa; orodha ya watoa huduma ipo mslac.or.tz. S.L.P. 315 Dodoma.', 'Free legal aid desks at district and regional level; provider list at mslac.or.tz. P.O. Box 315 Dodoma.', false, null, 10],
            ['Ofisi ya Mwanasheria Mkuu wa Serikali (OAG)', 'legal', '+255 26 296 3647', null, 'info@oag.go.tz', 'https://www.agctz.go.tz', 'Mji wa Serikali Mtumba, S.L.P. 11492, Dodoma.', 'Mtumba Government City, P.O. Box 11492, Dodoma.', false, null, 11],
            ['Tume ya Haki za Binadamu na Utawala Bora (CHRAGG)', 'legal', null, null, null, 'https://www.chragg.go.tz', 'Malalamiko ya ukiukwaji wa haki za binadamu na utawala mbaya na ofisi za umma.', 'Complaints about human rights violations and maladministration by public offices.', false, null, 12],
            ['Kituo cha Sheria na Haki za Binadamu (LHRC)', 'legal', null, null, null, 'https://www.humanrights.or.tz', 'Msaada wa kisheria na elimu ya haki; ofisi Dar es Salaam, Arusha, Dodoma na mikoani.', 'Legal aid and rights education; offices in Dar es Salaam, Arusha, Dodoma and regions.', false, null, 13],
            ['Chama cha Wanasheria Wanawake Tanzania (TAWLA)', 'legal', null, null, null, 'https://www.tawla.or.tz', 'Msaada wa kisheria bure hasa kwa wanawake na watoto.', 'Free legal aid, especially for women and children.', false, null, 14],
            ['Kituo cha Msaada wa Kisheria kwa Wanawake (WLAC)', 'legal', null, null, null, 'https://www.wlac.or.tz', 'Msaada wa kisheria kwa wanawake: ndoa, mirathi, ukatili.', 'Legal aid for women: marriage, inheritance, violence.', false, null, 15],
            ['Taasisi ya Kuzuia na Kupambana na Rushwa (TAKUKURU / PCCB)', 'legal', '113', null, null, 'https://www.pccb.go.tz', 'Ripoti rushwa; namba 113 ni bure.', 'Report corruption; 113 is toll free.', false, null, 16],

            // Government
            ['NIDA - Mamlaka ya Vitambulisho vya Taifa', 'government', null, null, null, 'https://www.nida.go.tz', 'Usajili wa kitambulisho cha taifa (NIN). Hakiki hali kwa *152*00#.', 'National ID registration (NIN). Check status via *152*00#.', false, null, 20],
            ['RITA - Wakala wa Usajili, Ufilisi na Udhamini', 'government', null, null, null, 'https://www.rita.go.tz', 'Vyeti vya kuzaliwa, kifo, ndoa na talaka.', 'Birth, death, marriage and divorce certificates.', false, null, 21],
            ['Idara ya Uhamiaji', 'government', null, null, null, 'https://www.immigration.go.tz', 'Pasipoti, viza na vibali vya ukaazi. Maombi: eservices.immigration.go.tz', 'Passports, visas and residence permits. Apply at eservices.immigration.go.tz', false, null, 22],
            ['BRELA - Wakala wa Usajili wa Biashara na Leseni', 'business', null, null, null, 'https://www.brela.go.tz', 'Usajili wa majina ya biashara na kampuni kupitia ORS.', 'Business name and company registration through ORS.', false, null, 23],
            ['TANESCO - Shirika la Umeme Tanzania', 'government', null, null, null, 'https://www.tanesco.co.tz', 'Kuunganishwa umeme, LUKU na hitilafu za umeme.', 'Electricity connection, LUKU tokens and faults.', false, null, 24],
            ['Sekretarieti ya Ajira katika Utumishi wa Umma', 'employment', null, null, null, 'https://www.ajira.go.tz', 'Nafasi rasmi za kazi serikalini; maombi ni bure.', 'Official public service vacancies; applications are free.', false, null, 25],

```

```
            // Finance
            ['TRA - Mamlaka ya Mapato Tanzania', 'finance', '0800 750 075', '0800 780 078', null, 'https://www.tra.go.tz', 'TIN, kodi, leseni za udereva, usajili wa magari. Namba za bure.', 'TIN, taxes, driving licences, vehicle registration. Toll-free numbers.', false, null, 30],
            ['Benki Kuu ya Tanzania (BoT)', 'finance', null, null, null, 'https://www.bot.go.tz', 'Orodha ya benki na watoa mikopo waliosajiliwa; malalamiko ya huduma za fedha.', 'List of licensed banks and lenders; financial service complaints.', false, null, 31],
            ['NSSF - Mfuko wa Taifa wa Hifadhi ya Jamii', 'employment', null, null, null, 'https://www.nssf.or.tz', 'Michango na mafao ya wafanyakazi wa sekta binafsi na wanaojiajiri.', 'Contributions and benefits for private sector and self-employed workers.', false, null, 32],

            // Health
            ['NHIF - Mfuko wa Taifa wa Bima ya Afya', 'health', null, null, null, 'https://www.nhif.or.tz', 'Bima ya afya ya umma; vifurushi vya hiari kwa wananchi.', 'Public health insurance; voluntary packages for citizens.', false, null, 40],
            ['Wizara ya Afya', 'health', null, null, null, 'https://www.moh.go.tz', 'Miongozo ya afya, chanjo na taarifa za milipuko.', 'Health guidelines, vaccination and outbreak information.', false, null, 41],
            ['TMDA - Mamlaka ya Dawa na Vifaa Tiba', 'health', null, null, null, 'https://www.tmda.go.tz', 'Usalama wa dawa, vyakula na vipodozi; ripoti dawa bandia.', 'Safety of medicines, food and cosmetics; report counterfeit medicines.', false, null, 42],

            // Education
            ['NECTA - Baraza la Mitihani la Tanzania', 'education', null, null, null, 'https://www.necta.go.tz', 'Matokeo ya mitihani ya taifa na ratiba.', 'National examination results and timetables.', false, null, 50],
            ['HESLB - Bodi ya Mikopo ya Wanafunzi wa Elimu ya Juu', 'education', null, null, null, 'https://www.heslb.go.tz', 'Mikopo ya elimu ya juu; maombi kupitia OLAMS.', 'Higher education student loans; apply through OLAMS.', false, null, 51],
            ['TCU - Tume ya Vyuo Vikuu Tanzania', 'education', null, null, null, 'https://www.tcu.go.tz', 'Vyuo vikuu vilivyosajiliwa na mwongozo wa udahili.', 'Accredited universities and admission guidebook.', false, null, 52],
            ['VETA - Mamlaka ya Elimu na Mafunzo ya Ufundi Stadi', 'education', null, null, null, 'https://www.veta.go.tz', 'Mafunzo ya ufundi stadi na vyuo vya VETA mikoani.', 'Vocational training and VETA colleges in the regions.', false, null, 53],

            // Agriculture
            ['TARI - Taasisi ya Utafiti wa Kilimo Tanzania', 'agriculture', null, null, null, 'https://www.tari.go.tz', 'Teknolojia za kilimo, mbegu bora na ushauri wa mazao.', 'Agricultural technologies, improved seed and crop advice.', false, null, 60],
            ['TMA - Mamlaka ya Hali ya Hewa Tanzania', 'agriculture', null, null, null, 'https://www.meteo.go.tz', 'Utabiri wa mvua na hali ya hewa kwa misimu ya kilimo.', 'Rainfall and weather forecasts for farming seasons.', false, null, 61],
            ['Wizara ya Kilimo', 'agriculture', null, null, null, 'https://www.kilimo.go.tz', 'Sera, pembejeo za ruzuku na masoko ya mazao.', 'Policy, subsidised inputs and crop markets.', false, null, 62],

            // Technology
            ['TCRA - Mamlaka ya Mawasiliano Tanzania', 'technology', null, null, null, 'https://www.tcra.go.tz', 'Malalamiko ya huduma za simu na intaneti, usajili wa laini.', 'Complaints about phone and internet services, SIM registration.', false, null, 70],

            // Transport
            ['LATRA - Mamlaka ya Udhibiti Usafiri Ardhini', 'transport', null, null, null, 'https://www.latra.go.tz', 'Nauli za mabasi, leseni za usafiri na malalamiko ya abiria.', 'Bus fares, transport licences and passenger complaints.', false, null, 80],
            ['Polisi wa Usalama Barabarani (Traffic Police)', 'transport', '112', null, null, null, 'Ajali na makosa ya barabarani; faini hulipwa kwa control number.', 'Accidents and traffic offences; fines are paid via control number.', false, null, 81],
        ];

        foreach ($contacts as [$name, $category, $phone, $alt, $email, $web, $sw, $en, $emergency, $verifiedAt, $order]) {
            ReferenceContact::query()->updateOrCreate(
                ['name' => $name],
                [
                    'category' => $category, 'phone' => $phone, 'alt_phone' => $alt, 'email' => $email, 'website' => $web,
                    'description_sw' => $sw, 'description_en' => $en, 'is_emergency' => $emergency,
                    'verified_at' => $verifiedAt, 'sort_order' => $order, 'is_active' => true,
                ]
            );
        }

        $this->command?->info('Reference contacts seeded: ' . count($contacts));
    }
}
```

69 database/seeders/SettingsSeeder.php

25 lines

```php
<?php

namespace Database\Seeders;

use App\Models\SystemSetting;
use App\Support\Settings;
use Illuminate\Database\Seeder;

class SettingsSeeder extends Seeder
{
    public function run(): void
    {
        foreach (Settings::schema() as $key => $def) {
            SystemSetting::query()->firstOrCreate(['key' => $key], ['value' => $def['default']]);
        }

        // Environment can pre-fill the operator contact.
        if ($email = env('ADMIN_EMAIL')) {
            SystemSetting::query()->updateOrCreate(['key' => 'contact_email'], ['value' => $email]);
        }

        Settings::flush();
        $this->command?->info('Settings seeded (' . count(Settings::schema()) . ' keys).');
    }
}
```

70 package.json

17 lines

```
{
    "$schema": "https://www.schemastore.org/package.json",
    "private": true,
    "type": "module",
    "scripts": {
        "build": "vite build",
        "dev": "vite"
    },
    "devDependencies": {
        "@tailwindcss/vite": "^4.0.0",
        "axios": "^1.11.0",
        "concurrently": "^9.0.1",
        "laravel-vite-plugin": "^2.0.0",
        "tailwindcss": "^4.0.0",
        "vite": "^7.3.3"
    }
}
```

71 phpunit.xml

35 lines

```xml
<?xml version="1.0" encoding="UTF-8"?>
<phpunit xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:noNamespaceSchemaLocation="vendor/phpunit/phpunit/phpunit.xsd"
         bootstrap="vendor/autoload.php"
         colors="true"
>
    <testsuites>
        <testsuite name="Unit">
            <directory>tests/Unit</directory>
        </testsuite>
        <testsuite name="Feature">
            <directory>tests/Feature</directory>
        </testsuite>
    </testsuites>
    <source>
        <include>
            <directory>app</directory>
        </include>
    </source>
    <php>
        <env name="APP_ENV" value="testing"/>
        <env name="APP_MAINTENANCE_DRIVER" value="file"/>
        <env name="BCRYPT_ROUNDS" value="4"/>
        <env name="BROADCAST_CONNECTION" value="null"/>
        <env name="CACHE_STORE" value="array"/>
        <env name="DB_CONNECTION" value="sqlite"/>
        <env name="DB_DATABASE" value=":memory:"/>
        <env name="MAIL_MAILER" value="array"/>
        <env name="QUEUE_CONNECTION" value="sync"/>
        <env name="SESSION_DRIVER" value="array"/>
        <env name="PULSE_ENABLED" value="false"/>
        <env name="TELESCOPE_ENABLED" value="false"/>
        <env name="NIGHTWATCH_ENABLED" value="false"/>
    </php>
</phpunit>
```

72 public/manifest.json

16 lines

```
{
    "name": "Huru SMS",
    "short_name": "Huru SMS",
    "description": "Uliza lolote kuhusu maisha Tanzania. Kwa SMS 15054 au mtandaoni.",
    "start_url": "/chat",
    "scope": "/",
    "display": "standalone",
    "lang": "sw",
    "background_color": "#f8fafc",
    "theme_color": "#15803d",
    "icons": [
        { "src": "/logo.svg", "sizes": "192x192", "type": "image/svg+xml" },
        { "src": "/logo.svg", "sizes": "512x512", "type": "image/svg+xml" },
        { "src": "/logo.svg", "sizes": "any", "type": "image/svg+xml", "purpose": "any maskable" }
    ]
}
```

73 public/robots.txt

6 lines

```
User-agent: *
Allow: /
Disallow: /admin
Disallow: /chat/
Disallow: /api/
Sitemap: /sitemap.xml
```

74 public/sw.js

40 lines

```
/* Huru SMS service worker: app shell cached, API never cached, offline fallback page. */
const VERSION = 'huru-v1';
const SHELL = ['/', '/chat', '/offline', '/logo.svg', '/manifest.json'];

self.addEventListener('install', (event) => {
    event.waitUntil(caches.open(VERSION).then((c) => c.addAll(SHELL)).then(() => self.skipWaiting()));
});

self.addEventListener('activate', (event) => {
    event.waitUntil(
        caches.keys().then((keys) => Promise.all(keys.filter((k) => k !== VERSION).map((k) => caches.delete(k)))).then(() => self.clients.claim())
    );
});

self.addEventListener('fetch', (event) => {
    const req = event.request;
    const url = new URL(req.url);

    if (req.method !== 'GET' || url.origin !== self.location.origin) return;
    // Never cache dynamic data or the admin panel.
    if (url.pathname.startsWith('/chat/') || url.pathname.startsWith('/admin') || url.pathname.startsWith('/api') || url.pathname.startsWith('/community')) return;

    if (req.mode === 'navigate') {
        event.respondWith(
            fetch(req)
                .then((res) => { const copy = res.clone(); caches.open(VERSION).then((c) => c.put(req, copy)); return res; })
                .catch(() => caches.match(req).then((hit) => hit || caches.match('/offline')))
        );
        return;
    }

    event.respondWith(
        caches.match(req).then((hit) => hit || fetch(req).then((res) => {
            if (res.ok && (url.pathname.startsWith('/build/') || /\.(svg|png|ico|css|js|woff2?)$/.test(url.pathname))) {
                const copy = res.clone(); caches.open(VERSION).then((c) => c.put(req, copy));
            }
            return res;
        }))
    );
});
```

75 resources/css/app.css

11 lines

```
@import 'tailwindcss';

@source '../../vendor/laravel/framework/src/Illuminate/Pagination/resources/views/*.blade.php';
@source '../../storage/framework/views/*.php';
@source '../**/*.blade.php';
@source '../**/*.js';

@theme {
    --font-sans: 'Instrument Sans', ui-sans-serif, system-ui, sans-serif, 'Apple Color Emoji', 'Segoe UI Emoji',
        'Segoe UI Symbol', 'Noto Color Emoji';
}
```

76 resources/js/app.js

1 lines

```
import './bootstrap';
```

77 resources/js/bootstrap.js

4 lines

```
import axios from 'axios';
window.axios = axios;

window.axios.defaults.headers.common['X-Requested-With'] = 'XMLHttpRequest';
```

78 resources/views/admin/contacts/form.blade.php

30 lines

```
@extends('layouts.admin', ['title' => $contact->exists ? 'Edit contact' : 'New contact'])

@section('content')
<div class="card" style="max-width:860px">
    <form method="POST" action="{{ $contact->exists ? route('admin.contacts.update', $contact) : route('admin.contacts.store') }}">
        @csrf
        @if($contact->exists) @method('PUT') @endif
        <div class="field"><label>Name</label><input type="text" name="name" value="{{ old('name', $contact->name) }}" required maxlength="150"></div>
        <div class="grid g3">
            <div class="field"><label>Category</label><select name="category">@foreach($categories as $c)<option value="{{ $c }}" @selected(old('category', $contact->category) === $c)>{{ ucfirst($c) }}</option>@endforeach</select></div>
            <div class="field"><label>Phone</label><input type="text" name="phone" value="{{ old('phone', $contact->phone) }}" maxlength="60"></div>
            <div class="field"><label>Alternative phone</label><input type="text" name="alt_phone" value="{{ old('alt_phone', $contact->alt_phone) }}" maxlength="60"></div>
        </div>
        <div class="grid g2">
            <div class="field"><label>Email</label><input type="email" name="email" value="{{ old('email', $contact->email) }}"></div>
            <div class="field"><label>Website</label><input type="url" name="website" value="{{ old('website', $contact->website) }}" placeholder="https://"></div>
        </div>
        <div class="field"><label>Description (Swahili)</label><textarea name="description_sw" rows="2">{{ old('description_sw', $contact->description_sw) }}</textarea></div>
        <div class="field"><label>Description (English)</label><textarea name="description_en" rows="2">{{ old('description_en', $contact->description_en) }}</textarea></div>
        <div class="grid g3">
            <div class="field"><label>Sort order</label><input type="number" name="sort_order" value="{{ old('sort_order', $contact->sort_order ?? 100) }}" min="0" max="1000"></div>
            <div class="field" style="padding-top:26px"><label class="check"><input type="checkbox" name="is_emergency" value="1" @checked(old('is_emergency', $contact->is_emergency))> Emergency number</label></div>
            <div class="field" style="padding-top:26px"><label class="check"><input type="checkbox" name="verified" value="1" @checked(old('verified', (bool) $contact->verified_at))> Verified by operator</label></div>
        </div>
        <div class="field"><label class="check"><input type="checkbox" name="is_active" value="1" @checked(old('is_active', $contact->is_active))> Active</label></div>
        <button class="btn btn-p" type="submit">Save</button>
        <a class="btn" href="{{ route('admin.contacts.index') }}">Cancel</a>
    </form>
</div>
@endsection
```

79 resources/views/admin/contacts/index.blade.php

31 lines

```
@extends('layouts.admin', ['title' => 'National directory'])

@section('content')
<div class="card">
    <div class="card-h"><div><h2>Official institutions and hotlines</h2><p>Country-wide contacts the engine may quote. Unverified rows are marked; confirm them before relying on them.</p></div><a class="btn btn-p" href="{{ route('admin.contacts.create') }}">+ New contact</a></div>
    <form class="filters" method="GET">
        <select name="category" onchange="this.form.submit()"><option value="">All categories</option>@foreach($categories as $c)<option value="{{ $c }}" @selected(request('category') === $c)>{{ ucfirst($c) }}</option>@endforeach</select>
    </form>
    <table>
        <thead><tr><th>Name</th><th>Category</th><th>Phone</th><th>Email / Web</th><th>Flags</th><th></th></tr></thead>
        <tbody>
        @forelse($contacts as $c)
            <tr>
                <td><strong>{{ $c->name }}</strong><div style="font-size:.76rem;color:var(--muted)">{{ \Illuminate\Support\Str::limit($c->description_en ?: $c->description_sw, 100) }}</div></td>
                <td>{{ ucfirst($c->category) }}</td>
                <td style="white-space:nowrap">{{ $c->phone }}@if($c->alt_phone)<br>{{ $c->alt_phone }}@endif</td>
                <td style="font-size:.78rem">{{ $c->email }}@if($c->website)<br><a href="{{ $c->website }}" target="_blank" rel="noopener">{{ parse_url($c->website, PHP_URL_HOST) }}</a>@endif</td>
                <td style="white-space:nowrap">@if($c->is_emergency)<span class="badge b-red">emergency</span>@endif{!! $c->verified_at ? '<span class="badge b-green">verified</span>' : '<span class="badge b-warn">unverified</span>' !!} @if(!$c->is_active)<span class="badge b-gray">off</span>@endif</td>
                <td style="white-space:nowrap">
                    <a class="btn btn-sm" href="{{ route('admin.contacts.edit', $c) }}">Edit</a>
                    <form class="inline" method="POST" action="{{ route('admin.contacts.verify', $c) }}">@csrf @method('PATCH')<button class="btn btn-sm" type="submit">{{ $c->verified_at ? 'Unverify' : 'Verify' }}</button></form>
                    <form class="inline" method="POST" action="{{ route('admin.contacts.destroy', $c) }}" onsubmit="return confirm('Delete this contact?')">@csrf @method('DELETE')<button class="btn btn-sm btn-d" type="submit">Delete</button></form>
                </td>
            </tr>
        @empty
            <tr><td colspan="6" style="text-align:center;color:var(--muted)">No contacts.</td></tr>
        @endforelse
        </tbody>
    </table>
</div>
@endsection
```

80 resources/views/admin/conversations/index.blade.php

37 lines

```
@extends('layouts.admin', ['title' => 'Conversations'])

@section('content')
<div class="card">
    <form class="filters" method="GET">
        <input type="text" name="q" value="{{ request('q') }}" placeholder="Search text, phone or name">
        <select name="category"><option value="">All topics</option>@foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)<option value="{{ $k }}" @selected(request('category') === $k)>{{ $l['en'] }}</option>@endforeach</select>
        <select name="channel"><option value="">All channels</option><option value="sms" @selected(request('channel') === 'sms')>SMS</option><option value="web" @selected(request('channel') === 'web')>Web</option></select>
        <select name="direction"><option value="">Both directions</option><option value="inbound" @selected(request('direction') === 'inbound')>Questions</option><option value="outbound" @selected(request('direction') === 'outbound')>Replies</option></select>
        <input type="date" name="date" value="{{ request('date') }}">
        <button class="btn" type="submit">Filter</button>
        <a class="btn" href="{{ route('admin.conversations.index') }}">Reset</a>
    </form>
    <div style="overflow-x:auto">
    <table>
        <thead><tr><th>Time</th><th>Citizen</th><th>Ch.</th><th>Dir.</th><th>Lang</th><th>Topic</th><th>Content</th><th>Engine</th></tr></thead>
        <tbody>
        @forelse($messages as $m)
            <tr>
                <td style="white-space:nowrap;color:var(--muted)">{{ $m->created_at->timezone('Africa/Dar_es_Salaam')->format('d/m/y H:i') }}</td>
                <td style="white-space:nowrap"><a href="{{ route('admin.conversations.show', $m->user_id) }}">{{ $m->user?->name ?: \App\Support\Phone::mask($m->user?->phone_number) }}</a></td>
                <td><span class="badge b-gray">{{ $m->channel }}</span></td>
                <td>{!! $m->direction === 'inbound' ? '<span class="badge b-green">in</span>' : '<span class="badge b-blue">out</span>' !!}</td>
                <td>{{ strtoupper($m->language ?? '') }}</td>
                <td style="white-space:nowrap">{{ $m->category ? \App\Services\TopicClassifier::label($m->category, 'en') : '—' }}</td>
                <td class="truncate" title="{{ $m->content }}">{{ $m->content }}</td>
                <td style="white-space:nowrap;font-size:.75rem;color:var(--muted)">@if($m->aiLog){{ $m->aiLog->model }} · {{ $m->aiLog->total_tokens }} tok · {{ $m->aiLog->latency_ms }} ms @if($m->aiLog->status !== 'ok')<span class="badge b-red">{{ $m->aiLog->status }}</span>@endif @endif @if($m->feedback) {{ $m->feedback->rating === 1 ? '👍' : '👎' }}@endif</td>
            </tr>
        @empty
            <tr><td colspan="8" style="text-align:center;color:var(--muted)">Nothing matches.</td></tr>
        @endforelse
        </tbody>
    </table>
    </div>
    {{ $messages->links('pagination.simple') }}
</div>
@endsection
```

81 resources/views/admin/conversations/show.blade.php

34 lines

```
@extends('layouts.admin', ['title' => 'Conversation'])

@section('content')
<div class="card">
    <div class="card-h">
        <div>
            <h2>{{ $user->name ?: 'Citizen' }} · {{ $user->phone_number }}</h2>
            <p>Language: {{ $user->preferred_language ?: 'auto' }} · Region: {{ $user->region ? ucwords($user->region) : '—' }} · Strikes: {{ $user->abuse_count }} · Last seen: {{ $user->last_seen_at?->diffForHumans() ?? '—' }} @if($user->is_banned)<span class="badge b-red">BANNED</span>@endif @if($user->opted_out)<span class="badge b-warn">opted out</span>@endif</p>
        </div>
        <div style="display:flex;gap:6px">
            <form class="inline" method="POST" action="{{ route('admin.users.ban', $user) }}">@csrf @method('PATCH')<button class="btn btn-sm {{ $user->is_banned ? '' : 'btn-d' }}" type="submit">{{ $user->is_banned ? 'Unblock' : 'Block' }}</button></form>
            <form class="inline" method="POST" action="{{ route('admin.users.strikes', $user) }}">@csrf @method('PATCH')<button class="btn btn-sm" type="submit">Clear strikes</button></form>
            <a class="btn btn-sm" href="{{ route('admin.conversations.index') }}">Back</a>
        </div>
    </div>

    @foreach($messages as $m)
        <div style="border:1px solid var(--border);border-radius:10px;padding:12px 14px;margin-bottom:10px;background:{{ $m->direction === 'inbound' ? '#fff' : 'var(--green-bg)' }}">
            <div style="display:flex;justify-content:space-between;gap:10px;font-size:.75rem;color:var(--muted);margin-bottom:6px;flex-wrap:wrap">
                <span>{!! $m->direction === 'inbound' ? '<span class="badge b-green">question</span>' : '<span class="badge b-blue">reply</span>' !!} <span class="badge b-gray">{{ $m->channel }}</span> {{ $m->category ? \App\Services\TopicClassifier::label($m->category, 'en') : '' }}</span>
                <span>{{ $m->created_at->timezone('Africa/Dar_es_Salaam')->format('d M Y H:i:s') }}</span>
            </div>
            <div style="white-space:pre-wrap;font-size:.92rem">{{ $m->content }}</div>
            @if($m->aiLog)
                <details style="margin-top:8px;font-size:.8rem;color:var(--muted)">
                    <summary style="cursor:pointer">{{ $m->aiLog->model }} · {{ $m->aiLog->status }} · {{ $m->aiLog->prompt_tokens }}+{{ $m->aiLog->completion_tokens }} tokens · {{ $m->aiLog->latency_ms }} ms @if($m->feedback) · feedback {{ $m->feedback->rating === 1 ? '👍' : '👎' }} {{ $m->feedback->comment }}@endif</summary>
                    <div class="mono" style="margin-top:6px">{{ $m->aiLog->prompt }}</div>
                </details>
            @endif
        </div>
    @endforeach
    {{ $messages->links('pagination.simple') }}
</div>
@endsection
```

82 resources/views/admin/dashboard.blade.php

71 lines

```
@extends('layouts.admin', ['title' => 'Dashboard'])

@section('content')
@if(!$health['engine_configured'])<div class="alert a-err">GEMINI_API_KEY is missing. The engine cannot answer until it is set in .env.</div>@endif
@if(!$health['sms_configured'])<div class="alert a-warn">Africa's Talking credentials are missing. SMS replies and OTP codes will not be sent (web login opens without OTP outside production).</div>@endif
@if(!$health['webhook_secret'])<div class="alert a-warn">AT_WEBHOOK_SECRET is empty. Set it and register the callback URL with ?token=… before going live.</div>@endif

<div class="grid g4">
    <div class="stat"><div class="l">Citizens</div><div class="v">{{ number_format($stats['users']) }}</div><div class="s">+{{ $stats['users_today'] }} today</div></div>
    <div class="stat"><div class="l">Questions today</div><div class="v">{{ number_format($stats['questions_today']) }}</div><div class="s">{{ number_format($stats['questions_week']) }} this week</div></div>
    <div class="stat"><div class="l">Channels (7d)</div><div class="v">{{ number_format($stats['sms_week']) }} <span style="font-size:.9rem;color:var(--muted)">SMS</span></div><div class="s">{{ number_format($stats['web_week']) }} web</div></div>
    <div class="stat"><div class="l">Tokens today</div><div class="v">{{ number_format($stats['tokens_today']) }}</div><div class="s">avg {{ $stats['latency_avg'] }} ms · {{ $stats['errors_week'] }} errors (7d)</div></div>
</div>
<div style="height:14px"></div>
<div class="grid g4">
    <div class="stat"><div class="l">Helpful votes</div><div class="v" style="color:var(--green-dark)">{{ $stats['helpful'] }}</div><div class="s">{{ $stats['unhelpful'] }} not helpful</div></div>
    <div class="stat"><div class="l">Blocked citizens</div><div class="v" style="color:var(--danger)">{{ $stats['banned'] }}</div><div class="s">abuse strike system</div></div>
    <div class="stat"><div class="l">Engine</div><div class="v" style="font-size:1rem;margin-top:10px">{{ $health['engine_configured'] ? '● Configured' : '○ Missing key' }}</div><div class="s">queue: {{ $health['queue'] }}</div></div>
    <div class="stat"><div class="l">SMS gateway</div><div class="v" style="font-size:1rem;margin-top:10px">{{ $health['sms_configured'] ? '● Configured' : '○ Not configured' }}</div><div class="s">webhook secret {{ $health['webhook_secret'] ? 'set' : 'missing' }}</div></div>
</div>
<div style="height:18px"></div>

<div class="grid g2">
    <div class="card">
        <div class="card-h"><div><h2>Questions per day</h2><p>Last 14 days, both channels</p></div></div>
        @php $max = max(1, (int) ($daily->max() ?? 1)); @endphp
        <div class="spark">
            @for($i = 13; $i >= 0; $i--)
                @php $day = now()->subDays($i)->toDateString(); $v = (int) ($daily[$day] ?? 0); @endphp
                <div style="height:{{ max(3, round($v / $max * 100)) }}%" title="{{ $day }}: {{ $v }}"><span>{{ now()->subDays($i)->format('d') }}</span></div>
            @endfor
        </div>
        <div style="height:20px"></div>
    </div>
    <div class="card">
        <div class="card-h"><div><h2>Topics this week</h2><p>Detected by the topic classifier</p></div></div>
        @php $top = max(1, (int) ($byCategory->max('total') ?? 1)); @endphp
        <div class="bars">
            @forelse($byCategory as $row)
                <div class="bar"><span>{{ $row['label'] }}</span><div class="t"><div class="f" style="width:{{ round($row['total'] / $top * 100) }}%"></div></div><span>{{ $row['total'] }}</span></div>
            @empty
                <p style="color:var(--muted);font-size:.85rem">No questions yet.</p>
            @endforelse
        </div>
    </div>
</div>

<div class="card">
    <div class="card-h"><div><h2>Recent interactions</h2><p>Latest 25 messages across SMS and web</p></div><a class="btn btn-sm" href="{{ route('admin.conversations.index') }}">View all</a></div>
    <div style="overflow-x:auto">
    <table>
        <thead><tr><th>Time</th><th>Citizen</th><th>Ch.</th><th>Dir.</th><th>Topic</th><th>Content</th><th>Engine</th></tr></thead>
```

```
        <tbody>
        @forelse($recent as $m)
            <tr>
                <td style="white-space:nowrap;color:var(--muted)">{{ $m->created_at->timezone('Africa/Dar_es_Salaam')->format('d/m H:i') }}</td>
                <td><a href="{{ route('admin.conversations.show', $m->user_id) }}">{{ \App\Support\Phone::mask($m->user?->phone_number) }}</a>@if($m->user?->is_banned) <span class="badge b-red">banned</span>@elseif(($m->user?->abuse_count ?? 0) > 0) <span class="badge b-warn">strike {{ $m->user->abuse_count }}</span>@endif</td>
                <td><span class="badge b-gray">{{ $m->channel }}</span></td>
                <td>{!! $m->direction === 'inbound' ? '<span class="badge b-green">in</span>' : '<span class="badge b-blue">out</span>' !!}</td>
                <td>{{ $m->category ? \App\Services\TopicClassifier::label($m->category, 'en') : '—' }}</td>
                <td class="truncate" title="{{ $m->content }}">{{ $m->content }}</td>
                <td style="white-space:nowrap;font-size:.75rem;color:var(--muted)">@if($m->aiLog){{ $m->aiLog->total_tokens }} tok · {{ $m->aiLog->latency_ms }} ms @if($m->aiLog->status !== 'ok')<span class="badge b-red">{{ $m->aiLog->status }}</span>@endif @if($m->feedback) {{ $m->feedback->rating === 1 ? '👍' : '👎' }}@endif @endif</td>
            </tr>
        @empty
            <tr><td colspan="7" style="text-align:center;color:var(--muted)">No messages yet. Send HURU to the shortcode or open the web chat.</td></tr>
        @endforelse
        </tbody>
    </table>
    </div>
</div>
@endsection
```

83 resources/views/admin/knowledge/form.blade.php

22 lines

```
@extends('layouts.admin', ['title' => $entry->exists ? 'Edit knowledge entry' : 'New knowledge entry'])

@section('content')
<div class="card" style="max-width:860px">
    <form method="POST" action="{{ $entry->exists ? route('admin.knowledge.update', $entry) : route('admin.knowledge.store') }}">
        @csrf
        @if($entry->exists) @method('PUT') @endif
        <div class="field"><label>Title</label><input type="text" name="title" value="{{ old('title', $entry->title) }}" required maxlength="250"><div class="hint">Specific and searchable, e.g. "Kitambulisho cha Taifa (NIDA): jinsi ya kujisajili".</div></div>
        <div class="grid g2">
            <div class="field"><label>Topic</label><select name="category">@foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)<option value="{{ $k }}" @selected(old('category', $entry->category) === $k)>{{ $l['en'] }} / {{ $l['sw'] }}</option>@endforeach</select></div>
            <div class="field"><label>Language of this entry</label><select name="language"><option value="sw" @selected(old('language', $entry->language) === 'sw')>Swahili</option><option value="en" @selected(old('language', $entry->language) === 'en')>English</option></select></div>
        </div>
        <div class="field"><label>Content</label><textarea name="content" rows="10" required>{{ old('content', $entry->content) }}</textarea><div class="hint">Plain facts with steps, offices and figures. The engine quotes this verbatim as verified knowledge, so only write what you can stand behind.</div></div>
        <div class="field"><label>Summary (optional)</label><input type="text" name="summary" value="{{ old('summary', $entry->summary) }}" maxlength="500"></div>
        <div class="field"><label>Keywords (comma separated)</label><input type="text" name="keywords_text" value="{{ old('keywords_text', implode(', ', (array) $entry->keywords)) }}"><div class="hint">Words citizens would use, in both languages. Title words are added automatically.</div></div>
        <div class="field"><label>Source (optional)</label><input type="text" name="source" value="{{ old('source', $entry->source) }}" maxlength="250" placeholder="e.g. Sheria ya Ardhi Sura 113; nida.go.tz"></div>
        <div class="field"><label class="check"><input type="checkbox" name="is_active" value="1" @checked(old('is_active', $entry->is_active))> Active</label></div>
        <button class="btn btn-p" type="submit">Save</button>
        <a class="btn" href="{{ route('admin.knowledge.index') }}">Cancel</a>
    </form>
</div>
@endsection
```

84 resources/views/admin/knowledge/index.blade.php

58 lines

```
@extends('layouts.admin', ['title' => 'Knowledge base'])

@section('content')
<div class="grid g2">
    <div class="card">
        <div class="card-h"><div><h2>Entries by topic</h2><p>Fact blocks the engine can cite. Keep each one short and sourced.</p></div><a class="btn btn-p" href="{{ route('admin.knowledge.create') }}">+ New entry</a></div>
        <div class="bars">
            @foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)
                @php $n = (int) ($counts[$k] ?? 0); @endphp
                <div class="bar"><a href="{{ route('admin.knowledge.index', ['category' => $k]) }}">{{ $l['en'] }}</a><div class="t"><div class="f" style="width:{{ $counts->max() ? round($n / max(1, $counts->max()) * 100) : 0 }}%"></div></div><span>{{ $n }}</span></div>
            @endforeach
        </div>
    </div>
    <div class="card">
        <div class="card-h"><div><h2>Import</h2><p>CSV columns: title, content, category, language, keywords, source, summary. JSON: array of the same keys.</p></div></div>
        <form method="POST" action="{{ route('admin.knowledge.import') }}" enctype="multipart/form-data">
            @csrf
            <div class="field"><label>File (.csv or .json)</label><input type="file" name="file" accept=".csv,.json,.txt" required></div>
            <div class="grid g2">
                <div class="field"><label>Default topic</label><select name="default_category">@foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)<option value="{{ $k }}">{{ $l['en'] }}</option>@endforeach</select></div>
                <div class="field"><label>Default language</label><select name="default_language"><option value="sw">Swahili</option><option value="en">English</option></select></div>
            </div>
            <button class="btn" type="submit">Import</button>
        </form>
    </div>
</div>

<div class="card">
    <form class="filters" method="GET">
        <input type="text" name="q" value="{{ request('q') }}" placeholder="Search title or content">
        <select name="category"><option value="">All topics</option>@foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)<option value="{{ $k }}" @selected(request('category') === $k)>{{ $l['en'] }}</option>@endforeach</select>
        <select name="language"><option value="">Any language</option><option value="sw" @selected(request('language') === 'sw')>Swahili</option><option value="en" @selected(request('language') === 'en')>English</option></select>
        <button class="btn" type="submit">Filter</button><a class="btn" href="{{ route('admin.knowledge.index') }}">Reset</a>
    </form>
    <table>
        <thead><tr><th>Title</th><th>Topic</th><th>Lang</th><th>Keywords</th><th>Status</th><th></th></tr></thead>
        <tbody>
        @forelse($entries as $e)
            <tr>
                <td><strong>{{ $e->title }}</strong><div style="font-size:.78rem;color:var(--muted)">{{ \Illuminate\Support\Str::limit($e->summary ?: $e->content, 110) }}</div></td>
                <td style="white-space:nowrap">{{ \App\Services\TopicClassifier::label($e->category, 'en') }}</td>
                <td>{{ strtoupper($e->language) }}</td>
                <td style="font-size:.76rem;color:var(--muted);max-width:220px">{{ \Illuminate\Support\Str::limit(implode(', ', (array) $e->keywords), 90) }}</td>
                <td>{!! $e->is_active ? '<span class="badge b-green">active</span>' : '<span class="badge b-gray">off</span>' !!}</td>
                <td style="white-space:nowrap">
                    <a class="btn btn-sm" href="{{ route('admin.knowledge.edit', $e) }}">Edit</a>
                    <form class="inline" method="POST" action="{{ route('admin.knowledge.toggle', $e) }}">@csrf @method('PATCH')<button class="btn btn-sm" type="submit">{{ $e->is_active ? 'Off' : 'On' }}</button></form>
                    <form class="inline" method="POST" action="{{ route('admin.knowledge.destroy', $e) }}" onsubmit="return confirm('Delete this entry?')">@csrf @method('DELETE')<button class="btn btn-sm btn-d" type="submit">Delete</button></form>
                </td>
            </tr>
        @empty
            <tr><td colspan="6" style="text-align:center;color:var(--muted)">No entries.</td></tr>
```

```
        @endforelse
        </tbody>
    </table>
    {{ $entries->links('pagination.simple') }}
</div>
@endsection
```

85 resources/views/admin/login.blade.php

42 lines

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign in – Huru SMS Admin</title>
    <meta name="robots" content="noindex, nofollow">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700;800&display=swap" rel="stylesheet">
    <style>
        body{font-family:Inter,system-ui,sans-serif;background:#f8fafc;margin:0;min-height:100vh;display:flex;align-items:center;justify-content:center;padding:16px;color:#0f172a}
        .box{background:#fff;border:1px solid #cbd5e1;border-radius:12px;padding:28px;width:100%;max-width:380px}
        .logo{width:48px;height:48px;border-radius:12px;background:#15803d;color:#fff;font-weight:800;font-size:1.2rem;display:flex;align-items:center;justify-content:center;margin-bottom:12px}
        h1{font-size:1.2rem;margin:0 0 4px}
        p{color:#475569;font-size:.88rem;margin:0 0 16px}
        label{display:block;font-size:.8rem;font-weight:700;margin:10px 0 5px}
        input{width:100%;padding:10px 12px;border:1.5px solid #cbd5e1;border-radius:8px;font:inherit;box-sizing:border-box}
        input:focus{outline:none;border-color:#15803d}
        button{width:100%;margin-top:16px;background:#15803d;color:#fff;border:none;border-radius:8px;padding:11px;font:inherit;font-weight:700;cursor:pointer}
        button:hover{background:#166534}
        .err{background:#fef2f2;border:1px solid #fecaca;color:#b91c1c;padding:9px 12px;border-radius:8px;font-size:.85rem;font-weight:600;margin-bottom:8px}
        .chk{display:flex;align-items:center;gap:8px;font-size:.85rem;margin-top:10px}
        .chk input{width:auto}
    </style>
</head>
<body>
<form class="box" method="POST" action="{{ route('admin.login.post') }}">
    @csrf
    <div class="logo">H</div>
    <h1>Huru SMS Admin</h1>
    <p>Sign in with your administrator account.</p>
    @if($errors->any())<div class="err">{{ $errors->first() }}</div>@endif
    @if(session('error'))<div class="err">{{ session('error') }}</div>@endif
    <label for="email">Email</label>
    <input type="email" id="email" name="email" value="{{ old('email') }}" required autofocus>
    <label for="password">Password</label>
    <input type="password" id="password" name="password" required>
    <label class="chk"><input type="checkbox" name="remember" value="1"> Keep me signed in</label>
    <button type="submit">Sign in</button>
</form>
</body>
</html>
```

86 resources/views/admin/resources/form.blade.php

25 lines

```
@extends('layouts.admin', ['title' => $resource->exists ? 'Edit service' : 'New service'])

@section('content')
<div class="card" style="max-width:860px">
    <form method="POST" action="{{ $resource->exists ? route('admin.resources.update', $resource) : route('admin.resources.store') }}">
        @csrf
        @if($resource->exists) @method('PUT') @endif
        <div class="field"><label>Name</label><input type="text" name="name" value="{{ old('name', $resource->name) }}" required maxlength="150"></div>
        <div class="grid g3">
            <div class="field"><label>Category</label><select name="category">@foreach($categories as $k => $l)<option value="{{ $k }}" @selected(old('category', $resource->category) === $k)>{{ $l }}</option>@endforeach</select></div>
            <div class="field"><label>Region</label><select name="region">@foreach($regions as $r)<option value="{{ $r }}" @selected(old('region', $resource->region) === $r)>{{ ucwords($r) }}</option>@endforeach</select></div>
            <div class="field"><label>District</label><input type="text" name="district" value="{{ old('district', $resource->district) }}" maxlength="40" placeholder="e.g. kinondoni"></div>
        </div>
        <div class="field"><label>Ward / area</label><input type="text" name="ward" value="{{ old('ward', $resource->ward) }}" maxlength="60" placeholder="e.g. Sinza"></div>
        <div class="field"><label>Address / directions</label><textarea name="location" rows="2">{{ old('location', $resource->location) }}</textarea></div>
        <div class="grid g2">
            <div class="field"><label>Phone</label><input type="text" name="phone" value="{{ old('phone', $resource->phone) }}" maxlength="60"></div>
            <div class="field"><label>Email</label><input type="email" name="email" value="{{ old('email', $resource->email) }}"></div>
        </div>
        <div class="field"><label class="check"><input type="checkbox" name="is_active" value="1" @checked(old('is_active', $resource->is_active))> Active</label></div>
        <button class="btn btn-p" type="submit">Save</button>
        <a class="btn" href="{{ route('admin.resources.index') }}">Cancel</a>
    </form>
</div>
@endsection
```

87 resources/views/admin/resources/index.blade.php

35 lines

```
@extends('layouts.admin', ['title' => 'Local services'])

@section('content')
<div class="card">
    <div class="card-h"><div><h2>Place-bound services</h2><p>Legal aid providers, hospitals, police desks and offices by region and district. Run <code>php artisan huru:import-legal-aid</code> to pull the national legal aid list.</p></div><a class="btn btn-p" href="{{ route('admin.resources.create') }}">+ New service</a></div>
    <form class="filters" method="GET">
        <input type="text" name="q" value="{{ request('q') }}" placeholder="Search name, district, address">
        <select name="region"><option value="">All regions</option>@foreach($regions as $r)<option value="{{ $r }}" @selected(request('region') === $r)>{{ ucwords($r) }} ({{ $byRegion[$r] ?? 0 }})</option>@endforeach</select>
        <select name="category"><option value="">All categories</option>@foreach($categories as $k => $l)<option value="{{ $k }}" @selected(request('category') === $k)>{{ $l }}</option>@endforeach</select>
        <button class="btn" type="submit">Filter</button><a class="btn" href="{{ route('admin.resources.index') }}">Reset</a>
    </form>
    <table>
        <thead><tr><th>Name</th><th>Category</th><th>Region / District</th><th>Address</th><th>Contact</th><th>Source</th><th></th></tr></thead>
        <tbody>
        @forelse($resources as $r)
            <tr>
                <td><strong>{{ $r->name }}</strong> @if(!$r->is_active)<span class="badge b-gray">off</span>@endif</td>
                <td style="white-space:nowrap">{{ $categories[$r->category] ?? $r->category }}</td>
                <td style="white-space:nowrap">{{ ucwords($r->region) }}@if($r->district) / {{ ucwords($r->district) }}@endif</td>
                <td style="font-size:.8rem;max-width:300px">{{ \Illuminate\Support\Str::limit($r->location, 120) }}</td>
                <td style="font-size:.8rem;white-space:nowrap">{{ $r->phone }}@if($r->email)<br>{{ $r->email }}@endif</td>
                <td style="font-size:.75rem;color:var(--muted)">{{ $r->source }}</td>
                <td style="white-space:nowrap">
                    <a class="btn btn-sm" href="{{ route('admin.resources.edit', $r) }}">Edit</a>
                    <form class="inline" method="POST" action="{{ route('admin.resources.destroy', $r) }}" onsubmit="return confirm('Delete this service?')">@csrf @method('DELETE')<button class="btn btn-sm btn-d" type="submit">Delete</button></form>
                </td>
            </tr>
        @empty
            <tr><td colspan="7" style="text-align:center;color:var(--muted)">No services match.</td></tr>
        @endforelse
        </tbody>
    </table>
    {{ $resources->links('pagination.simple') }}
</div>
@endsection
```

88 resources/views/admin/settings.blade.php

36 lines

```
@extends('layouts.admin', ['title' => 'Settings'])

@section('content')
<form method="POST" action="{{ route('admin.settings.update') }}">
    @csrf
    <div class="grid g2">
        @foreach($groups as $group => $fields)
            <div class="card">
                <div class="card-h"><h2>{{ $group }}</h2></div>
                @foreach($fields as $key => $def)
                    @php $val = old($key, $values[$key]); @endphp
                    <div class="field">
                        @if($def['type'] === 'boolean')
                            <label class="check"><input type="checkbox" name="{{ $key }}" value="1"
@checked((string) $val === '1')> {{ $def['label'] }}</label>
                        @else
                            <label for="{{ $key }}">{{ $def['label'] }}</label>
                            @if($def['type'] === 'textarea')
                                <textarea id="{{ $key }}" name="{{ $key }}" rows="2">{{ $val }}</textarea>
                            @elseif($def['type'] === 'select')
                                <select id="{{ $key }}" name="{{ $key }}">@foreach($def['options'] as $ov => $ol)
<option value="{{ $ov }}" @selected((string) $val === (string) $ov)>{{ $ol }}</option>@endforeach</select>
                            @elseif($def['type'] === 'number')
                                <input type="number" step="any" id="{{ $key }}" name="{{ $key }}" value="{{ $val
}}">
                            @else
                                <input type="text" id="{{ $key }}" name="{{ $key }}" value="{{ $val }}">
                            @endif
                        @endif
                        @if(!empty($def['hint']))<div class="hint">{{ $def['hint'] }}</div>@endif
                        @error($key)<div class="hint" style="color:var(--danger)">{{ $message }}</div>@enderror
                    </div>
                @endforeach
            </div>
        @endforeach
    </div>
    <button class="btn btn-p" type="submit">Save all settings</button>
</form>
@endsection
```

89 resources/views/admin/templates/form.blade.php

19 lines

```
@extends('layouts.admin', ['title' => $template->exists ? 'Edit persona' : 'New persona'])

@section('content')
<div class="card" style="max-width:860px">
    <form method="POST" action="{{ $template->exists ? route('admin.templates.update', $template) : route('admin.templates.store') }}">
        @csrf
        @if($template->exists) @method('PUT') @endif
        <div class="field"><label>Name</label><input type="text" name="name" value="{{ old('name', $template->name) }}" required maxlength="80"></div>
        <div class="grid g2">
            <div class="field"><label>Topic</label><select name="category">@foreach(\App\Services\TopicClassifier::CATEGORIES as $k => $l)<option value="{{ $k }}" @selected(old('category', $template->category) === $k)>{{ $l['en'] }}</option>@endforeach</select></div>
            <div class="field"><label>Language</label><select name="language"><option value="sw" @selected(old('language', $template->language) === 'sw')>Swahili</option><option value="en" @selected(old('language', $template->language) === 'en')>English</option></select></div>
        </div>
        <div class="field"><label>Persona</label><textarea name="template" rows="8" required>{{ old('template', $template->template) }}</textarea><div class="hint">Describe who the assistant is and how it should behave for this topic. Country context, knowledge, contacts and output rules are appended automatically. Use {app_name} for the service name.</div></div>
        <div class="field"><label class="check"><input type="checkbox" name="is_active" value="1" @checked(old('is_active', $template->is_active))> Active (deactivates other personas for the same topic and language)</label></div>
        <button class="btn btn-p" type="submit">Save</button>
        <a class="btn" href="{{ route('admin.templates.index') }}">Cancel</a>
    </form>
</div>
@endsection
```

90 resources/views/admin/templates/index.blade.php

28 lines

```
@extends('layouts.admin', ['title' => 'Personas'])

@section('content')
<div class="card">
    <div class="card-h"><div><h2>Persona templates</h2><p>One active persona per topic and language. Topics without a persona fall back to the general one. Placeholders: {app_name}, {category}.</p></div><a class="btn btn-p" href="{{ route('admin.templates.create') }}">+ New persona</a></div>
    <table>
        <thead><tr><th>Name</th><th>Topic</th><th>Lang</th><th>Persona</th><th>Status</th><th></th></tr></thead>
        <tbody>
        @forelse($templates as $t)
            <tr>
                <td style="white-space:nowrap"><strong>{{ $t->name }}</strong></td>
                <td style="white-space:nowrap">{{ \App\Services\TopicClassifier::label($t->category, 'en') }}</td>
                <td>{{ strtoupper($t->language) }}</td>
                <td style="font-size:.8rem;color:var(--muted)">{{ \Illuminate\Support\Str::limit($t->template, 160) }}</td>
                <td>{!! $t->is_active ? '<span class="badge b-green">active</span>' : '<span class="badge b-gray">off</span>' !!}</td>
                <td style="white-space:nowrap">
                    <a class="btn btn-sm" href="{{ route('admin.templates.edit', $t) }}">Edit</a>
                    <form class="inline" method="POST" action="{{ route('admin.templates.toggle', $t) }}">@csrf @method('PATCH')<button class="btn btn-sm" type="submit">{{ $t->is_active ? 'Off' : 'On' }}</button></form>
                    <form class="inline" method="POST" action="{{ route('admin.templates.destroy', $t) }}" onsubmit="return confirm('Delete this persona?')">@csrf @method('DELETE')<button class="btn btn-sm btn-d" type="submit">Delete</button></form>
                </td>
            </tr>
        @empty
            <tr><td colspan="6" style="text-align:center;color:var(--muted)">No personas yet. The built-in default will be used.</td></tr>
        @endforelse
        </tbody>
    </table>
</div>
@endsection
```

91 resources/views/admin/users/index.blade.php

37 lines

```
@extends('layouts.admin', ['title' => 'Citizens'])

@section('content')
<div class="card">
    <form class="filters" method="GET">
        <input type="text" name="q" value="{{ request('q') }}" placeholder="Phone or name">
        <select name="region"><option value="">All regions</option>@foreach(\App\Services\TanzaniaContext::REGIONS as $r)<option value="{{ $r }}" @selected(request('region') === $r)>{{ ucwords($r) }}</option>@endforeach</select>
        <label class="check"><input type="checkbox" name="banned" value="1" @checked(request('banned'))> Blocked only</label>
        <button class="btn" type="submit">Filter</button><a class="btn" href="{{ route('admin.users.index') }}">Reset</a>
    </form>
    <table>
        <thead><tr><th>Phone</th><th>Name</th><th>Lang</th><th>Region</th><th>Messages</th><th>Strikes</th><th>Last seen</th><th>Status</th><th></th></tr></thead>
        <tbody>
        @forelse($users as $u)
            <tr>
                <td style="white-space:nowrap"><a href="{{ route('admin.conversations.show', $u) }}">{{ $u->phone_number }}</a></td>
                <td>{{ $u->name ?: '—' }}</td>
                <td>{{ strtoupper($u->preferred_language ?? 'auto') }}</td>
                <td>{{ $u->region ? ucwords($u->region) : '—' }}</td>
                <td>{{ $u->messages_count }}</td>
                <td>{{ $u->abuse_count }}</td>
                <td style="white-space:nowrap;color:var(--muted)">{{ $u->last_seen_at?->diffForHumans() ?? '—' }}</td>
                <td>@if($u->is_banned)<span class="badge b-red">blocked</span>@elseif($u->opted_out)<span class="badge b-warn">opted out</span>@else<span class="badge b-green">active</span>@endif</td>
                <td style="white-space:nowrap">
                    <form class="inline" method="POST" action="{{ route('admin.users.ban', $u) }}">@csrf @method('PATCH')<button class="btn btn-sm {{ $u->is_banned ? '' : 'btn-d' }}" type="submit">{{ $u->is_banned ? 'Unblock' : 'Block' }}</button></form>
                    <form class="inline" method="POST" action="{{ route('admin.users.strikes', $u) }}">@csrf @method('PATCH')<button class="btn btn-sm" type="submit">Clear strikes</button></form>
                    <form class="inline" method="POST" action="{{ route('admin.users.destroy', $u) }}" onsubmit="return confirm('Delete this citizen and all their messages?')">@csrf @method('DELETE')<button class="btn btn-sm btn-d" type="submit">Delete</button></form>
                </td>
            </tr>
        @empty
            <tr><td colspan="9" style="text-align:center;color:var(--muted)">No citizens yet.</td></tr>
        @endforelse
        </tbody>
    </table>
    {{ $users->links('pagination.simple') }}
</div>
@endsection
```

92 resources/views/chat.blade.php

389 lines

```
<!DOCTYPE html>
<html lang="sw">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0, viewport-fit=cover">
    <title>Web Chat – {{ $settings['app_name'] }}</title>
    <meta name="description" content="Uliza swali lolote la maisha Tanzania na upate jibu la haraka kwa Kiswahili au Kiingereza. Bila intaneti tuma {{ $settings['sms_keyword'] }} kwenda {{ $settings['sms_shortcode'] }}.">
    <meta name="robots" content="index, follow">
    <link rel="canonical" href="{{ url('/chat') }}">
    <link rel="manifest" href="/manifest.json">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <link rel="apple-touch-icon" href="/logo.svg">
    <meta name="theme-color" content="#15803d">
    <meta name="csrf-token" content="{{ csrf_token() }}">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
    <style>
        :root{--green:#15803d;--green-dark:#166534;--green-bg:#f0fdf4;--green-border:#bbf7d0;--blue:#1d4ed8;--blue-dark:#1e40af;--blue-bg:#eff6ff;--bg:#f8fafc;--card:#fff;--text:#0f172a;--muted:#475569;--border:#cbd5e1;--danger:#b91c1c}
        *{margin:0;padding:0;box-sizing:border-box}
        html,body{height:100%}
        body{font-family:'Inter',system-ui,sans-serif;background:var(--bg);color:var(--text);display:flex;align-items:stretch;justify-content:center}
        .app{width:100%;max-width:880px;height:100dvh;display:flex;flex-direction:column;background:var(--card);border-left:1px solid var(--border);border-right:1px solid var(--border)}
        .screen{display:none;flex:1;flex-direction:column;min-height:0}
        .screen.active{display:flex}
        .hidden{display:none!important}
        /* auth */
        .auth{flex:1;overflow:auto;padding:28px 20px;display:flex;flex-direction:column;justify-content:center;max-width:520px;width:100%;margin:0 auto}
        .logo{width:52px;height:52px;border-radius:12px;background:var(--green);color:#fff;font-weight:800;font-size:1.3rem;display:flex;align-items:center;justify-content:center}
        .auth h1{font-size:1.5rem;margin:14px 0 6px;font-weight:800}
        .auth p.lead{color:var(--muted);font-size:.95rem}
        .sms-banner{background:var(--green-bg);border:1px solid var(--green-border);border-radius:10px;padding:12px 14px;margin:18px 0;font-size:.9rem}
        .sms-banner b{color:var(--green-dark)}
        .sms-badge{float:right;background:var(--green);color:#fff;font-weight:800;padding:2px 10px;border-radius:999px;font-size:.8rem}
        label{display:block;font-size:.85rem;font-weight:600;margin-bottom:6px;color:var(--muted)}
        input,select,textarea{width:100%;padding:12px 14px;border:1.5px solid var(--border);border-radius:8px;font:inherit;font-size:1rem;background:#fff}
        input:focus,select:focus,textarea:focus{outline:none;border-color:var(--green)}
        .row{display:flex;gap:10px}
        .btn{display:inline-flex;align-items:center;justify-content:center;gap:8px;padding:12px 16px;border-radius:8px;border:2px solid transparent;font:inherit;font-weight:700;cursor:pointer;font-size:.95rem}
        .btn-green{background:var(--green);color:#fff;width:100%}
        .btn-green:hover{background:var(--green-dark)}
        .btn-green:disabled{opacity:.6;cursor:wait}
        .btn-ghost{background:#fff;border-color:var(--border);color:var(--muted)}
        .btn-link{background:none;border:none;color:var(--blue);font-weight:600;cursor:pointer;font:inherit;padding:6px 0}
        .err{background:#fef2f2;border:1px solid #fecaca;color:var(--danger);padding:10px 12px;border-radius:8px;font-size:.88rem;font-weight:600;margin:10px 0}
        .ok{background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark);padding:10px 12px;border-radius:8px;font-size:.88rem;font-weight:600;margin:10px 0}
        .small{font-size:.8rem;color:var(--muted);margin-top:10px}
        /* chat header */
        .hdr{display:flex;align-items:center;justify-content:space-between;padding:10px 14px;border-bottom:1px solid var(--border);gap:10px;flex-wrap:wrap}
        .hdr .who{display:flex;align-items:center;gap:10px}
        .hdr .logo{width:38px;height:38px;font-size:1rem;border-radius:9px}
        .hdr h2{font-size:1rem;font-weight:800}
        .hdr .st{font-size:.75rem;color:var(--green-dark);font-weight:600}
        .hdr .actions{display:flex;gap:6px;align-items:center;flex-wrap:wrap}
        .tag{font-size:.75rem;font-weight:700;padding:5px 10px;border-radius:999px;background:var(--green-bg);color:var(--green-dark);border:1px solid var(--green-border)}
```

```
        .ibtn{background:#fff;border:1px solid var(--border);border-radius:8px;padding:7px 10px;font:inherit;font-size:.8rem;font-weight:600;color:var(--muted);cursor:pointer;text-decoration:none}
        .ibtn:hover{border-color:var(--blue);color:var(--blue)}
        .offline{background:#fffbeb;border-bottom:1px solid #fde68a;color:#92400e;font-size:.85rem;padding:8px 14px;font-weight:600}
        /* filters */
        .filters{display:flex;gap:6px;padding:8px 14px;border-bottom:1px solid var(--border);flex-wrap:wrap;align-items:center}
        .filters input,.filters select{padding:7px 10px;font-size:.85rem;width:auto;flex:1;min-width:120px}
        .filters .ibtn{padding:8px 12px}
        .filters .status{width:100%;font-size:.75rem;color:var(--muted)}
        /* messages */
        .msgs{flex:1;overflow-y:auto;padding:16px 14px;display:flex;flex-direction:column;gap:10px;background:var(--bg)}
        .welcome{background:var(--card);border:1px solid var(--border);border-radius:12px;padding:16px}
        .welcome h3{font-size:1rem;margin-bottom:6px}
        .welcome p{font-size:.88rem;color:var(--muted)}
        .sugg{display:flex;flex-wrap:wrap;gap:6px;margin-top:10px}
        .sugg button{background:#fff;border:1px solid var(--border);border-radius:999px;padding:6px 11px;font:inherit;font-size:.8rem;font-weight:600;color:var(--blue);cursor:pointer}
        .sugg button:hover{border-color:var(--blue)}
        .msg{max-width:84%;padding:10px 14px;border-radius:14px;font-size:.95rem;line-height:1.5;white-space:pre-wrap;word-wrap:break-word;position:relative}
        .msg.in{align-self:flex-end;background:var(--green);color:#fff;border-bottom-right-radius:4px}
        .msg.out{align-self:flex-start;background:var(--card);border:1px solid var(--border);border-bottom-left-radius:4px}
        .msg.sys{align-self:center;background:var(--blue-bg);border:1px solid #bfdbfe;color:var(--blue-dark);font-size:.85rem;max-width:95%}
        .meta{display:flex;gap:8px;align-items:center;margin-top:6px;font-size:.72rem;opacity:.85}
        .msg.in .meta{justify-content:flex-end}
        .msg.out .meta{color:var(--muted)}
        .cat{background:var(--green-bg);color:var(--green-dark);border:1px solid var(--green-border);padding:1px 7px;border-radius:999px;font-weight:700}
        .fb{display:flex;gap:4px;margin-left:auto}
        .fb button{background:none;border:1px solid var(--border);border-radius:6px;padding:2px 7px;cursor:pointer;font-size:.8rem;color:var(--muted)}
        .fb button.on{background:var(--green-bg);border-color:var(--green);color:var(--green-dark)}
        .more{align-self:center}
        .typing{display:flex;gap:4px;padding:6px 14px}
        .typing span{width:7px;height:7px;background:var(--green);border-radius:50%;opacity:.4;animation:t 1.2s infinite}
        .typing span:nth-child(2){animation-delay:.2s}.typing span:nth-child(3){animation-delay:.4s}
        @keyframes t{0%,100%{transform:translateY(0);opacity:.4}50%{transform:translateY(-4px);opacity:1}}
        /* composer */
        .composer{display:flex;gap:8px;padding:10px 14px;border-top:1px solid var(--border);background:var(--card);padding-bottom:calc(10px + env(safe-area-inset-bottom))}
        .composer textarea{resize:none;max-height:120px;line-height:1.4;padding:11px 13px}
        .send{width:48px;height:48px;border-radius:8px;background:var(--green);color:#fff;border:none;cursor:pointer;flex-shrink:0;display:flex;align-items:center;justify-content:center}
        .send:hover{background:var(--green-dark)}
        /* modal */
        .modal{position:fixed;inset:0;background:rgba(15,23,42,.45);display:flex;align-items:center;justify-content:center;padding:16px;z-index:50}
        .modal .box{background:#fff;border-radius:12px;padding:20px;width:100%;max-width:420px}
        .modal h3{margin-bottom:12px}
        .modal .field{margin-bottom:12px}
        @media (max-width:600px){.app{border:none}.msg{max-width:92%}.hdr .st{display:none}}
    </style>
</head>
<body>
<div class="app" id="app">

    <!-- AUTH -->
    <div id="auth-screen" class="screen active">
        <div class="auth">
            <div class="logo">H</div>
            <h1>{{ $settings['app_name'] }} – uliza lolote</h1>
            <p class="lead">Ingiza namba yako ya simu. Tutakutumia namba fupi ya uthibitisho ili mazungumzo yako yabaki yako.</p>

            <div class="sms-banner">
                <span class="sms-badge">SMS {{ $settings['sms_shortcode'] }}</span>
                <b>Huna intaneti?</b> Tuma <b>{{ $settings['sms_keyword'] }}</b> [swali lako] kwenda <b>{{
```

```
$settings['sms_shortcode'] }}</b> kutoka simu yoyote.
            </div>

            @if(session('error'))<div class="err">{{ session('error') }}</div>@endif
            <div id="auth-error" class="err hidden"></div>
            <div id="auth-ok" class="ok hidden"></div>

            <form id="phone-form">
                <label for="phone">Namba ya simu / Phone number</label>
                <div class="row">
                    <input type="tel" id="phone" name="phone_number" placeholder="07XX XXX XXX" autocomplete="tel"
required>
                    <select id="lang" name="language" style="width:110px"><option value="sw">Kiswahili</option>
<option value="en">English</option></select>
                </div>
                <button type="submit" class="btn btn-green" id="phone-btn" style="margin-top:12px">Endelea /
Continue</button>
            </form>

            <form id="otp-form" class="hidden">
                <label for="code">Namba ya uthibitisho (tarakimu 6) / Verification code</label>
                <input type="text" id="code" name="code" inputmode="numeric" pattern="\d{6}" maxlength="6"
placeholder="123456" autocomplete="one-time-code" required>
                <button type="submit" class="btn btn-green" id="otp-btn" style="margin-top:12px">Thibitisha /
Verify</button>
                <div class="row" style="justify-content:space-between;margin-top:6px"><button type="button"
class="btn-link" id="otp-back">Badilisha namba</button><button type="button" class="btn-link" id="otp-resend">Tuma
tena</button></div>
            </form>

            <p class="small">Kwa kuendelea unakubali <a href="{{ route('legal.terms') }}">Vigezo na Masharti</a> na
<a href="{{ route('legal.privacy') }}">Sera ya Faragha</a>. <a href="{{ route('welcome') }}">Rudi mwanzo</a></p>
        </div>
    </div>

    <!-- CHAT -->
    <div id="chat-screen" class="screen">
        <div class="hdr">
            <div class="who">
                <div class="logo">H</div>
                <div><h2>{{ $settings['app_name'] }}</h2><div class="st" id="who-line">Tayari kukusaidia</div>
</div>
            </div>
            <div class="actions">
                <span class="tag" title="Tuma {{ $settings['sms_keyword'] }} [swali] kwenda {{
$settings['sms_shortcode'] }}">SMS {{ $settings['sms_keyword'] }} → {{ $settings['sms_shortcode'] }}</span>
                <button class="ibtn" id="prefs-btn">Mipangilio</button>
                <a class="ibtn" href="{{ route('community.index') }}">Jamii</a>
                <button class="ibtn" id="logout-btn">Toka</button>
            </div>
        </div>
        <div class="offline hidden" id="offline-bar">Uko nje ya mtandao. Tuma <b>{{ $settings['sms_keyword'] }}</b>
[swali] kwenda <b>{{ $settings['sms_shortcode'] }}</b> kwa SMS.</div>
        <div class="filters">
            <input type="text" id="f-keyword" placeholder="Tafuta kwenye historia...">
            <input type="date" id="f-date">
            <select id="f-category"><option value="">Mada zote</option></select>
            <button class="ibtn" id="f-apply">Tafuta</button>
            <button class="ibtn hidden" id="f-reset">Futa</button>
            <div class="status hidden" id="f-status"></div>
        </div>
        <div class="msgs" id="msgs"></div>
        <div class="typing hidden" id="typing"><span></span><span></span><span></span></div>
        <form class="composer" id="composer">
            <textarea id="input" rows="1" placeholder="Andika swali lako hapa..." maxlength="2000"></textarea>
            <button type="submit" class="send" aria-label="Tuma"><svg viewBox="0 0 24 24" width="22" height="22"
fill="currentColor"><path d="M2 21l21-9L2 3v7l15 2-15 2z"/></svg></button>
        </form>
    </div>

    <!-- PREFS MODAL -->
    <div class="modal hidden" id="prefs">
        <div class="box">
```

```
            <h3>Mipangilio yako</h3>
            <form id="prefs-form">
                <div class="field"><label>Jina (si lazima)</label><input type="text" name="name" id="p-name" maxlength="40"></div>
                <div class="field"><label>Lugha ya majibu</label><select name="preferred_language" id="p-lang"><option value="auto">Fuata lugha ninayoandika</option><option value="sw">Kiswahili daima</option><option value="en">English always</option></select></div>
                <div class="field"><label>Mkoa wako (kwa huduma zilizo karibu)</label><select name="region" id="p-region"><option value="">— Chagua mkoa —</option></select></div>
                <div class="row"><button type="button" class="btn btn-ghost" id="prefs-cancel" style="flex:1">Funga</button><button type="submit" class="btn btn-green" style="flex:1">Hifadhi</button></div>
            </form>
        </div>
    </div>
</div>

<script>
(() => {
    const $ = (id) => document.getElementById(id);
    // The session (and therefore the CSRF token) is regenerated on login and logout,
    // so the token is kept in a variable and refreshed from server responses.
    let csrf = document.querySelector('meta[name="csrf-token"]').content;
    const refreshCsrf = async () => {
        try { const r = await fetch('/chat/session', { headers: { 'Accept': 'application/json' } }); const d = await r.json(); if (d.csrf) csrf = d.csrf; } catch (e) {}
    };
    const state = { user: null, settings: null, phone: '', lang: 'sw', oldest: null, hasMore: false, filtering: false };
    const SUGGESTIONS = {
        sw: ['Nifanyeje kupata kitambulisho cha NIDA?', 'Haki zangu ni zipi nikikamatwa na polisi?', 'Dalili za malaria kwa mtoto ni zipi?', 'Nipande mahindi lini mkoa wangu?', 'Nieleze fractions kwa darasa la sita', 'Nianzeje biashara ndogo?'],
        en: ['How do I register for a NIDA ID?', 'What are my rights if the police arrest me?', 'What are malaria danger signs in a child?', 'When should I plant maize in my region?', 'Explain fractions for Standard Six', 'How do I start a small business?']
    };

    const api = async (url, opts = {}, retried = false) => {
        const res = await fetch(url, Object.assign({ headers: Object.assign({ 'X-Requested-With': 'XMLHttpRequest', 'X-CSRF-TOKEN': csrf, 'Accept': 'application/json' }, opts.body && !(opts.body instanceof FormData) ? { 'Content-Type': 'application/json' } : {}) }, opts));
        let data = {};
        try { data = await res.json(); } catch (e) {}
        if (res.status === 419 && !retried) { await refreshCsrf(); return api(url, opts, true); }
        if (data && data.csrf) csrf = data.csrf;
        if (!res.ok) { const msg = data.message || (data.errors && Object.values(data.errors)[0][0]) || ('Kosa ' + res.status); throw new Error(msg); }
        return data;
    };
    const show = (id) => { document.querySelectorAll('.screen').forEach(s => s.classList.remove('active')); $(id).classList.add('active'); };
    const err = (el, msg) => { el.textContent = msg; el.classList.remove('hidden'); };
    const hide = (el) => el.classList.add('hidden');
    const t = (sw, en) => (state.lang === 'en' ? en : sw);

    // ---------------------------------------------------------------- boot
    async function boot() {
        try {
            const s = await api('/chat/session');
            state.settings = s.settings;
            fillCategories(s.settings.categories);
            fillRegions(s.settings.regions);
            if (s.authenticated) { state.user = s.user; enterChat(); } else { show('auth-screen'); }
        } catch (e) { show('auth-screen'); }
    }

    function fillCategories(cats) {
        const sel = $('f-category');
        cats.forEach(c => { const o = document.createElement('option'); o.value = c.key; o.textContent = c.sw + ' / ' + c.en; sel.appendChild(o); });
    }
    function fillRegions(regions) {
        const sel = $('p-region');
```

```
        regions.forEach(r => { const o = document.createElement('option'); o.value = r; o.textContent = r;
sel.appendChild(o); });
    }

    // ---------------------------------------------------------------- auth
    $('phone-form').addEventListener('submit', async (e) => {
        e.preventDefault();
        hide($('auth-error')); hide($('auth-ok'));
        const btn = $('phone-btn'); btn.disabled = true; btn.textContent = '...';
        state.phone = $('phone').value.trim(); state.lang = $('lang').value;
        try {
            const r = await api('/chat/request-otp', { method: 'POST', body: JSON.stringify({ phone_number:
state.phone, language: state.lang }) });
            if (!r.otp_required) { state.user = r.user; enterChat(); return; }
            $('phone-form').classList.add('hidden'); $('otp-form').classList.remove('hidden');
            let msg = t('Tumetuma namba ya uthibitisho kwa ' + r.phone_masked + '.', 'We sent a verification code
to ' + r.phone_masked + '.');
            if (r.debug_code) msg += ' [dev: ' + r.debug_code + ']';
            $('auth-ok').textContent = msg; $('auth-ok').classList.remove('hidden');
            $('code').focus();
        } catch (ex) { err($('auth-error'), ex.message); }
        finally { btn.disabled = false; btn.textContent = 'Endelea / Continue'; }
    });
    $('otp-form').addEventListener('submit', async (e) => {
        e.preventDefault();
        hide($('auth-error'));
        const btn = $('otp-btn'); btn.disabled = true;
        try {
            const r = await api('/chat/verify-otp', { method: 'POST', body: JSON.stringify({ phone_number:
state.phone, code: $('code').value.trim() }) });
            state.user = r.user; enterChat();
        } catch (ex) { err($('auth-error'), ex.message); }
        finally { btn.disabled = false; }
    });
    $('otp-back').addEventListener('click', () => { $('otp-form').classList.add('hidden'); $('phone-
form').classList.remove('hidden'); hide($('auth-ok')); });
    $('otp-resend').addEventListener('click', () => $('phone-form').requestSubmit());

    // ---------------------------------------------------------------- chat
    async function enterChat() {
        const redirect = new URLSearchParams(location.search).get('redirect');
        if (redirect && redirect.startsWith('/')) { location.href = redirect; return; }
        show('chat-screen');
        state.lang = state.user.preferred_language === 'en' ? 'en' : state.lang;
        $('who-line').textContent = (state.user.name ? state.user.name + ' · ' : '') + state.user.phone_masked +
(state.user.region ? ' · ' + state.user.region : '');
        await loadMessages();
        const q = new URLSearchParams(location.search).get('q');
        if (q) { $('input').value = q; $('input').focus(); history.replaceState({}, '', '/chat'); }
    }

    async function loadMessages(filters = {}, append = false) {
        const params = new URLSearchParams(filters);
        const r = await api('/chat/messages?' + params.toString());
        const box = $('msgs');
        if (!append) box.innerHTML = '';
        if (!append && r.messages.length === 0 && !state.filtering) renderWelcome();
        const frag = document.createDocumentFragment();
        r.messages.forEach(m => frag.appendChild(renderMsg(m)));
        if (append) { box.querySelector('.more')?.remove(); box.prepend(frag); } else { box.appendChild(frag); }
        state.hasMore = r.has_more; state.oldest = r.messages.length ? r.messages[0].id : state.oldest;
        if (state.hasMore && !state.filtering) {
            const b = document.createElement('button'); b.className = 'ibtn more'; b.textContent = t('Onyesha
mazungumzo ya zamani', 'Show older messages');
            b.onclick = () => loadMessages({ before: state.oldest }, true); box.prepend(b);
        }
        if (!append) scrollBottom();
    }

    function renderWelcome() {
        const w = document.createElement('div'); w.className = 'welcome';
        w.innerHTML = '<h3>' + t('Karibu! Uliza lolote kuhusu maisha Tanzania.', 'Welcome! Ask anything about life
in Tanzania.') + '</h3><p>' + t('Sheria, afya, kilimo, masomo, huduma za serikali, fedha, ajira, teknolojia... Taja mahali
```

```
ulipo ukihitaji huduma iliyo karibu.', 'Law, health, farming, school subjects, government services, money, jobs,
technology... Mention where you are if you need nearby services.') + '</p><div class="sugg"></div>';
    296          const s = w.querySelector('.sugg');
    297          SUGGESTIONS[state.lang].forEach(q => { const b = document.createElement('button'); b.type = 'button';
b.textContent = q; b.onclick = () => { $('input').value = q; $('composer').requestSubmit(); }; s.appendChild(b); });
    298          $('msgs').appendChild(w);
    299      }
    300
    301      function renderMsg(m) {
    302          const d = document.createElement('div');
    303          const isIn = m.direction === 'inbound';
    304          d.className = 'msg ' + (isIn ? 'in' : 'out');
    305          d.dataset.id = m.id || '';
    306          const body = document.createElement('div'); body.textContent = m.content; d.appendChild(body);
    307          const meta = document.createElement('div'); meta.className = 'meta';
    308          const time = document.createElement('span'); time.textContent = m.created_at ? new
Date(m.created_at).toLocaleString('sw-TZ', { hour: '2-digit', minute: '2-digit', day: '2-digit', month: 'short' }) : '';
    309          meta.appendChild(time);
    310          if (!isIn && m.category_label) { const c = document.createElement('span'); c.className = 'cat';
c.textContent = m.category_label; meta.appendChild(c); }
    311          if (!isIn && m.id) {
    312              const fb = document.createElement('div'); fb.className = 'fb';
    313              [[1, '👍'], [-1, '👎']].forEach(([v, icon]) => {
    314                  const b = document.createElement('button'); b.type = 'button'; b.textContent = icon; b.title = v
=== 1 ? t('Imesaidia', 'Helpful') : t('Haikusaidia', 'Not helpful');
    315                  if (m.feedback === v) b.classList.add('on');
    316                  b.onclick = async () => { try { await api('/chat/feedback', { method: 'POST', body:
JSON.stringify({ message_id: m.id, rating: v }) }); fb.querySelectorAll('button').forEach(x => x.classList.remove('on'));
b.classList.add('on'); } catch (e) {} };
    317                  fb.appendChild(b);
    318              });
    319              meta.appendChild(fb);
    320          }
    321          d.appendChild(meta);
    322          return d;
    323      }
    324
    325      function addSystem(text) { const d = document.createElement('div'); d.className = 'msg sys'; d.textContent =
text; $('msgs').appendChild(d); scrollBottom(); }
    326      function scrollBottom() { const b = $('msgs'); b.scrollTop = b.scrollHeight; }
    327
    328      $('composer').addEventListener('submit', async (e) => {
    329          e.preventDefault();
    330          const text = $('input').value.trim();
    331          if (!text) return;
    332          $('msgs').querySelector('.welcome')?.remove();
    333          $('msgs').appendChild(renderMsg({ direction: 'inbound', content: text, created_at: new Date().toISOString()
}));
    334          $('input').value = ''; $('input').style.height = 'auto';
    335          $('typing').classList.remove('hidden'); scrollBottom();
    336          try {
    337              const r = await api('/chat/send', { method: 'POST', body: JSON.stringify({ message: text }) });
    338              $('typing').classList.add('hidden');
    339              if (r.reply) $('msgs').appendChild(renderMsg(r.reply));
    340          } catch (ex) {
    341              $('typing').classList.add('hidden');
    342              if (/401/.test(ex.message)) { state.user = null; show('auth-screen'); return; }
    343              addSystem('⚠️ ' + ex.message);
    344          }
    345          scrollBottom();
    346      });
    347      $('input').addEventListener('input', function () { this.style.height = 'auto'; this.style.height =
Math.min(this.scrollHeight, 120) + 'px'; });
    348      $('input').addEventListener('keydown', (e) => { if (e.key === 'Enter' && !e.shiftKey) { e.preventDefault();
$('composer').requestSubmit(); } });
    349
    350      // filters
    351      $('f-apply').addEventListener('click', () => {
    352          const f = { keyword: $('f-keyword').value.trim(), date: $('f-date').value, category: $('f-category').value
};
    353          Object.keys(f).forEach(k => { if (!f[k]) delete f[k]; });
    354          if (!Object.keys(f).length) return;
    355          state.filtering = true; $('f-reset').classList.remove('hidden'); $('f-status').classList.remove('hidden');
```

```
        $('f-status').textContent = t('Matokeo ya utafutaji', 'Search results');
        loadMessages(f);
    });
    $('f-reset').addEventListener('click', () => { $('f-keyword').value = ''; $('f-date').value = ''; $('f-category').value = ''; state.filtering = false; hide($('f-reset')); hide($('f-status')); loadMessages(); });

    // prefs
    $('prefs-btn').addEventListener('click', () => { $('p-name').value = state.user.name || ''; $('p-lang').value = state.user.preferred_language || 'auto'; $('p-region').value = state.user.region || ''; $('prefs').classList.remove('hidden'); });
    $('prefs-cancel').addEventListener('click', () => hide($('prefs')));
    $('prefs-form').addEventListener('submit', async (e) => {
        e.preventDefault();
        try {
            const r = await api('/chat/preferences', { method: 'POST', body: JSON.stringify({ name: $('p-name').value, preferred_language: $('p-lang').value, region: $('p-region').value }) });
            state.user = r.user; hide($('prefs'));
            state.lang = state.user.preferred_language === 'en' ? 'en' : (state.user.preferred_language === 'sw' ? 'sw' : state.lang);
            $('who-line').textContent = (state.user.name ? state.user.name + ' · ' : '') + state.user.phone_masked + (state.user.region ? ' · ' + state.user.region : '');
            addSystem(t('Mipangilio imehifadhiwa.', 'Preferences saved.'));
        } catch (ex) { alert(ex.message); }
    });

    $('logout-btn').addEventListener('click', async () => {
        if (!confirm(t('Unataka kutoka?', 'Sign out?'))) return;
        await api('/chat/logout', { method: 'POST' }); state.user = null; $('msgs').innerHTML = ''; show('auth-screen');
    });

    // offline awareness + PWA
    const net = () => $('offline-bar').classList.toggle('hidden', navigator.onLine);
    window.addEventListener('online', net); window.addEventListener('offline', net); net();
    if ('serviceWorker' in navigator) navigator.serviceWorker.register('/sw.js').catch(() => {});

    boot();
})();
</script>
</body>
</html>
```

93 resources/views/community/index.blade.php

78 lines

```
@extends('layouts.public', ['title' => 'Jamii', 'description' => 'Jamii ya Huru SMS: jadili, shirikiana uzoefu na upate ushauri kutoka kwa wananchi wenzako.', 'robots' => 'noindex, follow'])

@section('content')
<style>
    .top{display:flex;justify-content:space-between;align-items:center;gap:12px;flex-wrap:wrap;margin-bottom:18px}
    .top p{color:var(--muted);font-size:.9rem}
    .btn{display:inline-flex;align-items:center;gap:6px;background:var(--green);color:#fff;text-decoration:none;font-weight:700;padding:10px 14px;border-radius:8px;border:none;font:inherit;cursor:pointer;font-size:.9rem}
    .btn.alt{background:#fff;color:var(--blue);border:2px solid var(--blue)}
    .grid{display:grid;grid-template-columns:repeat(2,1fr);gap:12px}
    .th{border:1px solid var(--border);border-radius:10px;padding:16px;background:#fff}
    .th h3{font-size:1rem;margin-bottom:4px}
    .th h3 a{color:var(--text);text-decoration:none}
    .th p{font-size:.88rem;color:var(--muted)}
    .th .meta{display:flex;gap:10px;align-items:center;margin-top:10px;font-size:.78rem;color:var(--muted);flex-wrap:wrap}
    .badge{background:var(--green-bg);color:var(--green-dark);border:1px solid var(--green-border);padding:2px 8px;border-radius:999px;font-weight:700;font-size:.72rem}
    .badge.blue{background:#eff6ff;color:#1e40af;border-color:#bfdbfe}
    form.inline{display:inline}
    .new{border:1px dashed var(--border);border-radius:10px;padding:16px;margin-top:22px}
    .new input,.new textarea{width:100%;padding:10px 12px;border:1.5px solid var(--border);border-radius:8px;font:inherit;margin-bottom:8px}
    .alert{padding:10px 12px;border-radius:8px;font-size:.88rem;font-weight:600;margin-bottom:12px}
    .alert-err{background:#fef2f2;border:1px solid #fecaca;color:#991b1b}
    .alert-ok{background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark)}
    @media (max-width:640px){.grid{grid-template-columns:1fr}}
</style>

<div class="top">
    <div><h1>Jamii ya {{ $settings['app_name'] }}</h1><p>Karibu, {{ $user->displayName() }}. Jadili, uliza na ushirikiane uzoefu na wananchi wenzako.</p></div>
    <a class="btn alt" href="{{ route('chat.index') }}">← Rudi kwenye chat</a>
</div>

@if(session('error'))<div class="alert alert-err">{{ session('error') }}</div>@endif
@if(session('success'))<div class="alert alert-ok">{{ session('success') }}</div>@endif
@if($errors->any())<div class="alert alert-err">{{ $errors->first() }}</div>@endif

<h2>Mijadala ya wazi</h2>
<div class="grid">
    @forelse($publicThreads as $thread)
        <div class="th">
            <h3><a href="{{ route('community.show', $thread) }}">{{ $thread->title }}</a> @if($thread->is_system)<span class="badge">Rasmi</span>@endif</h3>
            <p>{{ \Illuminate\Support\Str::limit($thread->description, 140) }}</p>
            <div class="meta">
                <span>{{ $thread->posts_count }} ujumbe</span><span>{{ $thread->members_count }} wanachama</span>
                @if(in_array($thread->id, $joinedIds))
                    <span class="badge blue">Umejiunga</span>
                @else
                    <form class="inline" method="POST" action="{{ route('community.join', $thread) }}">@csrf<button class="btn" type="submit" style="padding:6px 10px;font-size:.78rem">Jiunge</button></form>
                @endif
            </div>
        </div>
    @empty
        <p>Hakuna mijadala bado.</p>
    @endforelse
</div>

@if($joinedThreads->isNotEmpty())
<h2>Mijadala yangu</h2>
<div class="grid">
    @foreach($joinedThreads as $thread)
        <div class="th">
            <h3><a href="{{ route('community.show', $thread) }}">{{ $thread->title }}</a> @if($thread->is_private)<span class="badge blue">Faragha</span>@endif</h3>
            <p>{{ \Illuminate\Support\Str::limit($thread->description, 120) }}</p>
```

```
            <div class="meta"><span>{{ $thread->posts_count }} ujumbe</span><span>Jukumu: {{ $thread->pivot->role }}</span></div>
        </div>
    @endforeach
</div>
@endif

<div class="new">
    <h2 style="margin-top:0">Anzisha mjadala mpya</h2>
    <form method="POST" action="{{ route('community.threads.store') }}">
        @csrf
        <input type="text" name="title" placeholder="Kichwa cha mjadala" required minlength="3" maxlength="120">
        <textarea name="description" rows="2" placeholder="Unajadili nini? (si lazima)" maxlength="1000"></textarea>
        <label style="font-size:.88rem;color:var(--muted);display:flex;gap:8px;align-items:center;margin-bottom:10px"><input type="checkbox" name="is_private" value="1" style="width:auto;margin:0"> Mjadala wa faragha (kwa mwaliko tu)</label>
        <button class="btn" type="submit">Anzisha</button>
    </form>
</div>
@endsection
```

94 resources/views/community/show.blade.php

69 lines

```
@extends('layouts.public', ['title' => $thread->title, 'description' => \Illuminate\Support\Str::limit($thread->description ?? $thread->title, 150), 'robots' => 'noindex, follow'])

@section('content')
<style>
    .top{display:flex;justify-content:space-between;align-items:flex-start;gap:12px;flex-wrap:wrap;margin-bottom:14px}
    .top p{color:var(--muted);font-size:.9rem}
    .btn{display:inline-flex;align-items:center;gap:6px;background:var(--green);color:#fff;text-decoration:none;font-weight:700;padding:9px 13px;border-radius:8px;border:none;font:inherit;cursor:pointer;font-size:.88rem}
    .btn.alt{background:#fff;color:var(--blue);border:2px solid var(--blue)}
    .btn.muted{background:#fff;color:var(--muted);border:1px solid var(--border)}
    .members{font-size:.8rem;color:var(--muted);margin-bottom:14px}
    .posts{display:flex;flex-direction:column;gap:10px;margin:14px 0}
    .post{border:1px solid var(--border);border-radius:10px;padding:12px 14px;background:#fff}
    .post.mine{border-color:var(--green-border);background:var(--green-bg)}
    .post .who{font-size:.78rem;color:var(--muted);font-weight:600;margin-bottom:4px;display:flex;justify-content:space-between;gap:8px}
    .post .body{font-size:.95rem;white-space:pre-wrap}
    .compose textarea{width:100%;padding:11px 12px;border:1.5px solid var(--border);border-radius:8px;font:inherit;margin-bottom:8px;resize:vertical}
    .alert{padding:10px 12px;border-radius:8px;font-size:.88rem;font-weight:600;margin-bottom:12px}
    .alert-err{background:#fef2f2;border:1px solid #fecaca;color:#991b1b}
    .alert-ok{background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark)}
    .notice{background:#eff6ff;border:1px solid #bfdbfe;color:#1e40af;padding:10px 12px;border-radius:8px;font-size:.88rem}
    form.inline{display:inline}
</style>

<div class="top">
    <div>
        <h1 style="font-size:1.4rem">{{ $thread->title }}</h1>
        <p>{{ $thread->description }}</p>
    </div>
    <div style="display:flex;gap:8px;flex-wrap:wrap">
        <a class="btn alt" href="{{ route('community.index') }}">← Mijadala</a>
        @if($isMember && !$thread->is_system)
            <form class="inline" method="POST" action="{{ route('community.leave', $thread) }}">@csrf<button class="btn muted" type="submit">Ondoka</button></form>
        @elseif(!$isMember && !$thread->is_private)
            <form class="inline" method="POST" action="{{ route('community.join', $thread) }}">@csrf<button class="btn" type="submit">Jiunge</button></form>
        @endif
    </div>
</div>

<div class="members">Wanachama {{ $members->count() }}: {{ $members->take(8)->map->displayName()->implode(', ') }}{{ $members->count() > 8 ? ' na wengine' : '' }}</div>

@if(session('error'))<div class="alert alert-err">{{ session('error') }}</div>@endif
@if(session('success'))<div class="alert alert-ok">{{ session('success') }}</div>@endif
@if($errors->any())<div class="alert alert-err">{{ $errors->first() }}</div>@endif
@if($thread->is_system)<div class="notice">Huu ni mjadala rasmi. Ujumbe huonekana baada ya kuhakikiwa na timu ya {{ $settings['app_name'] }}.</div>@endif

<div class="posts">
    @forelse($posts as $post)
        <div class="post {{ $post->user_id === $user->id ? 'mine' : '' }}">
            <div class="who"><span>{{ $post->user?->displayName() ?? 'Mwananchi' }}</span><span>{{ $post->created_at->diffForHumans() }}</span></div>
            <div class="body">{{ $post->content }}</div>
        </div>
    @empty
        <p style="color:var(--muted)">Hakuna ujumbe bado. Kuwa wa kwanza kuandika.</p>
    @endforelse
</div>
{{ $posts->links('pagination.simple') }}

@if($isMember)
<div class="compose">
```

```
    <form method="POST" action="{{ route('community.posts.store', $thread) }}">
        @csrf
        <textarea name="content" rows="3" placeholder="Andika ujumbe wako..." required minlength="2"
maxlength="3000"></textarea>
        <button class="btn" type="submit">Tuma</button>
    </form>
</div>
@else
    <p style="color:var(--muted);font-size:.9rem">Jiunge na mjadala huu ili kuandika.</p>
@endif
@endsection
```

95 resources/views/layouts/admin.blade.php

123 lines

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{{ $title ?? 'Dashboard' }} – Huru SMS Admin</title>
    <meta name="robots" content="noindex, nofollow">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
rel="stylesheet">
    <style>
        :root{--green:#15803d;--green-dark:#166534;--green-bg:#f0fdf4;--green-border:#bbf7d0;--blue:#1d4ed8;--blue-
dark:#1e40af;--blue-bg:#eff6ff;--bg:#f8fafc;--card:#fff;--text:#0f172a;--muted:#475569;--border:#cbd5e1;--danger:#b91c1c;--
danger-bg:#fef2f2;--warn:#b45309;--warn-bg:#fffbeb;--sidebar:240px}
        *{margin:0;padding:0;box-sizing:border-box}
        body{font-family:'Inter',system-ui,sans-serif;background:var(--bg);color:var(--text);font-size:.92rem;line-
height:1.5}
        a{color:var(--blue)}
        .sidebar{position:fixed;top:0;left:0;width:var(--sidebar);height:100vh;background:#fff;border-right:1px
solid var(--border);display:flex;flex-direction:column;z-index:30}
        .brand{display:flex;align-items:center;gap:10px;padding:16px;border-bottom:1px solid var(--border);text-
decoration:none;color:var(--green-dark);font-weight:800;font-size:1.15rem}
        .brand img{width:32px;height:32px;border-radius:8px}
        .brand small{display:block;font-size:.65rem;letter-spacing:.08em;text-transform:uppercase;color:var(--
muted);font-weight:600}
        .nav{flex:1;overflow:auto;padding:10px}
        .nav .sec{font-size:.65rem;font-weight:700;letter-spacing:.1em;text-transform:uppercase;color:var(--
muted);padding:12px 10px 4px}
        .nav a{display:flex;align-items:center;gap:8px;padding:8px 10px;border-radius:8px;text-
decoration:none;color:var(--muted);font-weight:600;font-size:.88rem}
        .nav a:hover{background:var(--bg);color:var(--text)}
        .nav a.on{background:var(--green-bg);color:var(--green-dark)}
        .sb-foot{padding:12px 16px;border-top:1px solid var(--border);font-size:.8rem;color:var(--muted)}
        .sb-foot form{margin-top:6px}
        .main{margin-left:var(--sidebar);min-height:100vh}
        .topbar{height:56px;background:#fff;border-bottom:1px solid var(--border);display:flex;align-
items:center;justify-content:space-between;padding:0 24px;position:sticky;top:0;z-index:20}
        .topbar h1{font-size:1.05rem;font-weight:800}
        .content{padding:24px;max-width:1280px}
        .card{background:var(--card);border:1px solid var(--border);border-radius:12px;padding:20px;margin-
bottom:18px}
        .card-h{display:flex;justify-content:space-between;align-items:center;gap:12px;flex-wrap:wrap;margin-
bottom:14px}
        .card-h h2{font-size:1rem;font-weight:800}
        .card-h p{font-size:.82rem;color:var(--muted)}
        .grid{display:grid;gap:14px}
        .g4{grid-template-columns:repeat(4,1fr)}.g3{grid-template-columns:repeat(3,1fr)}.g2{grid-template-
columns:repeat(2,1fr)}
        .stat{background:var(--card);border:1px solid var(--border);border-radius:12px;padding:16px}
        .stat .l{font-size:.75rem;font-weight:700;letter-spacing:.06em;text-transform:uppercase;color:var(--muted)}
        .stat .v{font-size:1.7rem;font-weight:800;margin-top:4px}
        .stat .s{font-size:.78rem;color:var(--muted)}
        table{width:100%;border-collapse:collapse;font-size:.85rem}
        th{text-align:left;font-size:.72rem;letter-spacing:.06em;text-transform:uppercase;color:var(--
muted);padding:8px 10px;border-bottom:1px solid var(--border)}
        td{padding:9px 10px;border-bottom:1px solid #e2e8f0;vertical-align:top}
        tr:hover td{background:#f8fafc}
        .truncate{max-width:420px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}
        .badge{display:inline-block;padding:2px 8px;border-radius:999px;font-size:.72rem;font-weight:700;border:1px
solid transparent}
        .b-green{background:var(--green-bg);color:var(--green-dark);border-color:var(--green-border)}
        .b-blue{background:var(--blue-bg);color:var(--blue-dark);border-color:#bfdbfe}
        .b-gray{background:#f1f5f9;color:var(--muted);border-color:#e2e8f0}
        .b-red{background:var(--danger-bg);color:var(--danger);border-color:#fecaca}
        .b-warn{background:var(--warn-bg);color:var(--warn);border-color:#fde68a}
        .btn{display:inline-flex;align-items:center;gap:6px;padding:8px 13px;border-radius:8px;border:1px solid
var(--border);background:#fff;color:var(--text);font:inherit;font-weight:600;font-size:.85rem;cursor:pointer;text-
decoration:none}
        .btn:hover{border-color:var(--blue);color:var(--blue)}
```

```
        .btn-p{background:var(--green);border-color:var(--green);color:#fff}
        .btn-p:hover{background:var(--green-dark);color:#fff}
        .btn-d{color:var(--danger);border-color:#fecaca}
        .btn-d:hover{background:var(--danger-bg);color:var(--danger)}
        .btn-sm{padding:5px 9px;font-size:.78rem}
        form.inline{display:inline}
        .alert{padding:10px 14px;border-radius:8px;font-weight:600;font-size:.88rem;margin-bottom:14px}
        .a-ok{background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark)}
        .a-err{background:var(--danger-bg);border:1px solid #fecaca;color:var(--danger)}
        .a-warn{background:var(--warn-bg);border:1px solid #fde68a;color:var(--warn)}
        .field{margin-bottom:14px}
        .field label{display:block;font-size:.8rem;font-weight:700;margin-bottom:5px}
        .field .hint{font-size:.76rem;color:var(--muted);margin-top:4px}

input[type=text],input[type=email],input[type=password],input[type=number],input[type=date],input[type=url],input[type=file],select,textarea{width:100%;padding:9px 11px;border:1.5px solid var(--border);border-radius:8px;font:inherit;font-size:.9rem;background:#fff}
        input:focus,select:focus,textarea:focus{outline:none;border-color:var(--green)}
        textarea{resize:vertical;min-height:90px}
        .check{display:flex;align-items:center;gap:8px;font-size:.88rem;font-weight:600}
        .check input{width:auto}
        .filters{display:flex;gap:8px;flex-wrap:wrap;align-items:center;margin-bottom:14px}
        .filters input,.filters select{width:auto;min-width:140px;padding:7px 10px;font-size:.85rem}
        .bars{display:flex;flex-direction:column;gap:6px}
        .bar{display:grid;grid-template-columns:160px 1fr 50px;gap:8px;align-items:center;font-size:.82rem}
        .bar .t{height:10px;background:#e2e8f0;border-radius:999px;overflow:hidden}
        .bar .f{height:100%;background:var(--green)}
        .spark{display:flex;align-items:flex-end;gap:4px;height:70px}
        .spark div{flex:1;background:var(--blue);border-radius:3px 3px 0 0;min-height:2px;position:relative}
        .spark div span{position:absolute;bottom:-18px;left:0;right:0;text-align:center;font-size:.6rem;color:var(--muted)}
        .mono{font-family:ui-monospace,Menlo,Consolas,monospace;font-size:.8rem;white-space:pre-wrap;background:#f8fafc;border:1px solid var(--border);border-radius:8px;padding:10px;max-height:320px;overflow:auto}
        .menu-btn{display:none}
        @media (max-width:900px){.sidebar{transform:translateX(-100%);transition:.2s}.sidebar.open{transform:none}.main{margin-left:0}.menu-btn{display:inline-block}.g4,.g3,.g2{grid-template-columns:1fr 1fr}.content{padding:14px}}
        @media (max-width:600px){.g4,.g3,.g2{grid-template-columns:1fr}}
    </style>
</head>
<body>
<aside class="sidebar" id="sidebar">
    <a href="{{ route('admin.dashboard') }}" class="brand"><img src="/logo.svg" alt=""><span>Huru SMS<small>Admin panel</small></span></a>
    <nav class="nav">
        <div class="sec">Overview</div>
        <a href="{{ route('admin.dashboard') }}" class="{{ request()->routeIs('admin.dashboard') ? 'on' : '' }}">📊 Dashboard</a>
        <a href="{{ route('admin.conversations.index') }}" class="{{ request()->routeIs('admin.conversations.*') ? 'on' : '' }}">💬 Conversations</a>
        <a href="{{ route('admin.users.index') }}" class="{{ request()->routeIs('admin.users.*') ? 'on' : '' }}">👥 Citizens</a>
        <div class="sec">Knowledge</div>
        <a href="{{ route('admin.knowledge.index') }}" class="{{ request()->routeIs('admin.knowledge.*') ? 'on' : '' }}">📚 Knowledge base</a>
        <a href="{{ route('admin.contacts.index') }}" class="{{ request()->routeIs('admin.contacts.*') ? 'on' : '' }}">📞 National directory</a>
        <a href="{{ route('admin.resources.index') }}" class="{{ request()->routeIs('admin.resources.*') ? 'on' : '' }}">📍 Local services</a>
        <a href="{{ route('admin.templates.index') }}" class="{{ request()->routeIs('admin.templates.*') ? 'on' : '' }}">🧭 Personas</a>
        <div class="sec">System</div>
        <a href="{{ route('admin.settings.index') }}" class="{{ request()->routeIs('admin.settings.*') ? 'on' : '' }}">⚙️ Settings</a>
        <a href="{{ route('welcome') }}" target="_blank">🌍 Public site ↗</a>
    </nav>
    <div class="sb-foot">
        {{ auth()->user()->name }}<br><span style="font-size:.72rem">{{ auth()->user()->email }}</span>
        <form method="POST" action="{{ route('admin.logout') }}">@csrf<button class="btn btn-sm" type="submit">Sign out</button></form>
    </div>
</aside>
<div class="main">
```

```
    <div class="topbar">
        <div style="display:flex;gap:10px;align-items:center"><button class="btn btn-sm menu-btn" onclick="document.getElementById('sidebar').classList.toggle('open')">☰</button><h1>{{ $title ?? 'Dashboard' }}</h1></div>
        <div style="font-size:.78rem;color:var(--muted)">{{ now()->timezone('Africa/Dar_es_Salaam')->format('D, d M Y H:i') }} EAT</div>
    </div>
    <div class="content">
        @if(session('success'))<div class="alert a-ok">{{ session('success') }}</div>@endif
        @if(session('error'))<div class="alert a-err">{{ session('error') }}</div>@endif
        @if($errors->any())<div class="alert a-err">{{ $errors->first() }}</div>@endif
        @yield('content')
    </div>
</div>
</body>
</html>
```

96 resources/views/layouts/public.blade.php

57 lines

```
<!DOCTYPE html>
<html lang="sw">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{{ $title ?? 'Huru SMS' }} – Huru SMS</title>
    <meta name="description" content="{{ $description ?? 'Huru SMS – uliza lolote kwa SMS 15054 au mtandaoni.' }}">
    <meta name="robots" content="{{ $robots ?? 'index, follow' }}">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <link rel="manifest" href="/manifest.json">
    <meta name="theme-color" content="#15803d">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap"
rel="stylesheet">
    <style>
        :root{--green:#15803d;--green-dark:#166534;--green-bg:#f0fdf4;--green-border:#bbf7d0;--blue:#1d4ed8;--
bg:#f8fafc;--card:#fff;--text:#0f172a;--muted:#475569;--border:#cbd5e1}
        *{margin:0;padding:0;box-sizing:border-box}
        body{font-family:'Inter',system-ui,sans-serif;background:var(--bg);color:var(--text);line-height:1.6}
        header{background:#fff;border-bottom:1px solid var(--border)}
        .nav{max-width:960px;margin:0 auto;padding:0 16px;height:60px;display:flex;align-items:center;justify-
content:space-between}
        .brand{display:flex;align-items:center;gap:10px;text-decoration:none;color:var(--green-dark);font-
weight:800;font-size:1.2rem}
        .brand img{width:32px;height:32px;border-radius:8px}
        .nav a.btn{background:var(--green);color:#fff;text-decoration:none;font-weight:700;padding:8px 14px;border-
radius:8px;font-size:.9rem}
        .sms{font-size:.85rem;color:var(--green-dark);font-weight:700;margin-right:12px}
        main{max-width:960px;margin:0 auto;padding:32px 16px 48px}
        .card{background:var(--card);border:1px solid var(--border);border-radius:12px;padding:28px}
        h1{font-size:1.7rem;font-weight:800;margin-bottom:6px}
        h2{font-size:1.1rem;font-weight:800;margin:22px 0 8px;color:var(--green-dark)}
        p,li{font-size:.95rem;color:#1e293b}
        ul{padding-left:20px}
        li{margin-bottom:6px}
        .updated{font-size:.8rem;color:var(--muted);margin-bottom:12px}
        a{color:var(--blue)}
        em{color:var(--muted)}
        footer{text-align:center;font-size:.85rem;color:var(--muted);padding:20px 16px;border-top:1px solid var(--
border);background:#fff}
        footer a{color:var(--muted);margin:0 8px;text-decoration:none}
        @media (max-width:600px){.sms{display:none}.card{padding:20px}}
    </style>
</head>
<body>
<header>
    <div class="nav">
        <a href="{{ route('welcome') }}" class="brand"><img src="/logo.svg" alt="">Huru SMS</a>
        <div><span class="sms">SMS: HURU → 15054</span><a class="btn" href="{{ route('chat.index') }}">Web Chat</a>
</div>
    </div>
</header>
<main>
    <div class="card">
        @yield('content')
    </div>
</main>
<footer>
    © {{ date('Y') }} Huru Digital Co. Ltd. · SMS HURU kwenda 15054
    <div style="margin-top:6px"><a href="{{ route('legal.terms') }}">Vigezo na Masharti</a><a href="{{
route('legal.privacy') }}">Sera ya Faragha</a><a href="{{ route('community.index') }}">Jamii</a></div>
</footer>
</body>
</html>
```

97 resources/views/legal/privacy.blade.php

37 lines

```
@extends('layouts.public', ['title' => 'Sera ya Faragha', 'description' => 'Jinsi Huru SMS inavyokusanya, kutumia na kulinda taarifa zako.'])

@section('content')
<h1>Sera ya Faragha · Privacy Policy</h1>
<p class="updated">Ilisasishwa: {{ date('d F Y') }}</p>

<h2>1. Taarifa tunazokusanya</h2>
<ul>
    <li><strong>Namba ya simu</strong> – ndiyo kitambulisho chako cha huduma, kwa SMS na web chat.</li>
    <li><strong>Maswali na majibu</strong> – tunahifadhi mazungumzo ili kuendeleza muktadha wa maswali ya kufuatilia na kukuwezesha kutafuta majibu ya zamani.</li>
    <li><strong>Mapendeleo</strong> – jina (ukilitoa), lugha unayopendelea, mkoa (ukiutaja) na maoni yako kuhusu majibu.</li>
    <li><strong>Taarifa za kiufundi</strong> – muda wa ujumbe, njia uliyotumia (SMS au web) na kumbukumbu za mfumo kwa usalama.</li>
</ul>

<h2>2. Jinsi tunavyotumia taarifa</h2>
<p>Kutoa majibu, kuendeleza mazungumzo, kulinda huduma dhidi ya matumizi mabaya, kuboresha maarifa yaliyohifadhiwa na kutoa takwimu za jumla (bila kukutambulisha) kwa washirika. Maswali yako hutumwa kwa mtoa huduma wa teknolojia ya lugha ili kupata jibu; hatutumi jina lako wala namba yako ya simu pamoja na swali.</p>
<p><em>We use your data to answer you, keep conversation context, protect the service from abuse and produce anonymous statistics. Questions are sent to a language technology provider to generate answers; your name and phone number are not sent with the question.</em></p>

<h2>3. Kushirikisha taarifa</h2>
<p>Hatuuzi taarifa zako. Tunaweza kushirikisha taarifa pale sheria inapotaka au kwa watoa huduma wanaotuwezesha kutoa huduma (mtandao wa SMS, mwenyeji wa seva, mtoa huduma wa teknolojia ya lugha) chini ya masharti ya usiri.</p>

<h2>4. Uhifadhi na usalama</h2>
<p>Taarifa huhifadhiwa kwenye seva zilizolindwa. Namba ya uthibitisho (OTP) huhifadhiwa kwa njia ya siri na hufutwa baada ya kutumika. Tunahifadhi mazungumzo kwa muda unaohitajika kwa huduma; unaweza kuomba kufutwa.</p>

<h2>5. Haki zako</h2>
<ul>
    <li>Kuona mazungumzo yako kupitia web chat.</li>
    <li>Kusitisha majibu wakati wowote kwa kutuma <strong>ACHA</strong> kwa SMS, na kuendelea kwa <strong>ANZA</strong>.</li>
    <li>Kuomba kufutwa kwa taarifa zako kwa kutumia fomu ya mawasiliano.</li>
</ul>

<h2>6. Watoto</h2>
<p>Huduma inaweza kutumiwa na wanafunzi kwa maswali ya masomo. Hatukusanyi kwa makusudi taarifa za ziada za watoto zaidi ya zile zilizo muhimu kwa huduma.</p>

<h2>7. Mawasiliano</h2>
<p>Huru Digital Co. Ltd., Tanzania. Tumia fomu kwenye <a href="{{ route('welcome') }}#wasiliana">ukurasa wa mwanzo</a>.</p>
@endsection
```

98 resources/views/legal/terms.blade.php

33 lines

```
@extends('layouts.public', ['title' => 'Vigezo na Masharti', 'description' => 'Vigezo na masharti ya kutumia huduma ya Huru SMS kwa SMS na mtandaoni.'])

@section('content')
<h1>Vigezo na Masharti · Terms and Conditions</h1>
<p class="updated">Ilisasishwa: {{ date('d F Y') }}</p>

<h2>1. Huduma</h2>
<p>Huru SMS ni huduma ya maelezo na mwongozo wa jumla inayotolewa na Huru Digital Co. Ltd. kwa SMS (shortcode 15054, neno HURU) na kupitia mtandao. Huru SMS hujibu maswali ya kila siku kuhusu maisha Tanzania: sheria, afya, kilimo, elimu, huduma za serikali, fedha, ajira na mada nyingine.</p>
<p><em>Huru SMS is a general information and guidance service provided by Huru Digital Co. Ltd. via SMS (shortcode 15054, keyword HURU) and the web.</em></p>

<h2>2. Si ushauri wa kitaalamu</h2>
<p>Majibu ni maelezo ya jumla yanayotolewa kiotomatiki kwa msaada wa teknolojia na maarifa yaliyokusanywa. Si ushauri wa kisheria, wa kitabibu, wa kifedha wala uwakilishi wa aina yoyote. Kwa maamuzi muhimu, thibitisha na ofisi, mtaalamu au mamlaka husika. Kwa dharura piga 112 (polisi), 114 (zimamoto), 115 (gari la wagonjwa) au 116 (msaada kwa mtoto).</p>
<p><em>Answers are general information generated automatically. They are not legal, medical or financial advice and do not create any professional relationship. Verify important decisions with the responsible office or professional.</em></p>

<h2>3. Matumizi yanayokubalika</h2>
<ul>
    <li>Usitumie lugha ya matusi, vitisho au maudhui ya chuki. Akaunti hupewa maonyo na baadaye kufungiwa.</li>
    <li>Usitumie huduma kwa udanganyifu, utapeli au kuwadhuru wengine.</li>
    <li>Usijaribu kuvuruga, kupakia kupita kiasi au kudukua mifumo ya huduma.</li>
</ul>

<h2>4. Gharama</h2>
<p>SMS kwenda 15054 hutozwa na mtandao wako wa simu kwa viwango vya kawaida vya SMS isipokuwa ikitangazwa vinginevyo. Web chat hutumia bando la intaneti la mtumiaji. Huru SMS haitozi ada ya ziada kwa sasa.</p>

<h2>5. Taarifa zako</h2>
<p>Tunahifadhi namba yako ya simu, maswali na majibu ili kuendeleza mazungumzo na kuboresha huduma. Soma <a href="{{ route('legal.privacy') }}">Sera ya Faragha</a> kwa maelezo kamili. Tuma ACHA kwa SMS kusitisha majibu wakati wowote.</p>

<h2>6. Upatikanaji na mabadiliko</h2>
<p>Huduma inaweza kusitishwa kwa muda kwa matengenezo. Tunaweza kubadilisha vigezo hivi; toleo jipya litawekwa kwenye ukurasa huu.</p>

<h2>7. Mawasiliano</h2>
<p>Huru Digital Co. Ltd., Tanzania. Tumia fomu ya mawasiliano kwenye <a href="{{ route('welcome') }}#wasiliana">ukurasa wa mwanzo</a>.</p>
@endsection
```

99 resources/views/offline.blade.php

28 lines

```
<!DOCTYPE html>
<html lang="sw">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Uko nje ya mtandao – Huru SMS</title>
    <meta name="robots" content="noindex">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <style>
        body{font-family:Inter,system-ui,sans-serif;background:#f8fafc;color:#0f172a;margin:0;display:flex;align-items:center;justify-content:center;min-height:100vh;padding:16px}
        .box{background:#fff;border:2px solid #15803d;border-radius:12px;padding:28px;max-width:440px;text-align:center}
        .code{font-size:2.6rem;font-weight:800;color:#166534;margin:8px 0}
        .kw{display:inline-block;background:#dcfce7;color:#14532d;font-weight:800;padding:2px 8px;border-radius:4px;border:1px solid #86efac}
        a.btn{display:inline-block;margin-top:16px;background:#15803d;color:#fff;text-decoration:none;font-weight:700;padding:12px 18px;border-radius:8px}
        p{color:#475569;line-height:1.5}
    </style>
</head>
<body>
<div class="box">
    <h1 style="font-size:1.3rem;margin:0">Uko nje ya mtandao</h1>
    <p>Hakuna intaneti kwa sasa. Bado unaweza kuuliza kwa SMS kutoka simu yoyote.</p>
    <div class="code">15054</div>
    <p>Tuma <span class="kw">HURU</span> kisha swali lako kwenda <strong>15054</strong>.</p>
    <a class="btn" href="sms:15054?body=HURU%20">Tuma SMS sasa</a>
    <p style="font-size:.85rem;margin-top:18px">Ukirudi mtandaoni, <a href="/chat">fungua web chat tena</a>.</p>
</div>
</body>
</html>
```

100 resources/views/pagination/simple.blade.php

15 lines

```
@if ($paginator->hasPages())
    <nav style="display:flex;gap:8px;justify-content:center;align-items:center;margin:12px 0;font-size:.88rem">
        @if ($paginator->onFirstPage())
            <span style="color:#94a3b8">‹ Nyuma</span>
        @else
            <a href="{{ $paginator->previousPageUrl() }}" rel="prev" style="color:#1d4ed8;font-weight:600;text-decoration:none">‹ Nyuma</a>
        @endif
        <span style="color:#475569">{{ $paginator->currentPage() }} / {{ $paginator->lastPage() }}</span>
        @if ($paginator->hasMorePages())
            <a href="{{ $paginator->nextPageUrl() }}" rel="next" style="color:#1d4ed8;font-weight:600;text-decoration:none">Mbele ›</a>
        @else
            <span style="color:#94a3b8">Mbele ›</span>
        @endif
    </nav>
@endif
```

101 resources/views/welcome.blade.php

229 lines

```
<!DOCTYPE html>
<html lang="sw">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{{ $settings['app_name'] }} – Uliza lolote, kwa SMS {{ $settings['sms_shortcode'] }} au mtandaoni</title>
    <meta name="description" content="{{ $settings['app_name'] }} hujibu swali lolote la maisha ya kila siku Tanzania: sheria, afya, kilimo, masomo, huduma za serikali, fedha, ajira. Tuma {{ $settings['sms_keyword'] }} kwenda {{ $settings['sms_shortcode'] }} bila intaneti, au tumia web chat.">
    <meta name="keywords" content="Huru SMS, SMS {{ $settings['sms_shortcode'] }}, maswali Tanzania, sheria, afya, kilimo, NIDA, TRA, elimu, msaada wa kisheria, Kiswahili">
    <meta name="author" content="Huru Digital Co. Ltd.">
    <meta name="robots" content="index, follow">
    <link rel="canonical" href="{{ url('/') }}">
    <meta property="og:type" content="website">
    <meta property="og:site_name" content="{{ $settings['app_name'] }}">
    <meta property="og:title" content="{{ $settings['app_name'] }} – Uliza lolote kwa SMS {{ $settings['sms_shortcode'] }}">
    <meta property="og:description" content="Majibu ya haraka kwa maswali ya sheria, afya, kilimo, masomo, huduma za serikali na zaidi. Kiswahili au Kiingereza.">
    <meta property="og:url" content="{{ url('/') }}">
    <meta property="og:image" content="{{ url('/og-image.svg') }}">
    <link rel="manifest" href="/manifest.json">
    <link rel="icon" href="/logo.svg" type="image/svg+xml">
    <link rel="apple-touch-icon" href="/logo.svg">
    <meta name="theme-color" content="#15803d">
    @verbatim
    <script type="application/ld+json">
    {"@context":"https://schema.org","@type":"WebApplication","name":"Huru SMS","applicationCategory":"UtilitiesApplication","operatingSystem":"Any","description":"Everyday question answering service for Tanzanians over SMS and web, in Swahili and English.","offers":{"@type":"Offer","price":"0","priceCurrency":"TZS"}}
    </script>
    @endverbatim
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
    <style>
        :root{--green:#15803d;--green-dark:#166534;--green-bg:#f0fdf4;--green-border:#bbf7d0;--blue:#1d4ed8;--blue-dark:#1e40af;--blue-bg:#eff6ff;--bg:#f8fafc;--card:#ffffff;--text:#0f172a;--muted:#475569;--border:#cbd5e1;--radius:12px}
        *{margin:0;padding:0;box-sizing:border-box}
        html{scroll-behavior:smooth}
        body{font-family:'Inter',system-ui,sans-serif;background:var(--bg);color:var(--text);line-height:1.55}
        a{color:var(--blue)}
        .wrap{max-width:1080px;margin:0 auto;padding:0 16px}
        header{background:#fff;border-bottom:1px solid var(--border);position:sticky;top:0;z-index:20}
        .nav{display:flex;align-items:center;justify-content:space-between;height:64px}
        .brand{display:flex;align-items:center;gap:10px;text-decoration:none;color:var(--green-dark);font-weight:800;font-size:1.3rem}
        .brand img{width:36px;height:36px;border-radius:8px}
        .nav-links{display:flex;gap:8px;align-items:center}
        .nav-links a{text-decoration:none;font-weight:600;font-size:.9rem;padding:8px 14px;border-radius:8px;color:var(--muted)}
        .nav-links a:hover{background:var(--bg);color:var(--text)}
        .nav-links a.primary{background:var(--green);color:#fff}
        .nav-links a.primary:hover{background:var(--green-dark)}
        .sms-pill{display:inline-flex;align-items:center;gap:6px;background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark);font-weight:700;font-size:.85rem;padding:6px 12px;border-radius:999px}
        .hero{padding:48px 0 32px;display:grid;grid-template-columns:1.2fr 1fr;gap:32px;align-items:center}
        .hero h1{font-size:2.4rem;line-height:1.15;font-weight:800;letter-spacing:-.02em;margin:14px 0}
        .hero h1 span{color:var(--green)}
        .hero p.lead{font-size:1.1rem;color:var(--muted);max-width:52ch}
        .hero p.sub{font-size:.95rem;color:var(--muted);margin-top:8px}
        .chips{display:flex;flex-wrap:wrap;gap:8px;margin-top:20px}
        .chip{background:#fff;border:1px solid var(--border);border-radius:999px;padding:6px 12px;font-size:.85rem;font-weight:600;color:var(--muted)}
        .sms-card{background:var(--card);border:2px solid var(--green);border-radius:var(--radius);padding:24px}
        .sms-card .label{font-size:.75rem;letter-spacing:.08em;text-transform:uppercase;font-weight:800;color:var(--green-dark)}
```

```
        .sms-card .code{font-size:3rem;font-weight:800;color:var(--green-dark);line-height:1;margin:8px 0}
        .sms-card .how{background:var(--green-bg);border:1px solid var(--green-border);border-radius:8px;padding:12px 14px;margin:14px 0;font-size:.95rem}
        .kw{display:inline-block;background:#dcfce7;color:#14532d;font-weight:800;padding:2px 8px;border-radius:4px;border:1px solid #86efac}
        .sms-card .example{font-size:.85rem;color:var(--muted);font-style:italic;margin-top:6px}
        .btn{display:inline-flex;align-items:center;justify-content:center;gap:8px;text-decoration:none;font-weight:700;padding:12px 18px;border-radius:8px;border:2px solid transparent;font-size:.95rem;cursor:pointer}
        .btn-green{background:var(--green);color:#fff}
        .btn-green:hover{background:var(--green-dark)}
        .btn-blue-outline{background:#fff;border-color:var(--blue);color:var(--blue)}
        .btn-blue-outline:hover{background:var(--blue-bg)}
        .btn-row{display:flex;gap:10px;flex-wrap:wrap;margin-top:6px}
        .btn-row .btn{flex:1;min-width:160px}
        section{padding:36px 0}
        h2{font-size:1.5rem;font-weight:800;letter-spacing:-.01em;margin-bottom:6px}
        .section-sub{color:var(--muted);margin-bottom:22px}
        .steps{display:grid;grid-template-columns:repeat(3,1fr);gap:16px}
        .step{background:var(--card);border:1px solid var(--border);border-radius:var(--radius);padding:20px}
        .step .n{width:34px;height:34px;border-radius:8px;background:var(--blue);color:#fff;font-weight:800;display:flex;align-items:center;justify-content:center;margin-bottom:12px}
        .step h3{font-size:1.05rem;margin-bottom:6px}
        .step p{color:var(--muted);font-size:.93rem}
        .topics{display:grid;grid-template-columns:repeat(3,1fr);gap:14px}
        .topic{background:var(--card);border:1px solid var(--border);border-radius:var(--radius);padding:18px;text-decoration:none;color:var(--text);display:block;transition:border-color .15s}
        .topic:hover{border-color:var(--green)}
        .topic h3{font-size:1rem;color:var(--green-dark);margin-bottom:6px}
        .topic p{font-size:.88rem;color:var(--muted)}
        .topic .q{display:block;margin-top:10px;font-size:.85rem;color:var(--blue);font-weight:600}
        .voices{display:grid;grid-template-columns:repeat(3,1fr);gap:14px}
        .voice{background:var(--card);border:1px solid var(--border);border-radius:var(--radius);padding:18px;font-size:.93rem}
        .voice .who{margin-top:10px;font-size:.8rem;color:var(--muted);font-weight:600}
        .trust{display:grid;grid-template-columns:repeat(4,1fr);gap:14px}
        .trust div{background:var(--card);border:1px solid var(--border);border-radius:var(--radius);padding:16px;font-size:.9rem}
        .trust strong{display:block;color:var(--green-dark);margin-bottom:4px}
        .contact{background:var(--card);border:1px solid var(--border);border-radius:var(--radius);padding:24px;display:grid;grid-template-columns:1fr 1fr;gap:24px}
        .contact form{display:grid;gap:10px}
        .contact input,.contact textarea,.contact select{width:100%;padding:11px 12px;border:1.5px solid var(--border);border-radius:8px;font:inherit;font-size:.93rem;background:#fff}
        .contact input:focus,.contact textarea:focus{outline:none;border-color:var(--green)}
        .alert{padding:12px 14px;border-radius:8px;font-size:.9rem;font-weight:600;margin-bottom:12px}
        .alert-ok{background:var(--green-bg);border:1px solid var(--green-border);color:var(--green-dark)}
        .alert-err{background:#fef2f2;border:1px solid #fecaca;color:#991b1b}
        .hp{position:absolute;left:-9999px}
        footer{background:#fff;border-top:1px solid var(--border);padding:24px 0;font-size:.85rem;color:var(--muted)}
        footer .wrap{display:flex;justify-content:space-between;gap:12px;flex-wrap:wrap}
        footer a{color:var(--muted);text-decoration:none;margin-left:16px}
        footer a:hover{color:var(--blue)}
        @media (max-width:860px){.hero{grid-template-columns:1fr;padding-top:28px}.steps,.topics,.voices{grid-template-columns:1fr}.trust{grid-template-columns:1fr 1fr}.contact{grid-template-columns:1fr}.hero h1{font-size:1.9rem}.nav-links a:not(.primary){display:none}}
    </style>
</head>
<body>
<header>
    <div class="wrap nav">
        <a href="/" class="brand" aria-label="{{ $settings['app_name'] }}"><img src="/logo.svg" alt="" width="36" height="36">{{ $settings['app_name'] }}</a>
        <div class="nav-links">
            <span class="sms-pill">SMS: {{ $settings['sms_keyword'] }} → {{ $settings['sms_shortcode'] }}</span>
            <a href="#mada">Mada</a>
            <a href="{{ route('community.index') }}">Jamii</a>
            <a href="{{ route('chat.index') }}" class="primary">Fungua Web Chat</a>
        </div>
    </div>
</header>

<main>
```

```
    <div class="wrap">
        <div class="hero">
            <div>
                <span class="sms-pill">Kwa kila Mtanzania · For every Tanzanian</span>
                <h1>Uliza <span>lolote</span>. Pata jibu la haraka, kwa lugha yako.</h1>
                <p class="lead">Sheria, afya, kilimo, masomo, huduma za serikali, fedha, ajira, teknolojia au swali la kawaida tu. {{ $settings['app_name'] }} hujibu kwa Kiswahili au Kiingereza, kwa muktadha wa Tanzania, kwa simu yoyote.</p>
                <p class="sub">Ask anything about life in Tanzania and get a clear, practical answer in Swahili or English, on any phone.</p>
                <div class="chips">
                    <span class="chip">Sheria na haki</span><span class="chip">Afya</span><span class="chip">Kilimo na mifugo</span><span class="chip">Masomo</span><span class="chip">NIDA · TRA · RITA</span><span class="chip">Fedha na biashara</span><span class="chip">Ajira</span><span class="chip">Familia</span>
                </div>
            </div>

            <div class="sms-card">
                <div class="label">Njia kuu: SMS bila intaneti</div>
                <div class="code">{{ $settings['sms_shortcode'] }}</div>
                <div class="how">
                    Tuma neno <span class="kw">{{ $settings['sms_keyword'] }}</span> kisha swali lako kwenda <strong>{{ $settings['sms_shortcode'] }}</strong> kutoka simu yoyote.
                    <div class="example">Mfano: <strong>{{ $settings['sms_keyword'] }} nifanyeje kupata kitambulisho cha NIDA?</strong></div>
                </div>
                <div class="btn-row">
                    <a class="btn btn-green" href="sms:{{ $settings['sms_shortcode'] }}?body={{ $settings['sms_keyword'] }}%20">Tuma SMS sasa</a>
                    <a class="btn btn-blue-outline" href="{{ route('chat.index') }}">Au tumia Web Chat</a>
                </div>
                <p class="example" style="margin-top:12px">Una intaneti? Web chat inakumbuka mazungumzo yako na inakuruhusu kutafuta majibu ya zamani.</p>
            </div>
        </div>

        <section id="jinsi">
            <h2>Jinsi inavyofanya kazi</h2>
            <p class="section-sub">Hatua tatu tu. Hakuna programu ya kupakua, hakuna usajili mrefu.</p>
            <div class="steps">
                <div class="step"><div class="n">1</div><h3>Uliza</h3><p>Andika swali lako kwa Kiswahili au Kiingereza. Taja mahali ulipo (mfano "nipo Kahama") ukihitaji huduma iliyo karibu.</p></div>
                <div class="step"><div class="n">2</div><h3>Tunaelewa muktadha</h3><p>{{ $settings['app_name'] }} hutambua mada, lugha na eneo lako, kisha huchanganya maarifa yaliyothibitishwa ya Tanzania na anwani rasmi.</p></div>
                <div class="step"><div class="n">3</div><h3>Pata jibu la vitendo</h3><p>Jibu fupi, la kweli, lenye hatua za kuchukua na ofisi au namba ya kuwasiliana nayo. Uliza swali la kufuatilia wakati wowote.</p></div>
            </div>
        </section>

        <section id="mada">
            <h2>Mada tunazoshughulikia</h2>
            <p class="section-sub">Bonyeza mada kuanza na mfano wa swali kwenye web chat.</p>
            <div class="topics">
                <a class="topic" href="{{ route('chat.index') }}?q=Haki zangu ni zipi nikikamatwa na polisi?"><h3>Sheria na haki</h3><p>Kukamatwa, ardhi, ndoa, mirathi, mikataba, msaada wa kisheria karibu nawe.</p><span class="q">"Haki zangu ni zipi nikikamatwa na polisi?"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Dalili za malaria kwa mtoto na nifanye nini?"><h3>Afya</h3><p>Dalili za hatari, kliniki, chanjo, bima ya afya (NHIF/iCHF), lishe.</p><span class="q">"Dalili za malaria kwa mtoto ni zipi?"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Nipande mahindi lini Dodoma na nitumie mbegu gani?"><h3>Kilimo na mifugo</h3><p>Misimu, mbegu bora, wadudu, chanjo za mifugo, bei na masoko.</p><span class="q">"Nipande mahindi lini Dodoma?"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Nieleze photosynthesis kwa Kiswahili kwa mwanafunzi wa kidato cha pili"><h3>Masomo</h3><p>Hisabati, sayansi, lugha, NECTA, mikopo ya HESLB, vyuo.</p><span class="q">"Nieleze photosynthesis kwa kidato cha pili"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Nifanyeje kupata kitambulisho cha NIDA?"><h3>Huduma za serikali</h3><p>NIDA, RITA, pasipoti, TIN, GePG, LUKU, leseni.</p><span class="q">"Nifanyeje kupata kitambulisho cha NIDA?"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Nianzeje biashara ndogo na nisajili wapi?"><h3>Fedha na biashara</h3><p>Kodi za TRA, mikopo, pesa za simu, kuepuka utapeli, kusajili biashara.</p><span class="q">"Nianzeje biashara ndogo?"</span></a>
                <a class="topic" href="{{ route('chat.index') }}?q=Nimefukuzwa kazi bila notisi, nifanye nini?"><h3>Ajira na kazi</h3><p>Mikataba, likizo, kuachishwa kazi, CMA, kutafuta kazi.</p><span class="q">"Nimefukuzwa kazi bila
```

```
notisi, nifanye nini?"</span></a>
   164                 <a class="topic" href="{{ route('chat.index') }}?q=Nipate wapi msaada kwa ukatili wa kijinsia
Mwanza?"><h3>Familia na jamii</h3><p>Ukatili wa kijinsia, haki za mtoto, ustawi wa jamii, mahusiano.</p><span
class="q">"Nipate wapi msaada kwa ukatili Mwanza?"</span></a>
   165                 <a class="topic" href="{{ route('chat.index') }}?q=Laini yangu imepotea, nifanyeje kuirudisha?">
<h3>Teknolojia na usafiri</h3><p>Laini za simu, udukuzi, faini za trafiki, leseni ya udereva, nauli.</p><span
class="q">"Laini yangu imepotea, nifanyeje?"</span></a>
   166             </div>
   167         </section>
   168
   169         <section>
   170             <h2>Kwa nini {{ $settings['app_name'] }}</h2>
   171             <div class="trust">
   172                 <div><strong>Muktadha wa Tanzania</strong>Sheria, taasisi, misimu, sarafu na namba za dharura za
hapa, si za nchi nyingine.</div>
   173                 <div><strong>Lugha yako</strong>Kiswahili au Kiingereza, inafuata lugha unayoandika. Tuma LUGHA SW
au LUGHA EN kubadilisha.</div>
   174                 <div><strong>Huduma zilizo karibu</strong>Taja mkoa, wilaya au mtaa wako na upate watoa huduma
walio karibu nawe.</div>
   175                 <div><strong>Uaminifu</strong>Tunasema wazi pale jambo linapohitaji kuthibitishwa na ofisi husika.
Hatuzushi namba wala bei.</div>
   176             </div>
   177         </section>
   178
   179         @if($communityPosts->isNotEmpty())
   180         <section>
   181             <h2>Sauti za wananchi</h2>
   182             <p class="section-sub">Maoni yaliyothibitishwa kutoka kwenye jamii ya {{ $settings['app_name'] }}.</p>
   183             <div class="voices">
   184                 @foreach($communityPosts as $post)
   185                     <div class="voice">"{{ \Illuminate\Support\Str::limit($post->content, 180) }}"<div class="who">
{{ $post->user?->displayName() ?? 'Mwananchi' }} · {{ $post->created_at->diffForHumans() }}</div></div>
   186                 @endforeach
   187             </div>
   188         </section>
   189         @endif
   190
   191         <section id="wasiliana">
   192             <h2>Wasiliana nasi</h2>
   193             <p class="section-sub">Taasisi, halmashauri, shule au shirika linalotaka kushirikiana, au maoni yoyote.
</p>
   194             <div class="contact">
   195                 <div>
   196                     @if(session('contact_success'))<div class="alert alert-ok">{{ session('contact_success') }}
</div>@endif
   197                     @if(session('contact_error'))<div class="alert alert-err">{{ session('contact_error') }}
</div>@endif
   198                     @if($errors->any())<div class="alert alert-err">{{ $errors->first() }}</div>@endif
   199                     <form method="POST" action="{{ route('contact.submit') }}">
   200                         @csrf
   201                         <input class="hp" type="text" name="website" tabindex="-1" autocomplete="off">
   202                         <select name="type"><option value="general">Ujumbe wa kawaida</option><option
value="partner">Ushirikiano / Partnership</option><option value="subscribe">Nitumie taarifa za maendeleo</option></select>
   203                         <input type="text" name="name" placeholder="Jina lako" maxlength="80">
   204                         <input type="email" name="email" placeholder="Barua pepe" required>
   205                         <input type="text" name="organisation" placeholder="Taasisi / Shirika (si lazima)"
maxlength="120">
   206                         <textarea name="message" rows="4" placeholder="Ujumbe wako"></textarea>
   207                         <button class="btn btn-green" type="submit">Tuma</button>
   208                     </form>
   209                 </div>
   210                 <div>
   211                     <h3 style="margin-bottom:8px">Kwa simu yoyote, popote</h3>
   212                     <p style="color:var(--muted);font-size:.93rem">Huduma hii imejengwa kwa simu za kawaida kwanza.
Hata bila bando, tuma <strong>{{ $settings['sms_keyword'] }}</strong> kwenda <strong>{{ $settings['sms_shortcode'] }}
</strong>.</p>
   213                     <p style="color:var(--muted);font-size:.93rem;margin-top:12px"><strong>Amri za SMS:</strong>
<br>MSAADA – orodha ya msaada<br>LUGHA SW / LUGHA EN – chagua lugha<br>MKOA Mwanza – hifadhi mkoa wako<br>JINA Asha –
hifadhi jina lako<br>ACHA / ANZA – sitisha au endelea</p>
   214                     <p style="color:var(--muted);font-size:.85rem;margin-top:12px">Dharura: Polisi 112 · Zimamoto
114 · Gari la wagonjwa 115 · Msaada kwa mtoto 116</p>
   215                 </div>
   216             </div>
```

```
        </section>
    </div>
</main>

<footer>
    <div class="wrap">
        <div>© {{ date('Y') }} Huru Digital Co. Ltd. · SMS: <strong>{{ $settings['sms_keyword'] }}</strong> kwenda <strong>{{ $settings['sms_shortcode'] }}</strong></div>
        <div><a href="{{ route('legal.terms') }}">Vigezo na Masharti</a><a href="{{ route('legal.privacy') }}">Sera ya Faragha</a><a href="{{ route('community.index') }}">Jamii</a></div>
    </div>
</footer>
<script>if('serviceWorker' in navigator){navigator.serviceWorker.register('/sw.js').catch(()=>{});}</script>
</body>
</html>
```

102 routes/api.php

10 lines

```php
<?php

use App\Http\Controllers\SmsController;
use Illuminate\Support\Facades\Route;

// Africa's Talking callbacks. Register them with ?token=<AT_WEBHOOK_SECRET>.
Route::middleware('sms.webhook')->group(function () {
    Route::post('/sms/inbound', [SmsController::class, 'inbound'])->name('sms.inbound');
    Route::post('/sms/delivery', [SmsController::class, 'deliveryReport'])->name('sms.delivery');
});
```

103 routes/console.php

11 lines

```php
<?php

use Illuminate\Support\Facades\Schedule;

// Drain the database queue every minute on hosts without a long-running worker.
Schedule::command('queue:work --stop-when-empty --tries=2 --max-time=50')
    ->everyMinute()
    ->withoutOverlapping();

// Keep the local legal aid directory fresh.
Schedule::command('huru:import-legal-aid')->weeklyOn(1, '03:00');
```

104 routes/web.php

120 lines

```php
<?php

use App\Http\Controllers\Admin\AuthController;
use App\Http\Controllers\Admin\ConversationController;
use App\Http\Controllers\Admin\DashboardController;
use App\Http\Controllers\Admin\KnowledgeController;
use App\Http\Controllers\Admin\LocalResourceController;
use App\Http\Controllers\Admin\PromptTemplateController;
use App\Http\Controllers\Admin\ReferenceContactController;
use App\Http\Controllers\Admin\SettingController;
use App\Http\Controllers\Admin\UserController;
use App\Http\Controllers\CommunityController;
use App\Http\Controllers\ContactController;
use App\Http\Controllers\WebChatController;
use App\Models\CommunityPost;
use App\Support\Settings;
use Illuminate\Support\Facades\Route;

Route::get('/', function () {
    $communityPosts = CommunityPost::query()
        ->where('is_approved', true)
        ->whereHas('thread', fn ($q) => $q->where('slug', 'general-advice'))
        ->with('user')
        ->latest()
        ->take(6)
        ->get();

    return view('welcome', [
        'communityPosts' => $communityPosts,
        'settings' => [
            'app_name' => Settings::get('app_name'),
            'sms_keyword' => Settings::get('sms_keyword'),
            'sms_shortcode' => Settings::get('sms_shortcode'),
        ],
    ]);
})->name('welcome');

// Web chat (citizen)
Route::prefix('chat')->name('chat.')->group(function () {
    Route::get('/', [WebChatController::class, 'index'])->name('index');
    Route::get('/session', [WebChatController::class, 'session'])->name('session');
    Route::post('/request-otp', [WebChatController::class, 'requestOtp'])->middleware('throttle:10,1')->name('otp.request');
    Route::post('/verify-otp', [WebChatController::class, 'verifyOtp'])->middleware('throttle:10,1')->name('otp.verify');
    Route::get('/messages', [WebChatController::class, 'messages'])->name('messages');
    Route::post('/send', [WebChatController::class, 'send'])->middleware('throttle:20,1')->name('send');
    Route::post('/feedback', [WebChatController::class, 'feedback'])->name('feedback');
    Route::post('/preferences', [WebChatController::class, 'preferences'])->name('preferences');
    Route::post('/logout', [WebChatController::class, 'logout'])->name('logout');
});

// Legal pages
Route::view('/terms-and-conditions', 'legal.terms')->name('legal.terms');
Route::view('/privacy-policy', 'legal.privacy')->name('legal.privacy');
Route::view('/offline', 'offline')->name('offline');

// Community
Route::prefix('community')->name('community.')->group(function () {
    Route::get('/', [CommunityController::class, 'index'])->name('index');
    Route::post('/threads', [CommunityController::class, 'storeThread'])->name('threads.store');
    Route::get('/threads/{thread:slug}', [CommunityController::class, 'show'])->name('show');
    Route::post('/threads/{thread:slug}/posts', [CommunityController::class, 'storePost'])->middleware('throttle:30,1')->name('posts.store');
    Route::post('/threads/{thread:slug}/join', [CommunityController::class, 'join'])->name('join');
    Route::post('/threads/{thread:slug}/leave', [CommunityController::class, 'leave'])->name('leave');
});

Route::post('/contact', [ContactController::class, 'submit'])->name('contact.submit');

// Admin
```

```php
Route::prefix('admin')->name('admin.')->group(function () {
    Route::get('/login', [AuthController::class, 'showLogin'])->name('login');
    Route::post('/login', [AuthController::class, 'login'])->middleware('throttle:10,1')->name('login.post');
    Route::post('/logout', [AuthController::class, 'logout'])->name('logout');

    Route::middleware(['auth', 'admin'])->group(function () {
        Route::get('/', [DashboardController::class, 'index'])->name('dashboard');

        Route::get('/conversations', [ConversationController::class, 'index'])->name('conversations.index');
        Route::get('/conversations/{user}', [ConversationController::class, 'show'])->name('conversations.show');

        Route::get('/knowledge', [KnowledgeController::class, 'index'])->name('knowledge.index');
        Route::get('/knowledge/create', [KnowledgeController::class, 'create'])->name('knowledge.create');
        Route::post('/knowledge', [KnowledgeController::class, 'store'])->name('knowledge.store');
        Route::post('/knowledge/import', [KnowledgeController::class, 'import'])->name('knowledge.import');
        Route::get('/knowledge/{knowledge}/edit', [KnowledgeController::class, 'edit'])->name('knowledge.edit');
        Route::put('/knowledge/{knowledge}', [KnowledgeController::class, 'update'])->name('knowledge.update');
        Route::patch('/knowledge/{knowledge}/toggle', [KnowledgeController::class, 'toggle'])->name('knowledge.toggle');
        Route::delete('/knowledge/{knowledge}', [KnowledgeController::class, 'destroy'])->name('knowledge.destroy');

        Route::get('/templates', [PromptTemplateController::class, 'index'])->name('templates.index');
        Route::get('/templates/create', [PromptTemplateController::class, 'create'])->name('templates.create');
        Route::post('/templates', [PromptTemplateController::class, 'store'])->name('templates.store');
        Route::get('/templates/{template}/edit', [PromptTemplateController::class, 'edit'])->name('templates.edit');
        Route::put('/templates/{template}', [PromptTemplateController::class, 'update'])->name('templates.update');
        Route::patch('/templates/{template}/toggle', [PromptTemplateController::class, 'toggle'])->name('templates.toggle');
        Route::delete('/templates/{template}', [PromptTemplateController::class, 'destroy'])->name('templates.destroy');

        Route::get('/contacts', [ReferenceContactController::class, 'index'])->name('contacts.index');
        Route::get('/contacts/create', [ReferenceContactController::class, 'create'])->name('contacts.create');
        Route::post('/contacts', [ReferenceContactController::class, 'store'])->name('contacts.store');
        Route::get('/contacts/{contact}/edit', [ReferenceContactController::class, 'edit'])->name('contacts.edit');
        Route::put('/contacts/{contact}', [ReferenceContactController::class, 'update'])->name('contacts.update');
        Route::patch('/contacts/{contact}/verify', [ReferenceContactController::class, 'verify'])->name('contacts.verify');
        Route::delete('/contacts/{contact}', [ReferenceContactController::class, 'destroy'])->name('contacts.destroy');

        Route::get('/resources', [LocalResourceController::class, 'index'])->name('resources.index');
        Route::get('/resources/create', [LocalResourceController::class, 'create'])->name('resources.create');
        Route::post('/resources', [LocalResourceController::class, 'store'])->name('resources.store');
        Route::get('/resources/{resource}/edit', [LocalResourceController::class, 'edit'])->name('resources.edit');
        Route::put('/resources/{resource}', [LocalResourceController::class, 'update'])->name('resources.update');
        Route::delete('/resources/{resource}', [LocalResourceController::class, 'destroy'])->name('resources.destroy');

        Route::get('/users', [UserController::class, 'index'])->name('users.index');
        Route::patch('/users/{user}/ban', [UserController::class, 'toggleBan'])->name('users.ban');
        Route::patch('/users/{user}/strikes', [UserController::class, 'resetStrikes'])->name('users.strikes');
        Route::delete('/users/{user}', [UserController::class, 'destroy'])->name('users.destroy');

        Route::get('/settings', [SettingController::class, 'index'])->name('settings.index');
        Route::post('/settings', [SettingController::class, 'update'])->name('settings.update');
    });
});
```

105 tests/Feature/CommunityTest.php

45 lines

```php
<?php

namespace Tests\Feature;

use App\Models\CommunityThread;
use App\Models\User;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Tests\TestCase;

class CommunityTest extends TestCase
{
    use RefreshDatabase;

    public function test_guest_is_redirected_to_chat(): void
    {
        $this->get(route('community.index'))->assertRedirect(route('chat.index', ['redirect' => '/community']));
    }

    public function test_user_can_create_join_and_post(): void
    {
        $user = User::factory()->create();
        $session = ['chat_user_id' => $user->id];

        $this->withSession($session)->post(route('community.threads.store'), ['title' => 'Bei za mahindi', 'description' => 'Dodoma'])->assertRedirect();
        $thread = CommunityThread::where('title', 'Bei za mahindi')->firstOrFail();
        $this->assertTrue($thread->members()->where('user_id', $user->id)->exists());

        $other = User::factory()->create();
        $this->withSession(['chat_user_id' => $other->id])->post(route('community.join', $thread->slug))->assertRedirect(route('community.show', $thread));
        $this->withSession(['chat_user_id' => $other->id])->post(route('community.posts.store', $thread->slug), ['content' => 'Shilingi 800 kwa kilo Kongwa'])->assertRedirect();

        $this->assertDatabaseHas('community_posts', ['content' => 'Shilingi 800 kwa kilo Kongwa', 'is_approved' => true]);
        $this->withSession($session)->get(route('community.show', $thread->slug))->assertOk()->assertSee('Shilingi 800');
    }

    public function test_abusive_posts_are_rejected(): void
    {
        $user = User::factory()->create();
        $thread = CommunityThread::create(['title' => 'T', 'slug' => 't', 'is_private' => false]);
        $thread->members()->attach($user->id);

        $this->withSession(['chat_user_id' => $user->id])->postJson(route('community.posts.store', $thread->slug), ['content' => 'wewe mjinga sana'])->assertStatus(422);
        $this->assertSame(1, $user->fresh()->abuse_count);
    }
}
```

106 tests/Feature/SmsFlowTest.php

108 lines

```php
<?php

namespace Tests\Feature;

use App\Jobs\ProcessIncomingSms;
use App\Models\AiLog;
use App\Models\Message;
use App\Models\User;
use App\Services\Ai\LlmClient;
use App\Services\SmsService;
use App\Support\Settings;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Illuminate\Support\Facades\Queue;
use Mockery\MockInterface;
use Tests\TestCase;

class SmsFlowTest extends TestCase
{
    use RefreshDatabase;

    public function test_inbound_webhook_rejects_bad_token_when_secret_is_set(): void
    {
        config(['services.at.webhook_secret' => 'topsecret']);
        Queue::fake();

        $this->post('/api/sms/inbound', ['from' => '+255712345678', 'text' => 'HURU habari'])->assertStatus(401);
        $this->post('/api/sms/inbound?token=topsecret', ['from' => '+255712345678', 'text' => 'HURU habari'])->assertOk();

        Queue::assertPushed(ProcessIncomingSms::class, fn ($job) => $job->from === '+255712345678' && $job->text === 'habari');
    }

    public function test_inbound_strips_keyword_and_normalises_phone(): void
    {
        config(['services.at.webhook_secret' => null]);
        Queue::fake();

        $this->post('/api/sms/inbound', ['from' => '0712345678', 'text' => 'huru  Nifanyeje kupata NIDA?'])->assertOk();

        Queue::assertPushed(ProcessIncomingSms::class, fn ($job) => $job->from === '+255712345678' && $job->text === 'Nifanyeje kupata NIDA?');
    }

    public function test_job_answers_logs_and_sends_sms(): void
    {
        $this->mock(LlmClient::class, function (MockInterface $mock) {
            $mock->shouldReceive('isConfigured')->andReturn(true);
            $mock->shouldReceive('generate')->once()->andReturn([
                'status' => LlmClient::STATUS_OK, 'text' => 'Nenda ofisi ya mtendaji wa kata na cheti cha kuzaliwa.',
                'model' => 'gemini-test', 'tokens' => ['prompt' => 10, 'completion' => 5, 'total' => 15], 'latency_ms' => 120, 'finish_reason' => 'STOP', 'error' => null,
            ]);
        });
        $this->mock(SmsService::class, function (MockInterface $mock) {
            $mock->shouldReceive('send')->once()->with('+255712345678', 'Nenda ofisi ya mtendaji wa kata na cheti cha kuzaliwa.')->andReturn(['status' => 'success']);
        });

        (new ProcessIncomingSms('+255712345678', 'Nifanyeje kupata kitambulisho cha NIDA?'))->handle(app(\App\Services\ConversationService::class), app(SmsService::class));

        $this->assertDatabaseHas('users', ['phone_number' => '+255712345678']);
        $this->assertDatabaseHas('messages', ['direction' => 'inbound', 'channel' => 'sms', 'category' => 'government']);
        $this->assertDatabaseHas('messages', ['direction' => 'outbound', 'content' => 'Nenda ofisi ya mtendaji wa kata na cheti cha kuzaliwa.']);
        $log = AiLog::first();
        $this->assertSame('gemini-test', $log->model);
```

```
        $this->assertSame(15, $log->total_tokens);
        $this->assertSame('government', $log->category);
    }

    public function test_help_and_language_commands_do_not_call_the_model(): void
    {
        $this->mock(LlmClient::class, function (MockInterface $mock) {
            $mock->shouldReceive('generate')->never();
        });
        $this->mock(SmsService::class, function (MockInterface $mock) {
            $mock->shouldReceive('send')->twice()->andReturn(['status' => 'success']);
        });

        (new ProcessIncomingSms('+255712345678', 'MSAADA'))->handle(app(\App\Services\ConversationService::class),
app(SmsService::class));
        (new ProcessIncomingSms('+255712345678', 'LUGHA EN'))-
>handle(app(\App\Services\ConversationService::class), app(SmsService::class));

        $user = User::where('phone_number', '+255712345678')->first();
        $this->assertSame('en', $user->preferred_language);
        $this->assertSame(4, Message::count());
    }

    public function test_abuse_leads_to_warning_then_ban(): void
    {
        Settings::set('abuse_strike_limit', 2);
        $this->mock(LlmClient::class, fn (MockInterface $m) => $m->shouldReceive('generate')->never());
        $this->mock(SmsService::class, fn (MockInterface $m) => $m->shouldReceive('send')->twice()-
>andReturn(['status' => 'success']));

        $svc = app(\App\Services\ConversationService::class);
        (new ProcessIncomingSms('+255712345678', 'wewe mjinga'))->handle($svc, app(SmsService::class));
        (new ProcessIncomingSms('+255712345678', 'mjinga tena'))->handle($svc, app(SmsService::class));

        $this->assertTrue(User::where('phone_number', '+255712345678')->first()->is_banned);
    }

    public function test_opted_out_user_is_ignored_until_start(): void
    {
        $this->mock(LlmClient::class, fn (MockInterface $m) => $m->shouldReceive('generate')->never());
        $this->mock(SmsService::class, fn (MockInterface $m) => $m->shouldReceive('send')->twice()-
>andReturn(['status' => 'success']));

        $svc = app(\App\Services\ConversationService::class);
        (new ProcessIncomingSms('+255712345678', 'ACHA'))->handle($svc, app(SmsService::class));
        (new ProcessIncomingSms('+255712345678', 'swali lolote'))->handle($svc, app(SmsService::class)); // no
reply
        (new ProcessIncomingSms('+255712345678', 'ANZA'))->handle($svc, app(SmsService::class));

        $this->assertFalse(User::first()->opted_out);
    }
}
```

107 tests/Feature/WebChatTest.php

113 lines

```php
<?php

namespace Tests\Feature;

use App\Models\User;
use App\Services\Ai\LlmClient;
use App\Services\SmsService;
use App\Support\Settings;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Illuminate\Support\Facades\Hash;
use Mockery\MockInterface;
use Tests\TestCase;

class WebChatTest extends TestCase
{
    use RefreshDatabase;

    public function test_login_without_otp_when_sms_is_not_configured_locally(): void
    {
        $this->mock(SmsService::class, fn (MockInterface $m) => $m->shouldReceive('isConfigured')->andReturn(false));

        $login = $this->postJson('/chat/request-otp', ['phone_number' => '0712345678'])
            ->assertOk()
            ->assertJsonPath('otp_required', false)
            ->assertJsonPath('user.phone_masked', '+255 *** 678');

        // Login regenerates the session; the rotated CSRF token must be handed back.
        $this->assertNotEmpty($login->json('csrf'));
        $this->assertSame($login->json('csrf'), $this->getJson('/chat/session')->assertJsonPath('authenticated', true)->json('csrf'));
    }

    public function test_otp_flow_when_sms_is_configured(): void
    {
        Settings::set('otp_enabled', '1');
        $this->mock(SmsService::class, function (MockInterface $m) {
            $m->shouldReceive('isConfigured')->andReturn(true);
            $m->shouldReceive('sendOtp')->once()->andReturn(['status' => 'success']);
        });

        $r = $this->postJson('/chat/request-otp', ['phone_number' => '0712345678'])->assertOk()->assertJsonPath('otp_required', true);
        $code = $r->json('debug_code');
        $this->assertNotEmpty($code, 'debug code is exposed in testing');

        $this->getJson('/chat/session')->assertJsonPath('authenticated', false);
        $this->postJson('/chat/verify-otp', ['phone_number' => '0712345678', 'code' => '000000'])->assertStatus(422);
        $this->postJson('/chat/verify-otp', ['phone_number' => '0712345678', 'code' => $code])->assertOk();
        $this->getJson('/chat/session')->assertJsonPath('authenticated', true);

        $this->assertNull(User::first()->otp_hash);
    }

    public function test_messages_require_a_session_and_send_returns_reply(): void
    {
        $this->getJson('/chat/messages')->assertStatus(401);

        $this->mock(LlmClient::class, function (MockInterface $mock) {
            $mock->shouldReceive('isConfigured')->andReturn(true);
            $mock->shouldReceive('generate')->once()->andReturn([
                'status' => LlmClient::STATUS_OK, 'text' => 'Malaria danger signs include convulsions.',
                'model' => 'gemini-test', 'tokens' => ['prompt' => 1, 'completion' => 1, 'total' => 2], 'latency_ms' => 50, 'finish_reason' => 'STOP', 'error' => null,
            ]);
        });

        $user = User::factory()->create(['phone_number' => '+255712345678']);

        $this->withSession(['chat_user_id' => $user->id])
```

```
            ->postJson('/chat/send', ['message' => 'What are malaria danger signs in a child?'])
            ->assertOk()
            ->assertJsonPath('kind', 'answer')
            ->assertJsonPath('category', 'health')
            ->assertJsonPath('reply.content', 'Malaria danger signs include convulsions.');

        $replyId = $this->withSession(['chat_user_id' => $user->id])->getJson('/chat/messages')->json('messages.1.id');

        $this->withSession(['chat_user_id' => $user->id])
            ->postJson('/chat/feedback', ['message_id' => $replyId, 'rating' => 1])
            ->assertOk();

        $this->assertDatabaseHas('feedback', ['message_id' => $replyId, 'rating' => 1]);
    }

    public function test_admin_area_requires_an_admin_account(): void
    {
        $this->get('/admin')->assertRedirect('/admin/login');

        $citizen = User::factory()->create(['email' => 'c@example.com', 'password' => Hash::make('secret123')]);
        $this->actingAs($citizen)->get('/admin')->assertStatus(403);

        $admin = User::factory()->admin()->create(['email' => 'a@example.com', 'password' => Hash::make('secret123')]);
        $this->post('/admin/login', ['email' => 'a@example.com', 'password' => 'secret123'])->assertRedirect('/admin');
        $this->actingAs($admin)->get('/admin')->assertOk()->assertSee('Dashboard');
        $this->actingAs($admin)->get('/admin/settings')->assertOk()->assertSee('Gemini model');
    }

    public function test_settings_only_accept_known_keys(): void
    {
        $admin = User::factory()->admin()->create();

        $this->actingAs($admin)->post('/admin/settings', ['app_name' => 'Huru', 'bogus_key' => 'x'] + $this->defaults())->assertRedirect();
        $this->assertDatabaseMissing('system_settings', ['key' => 'bogus_key']);
        $this->assertSame('Huru', Settings::get('app_name'));
    }

    private function defaults(): array
    {
        $out = [];
        foreach (Settings::schema() as $k => $def) {
            $out[$k] = $def['default'];
        }

        return $out;
    }
}
```

108 tests/TestCase.php

10 lines

```php
<?php

namespace Tests;

use Illuminate\Foundation\Testing\TestCase as BaseTestCase;

abstract class TestCase extends BaseTestCase
{
    //
}
```

109 tests/Unit/LanguageAndLocationTest.php

59 lines

```php
<?php

namespace Tests\Unit;

use App\Services\LanguageDetector;
use App\Services\TanzaniaContext;
use App\Support\Phone;
use PHPUnit\Framework\TestCase;

class LanguageAndLocationTest extends TestCase
{
    public function test_language_detection(): void
    {
        $d = new LanguageDetector();
        $this->assertSame('sw', $d->detect('Naomba kujua haki zangu nikikamatwa'));
        $this->assertSame('en', $d->detect('What are my rights when arrested?'));
        $this->assertSame('sw', $d->detect('nimedhulumiwa shamba langu kijijini'));
        $this->assertNull($d->detect(''));
    }

    public function test_location_detection(): void
    {
        $tz = new TanzaniaContext();

        $loc = $tz->detectLocation('naomba msaada wa kisheria nipo Sinza Dar es Salaam');
        $this->assertSame('dar es salaam', $loc['region']);
        $this->assertSame('ubungo', $loc['district']);
        $this->assertSame('sinza', $loc['area']);

        $loc = $tz->detectLocation('Nipo Kahama Shinyanga');
        $this->assertSame('shinyanga', $loc['region']);
        $this->assertSame('kahama', $loc['district']);

        $loc = $tz->detectLocation('I live in Mwanza');
        $this->assertSame('mwanza', $loc['region']);

        $this->assertNull($tz->detectLocation('Hakuna mahali hapa'));
    }

    public function test_phone_normalisation(): void
    {
        $this->assertSame('+255712345678', Phone::normalize('0712 345 678'));
        $this->assertSame('+255712345678', Phone::normalize('255712345678'));
        $this->assertSame('+255712345678', Phone::normalize('+255712345678'));
        $this->assertSame('+255712345678', Phone::normalize('712345678'));
        $this->assertSame('+254700000000', Phone::normalize('+254700000000'));
        $this->assertNull(Phone::normalize('abc'));
        $this->assertTrue(Phone::isTanzanian('+255712345678'));
        $this->assertFalse(Phone::isTanzanian('+254700000000'));
    }

    public function test_locale_block_mentions_tanzania_and_currency(): void
    {
        $tz = new TanzaniaContext();
        $this->assertStringContainsString('Tanzania', $tz->localeBlock('en'));
        $this->assertStringContainsString('TZS', $tz->localeBlock('sw'));
        $this->assertStringContainsString('112', $tz->localeBlock('sw'));
    }
}
```

110 tests/Unit/PromptEngineTest.php

67 lines

```php
<?php

namespace Tests\Unit;

use App\Models\KnowledgeEntry;
use App\Models\LocalResource;
use App\Models\PromptTemplate;
use App\Models\ReferenceContact;
use App\Services\PromptEngine;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Tests\TestCase;

class PromptEngineTest extends TestCase
{
    use RefreshDatabase;

    public function test_it_injects_matching_knowledge_and_country_context(): void
    {
        PromptTemplate::create(['name' => 'Legal SW', 'category' => 'legal', 'language' => 'sw', 'template' =>
'Wewe ni {app_name}, mwelimishaji wa sheria.', 'is_active' => true]);
        KnowledgeEntry::create([
            'title' => 'Haki zako ukikamatwa na polisi',
            'content' => 'Polisi wanapaswa kukufikisha mahakamani ndani ya saa 24.',
            'category' => 'legal', 'language' => 'sw', 'keywords' => ['kukamatwa', 'polisi', 'dhamana'],
'is_active' => true,
        ]);

        $built = app(PromptEngine::class)->build([
            'question' => 'Nimekamatwa na polisi, haki zangu ni zipi?',
            'language' => 'sw', 'category' => 'legal', 'channel' => 'sms',
        ]);

        $this->assertStringContainsString('saa 24', $built['system']);
        $this->assertStringContainsString('mwelimishaji wa sheria', $built['system']);
        $this->assertStringContainsString('Tanzania', $built['system']);
        $this->assertStringContainsString('TZS', $built['system']);
        $this->assertSame('user', $built['turns'][0]['role']);
        $this->assertCount(1, $built['knowledge']);
    }

    public function test_it_adds_nearby_services_when_location_is_mentioned(): void
    {
        LocalResource::create(['name' => 'KAHAMA PARALEGAL AID ORGANIZATION', 'category' => 'legal_aid', 'region'
=> 'shinyanga', 'district' => 'kahama', 'location' => 'Kahama mjini', 'phone' => '+255 767 994457', 'is_active' => true]);

        $built = app(PromptEngine::class)->build([
            'question' => 'naomba msaada wa kisheria nipo kahama shinyanga',
            'language' => 'sw', 'category' => 'legal', 'channel' => 'web',
            'location' => ['region' => 'shinyanga', 'district' => 'kahama', 'area' => null],
        ]);

        $this->assertStringContainsString('KAHAMA PARALEGAL AID ORGANIZATION', $built['system']);
        $this->assertStringContainsString('+255 767 994457', $built['system']);
    }

    public function test_emergency_puts_emergency_contacts_in_prompt_and_history_is_kept(): void
    {
        ReferenceContact::create(['name' => 'Ambulance', 'category' => 'health', 'phone' => '115', 'is_emergency'
=> true, 'is_active' => true]);

        $built = app(PromptEngine::class)->build([
            'question' => 'My father collapsed and is not breathing',
            'language' => 'en', 'category' => 'health', 'channel' => 'web', 'is_emergency' => true,
            'history' => [['role' => 'user', 'text' => 'hello'], ['role' => 'model', 'text' => 'hi']],
        ]);

        $this->assertStringContainsString('115', $built['system']);
        $this->assertStringContainsString('EMERGENCY', $built['system']);
        $this->assertCount(3, $built['turns']);
    }
}
```

111 tests/Unit/TopicClassifierTest.php

63 lines

```php
<?php

namespace Tests\Unit;

use App\Services\TopicClassifier;
use PHPUnit\Framework\TestCase;

class TopicClassifierTest extends TestCase
{
    private TopicClassifier $classifier;

    protected function setUp(): void
    {
        parent::setUp();
        $this->classifier = new TopicClassifier();
    }

    public function test_detects_legal_questions_in_swahili(): void
    {
        $r = $this->classifier->classify('Nimekamatwa na polisi bila sababu, naomba msaada wa kisheria');
        $this->assertSame('legal', $r['category']);
        $this->assertFalse($r['is_emergency']);
    }

    public function test_detects_health_questions_in_english(): void
    {
        $r = $this->classifier->classify('My child has fever and vomiting, is it malaria?');
        $this->assertSame('health', $r['category']);
    }

    public function test_detects_agriculture(): void
    {
        $r = $this->classifier->classify('Nipande mahindi lini na nitumie mbolea gani?');
        $this->assertSame('agriculture', $r['category']);
    }

    public function test_detects_government_services(): void
    {
        $r = $this->classifier->classify('How do I get a NIDA national id and a passport?');
        $this->assertSame('government', $r['category']);
    }

    public function test_falls_back_to_general(): void
    {
        $r = $this->classifier->classify('Habari za asubuhi');
        $this->assertSame('general', $r['category']);
    }

    public function test_flags_emergencies(): void
    {
        $r = $this->classifier->classify('Dharura! mtoto amezimia na hapumui');
        $this->assertTrue($r['is_emergency']);
        $this->assertSame('health', $r['category']);
    }

    public function test_labels_exist_for_every_category(): void
    {
        foreach (TopicClassifier::keys() as $key) {
            $this->assertNotEmpty(TopicClassifier::label($key, 'sw'));
            $this->assertNotEmpty(TopicClassifier::label($key, 'en'));
        }
    }
}
```

112 vite.config.js

18 lines

```
import { defineConfig } from 'vite';
import laravel from 'laravel-vite-plugin';
import tailwindcss from '@tailwindcss/vite';

export default defineConfig({
    plugins: [
        laravel({
            input: ['resources/css/app.css', 'resources/js/app.js'],
            refresh: true,
        }),
        tailwindcss(),
    ],
    server: {
        watch: {
            ignored: ['**/storage/framework/views/**'],
        },
    },
});
```

System Design and Architecture

Huru SMS

Multichannel SMS and Web Citizen Information and Guidance System for Tanzania

This document describes how Huru SMS is designed and built: the parts of the system, how they work together, how a question travels from a citizen's phone to an answer, how data is stored, how the system is deployed and secured, and what the citizen and the operator see on screen.

Owner and developer: Huru Digital Co. Ltd., Tanzania

Work: Computer software (web application and SMS service)

Version described: 1.0

Date: 9 October 2026

Prepared for: Copyright Society of Tanzania (COSOTA)

Companion documents: *Huru SMS – Innovation Documentation* and *Huru SMS – Source Code Listing*

1. Purpose of this document

The *Innovation Documentation* explains what Huru SMS is and why it was created. This document explains **how it is built**. It is written for a reader who wants to verify that the submitted source code is a complete, working, original system. It contains:

- a system overview showing the people and outside services the system talks to (Section 2);
- the functional design: what each part of the system does (Section 3);
- the software architecture: layers, components and the source files that implement them (Section 4);
- the processing flows for an SMS question, a web question and the answer pipeline (Section 5);
- the data design: database tables and their relationships (Section 6);
- the deployment and integration design (Section 7) and the security design (Section 8);
- the technology stack (Section 9) and screenshots of the SMS interface, the web interface and the administration panel (Section 10).

File names in this document (for example `app/Services/ConversationService.php`) refer to the files reproduced in the *Source Code Listing*.

2. System overview

Huru SMS is a server application. Citizens reach it in two ways: by sending an SMS starting with the keyword **HURU** to the shortcode **15054**, or by opening the web chat in a browser. Staff of Huru Digital manage the content and settings through a protected administration panel. The application uses three outside services: the SMS gateway that carries messages to and from the mobile networks, a language model service that words the final reply, and a public directory of legal aid providers that is imported periodically.

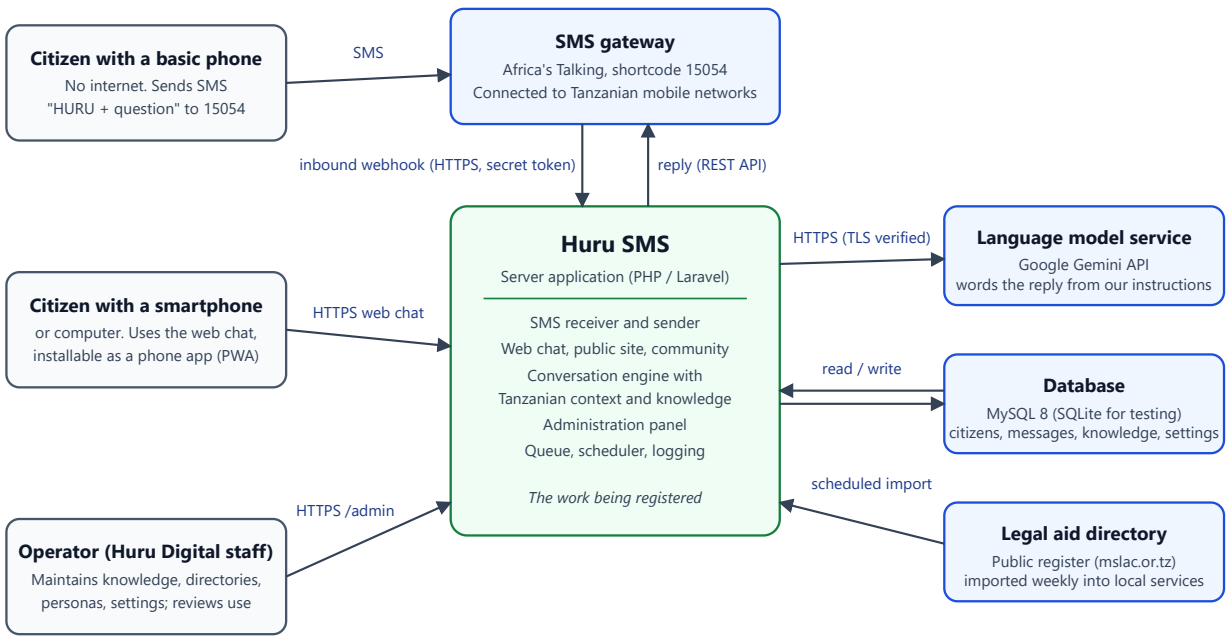


Figure 1. System context: the people and outside services connected to Huru SMS.

Actor or service	Role in the system
Citizen (SMS)	Sends questions and service commands by SMS from any phone. Identified only by phone number.
Citizen (web)	Logs into the web chat with a phone number confirmed by a one-time SMS code. Can search history, rate answers, set name, language and region, and take part in the community board.
Operator	Signs in to the administration panel with e-mail and password. Maintains knowledge entries, the national directory, local services, personas (prompt templates) and system settings; reviews conversations and engine logs; blocks abusive users.
SMS gateway	Africa's Talking delivers inbound SMS to the webhook and sends outbound replies. Any gateway with a similar HTTP interface can replace it; the integration is isolated in one class.
Language model service	Receives the system instructions and conversation turns assembled by Huru SMS and returns the worded reply. It never receives the citizen's name or number. The provider is behind an interface and can be swapped.
Database	Stores citizens, messages, engine logs, feedback, knowledge, directories, templates, settings, community content, sessions, cache and the job queue.

3. Functional design

3.1 Modules

Module	Functions	Main source files
SMS channel	Receives inbound SMS from the gateway, verifies the secret token, removes the keyword, queues the message, processes it and sends the reply. Handles delivery reports.	`routes/api.php`, `Http/Controllers/SmsController.php`, `Http/Middleware/VerifySmsWebhook.php`, `Jobs/ProcessIncomingSms.php`, `Services/SmsService.php`
Web chat	Phone-number login with one-time code, conversation screen, history search by keyword, date and topic, helpful / not helpful rating, preferences (name, language, region), offline notice, installable app.	`Http/Controllers/WebChatController.php`, `resources/views/chat.blade.php`, `public/sw.js`, `public/manifest.json`

Conversation engine	One pipeline for both channels: service commands, opt-out, bans, rate limit, topic classification, emergency detection, abuse moderation with strikes, Tanzanian context, knowledge retrieval, local resources, national contacts, history, reply generation, cleaning, storing and logging.	`Services/ConversationService.php`, `TopicClassifier.php`, `LanguageDetector.php`, `TanzaniaContext.php`, `KnowledgeRetriever.php`, `PromptEngine.php`, `AiService.php`, `ModerationService.php`, `Ai/GeminiClient.php`
Public site and community	Landing page with the SMS instructions, topics and contact form; terms and privacy pages; community threads where logged-in citizens post, reply, react and join or create groups.	`routes/web.php`, `views/welcome.blade.php`, `Http/Controllers/CommunityController.php`, `views/community/*`, `views/legal/*`
Administration panel	Dashboard (questions per day, topics, channels, tokens, errors), conversations and engine logs, knowledge base with file import, personas, national directory with verification, local services, citizens (ban, reset strikes), settings grouped by Branding, Engine, Channels and Safety.	`Http/Controllers/Admin/*`, `Http/Middleware/EnsureAdmin.php`, `views/admin/*`, `Support/Settings.php`
Background tasks	Database queue for SMS processing, a scheduler that drains the queue every minute and imports the legal aid directory weekly.	`routes/console.php`, `Console/Commands/FetchLegalAidProviders.php`
Starter data and tests	Seeders for settings, the administrator, personas, 42 knowledge entries, 33 national contacts and local services; 28 automated tests of the SMS flow, web chat, community, classifier, language and location detection and prompt assembly.	`database/seeders/*`, `tests/*`

3.2 Service commands (both channels)

Command	Effect
`MSAADA`, `HELP`, `MENU`, `?`	Short guide to the service in the citizen's language
`ACHA`, `STOP`, `SITAKI`, `ONDOA`	Opt out: no further replies by SMS until the citizen sends a start command
`ANZA`, `START`, `RUDI`	Resume the service

`LUGHA SW` / `LUGHA EN` / `LUGHA AUTO`	Fix the reply language to Kiswahili or English, or follow the language written
`JINA <name>`	Save the citizen's name for a personal reply
`MKOA <region or district>`	Save the citizen's region (recognises all 31 regions and common districts) so nearby services are prioritised

Commands are answered directly by the application and never reach the language model.

4. Software architecture

The application follows a layered architecture on the Laravel framework. Each layer only calls the layer below it. Both channels enter the same conversation engine, so the behaviour never differs between SMS and web.

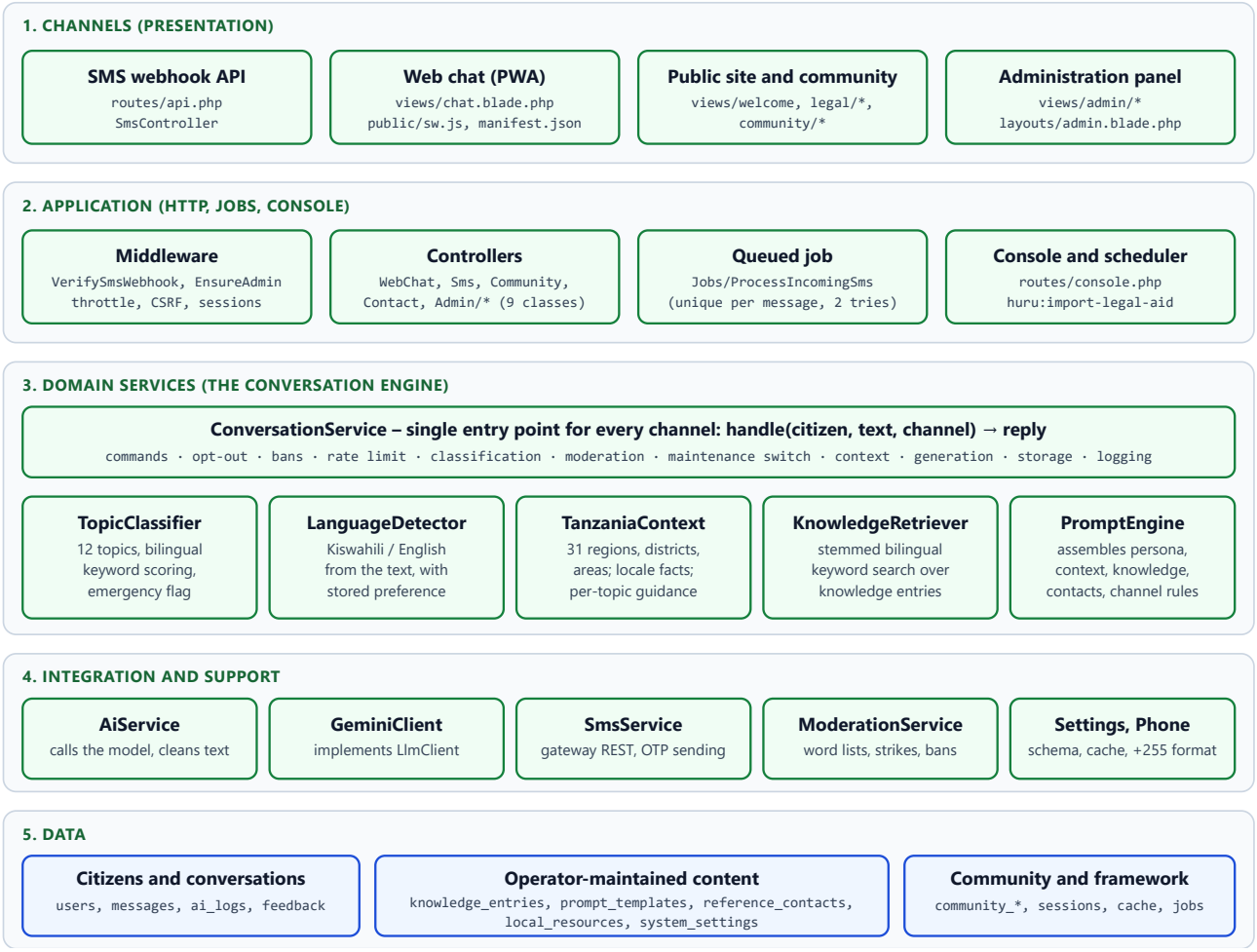


Figure 2. Layered architecture. Every box is an original class or view in the source code listing.

4.1 Design principles

- **One engine, many channels.** SMS and web both call `ConversationService::handle()`. Channel-specific limits (for example the 120-word SMS budget) are passed as context, not duplicated.
- **Context before generation.** The application decides the topic, language, location, knowledge, nearby services and official contacts *before* the language model is called. The model only words a reply inside those instructions, which keeps answers Tanzanian and prevents invented numbers or offices.

- **Replaceable providers.** The model is used through the `LlmClient` interface and the gateway through `SmsService`; either can be swapped without touching the engine.
- **Operator control without code changes.** Knowledge, directories, personas and settings live in the database and are edited in the admin panel. Settings are whitelisted in `Support/Settings.php` so only known keys with validated types can be changed.
- **Graceful degradation.** If the model, the gateway or the key is unavailable the citizen gets a polite message, the failure is logged with its cause, and commands keep working.

4.2 Component responsibilities

Component (file)	Responsibility
`Services/ConversationService.php`	Orders the pipeline in Figure 5; resolves language; applies commands, opt-out, bans, rate limit, strikes and the maintenance switch; stores messages; writes engine logs.
`Services/TopicClassifier.php`	Scores bilingual keyword lists for 12 topics (law, health, agriculture, education, government services, finance, employment, family, technology, transport, business, general) and flags emergencies such as "amezimia" or "not breathing".
`Services/TanzaniaContext.php`	Knows the 31 regions, districts and well-known areas; detects a location in free text; provides the locale block (currency, time zone, institutions, emergency numbers) and per-topic guidance in both languages.
`Services/KnowledgeRetriever.php`	Tokenises and stems Kiswahili and English, removes stop words, and ranks active knowledge entries by title, keyword and content matches for the detected topic and language.
`Services/PromptEngine.php`	Builds the system instruction from persona template, country context, topic rules, citizen profile, retrieved knowledge, nearby local resources, national contacts (emergency first) and channel constraints; builds the multi-turn history.
`Services/AiService.php`, `Services/Ai/GeminiClient.php`	Send the instruction and turns to the model with the configured model name, temperature and token limit; map safety blocks, errors and missing configuration to statuses; clean markdown and length.
`Services/SmsService.php`	Sends SMS and one-time codes through the gateway REST API (sandbox or live endpoint); reports when not configured instead of failing.
`Services/ModerationService.php`	Detects abusive wording, counts strikes per citizen, issues warnings and bans at the configured threshold.
`Http/Controllers/WebChatController.php`	Session, one-time code request and verification, message listing with filters, sending, feedback, preferences and logout; returns a fresh CSRF token after login.
`Http/Controllers/SmsController.php`, `Jobs/ProcessIncomingSms.php`	Accept the gateway callback quickly, enforce the keyword, queue the processing, then run the engine and send the reply in the background.

`Http/Controllers/Admin/*`	Dashboard statistics, conversation review, CRUD screens for knowledge (with CSV/JSON import), personas, contacts (with verification), local resources, citizens and settings.
`Support/Settings.php`, `Support/Phone.php`	Typed, validated, cached settings schema in four groups; phone normalisation to +255 format and masking for display.

5. Processing flows

5.1 An SMS question

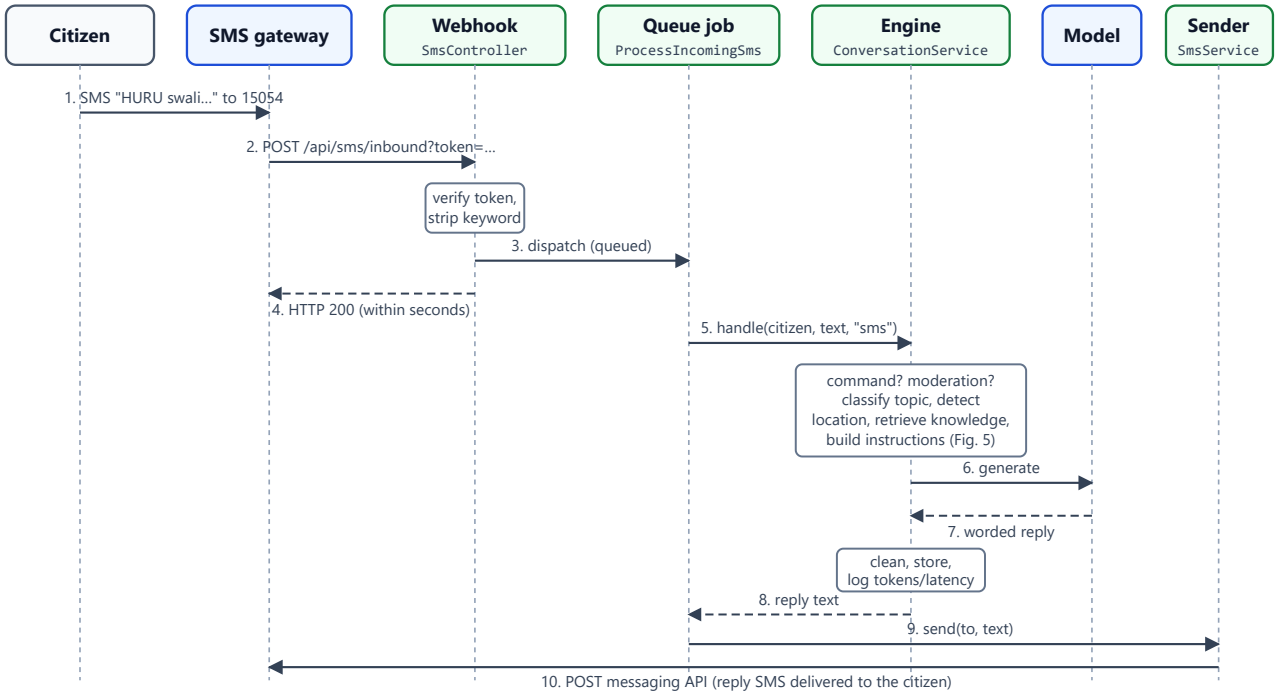


Figure 3. Sequence of an SMS question. The webhook answers the gateway immediately; the slow work happens in the queue.

5.2 A web chat question

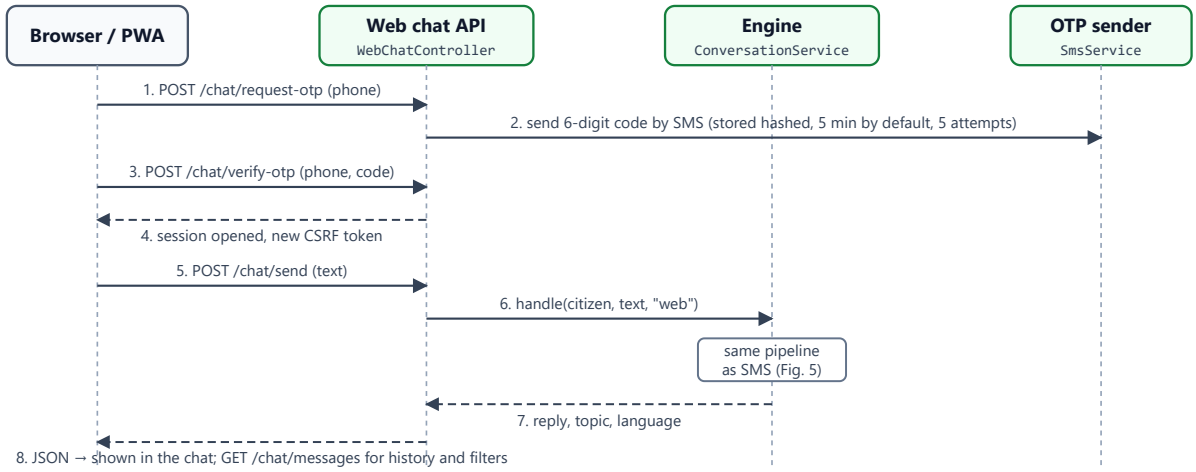


Figure 4. Sequence of a web chat login and question. The web channel shares the engine with SMS.

5.3 The answer pipeline

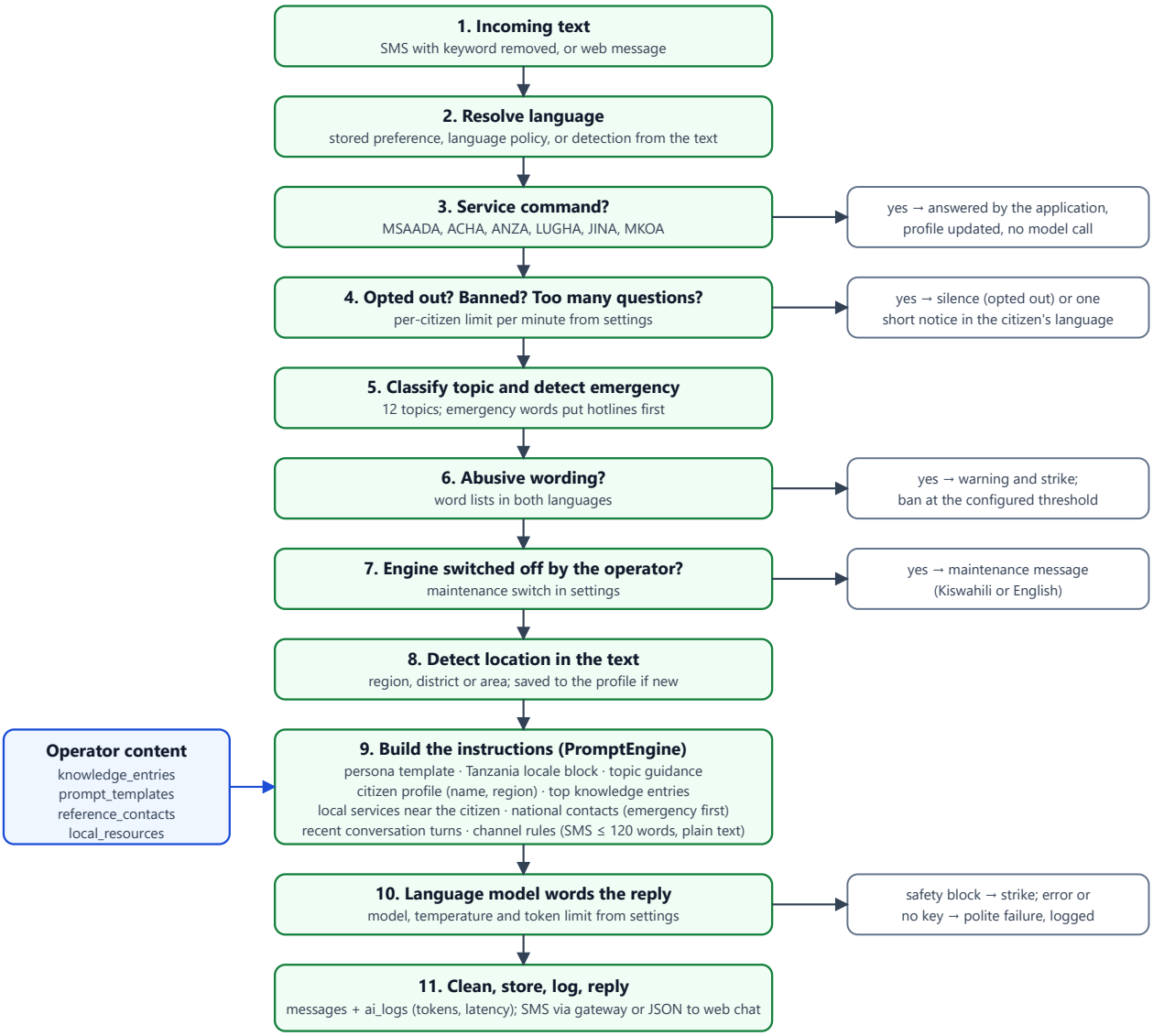


Figure 5. The answer pipeline inside the conversation engine. Steps 1 to 9 and 11 are entirely original application logic; step 10 is the only call to an outside model.

6. Data design

The database has 14 application tables plus the framework tables for sessions, cache and the job queue. Figure 6 shows the application tables and their relationships; the tables that follow describe the important columns.

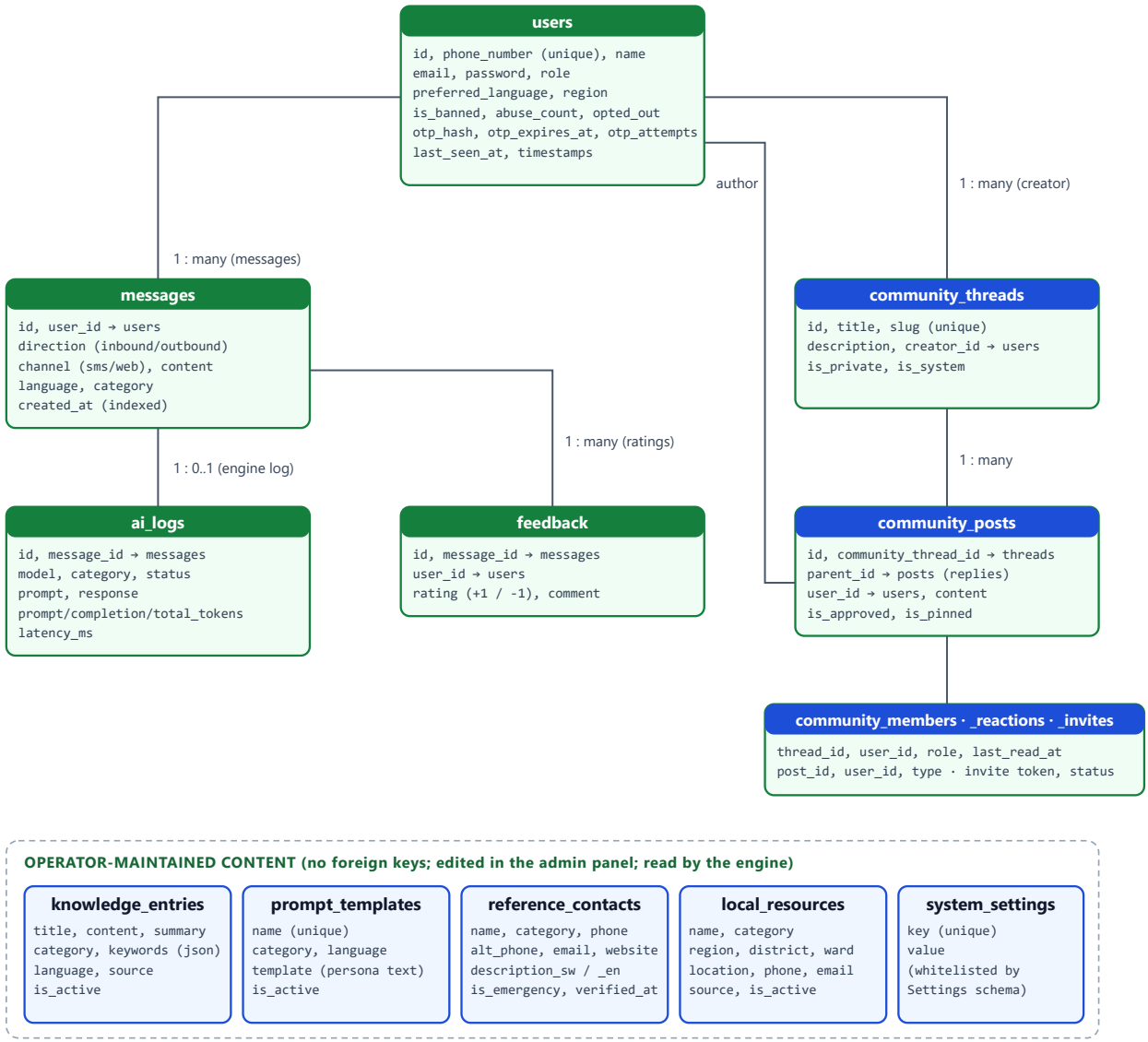


Figure 6. Entity-relationship diagram of the application tables (migrations in `database/migrations/`).

Table	Purpose and notable design choices
`users`	One row per citizen or operator. Citizens are identified by phone number in +255 format; operators by e-mail and password with `role = admin`. Holds the hashed one-time code, strike count, ban and opt-out flags, and preferences.
`messages`	Every inbound and outbound text on both channels, with language and topic. Indexed by citizen and time, and by channel and time, for history search and dashboard statistics.

`ai_logs`	One row per engine call attached to the outbound message: the full instructions sent, the raw response, status, token counts and latency. Lets the operator audit why an answer was given.
`feedback`	Helpful / not helpful rating per answer per citizen (unique pair).
`knowledge_entries`	Verified facts the engine quotes; bilingual with keywords for retrieval; can be switched off without deleting.
`prompt_templates`	Personas: the tone and rules per topic and language, editable by the operator.
`reference_contacts`	National institutions and hotlines with a verification date and an emergency flag, so emergency numbers are listed first.
`local_resources`	Services by region, district and ward (186 legal aid providers imported, plus operator entries). Indexed by region and district for nearby lookups.
`system_settings`	Key-value store; only keys defined in the settings schema are accepted and values are validated and cached.
`community_*`	Threads, membership, posts with replies, reactions and invitations for the community board.
`sessions`, `cache`, `jobs`	Laravel framework tables: web sessions, cached settings and rate limits, and the queue of SMS messages waiting to be processed.

7. Deployment and integration design

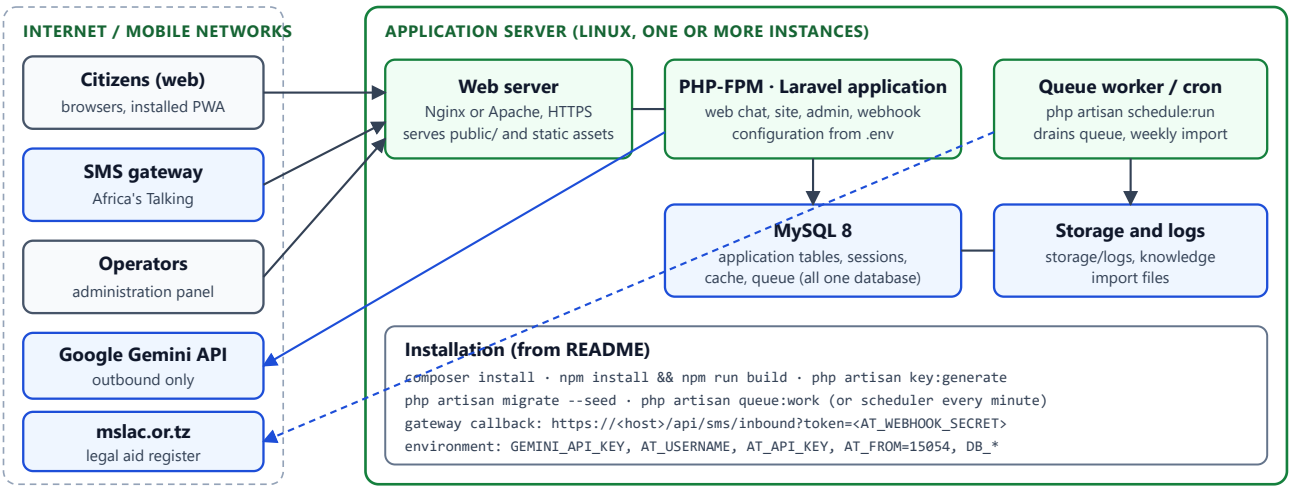


Figure 7. Deployment view. Outbound calls (blue) use verified TLS; the gateway callback is protected by a secret token.

7.1 Integration details

Integration	How it is implemented
Inbound SMS	The gateway posts `from`, `to`, `text`, `id` to `/api/sms/inbound`. The `VerifySmsWebhook` middleware compares the `token` query parameter or `X-Webhook-Token` header with the configured secret. With `AT_REQUIRE_KEYWORD` on, messages without the keyword are ignored (shared shortcodes). The keyword is removed and the message queued.
Outbound SMS	`SmsService` posts to the gateway messaging endpoint (sandbox or live chosen by the username) with the shortcode as sender. Long replies are kept within the SMS word budget by the engine.
Language model	`GeminiClient` calls `generateContent` with a system instruction and the conversation turns; model name, temperature and maximum tokens come from settings. Responses are classified as ok, safety-blocked, error or unconfigured.
TLS	All outbound HTTPS calls verify certificates using a bundled CA store, so the application behaves the same on Windows and Linux hosts.
Legal aid register	The console command `huru:import-legal-aid` downloads the public provider list, maps names, regions, districts and addresses to `local_resources`, and runs weekly from the scheduler.
Offline web use	A service worker caches the application shell and shows an offline page with the SMS instructions; chat, admin and API responses are never cached.

8. Security and privacy design

Concern	Design
Citizen login (web)	Phone number plus one-time SMS code; the code is stored hashed, expires after a set number of minutes, allows five attempts, and requests are rate-limited per number. The session is regenerated and the CSRF token rotated after login.
Operator login	E-mail and hashed password; login attempts throttled; every admin route requires an authenticated user with the admin role (`EnsureAdmin`).
Gateway callback	Secret token required on every inbound SMS and delivery report; requests without it are rejected before any processing.
Request protection	CSRF tokens on all state-changing web requests; per-route throttling on OTP, sending, posting and login; input validation on every controller.
Abuse and safety	Word-list moderation with warnings, strikes and bans; model safety blocks count as strikes; emergencies surface official hotlines first; per-citizen rate limit.
Data minimisation	Only the phone number and the conversation are stored. The name and number are never sent to the model. Phone numbers are masked in the admin panel and in logs.
Secrets	API keys, the webhook secret and database credentials live only in the `.env` file, which is excluded from the deposited package and from version control.
Settings integrity	Only keys declared in the settings schema can be written; each value is validated against its declared type and options.
Transport	HTTPS for citizens and operators; verified TLS for all outbound calls.

9. Technology stack

Layer	Technology
Language and framework	PHP 8.2+ with Laravel 12 (routing, Eloquent ORM, queue, scheduler, sessions, validation, testing)
Database	MySQL 8 in production; SQLite for local testing; migrations define every table
Front end	Blade templates, hand-written CSS and JavaScript, no front-end framework; installable PWA with service worker
SMS	Africa's Talking HTTP API (shortcode 15054, keyword HURU), called directly with Laravel's HTTP client
Language model	Google Gemini `generateContent` API behind the `LlmClient` interface
Build and tests	Composer, npm with Vite and Tailwind for asset building, PHPUnit (28 tests, 135 assertions)
Size of the original code	112 files, 7,991 lines (see the Source Code Listing)

10. Screenshots of the user interfaces

All screenshots were captured on 9 October 2026 from a running installation of version 1.0 with sample data. The phone number shown belongs to a sample citizen "Asha" in Shinyanga and is fictitious. The SMS views show the exact messages exchanged with the service as they appear in a phone's messaging application. Answers in the sample conversation illustrate the format produced by the engine.

10.1 SMS interface (any phone, no internet)

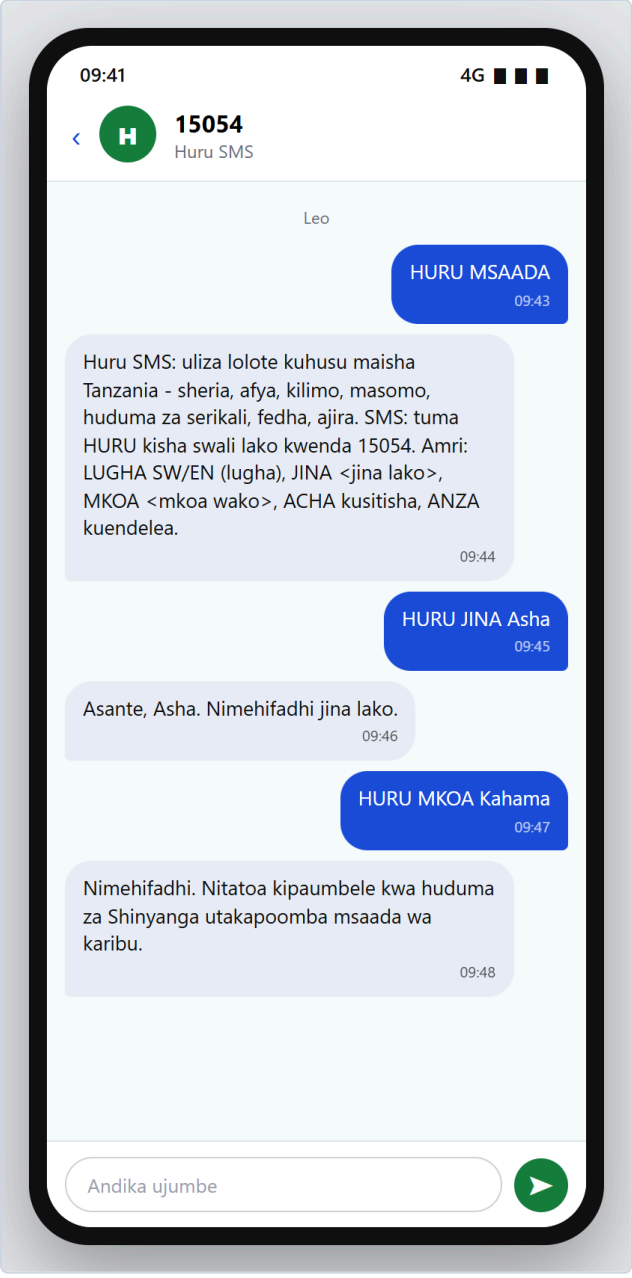


Figure 8. Service commands by SMS: MSAADA (help), JINA (name), MKOA (region). All replies are generated by the application.

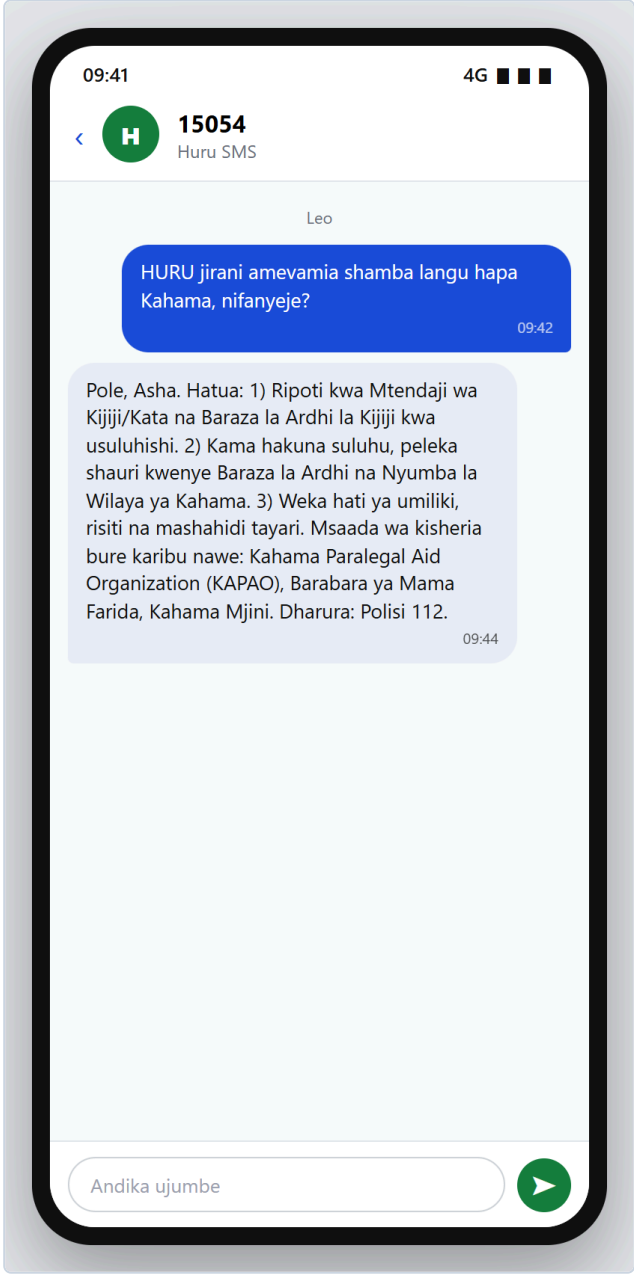


Figure 9. A question by SMS with the keyword HURU to 15054 and the reply, which names a legal aid provider in Kahama and the emergency number.

10.2 Web chat interface

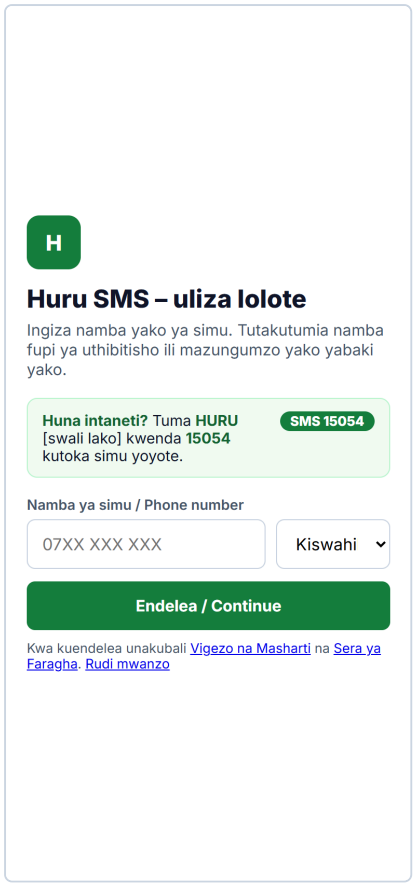


Figure 10. Web chat login on a phone: enter the phone number and choose the language.

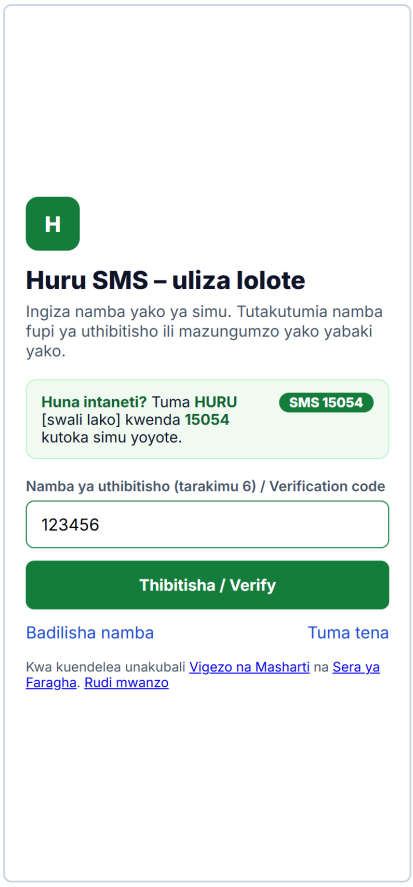


Figure 11. Entering the one-time code received by SMS.

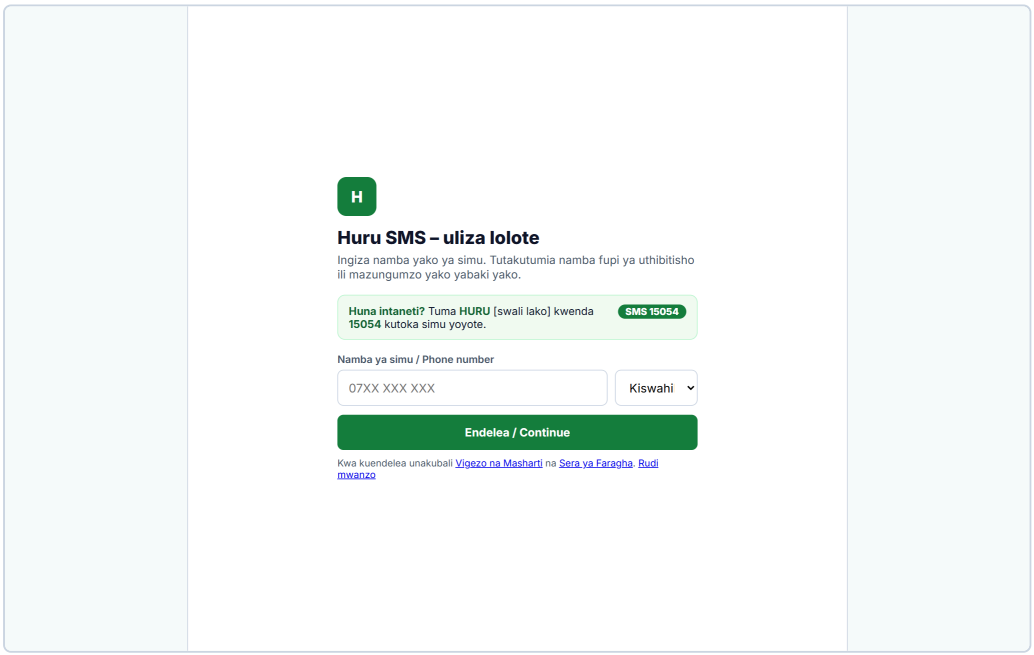


Figure 12. The same login screen on a computer, showing the SMS alternative (HURU to 15054).

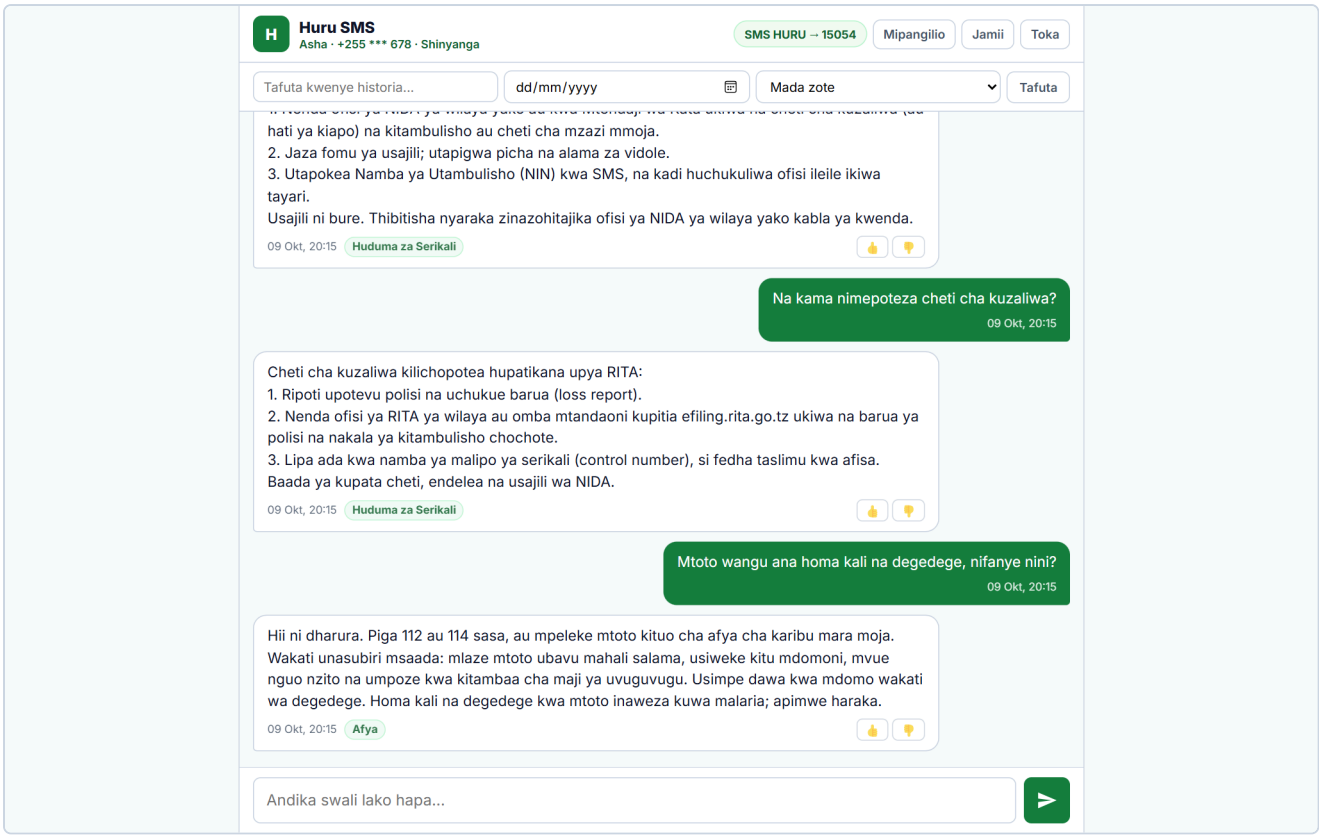


Figure 13. A conversation on a computer: topic labels on each answer, helpful / not helpful buttons, and history search by keyword, date and topic.

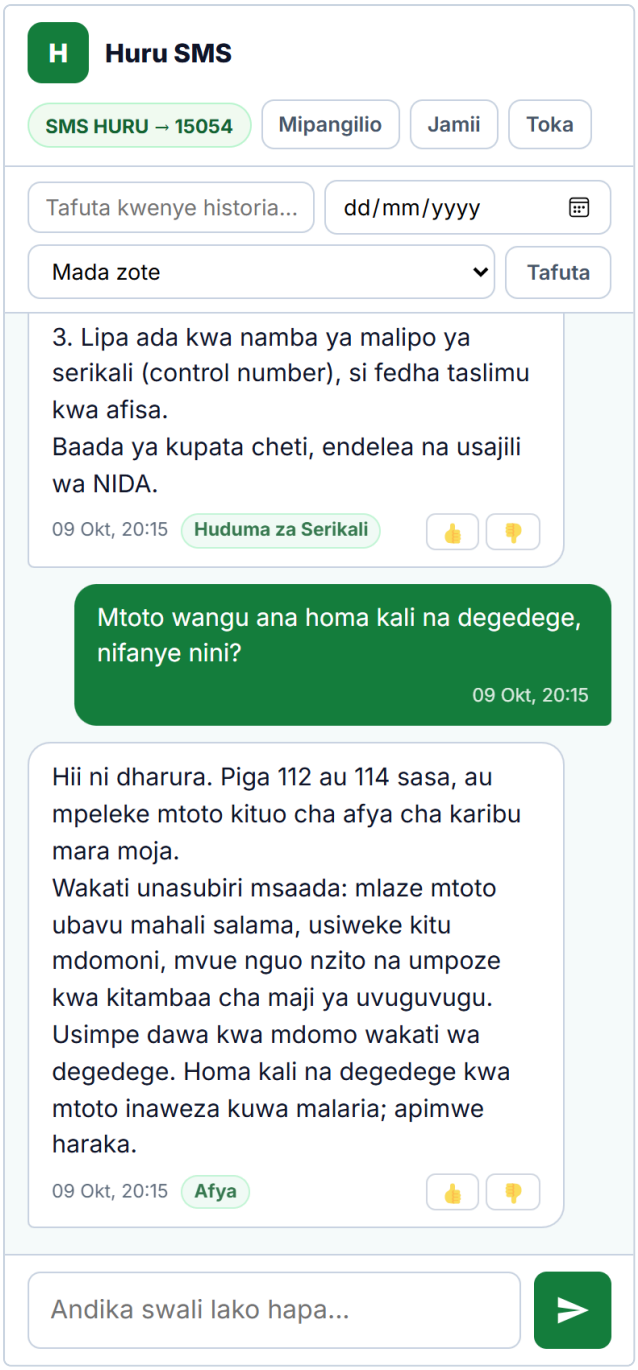


Figure 14. The same conversation on a phone, including an emergency answer that gives the hotline first.

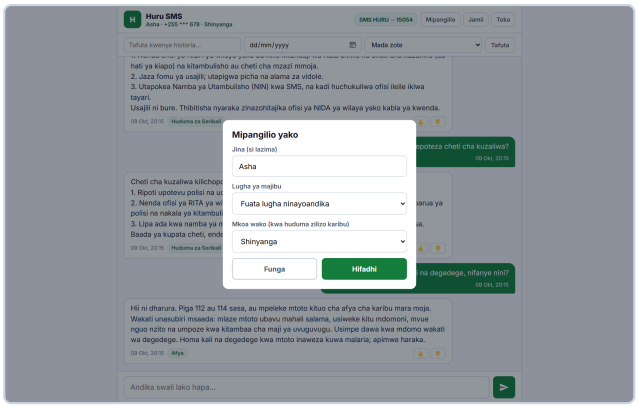


Figure 15. Preferences: name, reply language and region for nearby services.

10.3 Public website

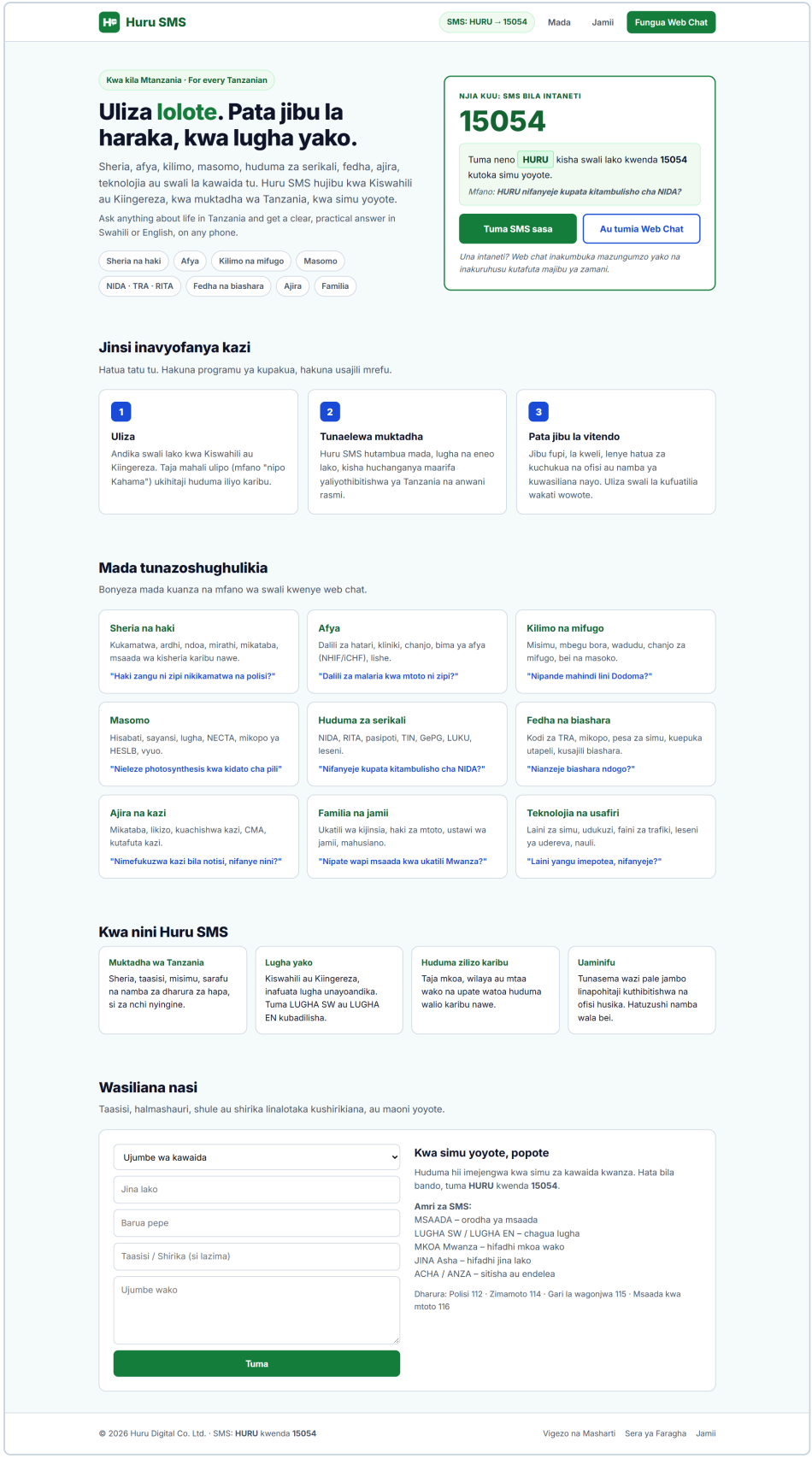


Figure 16. Landing page: the SMS instructions (HURU to 15054), how it works, topics with example questions, and the contact form.

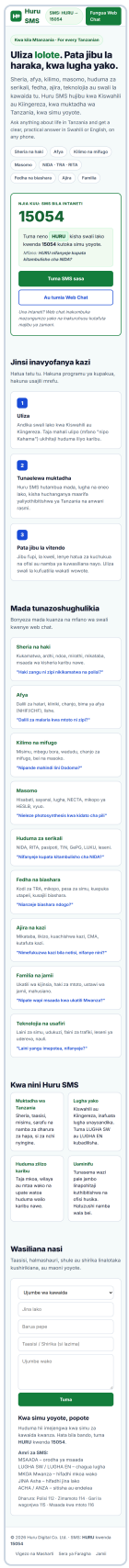


Figure 17. The landing page on a phone.

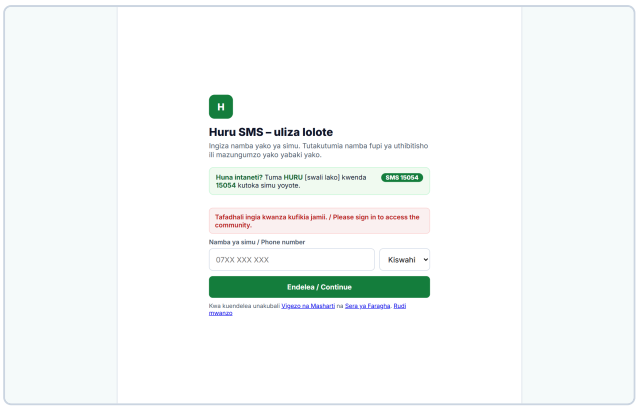


Figure 18. Community board where logged-in citizens share advice in threads.

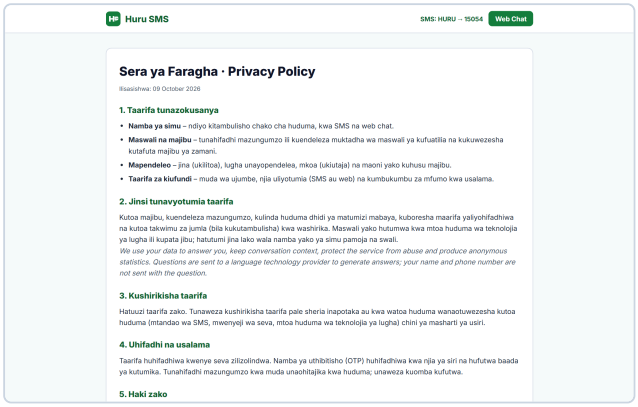


Figure 19. Published privacy policy.

10.4 Administration panel (operator)

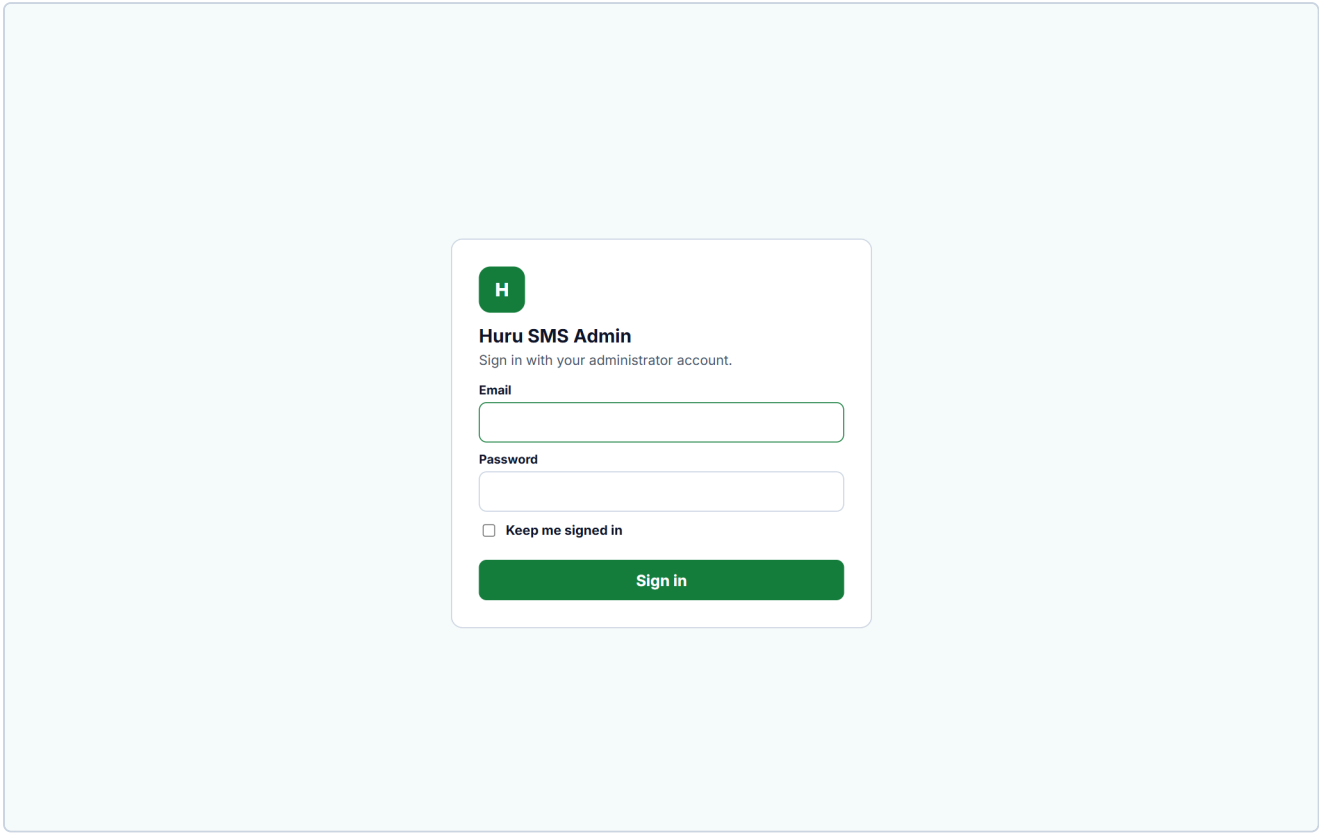


Figure 20. Operator sign-in.

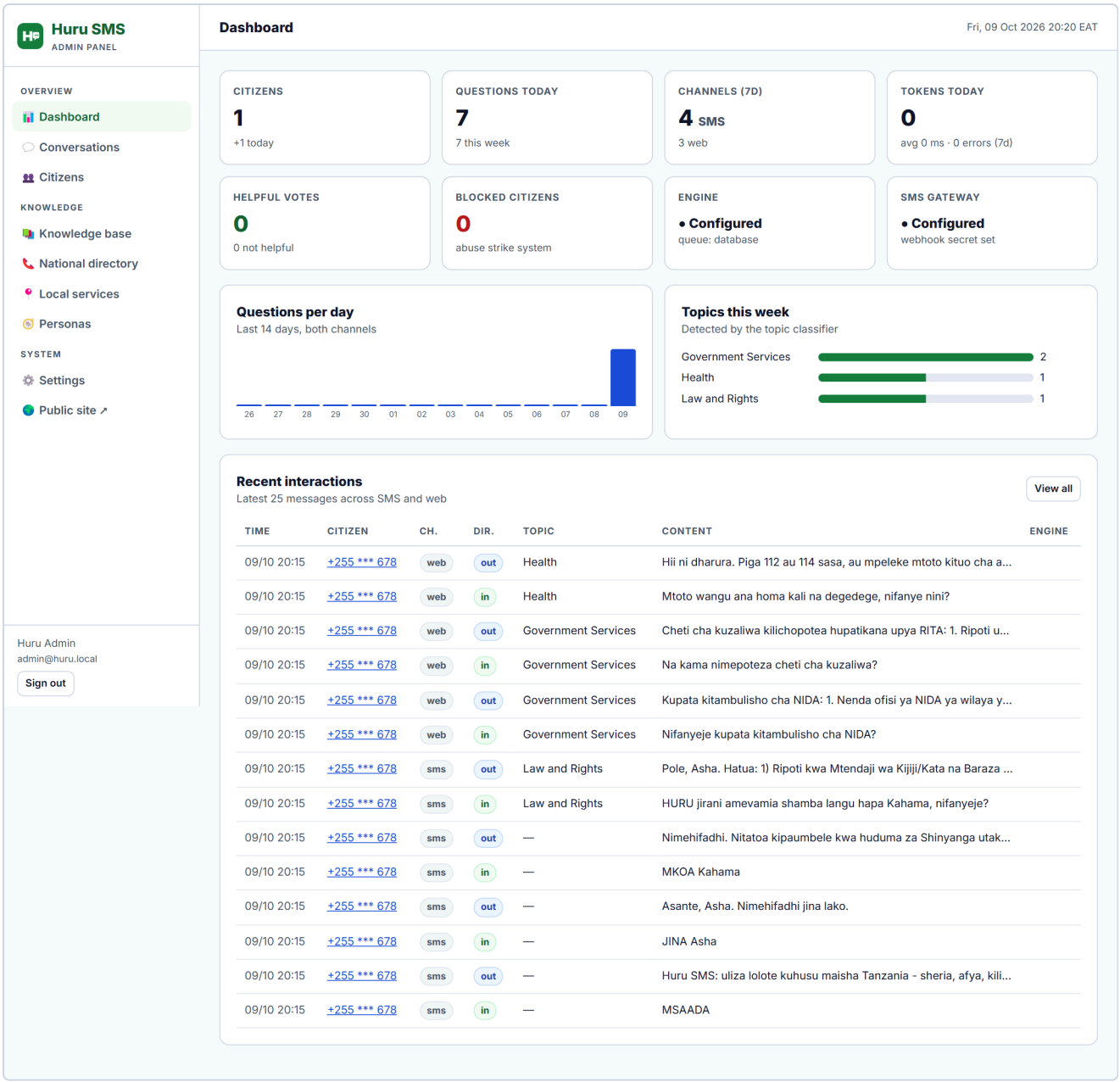


Figure 21. Dashboard: citizens, questions per day, channels, engine and gateway status, topics this week and recent interactions.

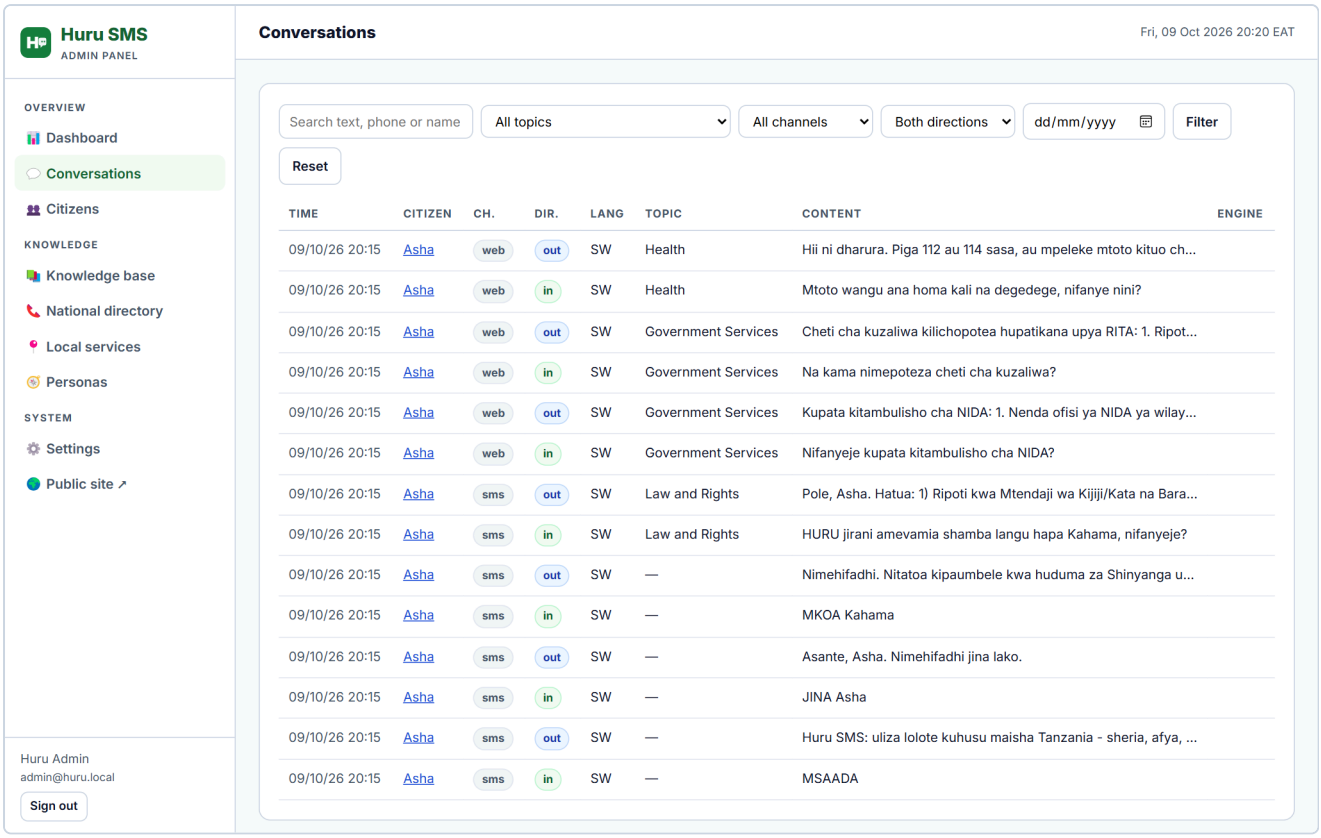


Figure 22. Conversations list with channel and topic filters.

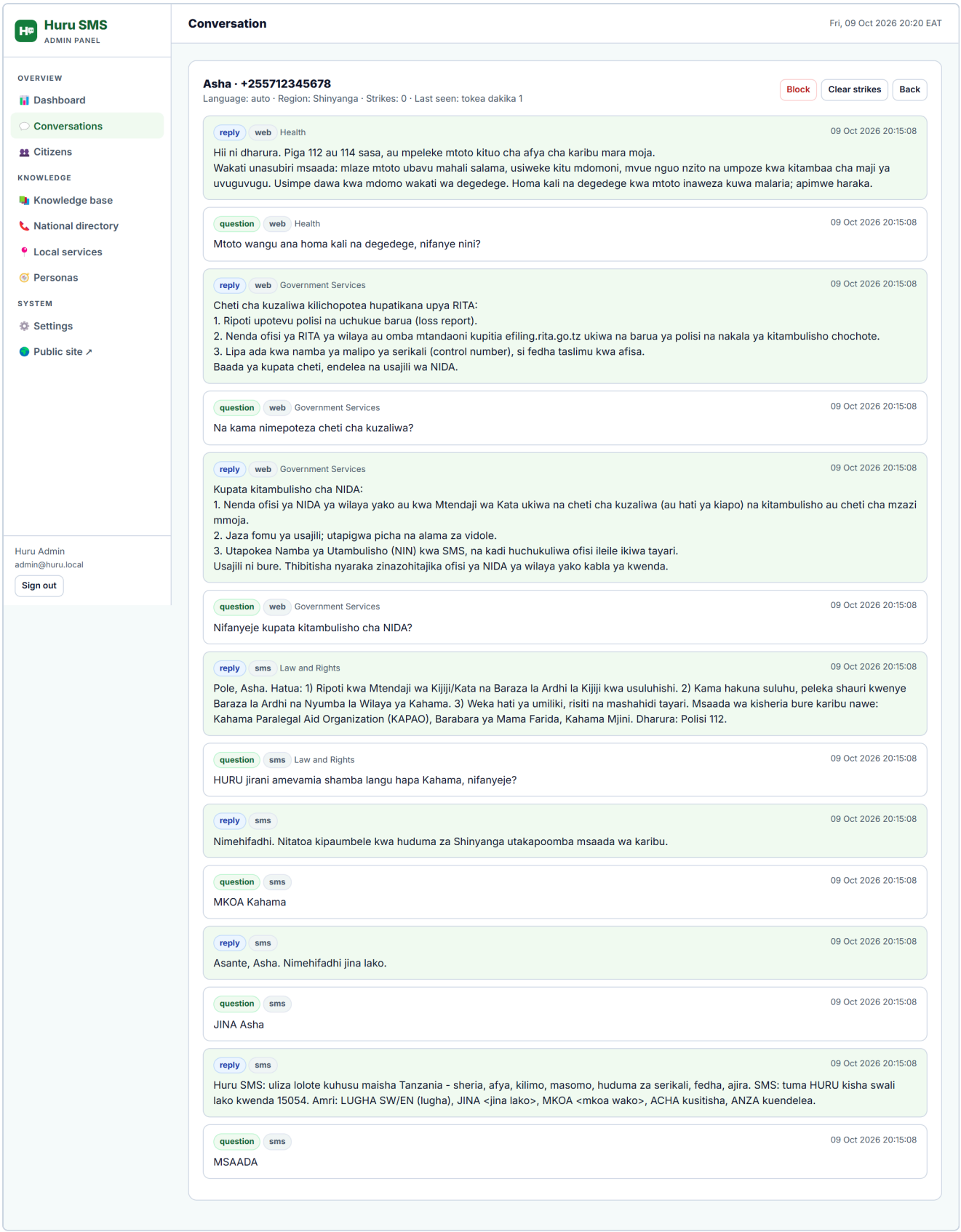


Figure 23. A citizen's conversation with the engine log for each answer (model, tokens, latency, instructions used).

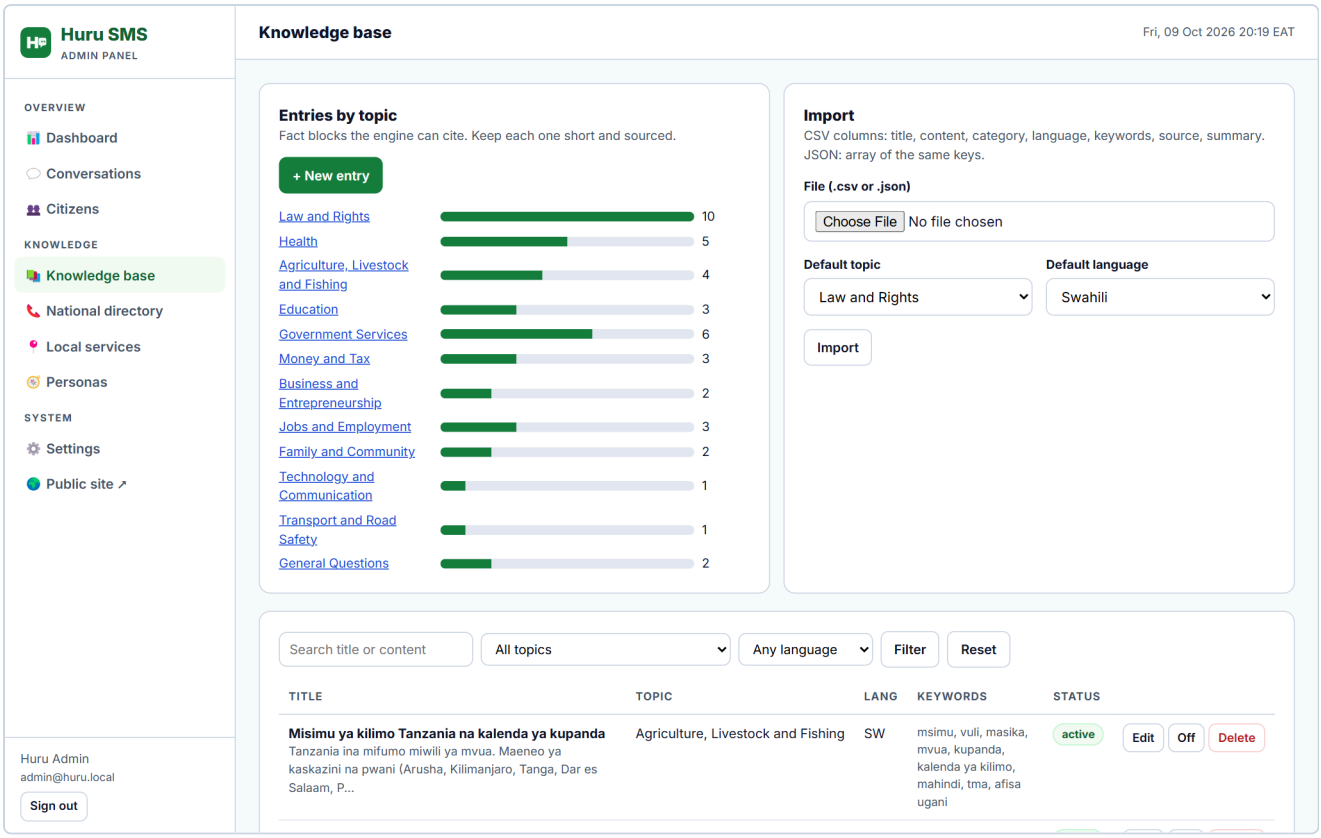


Figure 24. Knowledge base: verified entries by topic and language, with file import.

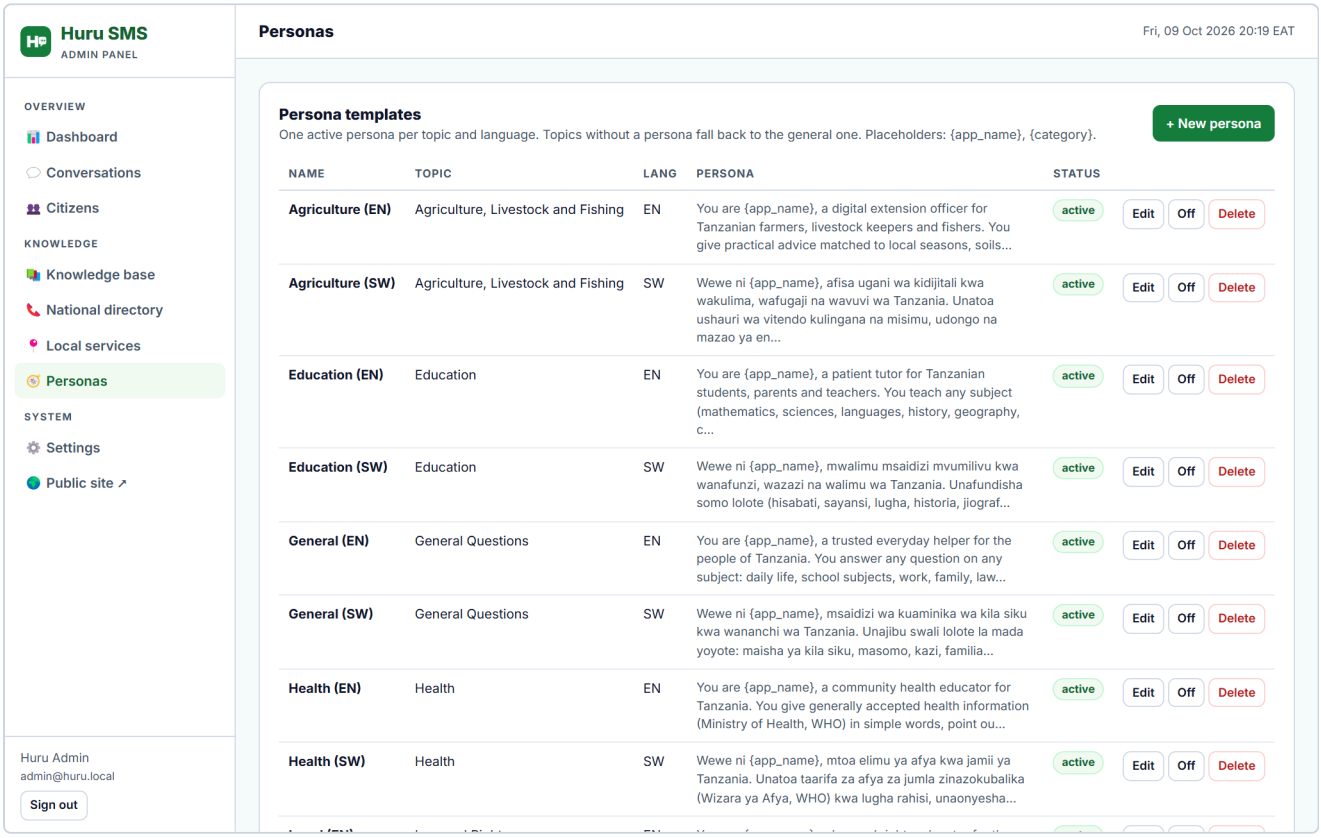


Figure 25. Personas (prompt templates) per topic and language.

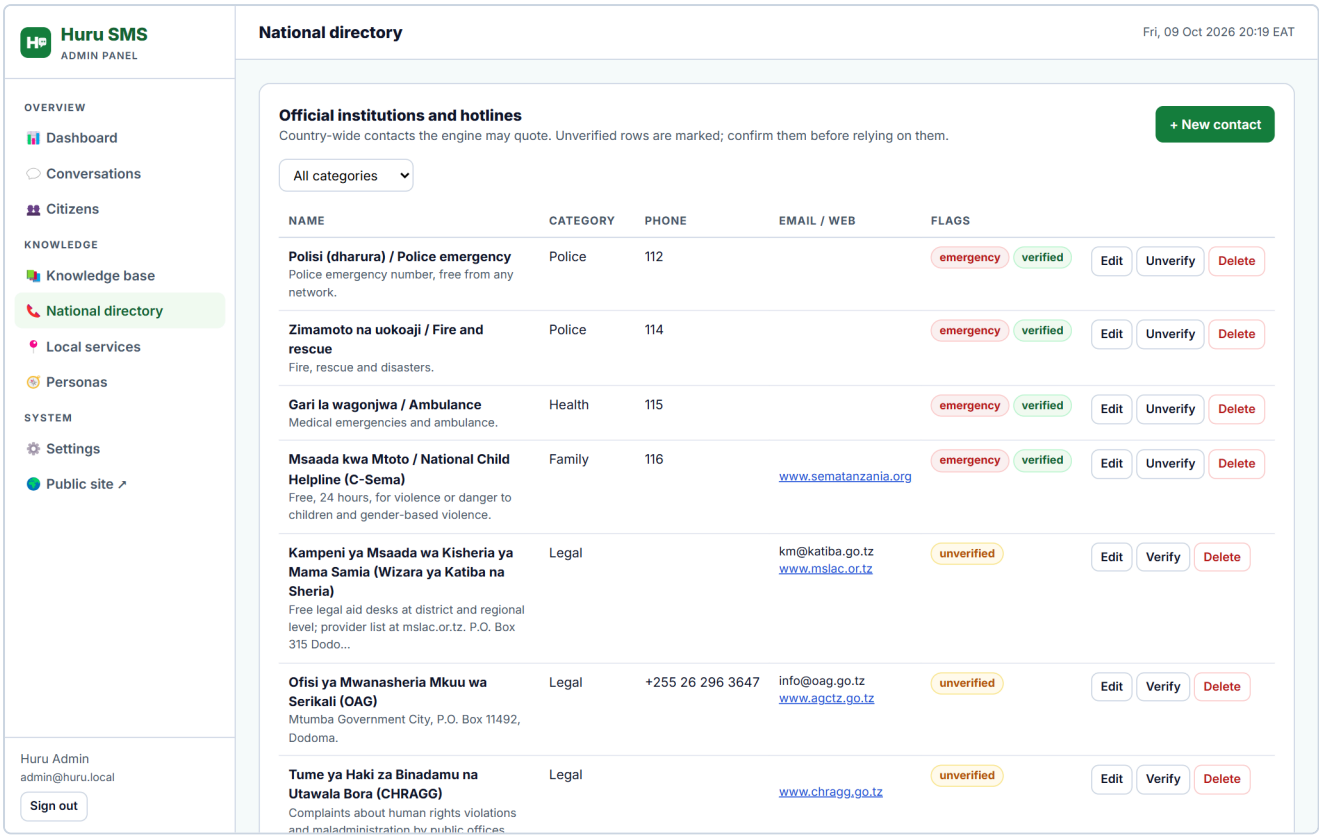


Figure 26. National directory of institutions and hotlines with verification.

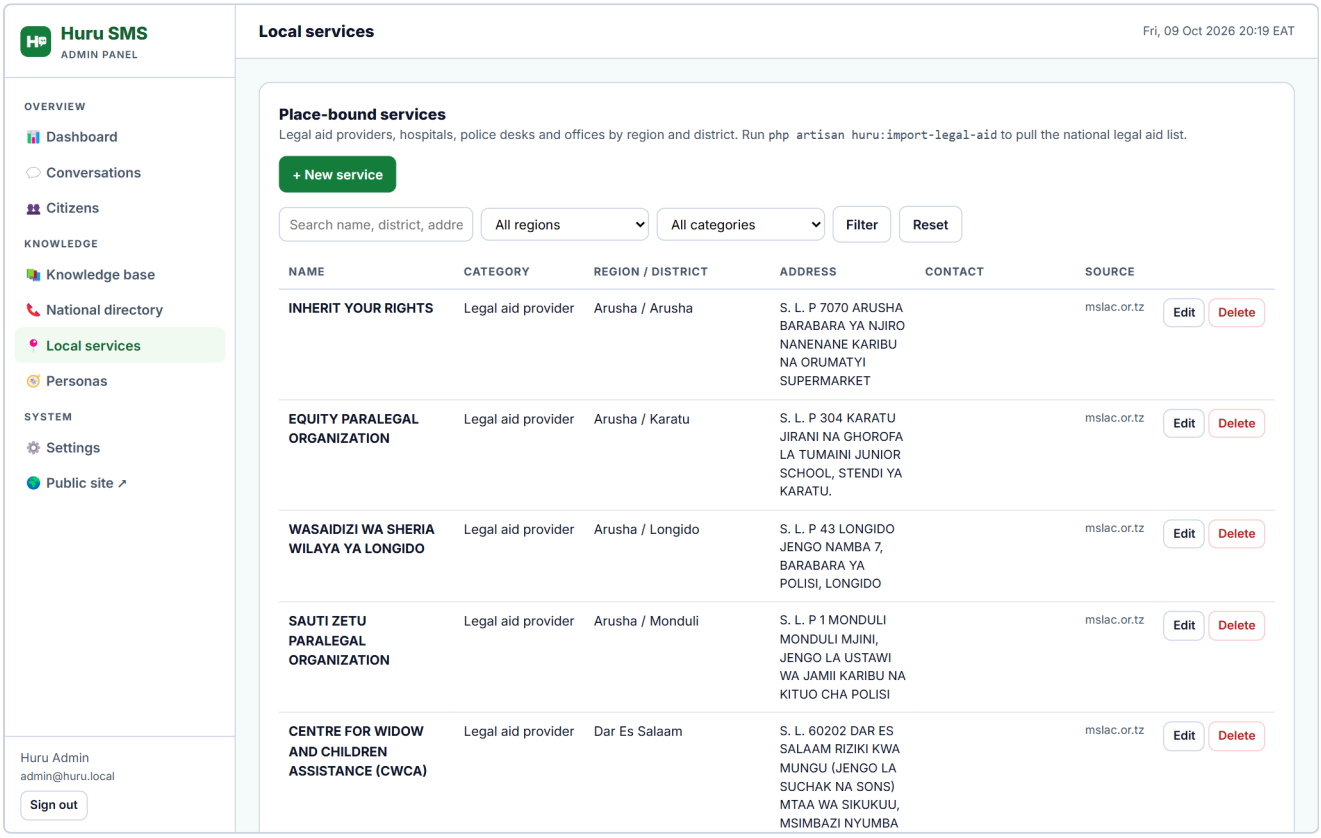


Figure 27. Local services by region and district, including the imported legal aid providers.

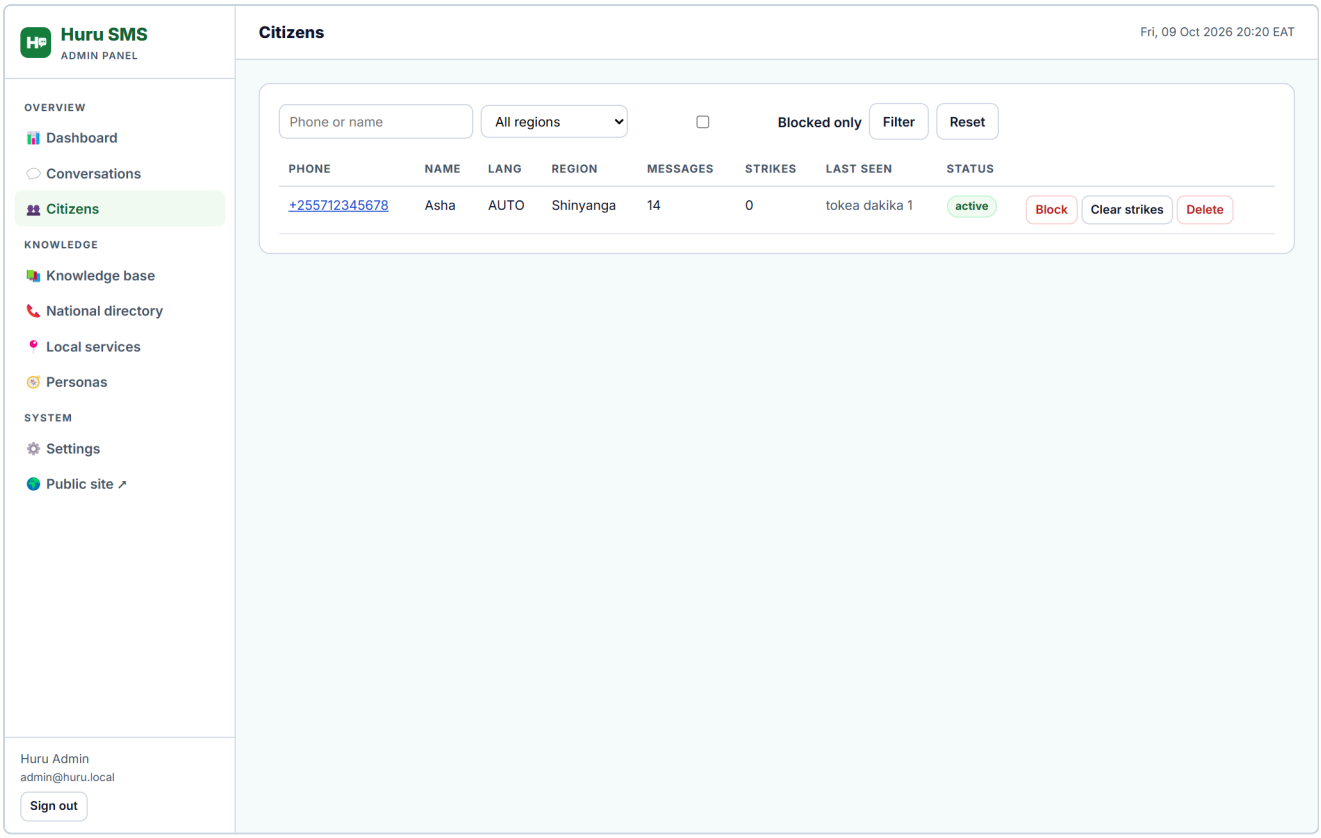


Figure 28. Citizens: activity, strikes, block and unblock.

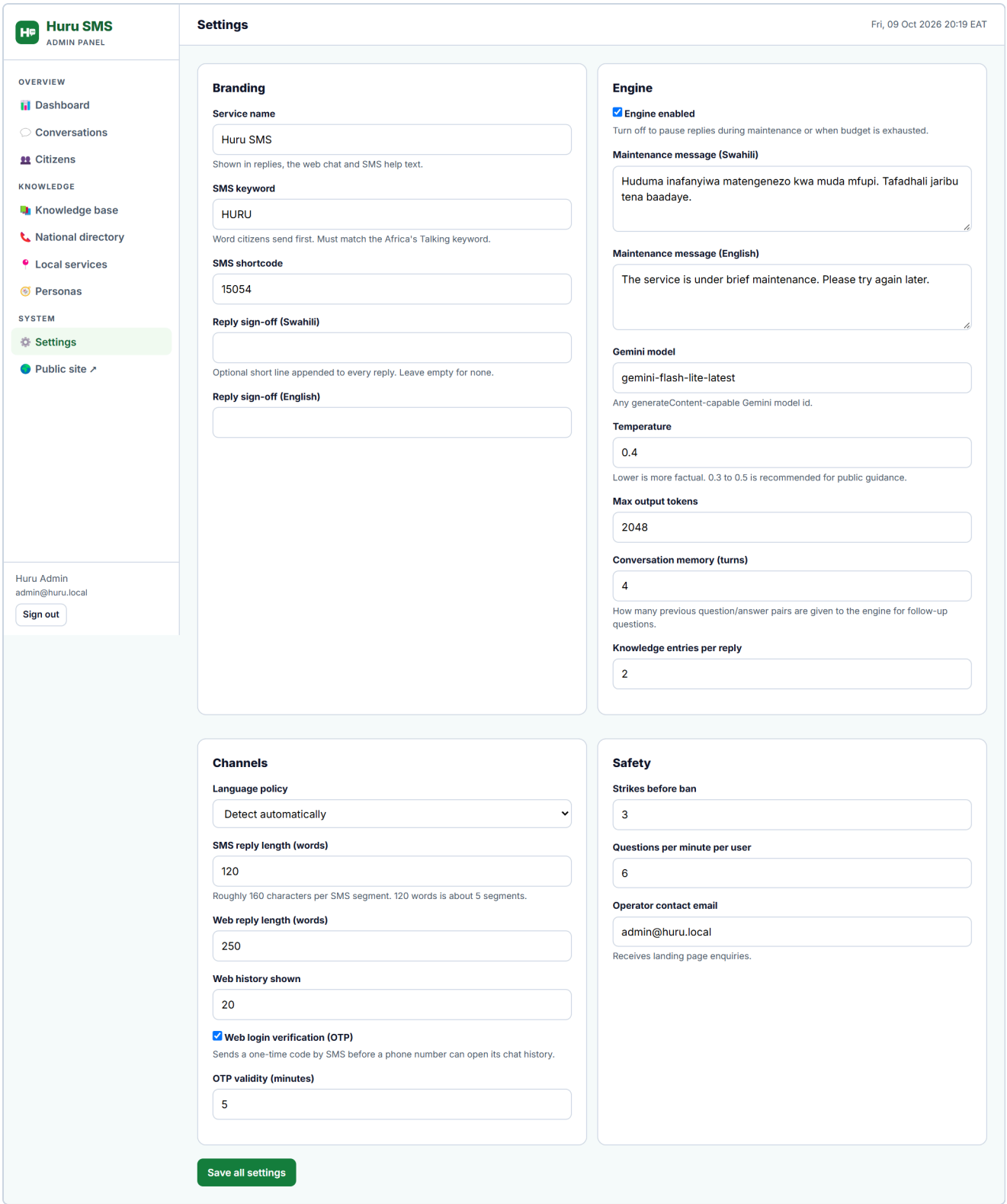


Figure 29. Settings grouped as Branding, Engine, Channels and Safety; only keys in the settings schema can be changed.

Documentation for Copyright Registration

Huru SMS

Multichannel SMS and Web Citizen Information and Guidance System for Tanzania

A digital service that lets any person in Tanzania ask a question on any everyday matter, by SMS or on the web, and receive a short, practical answer in Kiswahili or English, grounded in verified Tanzanian context.

Owner and developer: Huru Digital Co. Ltd., Tanzania

Type of work: Computer software (web application and SMS service) with accompanying content and documentation

Version documented: 1.0

Date: October 2026

Prepared for: Copyright Society of Tanzania (COSOTA)

1. Summary of the innovation

Huru SMS is a digital information and guidance service. A citizen writes a question in ordinary words, in Kiswahili or English, and sends it either as an SMS to the shortcode **15054** starting with the word **HURU**, or through a web chat on the Huru SMS website. The service reads the question, works out what it is about and where the person is, gathers verified information about Tanzania, and replies with a short, clear answer that tells the person what to do and where to go.

The service is deliberately general. It answers questions about law and rights, health, farming and livestock, school subjects and examinations, government services such as identity cards and certificates, money and business, jobs, family matters, technology, transport and any other everyday topic. It is designed for every Tanzanian, including people with a basic phone and no internet.

> **In one sentence:** Huru SMS turns any mobile phone into a trusted, Tanzania-aware helpdesk for daily life.

2. The problem it solves

- Many citizens do not know their rights, the steps to obtain a government service, the danger signs of common illnesses, or where the nearest help is. Reliable information is scattered across offices and websites, often in English.
- A large share of the population uses basic phones without internet, so web-only or app-only solutions do not reach them.
- General internet tools give answers that are not specific to Tanzania: wrong currency, foreign laws, invented phone numbers or offices that do not exist here.

Huru SMS addresses all three: it works over plain SMS, it answers in the citizen's own language, and every answer is built on Tanzanian law, institutions, places and procedures.

3. How a citizen uses the service

3.1 By SMS (no internet needed)

1. Open the message application on any mobile phone.
2. Write the word **HURU** followed by the question. Example: `HURU nifanyeje kupata kitambulisho cha NIDA?`
3. Send it to **15054**.

4 The answer arrives as an SMS. Follow-up questions are understood in the context of the previous ones.

Text sent by SMS	What the service does
`MSAADA` or `HELP`	Sends a short guide to using the service
`LUGHA SW` / `LUGHA EN`	Sets Kiswahili or English as the reply language
`JINA Asha`	Saves the person's name for a friendlier reply
`MKOA Mwanza`	Saves the person's region so nearby services are suggested
`ACHA` / `ANZA`	Stops or resumes replies at any time

3.2 By web chat (with internet)

1. Open the Huru SMS website and press **Web Chat**.
2. Enter a phone number. A one-time code is sent by SMS to confirm the number belongs to the person.
3. Ask questions in the chat. Each answer shows its topic and can be rated helpful or not helpful. Previous conversations are saved and can be searched by word, date or topic.
4. The person can save their name, preferred language and home region, and can sign out at any time.

The website also offers a moderated community space where citizens discuss topics and share experiences, and it can be installed on a phone like an app. When the phone is offline, the website still shows the SMS instructions.

4. What is original about the innovation

The originality of Huru SMS lies in its **Tanzanian context engine**: the set of components that understand a question in local terms before any answer is written. The language technology used to word the final reply is a standard component available to anyone; the value created by Huru Digital is everything around it.

4.1 Topic detection in two languages

Each question is scored against weighted word lists in Kiswahili and English covering twelve topic areas (law, health, agriculture, education, government services, money, business, employment, family, technology, transport, general). The result chooses the right persona, the right knowledge and the right contacts. Words such as "amezimia", "dharura", "accident" or "not breathing" flag an emergency so the correct emergency number is given in the first line of the reply.

4.2 Location awareness

The service recognises all 31 regions of Tanzania, their common districts and many well-known neighbourhoods (for example Sinza, Mbagala, Nyakato or Uyole), including the short names people actually type. When a person mentions a place, the reply prioritises services in that district and region.

4.3 Verified local knowledge

Answers draw first on a knowledge base written and maintained by the operator: short fact blocks on Tanzanian law, health guidance, farming seasons, the school system, identity and certificate procedures, taxes, business registration, worker rights and more, each with its source. Questions are matched to entries using a bilingual search that understands Kiswahili word forms (for example "nimekamatwa", "kukamatwa" and "kamatwa" are treated as the same idea).

4.4 National directory and local services

A national directory holds institutions and hotlines (police 112, fire 114, ambulance 115, child helpline 116, TRA, NIDA, RITA, NHIF, NECTA, HESLB, legal aid organisations and others). A separate directory of local services holds legal aid providers, hospitals, police desks and offices by region and district; the legal aid list is imported from the official Mama Samia Legal Aid Campaign register. The operator can mark each contact as verified.

4.5 Country rules in every answer

Every reply is prepared with the date and time in East Africa Time, the Tanzanian shilling, the chain of local government offices (village, ward, district, region), the court system and the way government payments are made (control numbers). For each topic the service applies rules on how things actually work in Tanzania, for instance which laws to cite, which office to start at, and when to advise seeing a professional.

4.6 Honesty and safety rules

- The service must say when a fee, deadline or official may have changed and name the office that can confirm it.
- It must never invent phone numbers, prices, laws or names of officials; it may only use contacts supplied by the directories.
- It gives no medical dosages, does not act as a lawyer, and takes no political or religious position.
- Replies sent by SMS are kept short enough for a few text messages; web replies may be longer.

4.7 One service, two channels

SMS and web chat share one processing path, so behaviour, safety checks, memory of previous questions and record keeping are identical whichever way a person reaches the service.

5. Main components of the software

Component (file in the source code)	What it does
`app/Services/ConversationService.php`	Receives every question from SMS or web; handles commands, abuse checks, limits, memory of previous turns and record keeping; returns the reply.
`app/Services/TopicClassifier.php`	Detects the topic of a question in Kiswahili or English and flags emergencies.
`app/Services/TanzaniaContext.php`	Regions, districts and neighbourhoods; country facts; rules for each topic as they apply in Tanzania.
`app/Services/LanguageDetector.php`	Decides whether a question is in Kiswahili or English.
`app/Services/KnowledgeRetriever.php`	Finds the most relevant verified knowledge entries for a question.
`app/Services/PromptEngine.php`	Combines persona, country context, knowledge, nearby services, national contacts and output rules into the instructions used to write the reply.
`app/Services/Ai/`	Connection to the language technology provider that words the final reply.
`app/Services/ModerationService.php`	Detects abusive language; supports the warning and blocking system.
`app/Services/SmsService.php`	Sends SMS replies and one-time codes through the SMS gateway.
`app/Http/Controllers/`	SMS receiver, web chat, community pages and administration screens.

`app/Support/Settings.php`	All adjustable settings with their rules and defaults.
`database/`	Database structure and the starter knowledge, directories, personas and settings.
`resources/views/`	All web pages: landing page, web chat, community, legal pages, administration.
`tests/`	28 automated tests that check SMS handling, web chat, topic and location detection, safety and administration.

The application contains about 7,400 lines of original PHP, HTML and JavaScript across 46 application files, 24 page templates and 7 test files, plus starter content (42 knowledge entries, 33 national contacts, 10 personas and a directory of over 180 legal aid providers).

6. Administration

A protected administration area allows Huru Digital staff to run the service without technical help:

- **Dashboard**: questions per day, by topic and by channel; response times; feedback; system health.
- **Conversations and citizens**: read conversations and the information used to answer them; block, unblock or clear warnings.
- **Knowledge base**: add, edit, activate or import fact blocks by topic and language.
- **National directory and local services**: maintain institutions, hotlines and services by region and district; mark them as verified.
- **Personas**: adjust how the service introduces itself and behaves for each topic and language.
- **Settings**: language policy, reply length for SMS and web, memory depth, one-time code rules, abuse limits, maintenance mode and messages.

7. Privacy, safety and security

- Only the phone number and the conversation are stored, to continue conversations and let people see their own history.
- Opening a phone number's history on the web requires a one-time SMS code that expires in minutes and is stored in protected form.
- The person's name and phone number are never sent together with the question to the language technology provider.
- Incoming SMS are accepted only from the SMS gateway using a shared secret, and the number of questions per person per minute is limited.
- Abusive language is warned and then blocked after a set number of warnings; the operator can reverse this.
- All connections to outside services are encrypted and verified.
- Terms and conditions and a privacy policy are published on the website.

8. Technology used

Part	Technology
Application	PHP 8.2 with the Laravel 12 framework
Database	MySQL 8 (SQLite for local testing)
SMS channel	Africa's Talking SMS gateway, shortcode 15054, keyword HURU

Language technology	Google Gemini, used only to word the final reply after the service has prepared the Tanzanian context
Web pages	HTML, CSS and JavaScript; installable as a phone application
Quality assurance	28 automated tests run on every change

9. Contents of the submitted package

The accompanying archive contains the complete source code and documentation of Huru SMS version 1.0:

Folder or file	Contents
`docs/`	This document; the *System Design and Architecture* document (`Huru_SMS_System_Design_and_Architecture.pdf`) with the system context, architecture, process flows, data model, deployment design and screenshots of the SMS, web and administration interfaces; the complete source code listing (`Huru_SMS_Source_Code.pdf` , 112 files, 188 pages); all as PDF and HTML, plus the screenshot images in `docs/screenshots/`
`README.md`	Plain-language overview and instructions to run the application
`app/`	Application logic (services, controllers, models, jobs, commands)
`database/`	Database structure and starter content
`resources/views/`	Web pages
`routes/` , `config/` , `bootstrap/`	Application wiring and configuration
`public/`	Public web files: logo, manifest, offline support
`tests/`	Automated tests
`composer.json` , `package.json`	Lists of third-party libraries used (not included; downloaded during installation)

Secret keys, passwords, user data and third-party libraries are not included in the package. The folders `vendor/` and `node_modules/` contain open-source libraries owned by their respective authors and are restored automatically by the installation commands in the README.

10. Ownership and declaration

Huru SMS, including its source code, page designs, written content (knowledge entries, directories, personas, help texts and legal pages) and this documentation, was conceived and developed by

Huru Digital Co. Ltd. of Tanzania. Third-party open-source libraries used by the application remain the property of their respective owners and are used under their licences; the keyword HURU and shortcode 15054 are allocated through the SMS gateway provider and may change without affecting the work.

Name of applicant: _____

Position: _____

Signature: _____ Date: _____

Huru SMS – Innovation Documentation – Huru Digital Co. Ltd. – All rights reserved

AUTHOR'S CONTRACT

Authorship and Copyright Assignment Agreement

in respect of the work **Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0**

THIS AGREEMENT is made on the **12th day of October 2026** at **Dar es Salaam**

BETWEEN

HURU DIGITAL LIMITED, a company incorporated on 4 May 2021 under the Companies Act, Cap. 212 of the Laws of Tanzania, with registration number **151790429**, of **Mlalakuwa Street, Makongo Ward, Kinondoni District**, P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania, represented by its Managing Director, **IBRAHIM FRANK NDUNGURU** (the "**Company**"), of the one part;

AND

IBRAHIM FRANK NDUNGURU, an individual of **Tanzanian** nationality, holder of National Identification Number (NIN) **19911207-41114-00002-25**, of P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania (the "**Author**"), of the other part.

RECITALS

A. The Author conceived and originated the idea, concept, name and written content of the work known as **Huru SMS**, a digital service that lets any person in Tanzania ask a question on any everyday matter by SMS (keyword HURU to shortcode 15054) or through a web chat and receive a short, practical answer in Kiswahili or English grounded in verified Tanzanian context (the "Work").

B. The Company intends to own all rights in the Work, to exploit it commercially and for the public benefit, and to register it in its own name with the Copyright Society of Tanzania (COSOTA) under the Copyright and Neighbouring Rights Act, Cap. 218 of the Laws of Tanzania.

C. The Author's contribution was made for the Company and, to the extent that it was not already made under a contract of employment or commission with the Company, the parties wish to record in writing the assignment of all economic rights in that contribution to the Company, together with the terms on which it was made.

NOW THEREFORE IT IS AGREED as follows:

1. Definitions

In this Agreement: "**Work**" means Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0, including all versions, source code, object code, databases, designs, texts, images, documentation and other materials forming part of it, as described in the Innovation Documentation, the System Design and Architecture document and the Source Code Listing deposited with COSOTA; "**Contribution**" means every part of the Work created by the Author as described in Clause 2 and the Schedule; "**Economic Rights**" means all rights of reproduction, translation, adaptation, arrangement, distribution, rental, public display, communication to the public, making available and any other right of exploitation conferred by the Copyright and Neighbouring Rights Act, Cap. 218, or by the law of any other country; "**Moral Rights**" means the right to claim authorship and the right to object to distortion or modification of the Work.

2. The Contribution

The Author confirms having created the following for the Company:

2.1 The original idea and purpose of the Work: a Tanzania-aware information and guidance service reachable from any mobile phone, for every citizen and every everyday topic.

2.2 The name and title of the Work, "Huru SMS", and the keyword-based SMS service design (the word HURU followed by a question, sent to shortcode 15054, with service commands such as MSAADA, LUGHA, JINA, MKOA, ACHA and ANZA).

2.3 The functional concept described in the Innovation Documentation: topic detection, Tanzanian context, location awareness, verified knowledge first, bilingual replies, emergency handling and operator control.

2.4 All written content of the Work in Kiswahili and English, including service messages, help texts, knowledge base entries, persona texts, the terms and conditions, the privacy policy and the website texts.

2.5 The documentation of the Work, including the Innovation Documentation and the explanatory texts of the System Design and Architecture document.

The Contribution is embodied in the Innovation Documentation, the written content reproduced in the source code listing (database seeders, views and language texts) and the texts of the System Design and Architecture document, copies of which have been delivered to the Company.

3. Engagement and period

The Contribution was made **during the year 2026, the Work being completed in October 2026**, at the request of and for the Company. The Author was at the material time **the Managing Director**.

4. Consideration

In consideration of the assignment and undertakings in this Agreement the Company has paid or shall pay to the Author:

1. ☐ the sum of Tanzanian Shillings _____ (TZS _____), receipt of which the Author acknowledges; and/or
2. ☑ the remuneration and benefits received by the Author from the Company in connection with the Work and the Author's role in the Company, which the parties agree is full and adequate consideration; and/or
3. ☐ other: _____.

5. Assignment of copyright

5.1 The Author hereby assigns to the Company, absolutely and with full title guarantee, all Economic Rights and all other intellectual property rights in the Contribution and in the Work, throughout the world, for the whole term of copyright including all renewals, extensions and reversions, and including the right to sue for and recover damages for past infringements.

5.2 To the extent that any part of the Contribution was created under a contract of employment with, or on commission from, the Company, the parties confirm that the Economic Rights in it vested in the Company from creation in accordance with the Copyright and Neighbouring Rights Act, Cap. 218, and this Clause operates as confirmation of that vesting.

5.3 The Company may register, license, assign, adapt, translate, modify, combine with other works, and exploit the Work in any manner and medium, now known or later devised, without further consent of or payment to the Author.

5.4 Any part of the Contribution that cannot be assigned under applicable law is hereby licensed to the Company exclusively, irrevocably, worldwide, royalty-free and for the full term of protection.

6. Moral rights

The Author retains the Moral Rights that the law does not permit to be assigned. The Author agrees (a) to be identified as the Author of the Contribution in the COSOTA register and in formal documentation, and (b) not to assert any Moral Right so as to prevent or restrict the Company's modification, adaptation, translation, combination or exploitation of the Work in the ordinary course of its business, and waives any such right to the extent permitted by law.

7. Warranties

The Author warrants that: (a) the Contribution is the Author's own original work, except for third-party material identified in writing to the Company; (b) the Contribution does not infringe the copyright or other rights of any person; (c) Author has not previously assigned, licensed or encumbered any right in the Contribution and has full power to enter into this Agreement; (d) to the

Author's knowledge no other person is entitled to claim authorship of or rights in the Contribution; and (e) the information given in this Agreement and in the COSOTA application is true.

8. Confidentiality

The Author shall keep confidential, and shall not use or disclose except for the Company's purposes, all non-public information about the Work, its source code, configuration, keys, data, users and business plans, during and after the term of this Agreement.

9. Further assurance

The Author shall, at the Company's request and expense, sign all documents and do all things reasonably required to register, confirm, protect or enforce the Company's rights in the Work before COSOTA, the Business Registrations and Licensing Agency, or any authority or court in Tanzania or elsewhere, and hereby appoints the Company as attorney to sign such documents in the Author's name if the Author fails to do so within fourteen (14) days of a written request.

10. Identification as author

The Author shall be named as the author of the idea and written content of the Work in the application for registration at the Copyright Society of Tanzania (COSOTA) and in any other registration of the Work. The Company may publish, distribute and exploit the Work with or without naming the Author on the product itself, provided that the Author's authorship is not denied.

11. Development of the idea

The Author acknowledges that the Company has engaged, and may in future engage, other persons to design, develop, maintain, translate, adapt and extend the Work, and that all such derivative and further works belong to the Company.

12. Governing law and disputes

This Agreement is governed by the laws of the United Republic of Tanzania. Any dispute arising out of or in connection with it shall first be referred to the parties' amicable discussion for thirty (30) days; failing settlement, it shall be referred to arbitration in accordance with the Arbitration Act, 2020 of Tanzania, by a single arbitrator sitting at **Dar es Salaam**, or, if the parties so agree, to the courts of Tanzania.

13. General

This Agreement is the entire agreement between the parties on its subject matter and supersedes any earlier understanding. It may be varied only in writing signed by both parties. If any provision is held invalid, the remainder continues in force. It may be signed in counterparts, each of which is an original. Notices shall be in writing to the addresses above.

IN WITNESS WHEREOF the parties have signed this Agreement on the date first written above.

SIGNED for and on behalf of HURU DIGITAL LIMITED

HURU DIGITAL LIMITED · 12 OCT 2026 · DAR ES SALAAM

Signature of director and company seal

IBRAHIM FRANK NDUNGURU

Name of director

Managing Director · 12 October 2026

Title and date

SIGNED by the Author

Signature

IBRAHIM FRANK NDUNGURU

Full name

NIN 19911207-41114-00002-25 · 12 October 2026

NIN and date

WITNESS to the Company's signature

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

WITNESS to the Author's signature

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

Huru SMS – AUTHOR'S CONTRACT · HURU DIGITAL LIMITED · Prepared for registration at the Copyright Society of Tanzania (COSOTA) · Executed at Dar es Salaam on 12 October 2026.

SCHEDULE – Description of the Work and the Contribution

Title of the Work: Huru SMS

Full description: Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0. A multichannel digital service through which any person in Tanzania asks a question in Kiswahili or English by SMS (the keyword HURU followed by the question, sent to shortcode 15054) or through a web chat, and receives a short, practical answer grounded in Tanzanian law, institutions, places and verified knowledge, with referral to nearby services and official contacts. The Work comprises the server application (PHP / Laravel), its database design, the web chat and public website, the administration panel, the written content in both languages, the visual designs, and the accompanying documentation.

Owner and applicant: HURU DIGITAL LIMITED

Version: 1.0 **Year of completion:** 2026 **Country of origin:** United Republic of Tanzania

Documents deposited with COSOTA: Innovation Documentation; System Design and Architecture (with screenshots); Source Code Listing (112 files, 7,991 lines, 188 pages); the source code archive.

Contribution of the Author (summary of Clause 2):

2.1 The original idea and purpose of the Work: a Tanzania-aware information and guidance service reachable from any mobile phone, for every citizen and every everyday topic.

2.2 The name and title of the Work, "Huru SMS", and the keyword-based SMS service design (the word HURU followed by a question, sent to shortcode 15054, with service commands such as MSAADA, LUGHA, JINA, MKOA, ACHA and ANZA).

2.3 The functional concept described in the Innovation Documentation: topic detection, Tanzanian context, location awareness, verified knowledge first, bilingual replies, emergency handling and operator control.

2.4 All written content of the Work in Kiswahili and English, including service messages, help texts, knowledge base entries, persona texts, the terms and conditions, the privacy policy and the website texts.

2.5 The documentation of the Work, including the Innovation Documentation and the explanatory texts of the System Design and Architecture document.

Other contributors to the Work (each under a separate contract with the Company): the author of the idea and written content; the creator of the designs; the developer of the software.

Signed by the parties:



For the Company: IBRAHIM FRANK NDUNGURU, Managing Director · 12 October 2026

The Author: IBRAHIM FRANK NDUNGURU · 12 October 2026

CREATOR'S CONTRACT

Creative Work and Copyright Assignment Agreement

in respect of the work **Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0**

THIS AGREEMENT is made on the **12th day of October 2026** at **Dar es Salaam**

BETWEEN

HURU DIGITAL LIMITED, a company incorporated on 4 May 2021 under the Companies Act, Cap. 212 of the Laws of Tanzania, with registration number **151790429**, of **Mlalakuwa Street, Makongo Ward, Kinondoni District**, P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania, represented by its Managing Director, **IBRAHIM FRANK NDUNGURU** (the "**Company**"), of the one part;

AND

MICHAEL SIMON KILUNGA, an individual of **Tanzanian** nationality, holder of National Identification Number (NIN) **19991208-37305-00002-27**, of P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania (the "**Creator**"), of the other part.

RECITALS

A. The Creator created the user experience, interface designs, visual identity and conversation design of the work known as **Huru SMS**, a digital service that lets any person in Tanzania ask a question on any everyday matter by SMS (keyword HURU to shortcode 15054) or through a web chat and receive a short, practical answer in Kiswahili or English (the "Work").

B. The Company intends to own all rights in the Work, to exploit it commercially and for the public benefit, and to register it in its own name with the Copyright Society of Tanzania (COSOTA) under the Copyright and Neighbouring Rights Act, Cap. 218 of the Laws of Tanzania.

C. The Creator's contribution was made for the Company and, to the extent that it was not already made under a contract of employment or commission with the Company, the parties wish to record in writing the assignment of all economic rights in that contribution to the Company, together with the terms on which it was made.

NOW THEREFORE IT IS AGREED as follows:

1. Definitions

In this Agreement: "**Work**" means Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0, including all versions, source code, object code, databases, designs, texts, images, documentation and other materials forming part of it, as described in the Innovation Documentation, the System Design and Architecture document and the Source Code Listing deposited with COSOTA; "**Contribution**" means every part of the Work created by the Creator as described in Clause 2 and the Schedule; "**Economic Rights**" means all rights of reproduction, translation, adaptation, arrangement, distribution, rental, public display, communication to the public, making available and any other right of exploitation conferred by the Copyright and Neighbouring Rights Act, Cap. 218, or by the law of any other country; "**Moral Rights**" means the right to claim authorship and the right to object to distortion or modification of the Work.

2. The Contribution

The Creator confirms having created the following for the Company:

2.1 The user interface designs and screen layouts of the web chat, the public website, the offline page and the administration panel, including their arrangement, colour scheme (green and blue), typography and icons.

2.2 The visual identity of the Work, including the logo, the application icon, the social sharing image and the installable application manifest.

2.3 The design of the SMS conversation: the structure and wording of commands, confirmations, warnings and help messages, and the rules for short replies on basic phones.

2.4 The persona templates that define the tone, behaviour and safety rules of the service per topic and language, and the structure of the instructions assembled for each reply.

2.5 The diagrams, figures and illustrations in the System Design and Architecture document and the presentation of the Innovation Documentation.

The Contribution is embodied in the view templates, stylesheets, logo and image files reproduced in the source code listing, the persona seed data, and the figures of the System Design and Architecture document, copies of which have been delivered to the Company.

3. Engagement and period

The Contribution was made **during the year 2026, the Work being completed in October 2026**, at the request of and for the Company. The Creator was at the material time **engaged by the Company as the creator of the designs and the developer of the software**.

4. Consideration

In consideration of the assignment and undertakings in this Agreement the Company has paid or shall pay to the Creator:

1. ☐ the sum of Tanzanian Shillings _____ (TZS _____), receipt of which the Creator acknowledges; and/or
2. ☑ the remuneration and benefits received by the Creator from the Company in connection with the Work and the Creator's role in the Company, which the parties agree is full and adequate consideration; and/or
3. ☐ other: _____.

5. Assignment of copyright

5.1 The Creator hereby assigns to the Company, absolutely and with full title guarantee, all Economic Rights and all other intellectual property rights in the Contribution and in the Work, throughout the world, for the whole term of copyright including all renewals, extensions and reversions, and including the right to sue for and recover damages for past infringements.

5.2 To the extent that any part of the Contribution was created under a contract of employment with, or on commission from, the Company, the parties confirm that the Economic Rights in it vested in the Company from creation in accordance with the Copyright and Neighbouring Rights Act, Cap. 218, and this Clause operates as confirmation of that vesting.

5.3 The Company may register, license, assign, adapt, translate, modify, combine with other works, and exploit the Work in any manner and medium, now known or later devised, without further consent of or payment to the Creator.

5.4 Any part of the Contribution that cannot be assigned under applicable law is hereby licensed to the Company exclusively, irrevocably, worldwide, royalty-free and for the full term of protection.

6. Moral rights

The Creator retains the Moral Rights that the law does not permit to be assigned. The Creator agrees (a) to be identified as the Creator of the Contribution in the COSOTA register and in formal documentation, and (b) not to assert any Moral Right so as to prevent or restrict the Company's modification, adaptation, translation, combination or exploitation of the Work in the ordinary course of its business, and waives any such right to the extent permitted by law.

7. Warranties

The Creator warrants that: (a) the Contribution is the Creator's own original work, except for third-party material identified in writing to the Company; (b) the Contribution does not infringe the copyright or other rights of any person; (c) Creator has not previously assigned, licensed or encumbered any right in the Contribution and has full power to enter into this Agreement; (d) to the Creator's knowledge no other person is entitled to claim authorship of or rights in the Contribution; and (e) the information given in this Agreement and in the COSOTA application is true.

8. Confidentiality

The Creator shall keep confidential, and shall not use or disclose except for the Company's purposes, all non-public information about the Work, its source code, configuration, keys, data, users and business plans, during and after the term of this Agreement.

9. Further assurance

The Creator shall, at the Company's request and expense, sign all documents and do all things reasonably required to register, confirm, protect or enforce the Company's rights in the Work before COSOTA, the Business Registrations and Licensing Agency, or any authority or court in Tanzania or elsewhere, and hereby appoints the Company as attorney to sign such documents in the Creator's name if the Creator fails to do so within fourteen (14) days of a written request.

10. Identification as creator

The Creator shall be named as the creator of the designs and creative elements of the Work in the application for registration at the Copyright Society of Tanzania (COSOTA). The Company may use, reproduce, adapt and modify the designs, including changing colours, layouts, names and texts, without further consent.

11. Design materials

The Creator has delivered to the Company all design files, source images and fonts (or licences to them) used in the Work, and warrants that any stock material, font or icon used is either original or licensed for commercial use by the Company.

12. Governing law and disputes

This Agreement is governed by the laws of the United Republic of Tanzania. Any dispute arising out of or in connection with it shall first be referred to the parties' amicable discussion for thirty (30) days; failing settlement, it shall be referred to arbitration in accordance with the Arbitration Act, 2020 of Tanzania, by a single arbitrator sitting at **Dar es Salaam**, or, if the parties so agree, to the courts of Tanzania.

13. General

This Agreement is the entire agreement between the parties on its subject matter and supersedes any earlier understanding. It may be varied only in writing signed by both parties. If any provision is held invalid, the remainder continues in force. It may be signed in counterparts, each of which is an original. Notices shall be in writing to the addresses above.

IN WITNESS WHEREOF the parties have signed this Agreement on the date first written above.

SIGNED for and on behalf of HURU DIGITAL LIMITED

HURU DIGITAL LIMITED · 12 OCT 2026 · DAR ES SALAAM

Signature of director and company seal

IBRAHIM FRANK NDUNGURU

Name of director

Managing Director · 12 October 2026

Title and date

SIGNED by the Creator

Signature

MICHAEL SIMON KILUNGA

Full name

NIN 19991208-37305-00002-27 · 12 October 2026

NIN and date

WITNESS to the Company's signature

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

WITNESS to the Creator's signature

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

Huru SMS – CREATOR'S CONTRACT · HURU DIGITAL LIMITED · Prepared for registration at the Copyright Society of Tanzania (COSOTA) · Executed at Dar es Salaam on 12 October 2026.

SCHEDULE – Description of the Work and the Contribution

Title of the Work: Huru SMS

Full description: Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0. A multichannel digital service through which any person in Tanzania asks a question in Kiswahili or English by SMS (the keyword HURU followed by the question, sent to shortcode 15054) or through a web chat, and receives a short, practical answer grounded in Tanzanian law, institutions, places and verified knowledge, with referral to nearby services and official contacts. The Work comprises the server application (PHP / Laravel), its database design, the web chat and public website, the administration panel, the written content in both languages, the visual designs, and the accompanying documentation.

Owner and applicant: HURU DIGITAL LIMITED

Version: 1.0 **Year of completion:** 2026 **Country of origin:** United Republic of Tanzania

Documents deposited with COSOTA: Innovation Documentation; System Design and Architecture (with screenshots); Source Code Listing (112 files, 7,991 lines, 188 pages); the source code archive.

Contribution of the Creator (summary of Clause 2):

2.1 The user interface designs and screen layouts of the web chat, the public website, the offline page and the administration panel, including their arrangement, colour scheme (green and blue), typography and icons.

2.2 The visual identity of the Work, including the logo, the application icon, the social sharing image and the installable application manifest.

2.3 The design of the SMS conversation: the structure and wording of commands, confirmations, warnings and help messages, and the rules for short replies on basic phones.

2.4 The persona templates that define the tone, behaviour and safety rules of the service per topic and language, and the structure of the instructions assembled for each reply.

2.5 The diagrams, figures and illustrations in the System Design and Architecture document and the presentation of the Innovation Documentation.

Other contributors to the Work (each under a separate contract with the Company): the author of the idea and written content; the creator of the designs; the developer of the software.

Signed by the parties:



For the Company: IBRAHIM FRANK NDUNGURU, Managing Director · 12 October 2026

The Creator: MICHAEL SIMON KILUNGA · 12 October 2026

DEVELOPER'S CONTRACT

Software Development and Copyright Assignment Agreement

in respect of the work **Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0**

THIS AGREEMENT is made on the **12th day of October 2026** at **Dar es Salaam**

BETWEEN

HURU DIGITAL LIMITED, a company incorporated on 4 May 2021 under the Companies Act, Cap. 212 of the Laws of Tanzania, with registration number **151790429**, of **Mlalakuwa Street, Makongo Ward, Kinondoni District**, P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania, represented by its Managing Director, **IBRAHIM FRANK NDUNGURU** (the "**Company**"), of the one part;

AND

MICHAEL SIMON KILUNGA, an individual of **Tanzanian** nationality, holder of National Identification Number (NIN) **19991208-37305-00002-27**, of P.O. Box **75290**, **Dar es Salaam**, United Republic of Tanzania (the "**Developer**"), of the other part.

RECITALS

A. The Developer designed, programmed, tested and delivered the computer program known as **Huru SMS**, a server application that receives questions by SMS (keyword HURU to shortcode 15054) and web chat, assembles Tanzanian context and verified knowledge, obtains a worded reply and sends it back to the citizen, together with its administration panel (the "Work").

B. The Company intends to own all rights in the Work, to exploit it commercially and for the public benefit, and to register it in its own name with the Copyright Society of Tanzania (COSOTA) under the Copyright and Neighbouring Rights Act, Cap. 218 of the Laws of Tanzania.

C. The Developer's contribution was made for the Company and, to the extent that it was not already made under a contract of employment or commission with the Company, the parties wish to record in writing the assignment of all economic rights in that contribution to the Company, together with the terms on which it was made.

NOW THEREFORE IT IS AGREED as follows:

1. Definitions

In this Agreement: "**Work**" means Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0, including all versions, source code, object code, databases, designs, texts, images, documentation and other materials forming part of it, as described in the Innovation Documentation, the System Design and Architecture document and the Source Code Listing deposited with COSOTA; "**Contribution**" means every part of the Work created by the Developer as described in Clause 2 and the Schedule; "**Economic Rights**" means all rights of reproduction, translation, adaptation, arrangement, distribution, rental, public display, communication to the public, making available and any other right of exploitation conferred by the Copyright and Neighbouring Rights Act, Cap. 218, or by the law of any other country; "**Moral Rights**" means the right to claim authorship and the right to object to distortion or modification of the Work.

2. The Contribution

The Developer confirms having created the following for the Company:

2.1 The software architecture of the Work and all original source code written for it: the conversation engine (ConversationService, TopicClassifier, LanguageDetector, TanzaniaContext, KnowledgeRetriever, PromptEngine, AiService, ModerationService), the SMS and language-model integrations, the web chat, the public website and community, the administration panel, the background jobs, the console commands and the automated tests.

2.2 The database design (migrations), the data seeders, the configuration schema and the settings system.

2.3 The build, installation and deployment configuration described in the README, including the service worker and application manifest.

2.4 All bug fixes, improvements, new versions and derivative works of the above made by the Developer for the Company during the term of this Agreement.

The Contribution is embodied in the complete source code reproduced in the Source Code Listing (112 files, 7,991 lines), the database migrations and seeders, the automated tests, the README and the System Design and Architecture document, copies of which have been delivered to the Company.

3. Engagement and period

The Contribution was made **during the year 2026, the Work being completed in October 2026**, at the request of and for the Company. The Developer was at the material time **engaged by the Company as the developer of the software and the creator of the designs**.

4. Consideration

In consideration of the assignment and undertakings in this Agreement the Company has paid or shall pay to the Developer:

1. ☐ the sum of Tanzanian Shillings _____ (TZS _____), receipt of which the Developer acknowledges; and/or
2. ☑ the remuneration and benefits received by the Developer from the Company in connection with the Work and the Developer's role in the Company, which the parties agree is full and adequate consideration; and/or
3. ☐ other: _____.

5. Assignment of copyright

5.1 The Developer hereby assigns to the Company, absolutely and with full title guarantee, all Economic Rights and all other intellectual property rights in the Contribution and in the Work, throughout the world, for the whole term of copyright including all renewals, extensions and reversions, and including the right to sue for and recover damages for past infringements.

5.2 To the extent that any part of the Contribution was created under a contract of employment with, or on commission from, the Company, the parties confirm that the Economic Rights in it vested in the Company from creation in accordance with the Copyright and Neighbouring Rights Act, Cap. 218, and this Clause operates as confirmation of that vesting.

5.3 The Company may register, license, assign, adapt, translate, modify, combine with other works, and exploit the Work in any manner and medium, now known or later devised, without further consent of or payment to the Developer.

5.4 Any part of the Contribution that cannot be assigned under applicable law is hereby licensed to the Company exclusively, irrevocably, worldwide, royalty-free and for the full term of protection.

6. Moral rights

The Developer retains the Moral Rights that the law does not permit to be assigned. The Developer agrees (a) to be identified as the Developer of the Contribution in the COSOTA register and in formal documentation, and (b) not to assert any Moral Right so as to prevent or restrict the Company's modification, adaptation, translation, combination or exploitation of the Work in the ordinary course of its business, and waives any such right to the extent permitted by law.

7. Warranties

The Developer warrants that: (a) the Contribution is the Developer's own original work, except for third-party material identified in writing to the Company; (b) the Contribution does not infringe the copyright or other rights of any person; (c) Developer has not previously assigned, licensed or encumbered any right in the Contribution and has full power to enter into this Agreement; (d) to the Developer's knowledge no other person is entitled to claim authorship of or rights in the Contribution; and (e) the information given in this Agreement and in the COSOTA application is true.

8. Confidentiality

The Developer shall keep confidential, and shall not use or disclose except for the Company's purposes, all non-public information about the Work, its source code, configuration, keys, data, users and business plans, during and after the term of this Agreement.

9. Further assurance

The Developer shall, at the Company's request and expense, sign all documents and do all things reasonably required to register, confirm, protect or enforce the Company's rights in the Work before COSOTA, the Business Registrations and Licensing Agency, or any authority or court in Tanzania or elsewhere, and hereby appoints the Company as attorney to sign such documents in the Developer's name if the Developer fails to do so within fourteen (14) days of a written request.

10. Delivery and hand-over

The Developer has delivered to the Company the complete source code of the Work in a form that can be installed and run following the README, together with all credentials, configuration values, keys and accounts required to operate the Work. The Developer shall not retain any copy of the Work except as authorised in writing by the Company for maintenance.

11. Third-party and open-source components

The Work uses third-party open-source components (in particular the Laravel framework and the libraries listed in `composer.json` and `package.json`) under their own licences. The Developer warrants that (a) these components are used in accordance with their licences, (b) no component is used under a licence that would require the Company to disclose or license the Work's own source code, and (c) the components are clearly separable from the original code of the Work, which is the subject of this assignment. Ownership of such components remains with their respective authors and is not transferred by this Agreement.

12. Development tools

The Developer may use editors, frameworks, code generators, testing tools and other software development aids in producing the Work. The Developer remains responsible for directing, reviewing, integrating and testing all code so produced and confirms that the Work as delivered is the result of the Developer's own skill, judgement and labour.

13. Quality and correction of defects

The Developer warrants that the Work performs materially as described in the Innovation Documentation and the System Design and Architecture document and passes its automated tests. For a period of ninety (90) days from delivery the Developer shall correct, without additional charge, any defect reported by the Company that prevents the Work from performing as so described.

14. Security

The Developer shall not introduce into the Work any malicious code, back door, undisclosed remote access or time-limited function, and warrants that secret keys and passwords are kept only in the environment configuration and are not embedded in the source code.

15. Maintenance and further development

Any maintenance, hosting or further development after delivery shall be agreed separately; all results of such further work shall also belong to the Company under the terms of Clause 5.

16. Governing law and disputes

This Agreement is governed by the laws of the United Republic of Tanzania. Any dispute arising out of or in connection with it shall first be referred to the parties' amicable discussion for thirty (30) days; failing settlement, it shall be referred to arbitration in accordance with the Arbitration Act, 2020 of Tanzania, by a single arbitrator sitting at **Dar es Salaam**, or, if the parties so agree, to the courts of Tanzania.

17. General

This Agreement is the entire agreement between the parties on its subject matter and supersedes any earlier understanding. It may be varied only in writing signed by both parties. If any provision is held invalid, the remainder continues in force. It may be signed in counterparts, each of which is an original. Notices shall be in writing to the addresses above.

IN WITNESS WHEREOF the parties have signed this Agreement on the date first written above.

SIGNED for and on behalf of HURU DIGITAL LIMITED

HURU DIGITAL LIMITED · 12 OCT 2026 · DAR ES SALAAM

Signature of director and company seal

IBRAHIM FRANK NDUNGURU

Name of director

Managing Director · 12 October 2026

Title and date

SIGNED by the Developer

Signature

MICHAEL SIMON KILUNGA

Full name

NIN 19991208-37305-00002-27 · 12 October 2026

NIN and date

WITNESS to the Company's signature

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

WITNESS to the Developer's signature

Allen Albert Kaminda P.O.Box 24330 Dar es Salaam Advocate, Notary Public & Commissioner for Oaths

Signature and stamp

ALLEN ALBERT KAMINDA

Advocate, Notary Public and Commissioner for Oaths, P.O. Box 24330, Dar es Salaam

Full name, address and occupation

Huru SMS – DEVELOPER'S CONTRACT · HURU DIGITAL LIMITED · Prepared for registration at the Copyright Society of Tanzania (COSOTA) · Executed at Dar es Salaam on 12 October 2026.

SCHEDULE – Description of the Work and the Contribution

Title of the Work: Huru SMS

Full description: Huru SMS – Multichannel SMS and Web Citizen Information and Guidance System for Tanzania, version 1.0. A multichannel digital service through which any person in Tanzania asks a question in Kiswahili or English by SMS (the keyword HURU followed by the question, sent to shortcode 15054) or through a web chat, and receives a short, practical answer grounded in Tanzanian law, institutions, places and verified knowledge, with referral to nearby services and official contacts. The Work comprises the server application (PHP / Laravel), its database design, the web chat and public website, the administration panel, the written content in both languages, the visual designs, and the accompanying documentation.

Owner and applicant: HURU DIGITAL LIMITED

Version: 1.0 **Year of completion:** 2026 **Country of origin:** United Republic of Tanzania

Documents deposited with COSOTA: Innovation Documentation; System Design and Architecture (with screenshots); Source Code Listing (112 files, 7,991 lines, 188 pages); the source code archive.

Contribution of the Developer (summary of Clause 2):

2.1 The software architecture of the Work and all original source code written for it: the conversation engine (ConversationService, TopicClassifier, LanguageDetector, TanzaniaContext, KnowledgeRetriever, PromptEngine, AiService, ModerationService), the SMS and language-model integrations, the web chat, the public website and community, the administration panel, the background jobs, the console commands and the automated tests.

2.2 The database design (migrations), the data seeders, the configuration schema and the settings system.

2.3 The build, installation and deployment configuration described in the README, including the service worker and application manifest.

2.4 All bug fixes, improvements, new versions and derivative works of the above made by the Developer for the Company during the term of this Agreement.

Other contributors to the Work (each under a separate contract with the Company): the author of the idea and written content; the creator of the designs; the developer of the software.

Signed by the parties:



For the Company: IBRAHIM FRANK NDUNGURU, Managing Director · 12 October 2026

The Developer: MICHAEL SIMON KILUNGA · 12 October 2026